

Questions and Problems **in** High-School Mathematics

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MIR PUBLISHERS

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**СБОРНИК ВОПРОСОВ И ЗАДАЧ
ПО МАТЕМАТИКЕ
ДЛЯ ПОСТУПАЮЩИХ В ТЕХНИКУМЫ**

**ИЗДАТЕЛЬСТВО
«ВЫСШАЯ ШКОЛА»
МОСКВА**

Questions and Problems in High-School Mathematics

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Preface

In the age of scientific and technical progress mathematics penetrates many fields of economy. This book is intended for all those who wish to enter technical colleges and its aim is to help students to revise the material they studied at school and consolidate their skills at solving the problems and questions which they will encounter during the entrance examinations to the technical colleges.

We hope that the book will also be useful to high-school teachers who will thereby become acquainted with the requirements of the entrance examinations of technical colleges.

All the theoretical questions are followed with brief answers and the problems are given with detailed solutions and hints.

But we advise the reader only to turn to the answers and solutions after a long and fruitless consideration of the problem.

In this book we use the terminology usually accepted in textbooks.

In selecting the questions we used our long experience of teaching at technical colleges and at school and have included sample examination papers which contain almost all the questions and problems usually encountered at the entrance examinations.

We wish to express our gratitude to N. N. Khlebnikova and to our reviewers N. K. Bedenko and V. T. Lisichkin for their inestimable help.

We would be grateful if future readers would send us any advice or comment.

The authors

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Notation

- $M = \{a, b, c\}$ The set M consists of the elements a, b, c .
- $x \in A$ x is an element of the set A , or x belongs to the set A
- $\mathbf{N}$ The set of all natural numbers
- $\mathbf{Z}_0$ The set of all nonnegative integers
- $\mathbf{Z}$ The set of all integers
- $\mathbf{Q}$ The set of all rational numbers
- $\mathbf{R}$ The set of all real numbers (a number line)
- $\emptyset$ An empty set
- $A \subset B$ The set A is contained in the set B , or A is a subset of the set B
- $A \supset B$ The set A contains a set B
- $A \cap B$ An intersection of the sets A and B
- $A \cup B$ A union of the sets A and B
- $n \in \mathbf{N}$ The number n belongs to the set of natural numbers
- $[x]$ The integral part of a number
- $\{x\}$ A fractional part of a number
- $a \geq b$ A number a is greater than or equal to a number b , or the number a is not smaller than a number b
- $a \leq b$ The number a is smaller than or equal to a number b , or the number a is not larger than a number b
- $a \geq 0$ The number a is nonnegative
- $a \leq 0$ The number a is nonpositive
- $[a, b]$ A number interval from a to b , or the set of solutions of the inequality $a \leq x \leq b$
- $a, b)$ A number interval from a to b , the numbers a and b exclusive, or the set of solutions of the inequality $a < x < b$
- $(a, b]$ A number interval from a to b , the number a exclusive, or the set of solutions of the inequality $a < x \leq b$
- $(-\infty, a]$ A number interval from minus infinity to a , or the set of solutions of the inequality $x \leq a$
- $[a, +\infty)$ A number interval from a to plus infinity, or the set of solutions of the inequality $x \geq a$
- $(-\infty, +\infty)$ The set of all numbers (a number line)
- $|x|$ The absolute value of the number x
- f A symbol for a function
- $f(x)$ The value of a function at a point x
- X The domain of definition of a function
- Y The range of a function
- (a_n) A number sequence
- $\log a$ Common or decimal logarithm of a number a
- $[AB]$ A line segment with the endpoints A and B
- (AB) A straight line passing through the points A and B
- $\overrightarrow{AB}$ A ray AB with its origin at a point A
- $X \in [AB]$ A point X lies on the interval AB
- $X \notin (AB)$ A point X does not lie on the straight line AB , or the straight line AB does not pass through the point X

$\angle ABC$	An angle ABC
$\widehat{ABC}$	The value of the angle ABC
$\frown AC$	An arc AC
$\widehat{AC}$	The degree measure of the arc AC
$\text{Cir}(O, r)$	A circle with centre O and radius r
$\mathbf{a}, \overrightarrow{AB}$	A vector
$0, \overrightarrow{AA}$	A zero or null vector
$ \overrightarrow{AB} $	The length of the vector $\overrightarrow{AB}$
$F_1 = F_2$	The figures F_1 and F_2 coincide
$F_1 \cong F_2$	The figures F_1 and F_2 are congruent
$A_1 = S_l(A)$	A point A_1 is symmetric with respect to a point A about the l -axis
Z_0	Central symmetry with centre at O
$T(X) = X_1$	Under parallel displacement a point X passes into a point X_1
R_O^α	Rotation of a plane (a ray, a vector) through an angle α about a point O . If O is the origin, we write R^α
$R_O^\alpha \circ R_O^\beta$	A successive performance of rotation of a plane (a ray, a vector) about the same centre first through an angle α and then through an angle β ($\circ$ is the sign of composition)
$F_1 \overset{k}{\sim} F_2$	A figure F_1 is similar to a figure F_2 , the ratio of similitude is k ; it is sometimes written as $F_1 \sim F_2$
H_0^k	A homothetic transformation with centre at O and the ratio of similitude k ; it is sometimes written as H^k

CHAPTER 1

Elements of Arithmetic and Algebra

1.1. SETS

A *set* is one of the fundamental mathematical concepts. It is a **primary** concept and, like the concepts of a point, a straight line, a number, cannot be reduced to simpler concepts.

In everyday life, instead of saying a "set" we usually say a collection, a gathering, a team, a herd, etc. Sets are denoted by Latin capital letters; the elements of sets are written in braces. The order in which the elements are written is immaterial. Two identical elements are considered to be one element.

We say a "subset" rather than a "part of the set"; more precisely: *a set B is a subset of a set A if every element of the set B is an element of the set A .* This is written as $B \subset A$, or $A \supset B$. Read: B is contained in A or A contains B . An empty set is a subset of any set, i.e. $\emptyset \subset A$. Every set is a subset of itself, or $A \subset A$.

Examples: $\{a, b, c\} \subset \{a, b, c\}$, $\{1, 2, 3\} \subset \{1, 2, 3, 4\}$. The set of all squares is a subset of all rectangles. The set N of all natural numbers is a subset of the set Z of all integers $N \subset Z$. A segment AB is a subset of a straight line AB , or $[AB] \subset (AB)$.

1. What is (a) the intersection of sets A and B , the intersection of two figures; (b) the union of sets A and B , the union of two figures?

2. Assume that E is a set of divisors of the number 12 and F is a set of divisors of the number 18. Use braces to write $E \cap F$; $E \cup F$.

3. Find the union of the set A , which is the set of all integral-valued solutions of the inequality $-1 < x \leq 4$, and the set B , which is the set of all integral solutions of the inequality $-2 < x < 3$.

4. (a) Draw two triangles so that their intersection is a point; a line segment; a triangle; a quadrilateral; a trapezoid.

(b) Draw two triangles so that their union is a parallelogram; a trapezoid; a triangle.

1.2. INTEGERS

Integral numbers, or *integers*, are numbers which belong to a union of sets: the sets of natural numbers, of numbers opposite to natural numbers, and of a set consisting of one number zero. We denote the set of integers by a letter Z , $Z = \{ \dots, -2, -1, 0, 1, 2, \dots \}$.

Operation Involving a Zero

The following equalities hold true for any a :

$$a + 0 = a; \quad 0 + a = a; \quad a - 0 = a; \quad 0 - a = -a;$$

$$a - a = 0; \quad 0 \cdot a = 0; \quad a \cdot 0 = 0; \quad 0 : a = 0.$$

You cannot divide by a zero.

5. What numbers are known as natural numbers?

6. What do we call numbers which are added together, subtracted from one another, multiplied by one another, divided by one another, and how do we call the results of these operations?

7. How can we find (a) an unknown term, (b) an unknown minuend, (c) an unknown subtrahend, (d) an unknown factor, (e) an unknown dividend, (f) an unknown divisor?

8. Using the properties of arithmetic operations, calculate x : (a) $((2x - 18) 15 + 120) \div 45 = -4$.

(b) $-208 \div (112 - (96 - 3x) 4) = 2$.

9. Perform the indicated operations: (a) $123 \cdot 26 + (-12924) \div 36 - (-34248) \div 24$, (b) $(21344 \div 16 - (978 - 1540) - (-32) 74) \div 52$, (c) $(125972 - 409 \times 308) \div (-92)$, (d) $25 \div (2035 \div 5 - 11 \cdot 37) + 21$.

Operations on numbers obey the following laws.

Laws of Addition

Commutativity: *a change in the order of the addends does not affect the value of the sum, i.e. $a + b = b + a$.*

Associativity: *to add a third number to a sum of two numbers, we can add the sum of the second and the third number to the first number, i.e. $(a + b) + c = a + (b + c)$.*

Laws of Multiplication

Commutativity: *a change in the order of the factors does not affect the value of the product, i.e. $a \cdot b = b \cdot a$.*

Associativity: *to multiply a product of two numbers by a third number, we may multiply the first number by the product of the second and the third number, i.e. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*

Distributivity: to multiply a sum by a number, we may multiply each addend by that number and add the resulting products, i.e. $(a + b) \cdot c = ac + bc$.

10. Using the laws of addition and multiplication, calculate:

- (a) $1282 + 247 + 1318 + 353 + 800$, (b) $25 \cdot 52 \cdot 8 \cdot 125$,
(c) $75 \cdot 25 \cdot 16$, (d) $161 \cdot 27 + 139 \cdot 27$.

11. Calculate orally:

- (a) $248 \cdot 5$, (b) $244 \cdot 25$, (c) $72 \cdot 250$, (d) $35 \cdot 9$, (e) $34 \cdot 11$,
(f) $17 \cdot 101$.

Divisors and Multiples

A *divisor* of a number a is a quantity by which a can be divided without a remainder. Thus the numbers 1, 2, 3, 6, 9, 18 are divisors of the number 18.

Remark. The word "divisor" has another meaning, a component of the operation of division.

A *multiple* of a is a number which can be divided by a without a remainder. A set of numbers, which are multiples of, say, 12 is infinite: $A = \{0, 12, 24, \dots\}$.

12. What is a prime number; a composite number?

13. What are coprime numbers?

14. Which of the numbers 678, 1980, 65 425, 719 820 can be divided by 2; by 3; by 5; by 9?

15. What is the least common multiple of two or several natural numbers?

16. What is the least common divisor of several natural numbers equal to? Does the greatest common multiple of several natural numbers exist?

1.3. FRACTIONS

Common Fractions

Numbers of the form $\frac{a}{b}$ (or a/b), where a and b are integers and $b \neq 0$, are known as *common* (or *simple*) *fractions*.

Here are formulas which define operations on fractions:

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}, \quad \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd},$$

$$\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b}, \quad \frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd},$$

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \cdot n = \frac{a}{b} \cdot \frac{n}{1} = \frac{an}{b},$$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}, \quad \frac{a}{b} \div n = \frac{a}{b} \div \frac{n}{1} = \frac{a}{bn}, \quad n \div \frac{a}{b} = \frac{n}{1} \div \frac{a}{b} = \frac{bn}{a}.$$

Remark. When a mixed number is multiplied or divided by a mixed number, the numbers must be first turned into improper fractions. The same must be done when an integer is divided by a mixed number.

17. In what case do the fractions a/b and c/d are considered to be equal?

18. When is the fraction a/b called proper?

19. What is a mixed number?

20. What is the principal property of a fraction? What transformations of fractions are based on this property?

21. What two numbers are said to be reciprocal?

22. Formulate the addition laws for fractions with the same denominators, with different denominators. Give numerical examples.

23. How can we find a product of two or several fractions?

24. How can a fraction be divided by a fraction?

25. Perform the following operations: (a) $5 \cdot \frac{5}{9} +$

$(14 \frac{5}{6} - 11 \frac{1}{2}) \div 11 \frac{1}{4}$, (b) $(25 \frac{1}{3} - 135 \div 13 \frac{1}{2}) \div 5 \frac{1}{3}$,

(c) $\frac{4 \left(\frac{9}{10} - \frac{4}{15} \right) \div 3 \frac{1}{6}}{8 - 1 \frac{22}{23} \cdot \left(\frac{14}{15} - \frac{1}{6} \right)}$, (d) $\left(2 \frac{3}{5} \cdot 5 + 3 \frac{2}{5} \div \frac{1}{5} - \right.$

$15 \frac{4}{15} \cdot 15 - \left(1 - \frac{5}{9} \cdot \frac{3}{5} \right) \div 8$.

Decimal Fractions

A *decimal fraction* is a fraction whose denominator 10, 100, 1000, ... is a natural power of the number 10.

26. Formulate the laws of addition (subtraction) for decimal fractions. Give examples.

27. Formulate the multiplication law for decimal fractions. Give examples.

28. How can a decimal fraction be multiplied by 10, 100, 1000, ..., 10^n ?

29. How can a decimal fraction be divided by a natural number?

30. How can a decimal fraction be divided by 10, 100, 1000, . . . , 10^n ?

31. Formulate the law of division of a number by a decimal fraction.

32. Turn the following common fractions into decimal fractions:

$$\frac{3}{4}, \frac{5}{8}, 1\frac{4}{25}, \frac{6}{125}.$$

33. Indicate, without calculating, which of the following fractions can be changed to terminating decimal fractions and which can be changed to nonterminating fractions:

$$\frac{3}{50}, \frac{9}{18}, \frac{12}{15}, \frac{5}{48}, 1\frac{3}{25}, \frac{15}{45}, \frac{7}{16}.$$

Numerical Expressions

34. Find the values of the following expressions:

$$(a) \frac{5}{32} \div 0,25, \quad (b) 14.56 \div \frac{7}{50}, \quad (c) \frac{1\frac{1}{3}}{2\frac{1}{2}} + \frac{3.9}{4.2},$$

$$(d) \frac{4\frac{4}{25} + 2.59}{\left(\frac{3}{4} - 0.15\right) \div 4}, \quad (e) \frac{\frac{7}{24} \div 0.125 + 3.5}{\frac{2}{3} - 0.25},$$

$$(f) \frac{1.05 + \frac{4}{25}}{\left(7.5 - 3\frac{1}{3}\right) \cdot 0.6}, \quad (g) \frac{\left(4\frac{5}{11} \cdot 1\frac{2}{3} - 6.75\right) \cdot \frac{2}{3}}{\left(\frac{2}{3} + 3\frac{1}{3} \cdot 0.3\right) \div 2\frac{2}{3}},$$

$$(h) \frac{1\frac{1}{12} - 5.3 \div 2.4}{0.415 - 0.702 \div 1.3} + \frac{3.4 \div 1.8 - 1\frac{2}{3} + \frac{7}{9}}{0.45 + 0.918 \div 0.51},$$

$$(i) \frac{0.8 - 1.2 \cdot \frac{2}{3}}{10 - 5.4 \cdot \frac{6}{9}} + \frac{20.48 - 51.2 \cdot \frac{2}{5}}{12 - 6.3 \cdot \frac{7}{9}},$$

$$(j) \frac{(3.2 - 1.7) \div 0.003}{\left(1 - 6.3 \cdot \frac{7}{9}\right) \cdot \left(\frac{5}{16} \div 0.125 - 0.8 \cdot 3.125\right)}.$$

The Integral and Fractional Parts of a Number

An *integral part* of a number a is the greatest integer, which does not exceed a ; it is designated as $[a]$, e.g. $[6.8] = 6$.

A *fractional part* of a number a is the difference between the number a and its integral part; it is designated as $\{a\}$. By definition, $\{a\} = a - [a]$, e.g. $\{6.2\} = 6.2 - [6.0] = 0.2$.

35. Find the integral and fractional parts of the number (a) 25.3, (b) 0.25, (c) -6.3 , (d) -10.5 , (e) $-5/6$.

36. Calculate $y = [x] - \{x\}$ if $x = -0.1, -1.1, -2.1$.

1.4. PER CENT

A *per cent* of a number is a hundredth part of it. Per cents are used in economic, financial and other computations to obtain a numerical characteristic of the events and phenomena being studied and to compare them.

Numbers are as often written as per cents in the form of decimals or common fractions. For example,

$$1\% = 0.01 = \frac{1}{100}, \quad 75\% = \frac{3}{4}.$$

37. A half is what percentage of a whole, and what percentage of a quarter?

38. 1. What fractions of a number are 10%, 75%, 0.1%, 0.5%, $33\frac{1}{3}\%$, $16\frac{2}{3}\%$, $p\%$? 2. How do you express a percentage as a fraction?

39. What is the simplest way to find (a) 1%, (b) 50%, (c) 25%, (d) 12.5%, (e) 75%, (f) $33\frac{1}{3}\%$ of a number?

40. Prove that (a) to increase a number a by 40%, it is sufficient to multiply it by 1.4, (b) to decrease a given number a by 10 per cent, it is sufficient to multiply it by 0.9.

Problems in Per Cent

Problems can be encountered in which it is required to find (1) a percentage of a given number, (2) an unknown number x , if we know what a certain percentage of x is equal to, (3) what percentage of a number is another number. It is convenient to solve problems of the second and the third type by setting up an equation or a proportion.

41. In 1983 a factory produced articles worth 90 million roubles, and in 1985 its production increased by 28 per cent. What was its production worth in 1985?

42. Find the number 42% of which is 12.6.

43. A lathe operator turned 159 parts having overfulfilled his daily quota by 6%. How many parts per day must the operator turn?

44. Find the number 10% of which is 40% of 33.

45. Solve the problem orally. What percentage is: (a) 1 cm of 1 m, (b) 1 centner of 1 ton, (c) 10 of 5, (d) 37 roubles of 50 roubles?

46. Forty-two workers out of sixty in a shop are members of the Young Communist League. What percentage are they of all the workers in the shop?

47. An automatic machine-tool manufactured 240 parts per hour. It was reconditioned and then began to manufacture 288 parts of the same kind per hour. What was the per cent increase in the productivity of the tool?

48. What are (a) 10% of 50%, (b) 25% of 25% equal to?

49. In the second year of a Five-Year Plan a factory increased its production by 6% with respect to the first year and in the third year it increased its production by 10% with respect to the second year. What is the productivity increase in the third year in per cent with respect to the first year?

50. After 38 litres of petrol were poured into a tank, it was still 5% empty. How much petrol must be poured into the tank in order to fill it to the brim?

51. Five litres of water were added to ten litres of the 30% solution of sulphuric acid. Find the per cent concentration of the resulting solution.

52. 1. A number is 50% of another number. By how many per cent is the second number greater than the first?

2. The price of vegetables was decreased by 10% and then again by 10%. Would the price of the vegetables become lower if it had been reduced once by 20% rather than twice by 10%?

1.5. RATIOS AND PROPORTIONS

A *ratio* of two numbers is the quotient resulting from the division of the first number by the second. Thus the ratio of 2 to 5 is $2/5$, the ratio of -0.6 to 0.2 is -3 .

An equality between two ratios is known as a *proportion*. We can use letters to write a proportion as follows:

$$a : b = c : d, \text{ or } \frac{a}{b} = \frac{c}{d}, \text{ or } a/b = c/d.$$

Here a and d are extreme terms and b and c are middle terms of the proportion.

Basic Property of a Proportion

The product of the extremes of any proportion is equal to that of its means, i.e. if $a : b = c : d$, then $ad = bc$.

53. 1. Prove the principal property of a proportion.

2. What can you infer from the proportion $\frac{a}{b} = \frac{a}{c}$?

3. Use two proportions $\frac{a}{b} = \frac{c}{d}$ and $\frac{a}{b} = \frac{m}{n}$ to form a new proportion.

54. Form proportions from two equal products: (a) $25 \cdot 3 = 15 \cdot 5$, (b) $2a = 3b$, (c) $ab = mn$, (d) $x^2 = ab$.

55. Find x from the following proportions: (a) $13 :$

$x = 19 : 9.5$, (b) $7.5 : 20.8 = 3 \frac{11}{13} : x$, (c) $\frac{x+1}{8} =$

$\frac{3.75}{7}$, (d) $\left(2.3 + 1 : 3 \frac{1}{3} \right) : \left(1.17 \cdot 1 \frac{1}{9} \right) = x : (3.2 +$

$0.8 (5.5 - 3.25))$.

1.6. RATIONAL NUMBERS

Rational numbers are all integral and fractional numbers, both positive and negative, and the number zero.

Any rational number can be represented in the form p/q , where $q \neq 0$ and the integers p and q are coprime. Thus the numbers $3 =$

$\frac{3}{1}$, $0 = \frac{0}{1}$, $-7 = -\frac{7}{1}$, $0.7 = \frac{7}{10}$, $\frac{1}{3} = 0.333 \dots = 0.\bar{3}$

are rational.

56. 1. What is a coordinate axis?

2. What is the coordinate of a point on a straight line?

3. Where, on a coordinate axis, do negative numbers lie and where do positive numbers lie?

4. What two numbers are said to be opposite?

57. What decimal fraction is known as periodic?

Absolute Value of a Number

An absolute value (or modulus) of a number x is a nonnegative number which satisfies the condition

$$\begin{array}{ll} |x| = x & \text{if } x \geq 0, \\ |x| = -x & \text{if } x < 0, \end{array}$$

In terms of geometry, the absolute value of a number expresses the distance from the representation of the number on the coordinate axis to the origin of the axis. For instance, the number 5 is five units distant from the origin O and, therefore, $|5| = 5$. The number -5 is at the same distance from the point O along the axis and, therefore, $|-5| = 5$.

58. What is the absolute value of a positive number equal to; that of a negative number, of a zero?

59. Solve the following equations: (a) $|x| = 6$, (b) $|y| = 8.5$, (c) $|a| = 0$, (d) $|m| = -2$, (e) $|x - 3| = 2$, (f) $|x + 3| = 2$.

60. Write the set of integral-valued solutions of the following inequalities: (a) $|x| < 2.5$, (b) $|a| \leq 3$.

61. Given $m = -23$, $n = 23.5$, find the values of the following expressions: (a) $|m| - |n|$, (b) $|m - n|$, (c) $|m| + |n|$, (d) $|m + n|$.

1.7. APPROXIMATE VALUES OF QUANTITIES

An *approximate value* of a number a is the value which differs from its exact (true) value x by some number.

If the approximate value of a is greater than x , then we say that we make an approximation *by excess*, and if a is smaller than x , then we say that we make an approximation *by defect*.

62. Formulate the rules of rounding off decimal fractions. Give examples.

63. 1. What is an error of approximation?

2. What is an absolute error of approximation?

3. What is a relative error of approximation?

64. Round off the following numbers to the nearest tenths and find the absolute and relative errors of approximation: (a) 4.36, (b) 0.825.

65. Find the bounds of the number x if (a) $x = 16.8 \pm 0.5$, (b) $x = 0.4 \pm 0.15$.

66. Assume that x is an exact value of a quantity, and a is its approximate value. When is the number a called an approximate value of the number x with an accuracy to within h ?

67. Prove that (a) the number 0.27 is an approximate value of the fraction $3/11$ with an accuracy to within 0.01, (b) the number 0.6 is an approximate value of the fraction $2/3$ to within 0.1.

68. What is the notation for approximate values of numbers?

69. All the digits in the notation of an approximate number are correct. Indicate the bound of the absolute error of that approximation if (a) $x \approx 1.65$, (b) $y \approx 2.864$, (c) $z \approx 8.4$, (d) $m \approx 3.20$, (e) $n \approx 165$.

70. What is the difference between the results of the following three measurements: 21 mm, 21.0 mm, 21.00 mm?

71. Find the approximate values of the perimeter and the area of a rectangle which is x cm long and y cm wide if $x = 6.4 \pm 0.1$ and $y = 2.4 \pm 0.1$.

72. How many decimal places and significant digits are there in the numbers 2.6, 26, 0.26, 0.206, 2.060, 0.0026?

73. Using the rules for approximation by excess, calculate: (a) $x + y + z$, where $x \approx 1.3$, $y \approx 12.341$, $z \approx 0.4764$, (b) ab if $a \approx 12.26$, $b \approx 0.64$, (c) a/b if $a \approx 3.6$, $b \approx 1.386$.

1.8. EXPRESSIONS WITH VARIABLES

An *expression with variable quantities* consists of numbers, letters, and signs which indicate what operations and in what order must be carried out on those numbers and letters. A single letter (number) is considered to be an expression. Examples: $9a^2 - 8$ is an expression in one variable a , $\frac{9a}{b-5}$ is an expression in the variables a and b .

74. 1. What is the domain of definition of an expression in one variable?

2. Find the domains of definition of the following expressions:

$$(a) \frac{x}{5-x}, (b) \frac{2a-4}{2a+4}, (c) \frac{20}{x-y}.$$

A Power with a Natural Exponent

An expression a^n for natural $n > 1$ is a product of n factors each of which equals a , i.e.

$$a^n = aa \dots a \text{ (} n \text{ factors)}.$$

For $n = 1$ this definition is meaningless since there are no products consisting of one factor; in this case the definition must be formulated as $a^1 = a$.

In the notation a^n the number a is the *base of the power*, the number n is an *exponent*, and the number a^n is a *power*.

75. 1. What does the expression a^0 mean if $a \neq 0$?
 2. What does an expression a^{-n} with an integral negative exponent mean?

The principal property of a power. *A product of two powers with the same bases is equal to a power with the same base and an exponent equal to the sum of the exponents of those powers, i.e.*

$$a^m \cdot a^n = a^{m+n},$$

where $a \neq 0$, $m \in \mathbb{Z}$ and $n \in \mathbb{Z}$.

76. 1. Prove the principal property of a power for the exponents $m = 3$ and $n = 2$.

2. Formulate the rules of transformation of powers which can be expressed by the formulas (a) $a^m \cdot a^n = a^{m+n}$, (b) $a^m : a^n = a^{m-n}$, (c) $(a^m)^n = a^{mn}$, (d) $(ab)^n = a^n b^n$, (e) $(a/b)^n = a^n/b^n$.

77. Calculate the values of the following expressions:
 (a) $4a - b^3 c^2$, (b) $(4a - b^3 c)^2$, (c) $(4a - b^3) c^2$,
 (d) $4(a - b)^3 c^2$, (e) $4(a - b^3 c^2)$, (f) $4(a - b^3 c)^2$,
 (g) $4(a - b^3) c^2$ for $a = 3$, $b = -1$ and $c = 2$.

78. Find the numerical values of the following expressions: (a) $\frac{0.5a^2 + (0.5a)^2}{5(a+b)^2 - (2a+b)^2}$ for $a = 4$, $b = -1$,

(b) $\frac{mn^2 + (mn)^2 + m^2 n}{(m/2 - 2n)^2 \div (m + 4n^2)}$ for $m = 2$, $n = -0.25$,

(c) $\frac{3(x-y)^2 + 3(x-y^2)}{xy^2 - (xy)^2}$ for $x = -0.5$, $y = -\frac{2}{3}$,

(d) $|x - 2.5| + |3.5 - x| - |x + 2|$ for $x = -3$, (e) $10a^2 - 5|a| + 1$ for $a = -0.4$.

Standard Form of a Number

Any positive number can be represented in the form of a product $a \cdot 10^n$ (this is a *standard form of the number*), where $1 \leq a < 10$ and $n \in \mathbb{Z}$. The exponent n is the *order of the number*.

79. 1. Represent the following numbers in standard form: (a) 0.00764, (b) 0.02, (c) 17 000, (d) $\frac{1}{3} \cdot 10^{-3}$.

2. Represent the following products in standard form: (a) $(4.8 \cdot 10^{-4}) \cdot (1.5 \cdot 10^8)$, (b) $(2.4 \cdot 10^5) \cdot (12.5 \cdot 10^{-3})$.

The Concept of an Identity

Two expressions are said to be *identically equal* on a given set if they have sense on that set and all their respective values are equal. Equations whose left-hand and right-hand sides are identically equal expressions are called *identities*.

For example, the identity $2(a + 3) = 2a + 6$ is valid on the set of all real numbers, and the identity $\sqrt{x^2} = x$ is valid on the set of all nonnegative numbers.

Valid numerical equalities are also called *identities*.

80. Are the following expressions identically equal on their domains: (a) $\frac{5}{x}$ and $\frac{5x}{x^2}$, (b) $\frac{2(x^2 - y^2)}{x + y}$ and $2(x - y)$?

81. Prove that the following expressions are not identities on the set of all numbers: (a) $x^2 + x = x^3$, (b) $(a + 5)^2 = a^2 + 25$, (c) $|-x|^3 = -x^3$.

1.9. MONOMIALS AND POLYNOMIALS

Monomials are expressions which are products of numbers, variables, powers of variables, such as $0.5x^3$, $6x^2y^2$, $-7c^2$.

Separate numbers, variables, powers of variables are also monomials, for example, -7 , a , x^3 .

Polynomials are sums of several monomials, for example, $5x^4 + 3x^3 - 0.1x^2 + 8x + 1$.

82. 1. What is the power of a monomial; polynomial?

2. What form of a monomial is known as standard; what polynomial is known as a polynomial of standard form?

83. Reduce the following polynomials to standard form: (a) $-(2a^2 - 3b) + (4b + 6a^2 - 1)$, (b) $-\left(\frac{3}{4}x^2 - 2\frac{1}{4}x\right) + \left(\frac{3}{8}x^2 - \frac{1}{2}x + 1\right) - \left(4 - \frac{5}{8}x^2\right)$.

84. Find the values of the following expressions:

(a) $(6.3x^2y - 2.4xy + 2y^3) - (-3.7x^2y + 6.6xy + 2y^3)$ for $x = 1$, $y = 5$, (b) $5a^2 - (3ab - a^2) + (5ab - 6a^2)$ for $a = 0.4$, $b = -0.5$, (c) $9x^3 - (8x^3 - (1 - 2x^3))$ for $x = -0.5$.

85. Carry out the indicated operations: (a) $5^{x+2} \cdot 5^{x-1} \div 5^{2x-1}$, (b) $2 \cdot 3^{x-3} \cdot 4^{x-1} \div 10 \cdot 3^{x-5} \cdot 4^{x-2}$, (c) $12 \cdot 5^6 \div 0.1 \cdot 5^4$.

Multiplying a Monomial by a Polynomial. Multiplying Polynomials

To multiply a monomial by a polynomial, it is sufficient to multiply the monomial by each term entering into the polynomial and add the resulting products together (retaining their signs).

Example: $-0.4xy (-2x^2 - 4xy + 5y^2) = 0.8x^3y + 1.6x^2y^2 - 2xy^3$.

To multiply a polynomial by a polynomial, it is sufficient to multiply each term entering into one polynomial by each term of the other and add the resulting products together.

Example: $(a - b)(c + d) = ac - bc + ad - bd$.

86. Find the values of the following expressions:

(a) $5x(x - 4y) - 10y \cdot (y - 2x)$ for $x = -0.2$, $y = -0.5$, (b) $a(a - b + 2) - b(b - a + 1)$ for $a = -1/3$, $b = -2/3$, (c) $(m - 2n) \cdot (2m + n) - m(2m - 4n)$ for $m = 0.4$, $n = -0.6$.

87. Prove that for any values of a and b the expression $(2a - 3b)(4a + 0.5b) - a(8a - 11b)$ assumes nonpositive values.

1.10. ABRIDGED MULTIPLICATION IDENTITIES

1. *The product of the difference of two expressions by their sum is equal to the difference of their squares:*

$$(a - b)(a + b) = a^2 - b^2.$$

2. *The square of a binomial is equal to the sum of three expressions, viz. the square of the first term, the doubled product of the first and the second term and the square of the second term:*

$$(a \pm b)^2 = (a \pm b)(a \pm b) = a^2 \pm 2ab + b^2.$$

3. *The cube of a binomial is equal to the sum of four expressions, viz. the cube of the first term, the trebled product of the square of the first term by the second term, the trebled product of the first term by the square of the second term and the cube of the second term:*

$$(a \pm b)^3 = (a \pm b)^2 (a \pm b) = a^3 \pm 3a^2b + 3ab^2 \pm b^3.$$

$$4. (a + b)(a^2 - ab + b^2) = a^3 + b^3.$$

$$5. (a - b)(a^2 + ab + b^2) = a^3 - b^3.$$

88. Reduce the following expressions to polynomials of standard form: (a) $(a - 0.1)(a + 0.1)$, (b) $(1.5x^2 - 3y^3)(1.5x^2 + 3y^3)$, (c) $\left(\frac{1}{2}a - 3b\right)\left(3b + \frac{1}{2}a\right)\left(\frac{1}{4}a^2 + 9b^2\right)$, (d) $(a^x + a^{-x})^2$, (e) $(3^m - 3^{m-1})^2$, (f) $(a + b)^2(a - b)^2$, (g) $(2a + 1)^3 + (2a - 1)^3$, (h) $(x - 1)(x^2 + x + 1)$.

89. Using abridged multiplication identities, find the values of the following expressions: (a) $99 \cdot 101$, (b) 20.1×19.9 , (c) $29.5 \cdot 30.5$, (d) 31^2 , (e) 98^2 , (f) 9.9^2 .

90. Find the values of the following expressions: (a) $(2a - 3)^2 - 2a(2a + 3)$ for $a = 2.5$, (b) $(3x - 1)^2 - (2x + 1)(4.5x - 1)$ for $x = 0.2$.

91. Prove that for any $n \in \mathbb{N}$ the value of the expression $(2n + 3)^2 - (2n - 3)^2$ is divisible by 24 (is a multiple of 24).

1.11. POLYNOMIAL FACTORIZATION

To simplify calculations, reduce fractions to their lowest terms or to a common denominator, it is sometimes convenient, when solving equations, to represent a polynomial as a product of two or several polynomials, some of which may be monomials. The following techniques are used to factor polynomials: (1) *putting the common factor before the brackets*, (2) *grouping*, (3) *applying abridged multiplication identities*.

92. Put the common factor before the brackets in the following expressions: (a) $12a^3b - 18a^2b^2 + 6a^2b$, (b) $15x^5 - 10x^5 + 5x^4$, (c) $3a(x - 2y) + 5b(x - 2y) + x - 2y$, (d) $5x(2a - 3) - 3y(2a - 3) - 2a + 3$.

93. 1. Factor out 5 in the trinomial $5a + 3b - 2c$.

2. Factor out a in the trinomial $ax^2 + bx + c$.

3. Factor out $1/2$ in the trinomial $9x - 3y + 2$.

94. By means of grouping factorize the polynomials (a) $ax + ay + bx + by$, (b) $ac + bc + a + b$, (c) $a^3 + a^2b - ab^2 - b^3$.

Identities 1, 2, 4 and 5 on p. 22 can be represented as

$$\begin{aligned} a^2 - b^2 &= (a - b)(a + b), \\ a^2 + 2ab + b^2 &= (a + b)(a + b) = (a + b)^2, \\ a^3 - b^3 &= (a - b)(a^2 + ab + b^2), \\ a^3 + b^3 &= (a + b)(a^2 - ab + b^2). \end{aligned}$$

In this form it is convenient to use the identities for factorization.

95. Factorize the following expressions: (a) $0.04a^2 - 9$, (b) $\frac{1}{9}x^2 - 2\frac{1}{4}$, (c) $(a + 4)^2 - 25$, (d) $x^2(3x - 1) + 9(1 - 3x)$, (e) $4a^2 - (a - 1)^2$, (f) $(a^2 + 2ab + b^2) - (a^2 + 2ac + c^2)$, (g) $x^3 + 27$, (h) $a^3 - 0.125$.

96. Simplify the following expressions and then find their values: (a) $16a^2 + 24a + 9$ for $a = -3/4$, (b) $(x + 6)^2 + 2(x - 1)(x + 6) + (x - 1)^2$ for $x = -0.5$.

97. Find the values of the following expression:

- (a) $113^2 - 112^2$, (b) $0.727^2 - 0.273^2$, (c) $\left(3\frac{2}{3}\right)^2 - \left(3\frac{1}{3}\right)^2$, (d) $26^2 + 2 \cdot 26 \cdot 14 + 14^2$, (e) $81.1^2 - 2 \cdot 81.1 \times 61.1 + 61.1^2$.

1.12. FRACTIONAL EXPRESSIONS

An expression of the form $\frac{a}{b}$, where a and b are expressions containing variables, $b \neq 0$, is known as a *fractional expression (fraction)*.

Fractional expressions are transformed according to the same rules as common fractions (see 1.3).

98. For what values of the variable are the following fractions meaningless: (a) $\frac{x+5}{x(x+3)}$, (b) $\frac{a^2-1}{4a^2-1}$?

Warning! Sometimes students make a mistake in reducing a fraction whose numerator and denominator are polynomials: they cancel out not a common factor but a common summand. For example, we cannot reduce the fraction $\frac{2a+c}{2a+d}$ since there is no common factor in it, if we divide the numerator and denominator by $2a$, we get $\frac{1+c \div 2a}{1+d \div 2a}$. For the same reason we cannot reduce the fractions $\frac{a^2+b^2}{a+b}$ and $\frac{(x+y)^2-(x-y)^2}{(x-y)(x+y)}$.

99. Reduce the following fractions: (a) $\frac{a-2}{2-a}$, (b) $\frac{(a-b)^2}{b^2-a^2}$, (c) $\frac{x+1}{1-x^2}$, (d) $\frac{m^3-m}{m^2-2m+1}$, (e) $\frac{a^2-b^2-c^2-2bc}{a^2+b^2-2ab-c^2}$, (f) $\frac{ab+1-a-b}{b+c-1-bc}$.

100. Simplify the following expressions and find their values: (a) $\frac{x^2-2xy+y^2}{xy-y^2}$ for $x = -0.5$; $y = -1.5$, (b) $\frac{a^2+b^2-c^2+2ab}{a+b+c}$ for $a = 0.25$, $b = 0.5$, $c = -0.5$.

101. Find the values of the following expressions:

$$(a) \quad \frac{1}{x^2-3x} + \frac{1}{9-3x} \text{ for } x = -1\frac{1}{3}, \quad (b) \quad \frac{2}{4a-a^3} + \frac{a}{2a^2-8} \text{ for } a = -0.1, \quad (c) \quad \frac{x-x^3}{x^2-1} - \frac{x^2}{0.2} - \frac{x+x \div 0.1}{x} \text{ for } x = -0.2.$$

Warning! Sometimes, when subtracting fractions with polynomial numerators, the students relate the minus sign, which stands before the fraction, not to the whole numerator but to its first term alone, overlooking the fact that the fraction bar is a substitute for brackets. To avoid this mistake, we recommend the reader to write the numerators in brackets when multiplying by a complementary factor.

$$102. \text{ Simplify the following expressions: } (a) \quad \frac{5a}{a+b} - \frac{2a-3b}{a-b} - \frac{2a^2-6ab+2b^2}{a^2-b^2}, \quad (b) \quad \left(\frac{xy}{x^2-y^2} - \frac{y}{2x-2y} \right) \div \frac{2y}{x^2-y^2} - \frac{x-5y}{4}, \quad (c) \quad \left(\frac{a}{(b-a)^2} - \frac{a}{b^2-a^2} \right) \frac{(a-b)^2}{a} + \frac{2b}{a+b}, \quad (d) \quad \left(\frac{2x}{x+1} + \frac{2}{x-1} + \frac{4x}{1-x^2} \right) \div \left(\frac{x-3}{x+1} + 1 \right), \quad (e) \quad \left(\frac{x}{6x+6y} + \frac{y}{6x} \right) \cdot \left(\frac{1}{y^2} - \frac{1}{x^2} \right) - \frac{1}{6y^2} + \frac{y}{6x^3}, \quad (f) \quad \left(\frac{a-1}{a^2+a+1} - \frac{1}{a-1} \right) \left(a+1 + \frac{1}{a} \right), \quad (g) \quad \frac{20x}{1-5x} - \left(\frac{10x+2}{3-5x} \right)^2 \cdot \left(\frac{5x+9}{(5x+1)^2} - \frac{10x}{25x^2-1} \right), \quad (h) \quad \frac{|x-1|+x}{2x-1} - \frac{|x-1|}{x-1}, \quad (i) \quad \frac{a|a-3|}{a}.$$

1.13. ROOTS

The Arithmetic Square Root

The *arithmetic square root* of a number a is a nonnegative number whose square is a ; it is designated $\sqrt{a}$, where $\sqrt{}$ is the sign of the arithmetic root, or radical, and a is a radicand.

103. 1. For what values of the variable a does the expression $\sqrt{a}$ have sense?

2. Give examples of the values of a for which $\sqrt{a}$ is a rational number; an irrational number.

104. Why $(\sqrt{a})^2 = a$, where $a \geq 0$?

105. Is $\sqrt[n]{a^2}$ always equal to a^2 ?

106. Find the values of the variable for which the following expressions have sense: (a) $\sqrt{-a}$, (b) $\sqrt{-x^2}$, (c) $\sqrt{2a-3}$, (d) $\sqrt{-x-3}$.

107. Reduce the fraction $\frac{a^2-5}{a-\sqrt{5}}$.

The n th Root

The n th root of a number a is a number whose n th power is equal to a . Its designation is $\sqrt[n]{a}$, where n is a natural number greater than unity.

108. In what cases does the expression $\sqrt[n]{a}$ have sense?

109. Why $(\sqrt[n]{a})^n = a$? Is this equality always valid?

110. Find the values of x for which the following equalities hold true: (a) $\sqrt{x^2} = -x$, (b) $\sqrt[3]{x^3} = x$, (c) $\sqrt[4]{x^4} = -x$, (d) $\sqrt[6]{x^6} = |x|$.

111. Simplify the following expressions:

(a) $\sqrt{(\sqrt{5}-2)^2}$, (b) $\sqrt{(1-\sqrt{2})^2}$, (c) $\sqrt{(\sqrt{6}-\sqrt{7})^2}$.

112. Find the value of the expression $\sqrt{4a^2-4a+1}+3a$ for $a=0.2$.

The Properties of the n th Arithmetic Root ($n > 1$)

1°. If $a \geq 0$ and $b \geq 0$, then $\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$.

2°. If $a \geq 0$ and $b > 0$, then $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$.

3°. If $a \geq 0$, then $\sqrt[n]{\sqrt[k]{a}} = \sqrt[nk]{a}$.

4°. If $a \geq 0$, then $\sqrt[nk]{a^{mk}} = \sqrt[n]{a^m}$ (the basic property of the root).

113. Give formulas which express the properties of the arithmetic root.

114. Find the values of the following products:

(a) $\sqrt[3]{\sqrt{30}+\sqrt{3}} \sqrt[3]{\sqrt{30}-\sqrt{3}}$, (b) $\sqrt[5]{3\sqrt{2}-\sqrt{17}} \times \sqrt[5]{3\sqrt{2}+\sqrt{17}}$, (c) $\sqrt[4]{(\sqrt{5}-1)^2} \sqrt[4]{6+2\sqrt{5}}$.

115. Find the values of the following expressions:

- (a) $\sqrt{21 \cdot 28 \cdot 12}$, (b) $\sqrt[3]{16 \cdot 36 \cdot 81}$, (c) $\sqrt[3]{313^2 - 312^2} \times \sqrt[3]{13^2 - 12^2}$, (d) $\sqrt[3]{27a} - \sqrt[4]{a} \sqrt[3]{a}$ for $a = 8$, (e) $\sqrt[5]{a} \sqrt[4]{a} - \sqrt[4]{16a}$ for $a = 81$.

116. Put the multiplier before the root sign in the following expressions: (a) $\sqrt{-5y^3}$, where $y < 0$, (b) $\sqrt[3]{27a}$, where $a \geq 0$, (c) $\sqrt[4]{81x^6}$, where $x < 0$.

117. Put the multiplier under the root sign in the following expressions: (a) $x\sqrt{2}$, where $x < 0$, (b) $a\sqrt{-3/a}$.

118. Find the values of the following expressions: (a) $\sqrt[4]{16a^4b}$ for $a = -0.5$ and $b = 1$, (b) $\sqrt[6]{10a^9b^6}$ for $a = 2$ and $b = -2$.

119. Compare the following numbers: (a) $\sqrt{2}$ and $\sqrt[3]{3}$, (b) $\sqrt[4]{4}$ and $\sqrt{2}$, (c) $2\sqrt[3]{2}$ and $\sqrt[3]{7\sqrt{3}}$, (d) $\sqrt{14}$ and $\sqrt{3} + 2$.

120. Reduce the following fractions: (a) $\frac{2 + \sqrt{2}}{\sqrt{6} + \sqrt{3}}$, (b) $\frac{x - \sqrt{xy}}{y - \sqrt{xy}}$.

121. Reducing the denominator to a rational number, simplify the following expressions: (a) $\frac{15}{\sqrt{3}}$,

(b) $\frac{2}{\sqrt[3]{4}}$, (c) $\frac{2}{\sqrt{3} - \sqrt{2}}$, (d) $\frac{14}{\sqrt{10} - \sqrt{3}}$, (e) $\frac{1}{3\sqrt{2} - 4}$.

122. Carry out the following operations: (a) $\frac{1}{\sqrt{2} - 1} - \frac{1}{\sqrt{2} + 1}$, (b) $\frac{1}{9 - 4\sqrt{5}} - \frac{1}{9 + 4\sqrt{5}}$, (c) $(4 - \sqrt{3})^2(19 + 8\sqrt{3}) - (9 + 4\sqrt{2})(1 - 2\sqrt{2})^2$, (d) $\frac{(\sqrt{2} - 3)^2(11 + 6\sqrt{2})}{2(9 + 4\sqrt{2})(\sqrt{2} - 4)^2}$.

1.14. A POWER WITH A RATIONAL EXPONENT

If $a > 0$, m is an integer, and n is a natural number, then

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}.$$

If m and n are natural numbers, then

$$0^{\frac{m}{n}} = 0.$$

Properties of Powers with Rational Exponents

The properties of powers with integral exponents are also valid for powers with *any rational* exponents and *positive* bases.

The following equalities are valid if $a > 0$, $b > 0$, and $\frac{m}{n}$ and $\frac{p}{q}$ are rational numbers.

$$\begin{aligned} 1^\circ. \quad a^{\frac{m}{n}} \cdot a^{\frac{p}{q}} &= a^{\frac{m}{n} + \frac{p}{q}}. \quad 2^\circ. \quad a^{\frac{m}{n}} \div a^{\frac{p}{q}} = a^{\frac{m}{n} - \frac{p}{q}}. \quad 3^\circ. \quad (a^{\frac{m}{n}})^{\frac{p}{q}} = a^{\frac{m}{n} \cdot \frac{p}{q}}. \\ 4^\circ. \quad (ab)^{\frac{m}{n}} &= a^{\frac{m}{n}} \cdot b^{\frac{m}{n}}. \quad 5^\circ. \quad \left(\frac{a}{b}\right)^{\frac{m}{n}} = \frac{a^{\frac{m}{n}}}{b^{\frac{m}{n}}}. \end{aligned}$$

123. Calculate: (a) $1000^{1/3} \cdot 0.0001^{1/4} + (0.027)^{1/3} \cdot 5^0 - \left(\frac{10}{13}\right)^{-1}$, (b) $\left(2 \frac{10}{27}\right)^{-2/3} \div \left(11 \frac{1}{9}\right)^{-0.5} + (6.25)^{0.5} \div (-4)^{-1}$.

124. Represent the following expressions as powers with the base a : (a) $a^{\frac{5}{6}} \cdot a^{\frac{2}{3}} \div a^{\frac{1}{6}}$, (b) $\left(a^{\frac{4}{9}}\right)^{-3} \left(a^{\frac{1}{6}}\right)^{-2}$.

125. Find the values of the following expressions: (a) $(2a^{1/2} - 5)^2 + 20a^{1/2}$ for $a = 1.2$, (b) $(6 - 1.5x^{1/2})^2 + 18x^{1/2}$ for $x = 4$.

126. Reduce the following fractions: (a) $\frac{a - 9a^{1/2}}{a^{3/4} - 3a^{1/2}}$, (b) $\frac{x - 36x^{1/3}}{x^{2/3} - 6x^{1/3}}$.

127. Simplify the following expressions:

$$\begin{aligned}
 & \text{(a) } \left(\frac{b^{\frac{1}{2}}}{a + a^{\frac{1}{2}} b^{\frac{1}{2}}} - \frac{b^{\frac{1}{2}}}{a - a^{\frac{1}{2}} b^{\frac{1}{2}}} \right) \frac{b-a}{2a^{\frac{1}{2}} b^{\frac{1}{2}}}, \quad \text{(b) } \left(\frac{a-b}{a^{\frac{3}{4}} + a^{\frac{1}{2}} b^{\frac{1}{4}}} - \right. \\
 & \left. \frac{a^{\frac{1}{2}} - b^{\frac{1}{2}}}{a^{\frac{1}{4}} + b^{\frac{1}{4}}} \right) \left(a^{\frac{1}{4}} b^{-\frac{1}{2}} - b^{-\frac{1}{4}} \right)^{-1}, \quad \text{(c) } \frac{a-b}{a^{\frac{1}{3}} - b^{\frac{1}{3}}} - \frac{a+b}{a^{\frac{1}{3}} + b^{\frac{1}{3}}}, \\
 & \text{(d) } \frac{\frac{x-y}{x^{\frac{2}{3}} + x^{\frac{1}{3}} y^{\frac{1}{3}} + y^{\frac{2}{3}}} - \frac{x+y}{x^{\frac{2}{3}} - x^{\frac{1}{3}} y^{\frac{1}{3}} + y^{\frac{2}{3}}}.
 \end{aligned}$$

1.15. ELEMENTARY FUNCTIONS

A relationship between two sets X and Y in which each element of the set X is associated with not more than one element of the set Y is a *function*.

Remark. The expression "not more than one element" means one element or no elements.

The set X is known as the *domain of definition of a function* and the set Y is the *range of the function*.

A function with the domain X and the range Y is also called a *mapping of the set X onto the set Y* .

The terms "function" and "mapping" are synonyms, they express the same notion, but in algebra we usually use the term "function" and in geometry, the term "mapping".

If x is an arbitrary element of the domain of a function f , and y is the corresponding element of the range of the function, then we

write $f(x) = y$ and read " f of x equals y ". The notation $x \xrightarrow{f} y$ is also used, which reads as "the correspondence f transforms x into y ".

In algebra we usually deal with numerical functions, i.e. functions whose domain of definition and range of values are number sets.

128. In what ways can a function be defined?

129. 1. Given the graph of a function, how can you find its domain and range?

2. Can we say that every correspondence defined by a graph is a function?

130. What is a root (zero) of a function? How can you find the roots of a function on its graph?

131. What function is said to be increasing (decreasing) on a given number set? Describe the movement of a point along the graphs of such functions.

132. Prove that (a) the function $f(x) = 2x - 3$ increases on the entire number axis, (b) the function $g(x) = x^2$ decreases if $x < 0$.

133. 1. What function is invertible? What two functions f and g are inverse to each other?

2. What is the peculiarity of the position of the graphs of two functions inverse to each other?

134. A function f is specified by the formula $y = 2x - 3$ on the set $A = \{-2, -1, 0, 1, 2\}$. Find the set B of the values of the function f . Use pairs of functions to write the inverse correspondence. Is the function f invertible?

135. What transformations must be done to give the inverse of the function $y = f(x)$?

136. A function is specified by the formula (a) $y = \frac{1}{2}x - 3$, (b) $y = -\frac{1}{3}x + 1$. Find a formula to specify the inverse of the given function.

Direct Proportionality

A function, which can be specified by the formula $y = kx$, where x and y are variables, and $k \neq 0$, is known as a *direct proportionality*. The number k is a *proportionality factor*; we say that the variable y is proportional to the variable x .

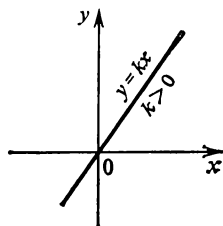


Fig. 1

The graph of the function specified by the formula $y = kx$, where $k \neq 0$, is a straight line passing through the origin (Fig. 1).

A Property of Direct Proportionality

If a function $x \xrightarrow{f} y$ is a direct proportionality and $(x_1, y_1), (x_2, y_2)$ are pairs of respective values of the variables x and y , with $x_2 \neq 0$ and $k \neq 0$, and, hence, $y_2 \neq 0$, then $x_1/x_2 = y_1/y_2$, i.e. the quotients resulting from the division of the respective values of x and y are equal.

If $x > 0$ and $y > 0$, then this property can be formulated as follows: when the value of x increases several times, the respective value of y increases the same number of times.

137. 1. The length of a circle is expressed by the formula $C = \pi D$, where C is the length of the circle, $\pi \approx 3.14$

is a constant number, and D is the diameter. How are C and D related? What is the proportionality factor equal to?

2. Express the length of the diameter D as a function of length of the circle C . In the resulting formula indicate the proportionality factor.

3. Write formulas you know from the course of physics which contain directly proportional quantities.

138. 1. What is the most convenient way of constructing the graph of the function $y = kx$?

2. Plot the graph of the function $y = kx$ for $k > 0$ and $k < 0$.

139. What angles do the graphs of the functions (a) $y = x$, (b) $y = -x$ make with the positive direction of the x -axis?

Inverse Proportionality

A function which can be specified by the formula $y = k/x$, where $k \neq 0$, is known as an *inverse proportionality*. The variable y is said to be in inverse proportion to x .

The graph of the function specified by the formula $y = k/x$, where $k \neq 0$, is a curve consisting of two branches. A curve of this shape is called a *hyperbola* (Fig. 2).

A Property of Inverse Proportionality

If the function $x \rightarrow y$ is an inverse proportionality, and (x_1, y_1) and (x_2, y_2) are pairs of respective values of the variables x and y ,

then $\frac{x_1}{x_2} = \frac{y_2}{y_1}$ ($x_2 \neq 0$ and $y_1 \neq 0$).

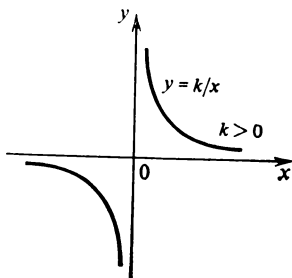


Fig. 2

If $x > 0$ and $y > 0$, then this property can be formulated as follows: when x increases several times the respective value of y decreases the same number of times.

140. 1. What is the domain of the function $y = k/x$?

2. In what quadrants does the graph of the function $y = k/x$ lie when $k > 0$, and in what quadrants does it lie when $k < 0$?

141. The area of a rectangle is expressed by the formula $S = ah$, where a is the base and h is the altitude. Are (a) S and h , (b) S and a , (c) a and h in a direct or in an

inverse proportion if we consider the third quantity to be constant?

142. Find the coordinates of the points of intersection of the graphs of the functions (a) $y = 12/x$ and $y = 3x$, (b) $y = -\frac{6}{x}$ and $y = -\frac{2}{3}x$.

A Linear Function

A function which can be specified by a formula of the form $y = kx + l$, where k and l are some numbers, is said to be *linear*.

The graph of a function specified by a formula of the form $y = kx + l$ is a straight line (Fig. 3).

143. The graph of the function specified by the formula $y = kx + 3$ passes through the point $A (-1, 5)$. Find the factor k . Is this function increasing or decreasing? Find its root.

144. Find the coordinates of the points of intersection

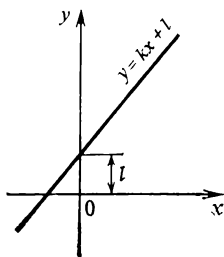


Fig. 3

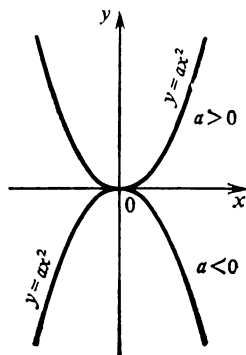


Fig. 4

of the following functions with the coordinate axes and construct their graphs: (a) $y = x - 2$, (b) $y = 0.5x - 3$, (c) $y = -2x - 5$.

145. How do the values of k and l affect the position of the graph of the function $y = kx + l$?

146. Construct the graph of the function specified by the formula $y = kx + l$ if (a) $k > 0$, $l > 0$, (b) $k < 0$, $l > 0$, (c) $k < 0$, $l < 0$, (d) $k = 0$, $l > 0$, (e) $k < 0$, $l = 0$.

The Function $y = ax^2$

The graph of the function specified by the formula $y = ax^2$ ($a \neq 0$) is a curve known as a *parabola*. The domain X of the function $y = ax^2$ is the set of all numbers, i.e. $X = (-\infty, +\infty)$. For $a > 0$ the graph of this function lies in the upper half-plane, and for $a < 0$ it lies in the lower half-plane (Fig. 4).

147. What is the similarity of and the difference between the graphs of the functions specified by the formulas (a) $y = x^2$ and $y = x^2/2$, (b) $y = x^2$ and $y = 2x^2$, (c) $y = 2x^2$ and $y = -2x^2$.

148. At what values of x does the function $y = ax^2$ increase for $a > 0$ and at what x does it increase for $a < 0$?

149. The graph of the function $y = ax^2$ passes through a point (a) $A(3, -9)$, (b) $B(-1, 3)$. Find a and plot the graphs of these functions.

The Power Function $y = x^n$ ($n \in \mathbb{N}$)

If a power function has $n = 1$, then $y = x$ and the graph of this function is a straight line (Fig. 5a); if $n = 2$, then $y = x^2$ and the graph of the function is a parabola (Fig. 5b); if $n = 3$, then $y = x^3$ and the graph of the function is a cubic parabola (Fig. 5c).

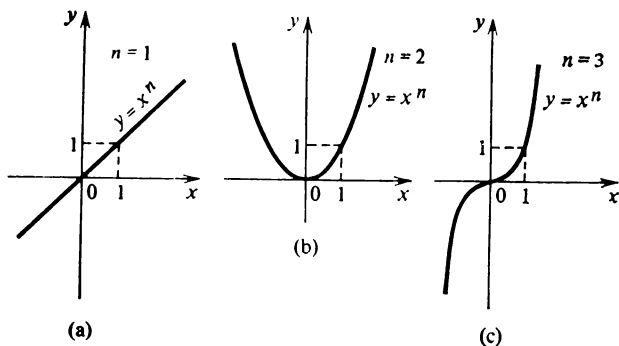


Fig. 5

Properties of the Function $y = x^n$ ($n \in \mathbb{N}$)

1°. The graph of the function passes through the origin and through the point $(1, 1)$. This follows from the fact that $0 = 0^n$ and $1 = 1^n$.

2°. If n is an even number, then the y -axis is the axis of symmetry of the graph (see Fig. 5b).

3°. If n is an odd number, then the origin is the centre of symmetry of the graph (see Fig. 5a and 5c).

The Function $y = \sqrt{x}$ and Its Graph

The ray $[0, +\infty)$ serves as the *domain and the range* of the function $y = \sqrt{x}$, and therefore the graph of the function lies in the first quadrant and the origin belongs to the graph (Fig. 6). The function $y = \sqrt{x}$ is increasing.

The function $y = \sqrt{x}$ is an inverse of the function $y = x^2$. The graphs of these functions are symmetric about the bisector of the first quadrant (Fig. 7).

150. Compare the values of the following expressions:

- (a) $\sqrt[3]{0.6}$ and $\sqrt[3]{2/3}$, (b) $\sqrt[3]{5/7}$ and $\sqrt[3]{0.7}$, (c) $\sqrt[8]{0.88}$ and 1, (d) $\sqrt[10]{6.2^2 - 6.1^2}$ and $\sqrt[10]{1.22}$.

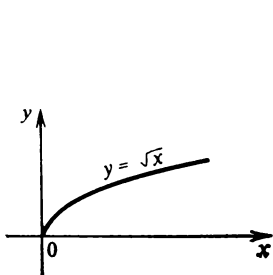


Fig. 6

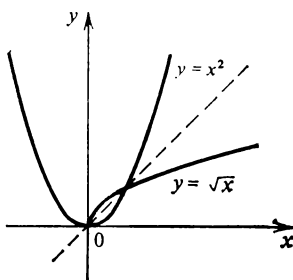


Fig. 7

1.16. EQUATIONS

An *equation* is an equality which contains one or several variables (unknowns).

151. 1. What is the root of an equation?

2. What do we mean by the expression to solve an equation?

3. How many elements can there be in the set of roots of an equation in one variable?

152. What set of roots can the linear equation $ax = b$ possess?

153. Solve the following equations: (a) $(x - 13)^2 - (x + 5)^2 = (x - 12)^2$, (b) $\frac{5x-1}{2} - \frac{x+5}{6} = 1$, (c) $\frac{x-3}{8} - 2 = \frac{3x+27}{20} - \frac{x+9}{12}$, (d) $\frac{x-9}{2} - \frac{x-1}{4} + \frac{1}{8} \left(\frac{x-5}{4} - \frac{14-2x}{5} \right) = \frac{7}{8}$, (e) $\frac{3x+2}{5} - \frac{4x-3}{7} - \frac{x-2}{35} = 4$, (f) $\frac{6(2-3x)}{0.03} -$

$$\frac{0.01-x}{0.01} = 22, \text{ (g) } (2x-1-\sqrt{3})(x+1-\sqrt{3})=0.$$

154. Solve the following equations which possess a variable in the denominator of the fraction: (a) $\frac{6x-37}{2(x-8)} - \frac{2(5x-39)}{3(x-8)} = \frac{7}{8}$, (b) $\frac{x}{x-4} - \frac{5}{(x-4)^2} = 1$, (c) $\frac{30}{x^2-49} + \frac{x-1}{x-7} = 1$, (d) $\frac{4}{x-2} - \frac{5}{x+2} = \frac{12}{x^2-4}$, (e) $\frac{x+3}{x-3} - \frac{12x-15}{x^2-9} - \frac{x-3}{x+3} = 0$.

Problem Solving

The following technique can be used to solve a problem by setting up an equation. 1. We choose one of the unknowns entering into the hypothesis, denoted it by some letter, and then express all the other unknowns entering into the hypothesis in terms of it. 2. Then we set up an equation and solve it. 3. And finally we verify whether the root obtained satisfies the hypothesis and indicate the answer.

155. A passenger train covers the distance between stations A and B 36 minutes faster than a goods train. Find this distance if the average speed of the passenger train is 60 km/h and that of the goods train is 48 km/h.

156. There are two different roads between two towns. The first is 10 km longer than the second. A car travels along the first road and covers the distance between the towns in 3.5 h. Another car travels along the second road and covers the distance in 2.5 h. What is the speed of each car if the speed of the first is 20 km/h lower than that of the second?

157. When some improvements in the technological process had been introduced, it took an hour less to machine one part than before. The same time is now needed to machine 30 parts as was needed earlier to machine 24 parts. How much time does it now take to machine one part?

158. A total of 1400 kg of potatoes were sold in three days. On the first day 100 kg less potatoes were sold than on the second day and on the third day, $\frac{3}{5}$ of the amount sold on the first day. How many kilograms of potatoes were sold each day?

159. One solution contains 30 per cent (by volume) of nitric acid and the other 55 per cent. How much of the first and the second solution must be taken to obtain 100 litres of a 50 per cent solution of nitric acid?

160. We have five litres of a 70 per cent solution of sulphuric acid. How many litres of a 80 per cent solution of sulphuric acid must we add to the original solution to obtain a 72 per cent solution of sulphuric acid?

161. Two excavating machines needed twelve days to dig out a foundation pit for an electric power station. The first excavating machine could have done the job twice as fast as the second. How many days would it take each machine to do the job if they worked separately?

1.17. QUADRATIC EQUATIONS

A *quadratic* equation is an equation of the form $ax^2 + bx + c = 0$, where x is a variable, a , b and c are some numbers, and $a \neq 0$.

A quadratic equation whose first coefficient is unity is said to be in a *reduced form*.

162. 1. Why can the first coefficient in a quadratic equation not be equal to zero?

2. What are the ways to solve incomplete quadratic equations of the forms $ax^2 + bx = 0$, $ax^2 + c = 0$, $ax^2 = 0$?

163. Solve the following equations orally: (a) $x^2 = x$, (b) $0.5x^2 = 4.5$, (c) $0.5x^2 = 0$.

164. Use graphical means to solve the following equations: (a) $x^2 - 2x = 0$, (b) $0.5x^2 = 2$.

165. 1. Find the value of R from the formula $S = \pi R^2$; find the value of t from the formula $s = \frac{gt^2}{2}$.

2. Solve the equation $\frac{1}{2}gt^2 - vt = 0$ for t ($t > 0$).

Formulas for the Roots of a Quadratic Equation

$$x = \frac{-b \pm \sqrt{D}}{2a}, \quad (1) \quad x = \frac{-\frac{b}{2} \pm \sqrt{\frac{D}{4}}}{a}. \quad (2)$$

It is convenient to use formula (2) when b is an even number and, hence, $b/2$ is an integer.

The expression $D = b^2 - 4ac$ is a *discriminant* of a quadratic equation.

If $D > 0$, then $x = \left\{ \frac{-b - \sqrt{D}}{2a}, \frac{-b + \sqrt{D}}{2a} \right\}$, and the quadratic equation has two roots.

If $D = 0$, then $x = \left\{ -\frac{b}{2a} \right\}$, and the quadratic equation has one root (we sometimes say that in this case a quadratic equation has two equal roots).

If $D < 0$, then $x = \emptyset$, and the quadratic equation has no roots.

166. For what values of m do the following equations have one root: (a) $mx^2 + 2x + m = 0$, (b) $25x^2 + mx + 9 = 0$, (c) $x^2 - 6x + m = 0$?

167. Solve the following equations: (a) $x^2 + x - 6 = 0$, (b) $(x-4)^2 + (x+2)^2 + (x-3)(x+3) = 1 - 15x$, (c) $x^2 - 4\sqrt{3}x + 9 = 0$, (d) $2x^2 + (3 - 2\sqrt{2})x - 3\sqrt{2} = 0$, (e) $\frac{5}{x^2+3x} = \frac{6}{x^2-9} + \frac{1}{3-x}$, (f) $\frac{6}{4x^2-1} + \frac{3}{2x+1} - \frac{2}{2x-1} = 1$.

168. Solve the following equations by introducing a new variable: (a) $4x^4 - 13x^2 + 9 = 0$, (b) $\sqrt{x} + x = 12$, (c) $\frac{2}{x^2+2} + \frac{3}{x^2+3} = 2$, (d) $\frac{x^2+1}{x} + \frac{x}{x^2+1} = 2.9$, (e) $2(x-2)^2 - 3(x-2) + 1 = 0$, (f) $x^2 - \frac{50}{2x^2+7} = 7$, (g) $\frac{21}{x^2-4x+10} = x^2 - 4x + 6$.

169. Solve the following equations by graphical means: (a) $x^2 - x - 2 = 0$, (b) $2x^2 + x - 3 = 0$, (c) $x^2 + 4x + 4 = 0$, (d) $x^2 - 2x + 5 = 0$.

The Vieta Theorem

The sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is $-b/a$ and the product of the roots is c/a . If this quadratic equation is in a reduced form ($a = 1$), then $x_1 + x_2 = -b$, $x_1x_2 = c$.

170. Is the converse of the Vieta theorem true?

171. Without solving the following equations, find the sum and the product of their roots (if any): (a) $2.25x^2 + 4x - 1 = 0$, (b) $3x^2 - 20 = 0$, (c) $x^2 - 48x = 0$.

172. One of the roots of the equation $x^2 + bx - 9 = 0$ is unity. Find the other root and the coefficient b .

173. Set up a quadratic equation being given its roots: (a) $x_1 = -0.6$ and $x_2 = 1.6$, (b) $x_1 = -\sqrt{3}$ and $x_2 = \sqrt{3}$, (c) $x_1 = 5 - 2\sqrt{6}$ and $x_2 = 5 + 2\sqrt{6}$, (d) $x_1 = \frac{k + \sqrt{k^2 - ac}}{a}$ and $x_2 = \frac{k - \sqrt{k^2 - ac}}{a}$.

174. Solve the following equations without using the formula for the roots of a quadratic equation: (a) $x^2 - (m + n)x + mn = 0$, (b) $x^2 - (\sqrt{5} + \sqrt{3})x + \sqrt{15} = 0$.

Factoring a Quadratic Trinomial into Linear Factors

An expression $ax^2 + bx + c$, where a , b and c are numbers, $a \neq 0$, and x is a variable is called a *quadratic trinomial*.

A *root of a quadratic trinomial* is the value of the variable for which the value of the trinomial is zero.

If x_1 and x_2 are roots of the quadratic trinomial, then

$$ax^2 + bx + c = a(x - x_1)(x - x_2).$$

This formula serves for *factoring a quadratic trinomial*.

175. 1. How can you find the roots of the quadratic trinomial $ax^2 + bx + c$?

2. How many roots can a quadratic trinomial possess?

176. 1. Write the formula for factoring the reduced quadratic trinomial $x^2 + px + q$.

2. Write the formula for factoring the quadratic trinomial $ax^2 + bx + c$ if its discriminant is zero.

177. Factor the following trinomials into linear factors: (a) $5x^2 - 3x - 2$, (b) $12x^2 - 11x + 2$.

178. Reduce the following fractions: (a) $\frac{a^2 + a - 12}{9 - a^2}$,

(b) $\frac{a^2 + 6a + 9}{6 - a - a^2}$, (c) $\frac{6a^2 - 7a - 3}{2a^2 - a - 3}$, (d) $\frac{5x^{n+2} + 4x^{n+1} - 9x^n}{x^{n+2} - x^n}$.

179. Carry out the indicated operations: (a) $\left(\frac{a-3}{a^2-2a-3} - \frac{a+3}{a^2+4a+3}\right)(a^2-1)$, (b) $\left(\frac{a^2-6a+5}{a^2-7a+10} - \frac{a^2-6a+8}{a^2-5a+4}\right) \times (a^2-3a+2)$. For what values of a is this expression meaningless?

180. Solve the following equations: (a) $\frac{x}{x+2} - \frac{3}{x+3} + \frac{5}{x^2+5x+6} = 0$, (b) $\frac{2}{2y+1} - \frac{1}{1-3y} + \frac{1}{6y^2+y-1} = 0$.

181. Plot the graphs of the following functions:
 (a) $y = \frac{x^2+x-2}{x-1}$, (b) $y = \frac{2x^2+3x+1}{x+1}$.

The Graph of a Quadratic Trinomial

A function of the form $y = ax^2 + bx + c$, where x is a variable, and a, b, c are given numbers, with $a \neq 0$, is a *quadratic function*.

By isolating a perfect square, we can give the quadratic trinomial in the form $y = a \times$

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a}.$$

The graph of the function $y = ax^2 + bx + c$ is a *parabola* which is congruent to the parabola $y = ax^2$, and the straight line $x = -\frac{b}{2a}$ is the axis of symmetry. The coordinates of the vertex O_1 of this parabola are $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$ (Fig. 8).

For $a > 0$ the parabola is convex downwards and for $a < 0$, upwards.

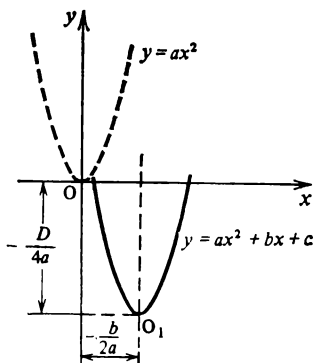


Fig. 8

182. Having isolated a perfect square, draw schematically the graphs of the following functions: (a) $y = x^2 - 8x + 10$, (b) $y = -x^2 - 5x + 4$.

183. Plot the graphs of the following functions: (a) $y = x^2 + 3x - 4$, (b) $y = 5 + 4x - x^2$, (c) $y = 2x^2 + 4x - 6$. Using the graphs, find the sets of values of the arguments on which the functions (1) decrease, increase, (2) are positive, negative.

Problems Involving Solution of Quadratic Equations

The general instructions concerning derivation of equations are given in 1.16.

184. The number of rows in a cinema equals the number of seats in each row. If we double the number of rows

and reduce the number of seats in a row by 10, then the number of seats in the cinema will increase by 300. How many rows are there in the cinema?

185. If we double the length of one leg of an isosceles right triangle and reduce the length of the other by 2 cm, then the area of the triangle will increase by 6 cm². Find the length of the legs of the triangle.

186. A train was delayed at the station for 6 min and made up for this delay at the stage of 36 km by increasing its speed by 4 km/h. Find the initial speed of the train.

187. A motor ship was to cover 72 km with a definite speed. Actually, the ship covered the first half of its trip with a speed 3 km/h lower and the second half with a speed 3 km/h higher than its preassigned speed. The ship covered the whole distance in 5 h. How many minutes was the ship late?

188. A collective farm gathered in the harvest of 8400 centners of wheat. The same harvest could have been gathered in from an area 20 hectares less if it were increased by 1 centner per hectare. How many hectares had been sown and what was the harvest per hectare?

189. Two cranes, working together, unloaded a barge in 6 h. How much time will it take each crane to unload the barge, if they work separately, and if one of them can do the job 5 h faster than the other?

190. There is an external tangency between two circles the sum of whose areas is 130π cm². Find the radii of the circles if the distance between their centres is 14 cm.

191. The population of a settlement trebled in two years. What is the average annual per cent increase of the population of the settlement?

1.18. SIMULTANEOUS EQUATIONS

An equation of the form $ax + by = c$, where a , b , c are some numbers, and x and y are variables is a *linear equation in two variables*.

Two simultaneous equations in two variables form a set of two equations

$$\begin{cases} a_1x + b_1y = c_1, \\ a_2x + b_2y = c_2, \end{cases}$$

where a_1 , b_1 , c_1 , a_2 , b_2 and c_2 are given numbers, and x and y are variables. We assume that in both equations one and the same letter denotes the same number.

192. 1. What is the shape of the graph of a linear equation in two variables?

2. What do we mean by saying that we solve an equation in two variables?

193. 1. What do we mean by saying that we solve simultaneous equations in two variables?

2. What do we mean by the expression "to solve simultaneous equations"?

194. Use graphical means to solve the following simultaneous equations:

$$(a) \begin{cases} 2x + y = 1, \\ 2y - x = 7, \end{cases} \quad (b) \begin{cases} 2x - y = 3, \\ 5x + 2y = 3. \end{cases}$$

195. Show by graphical means that the following simultaneous equations do not have solutions:

$$(a) \begin{cases} 2x + 3y = 6, \\ x + 1.5y = 5, \end{cases} \quad (b) \begin{cases} 2x - y = 1, \\ -x + 0.5y = 2y. \end{cases}$$

196. Show by graphical means that the following simultaneous equations have infinitely many solutions:

$$(a) \begin{cases} 2x - 6y = 3, \\ x - 3y = 1.5, \end{cases} \quad (b) \begin{cases} 3x - 2y = 6, \\ -x + \frac{2}{3}y = -2. \end{cases}$$

197. Choose values of a and c such that the simultaneous equations

$$\begin{cases} 2x - y = 4, \\ ax - y = c \end{cases}$$

have a unique solution, have no solutions, have infinitely many solutions.

198. Solve the following simultaneous equations:

$$(a) \begin{cases} 16x - 27y = 20, \\ 5x + 18y = 41.5, \end{cases} \quad (b) \begin{cases} x \div y = 5 \div 8, \\ 4x + 0.25y = 11, \end{cases}$$
$$(c) \begin{cases} \frac{15x+7y}{16} - \frac{3x-y}{4} = -1, \\ 3x - \frac{10-y}{6} = 2-y, \end{cases} \quad (d) \begin{cases} x - 10y = 10^{10}, \\ x + 10y = 10^{11}. \end{cases}$$

199. For what values of k and l does the straight line specified by the equation $y = kx + l$ pass through the points $A(2, -3)$ and $B(-1, 4.5)$? Write the equation of this straight line.

The Equation of a Circle

The graph of the equation $x^2 + y^2 = r^2$ is a *circle* of radius r whose centre is at the origin.

200. Use graphical means to solve the following simultaneous equations:

$$\begin{array}{ll} \text{(a)} \quad \begin{cases} x^2 + y^2 = 25, \\ 3x + 4y = 0, \end{cases} & \text{(b)} \quad \begin{cases} xy = 12, \\ x + y = 8, \end{cases} \\ \text{(c)} \quad \begin{cases} y - x^2 = 0, \\ x + y = 6, \end{cases} & \text{(d)} \quad \begin{cases} x^2 + y = 3, \\ x + y = 1, \end{cases} & \text{(e)} \quad \begin{cases} xy = 6, \\ y = x^2 + 5. \end{cases} \end{array}$$

201. Solve two simultaneous equations in two variables one of which is of the first degree and the other of the second:

$$\begin{array}{ll} \text{(a)} \quad \begin{cases} x + y = 4, \\ x^2 - 2y^2 - xy + 5x + 15 = 0, \end{cases} & \text{(b)} \quad \begin{cases} xy = 5, \\ x - 2y = 3, \end{cases} \\ \text{(c)} \quad \begin{cases} \frac{x+y}{x-y} = \frac{3}{7}, \\ xy + 2y = -14. \end{cases} \end{array}$$

202. Solve the following simultaneous equations:

$$\begin{array}{ll} \text{(a)} \quad \begin{cases} x^2 + x = 0, \\ x^3 + y = 5, \end{cases} & \text{(b)} \quad \begin{cases} y^2 + 2y - 3 = 0, \\ 2x + y = 3, \end{cases} \\ \text{(c)} \quad \begin{cases} \frac{3x-1}{x+1} + \frac{1}{y-1} = \frac{y+2}{(y-1)(x+1)}, \\ 4y - x = 7. \end{cases} \end{array}$$

Problems Which Involve Solving Simultaneous Equations

When solving problems which involve derivation of simultaneous equations, basically the same techniques are used as in deriving an equation in one variable (see 1.16).

203. If we reduce the numerator of a fraction by unity, then the fraction becomes $1/5$, and if we reduce its denominator by unity, then the fraction becomes $1/4$. Find this fraction.

204. The sum of the digits of a two-digit number is 6. The ratio of this number to the number which will result if we write the digits in the reverse order is $4/7$. Find these numbers.

205. To construct a sewerage system 160 m long 150 ceramic pipes 800 and 1200 mm long were used. How many pipes of each length were used?

206. Two mills were to manufacture 360 machine-tools in a month. However, the first mill fulfilled its plan by 110 per cent and the other mill by 112 per cent. The two mills together manufactured 400 tools in a month. How many more tools than required by the plan did each mill manufacture?

207. The difference of two numbers is 2 and the difference of their squares is 16. Find the numbers.

208. The perimeter of a rectangle is 20 cm and its area is 24 cm^2 . Find the lengths of the sides of the rectangle.

209. The perimeter of a right triangle is 30 cm and its hypotenuse is 13 cm long. Find the length of its legs.

210. The sum of the lengths of the legs of a right triangle is 14 cm and the area of the triangle is 24 cm^2 . Find the lengths of the legs.

211. A rectangular plot of land with an area of 600 m^2 is fenced, the length of the fence being 100 m. Find the length and the width of the plot.

212. If we divide a two-digit number by the sum of its digits, we obtain six as a quotient and two as a remainder, if we divide this number by the product of its digits, we get five as a quotient and two as a remainder. Find the number.

1.19. INEQUALITIES

An *inequality* consists of two expressions connected by a sign $>$ or $<$, $\geq$ or $\leq$.

If $a > b$, then the difference $a - b$ is a positive number;

if $a < b$, then the difference $a - b$ is a negative number. And conversely

if the difference $a - b$ is positive, then $a > b$,

if the difference $a - b$ is negative, then $a < b$.

Inequalities of the form $a > b$, $c < d$ are *strict inequalities*, inequalities of the form $a \geq b$, $c \leq d$ are *weak or slack inequalities*.

213. Prove that the following inequalities hold true for any value of the variable: (a) $a^2 + 2 \geq 2(2a - 1)$, (b) $2(p + 1)^2 + 3p > p(p + 5)$, (c) $2xy \leq x^2 + y^2$.

214. 1. Prove that the inequality $a + 1/a \geq 2$ holds true for $a > 0$. Formulate the resulting relation.

2. Prove that $b + 1/b \leq -2$ for $b < 0$.
 3. Prove that $(a + b)/2 \geq \sqrt{ab}$ for $a \geq 0$ and $b \geq 0$. Formulate the resulting relation.

Principal Properties of Numerical Inequalities

- 1°. If $a > b$, then $b < a$.
 2°. If $a > b$ and $b > c$, then $a > c$ (transitivity).
 3°. If $a > b$ and c is any number, then $a + c > b + c$ and $a - c > b - c$.
Corollary. If $a + b > c$, then $a > c - b$.
 4°. If $a > b$ and $c > 0$ (c is a positive number), then $ac > bc$.
 If $a > b$ and $c < 0$ (c is a negative number), then $ac < bc$.
 5°. If the numbers a and b are both positive or both negative and $a > b$, then $\frac{1}{a} < \frac{1}{b}$.

Operations on Inequalities

1. **Addition:** if $a > b$ and $c > d$, then $a + c > b + d$.
 2. **Multiplication:** if $a > b$, $c > d$ and a, b, c and d are positive numbers, then $ac > bd$.
 3. **Raising to a power:** if $a > b$, where the numbers a and b are positive, then $a^2 > b^2$, $a^3 > b^3$; in general, $a^n > b^n$ for any natural n .

215. Prove that if $a > 3$ and $b > 4$, then (a) $3a + 2.5b > 19$, (b) $2ab > 24$, (c) $a^2 + b^2 > 25$.

216. Find the bounds of the value $5a + 0.2b$ if $1.4 < a < 1.6$ and $5 < b < 6$.

217. Find the bounds of the value $\frac{1}{a} + \frac{1}{b}$ if $3 < a < 4$ and $5 < b < 6$.

218. Given the bounds $8.5 \leq a \leq 8.6$ and $2.5 \leq b \leq 2.6$ of length a and width b (in cm) of a rectangle. Estimate the area S of the rectangle.

Linear Inequalities in One Variable

Inequalities of the form $ax > b$ or $ax < b$ ($ax \geq b$ or $ax \leq b$, respectively), where a and b are some numbers, are linear inequalities.

A solution of an inequality in one variable is the value of the variable for which the inequality holds true.

219. What do we mean by saying that we solve an inequality?

220. Solve the following inequalities: (a) $\frac{7x+5}{6} - \frac{3x-9}{2} > 8$, (b) $\frac{2a-1}{4} - \frac{2a-2}{3} < 1$, (c) $\frac{5x-3}{10} > \frac{3x+1}{6}$.

221. Find the set of the natural values of x which satisfy the following inequalities: (a) $(2x+3)(2x-3) - (2x-1)^2 \leq 2$, (b) $(x-1)(x^2+x+1) - x(x^2-1) \leq 4$.

Simultaneous Linear Inequalities in One Variable

Simultaneous inequalities of the first degree in one variable form a set of inequalities which contain the same variable and for which the values of the variable must be found which turn each of the inequalities into a valid numerical inequality.

222. Solve the following simultaneous inequalities:

(a) $\begin{cases} x+2 > 0, \\ 2x-6 < 0, \end{cases}$ (b) $\begin{cases} 2x+1 > 6, \\ 3x-1 < 3.5, \end{cases}$
 (c) $\begin{cases} 2x-4 > 6, \\ -5x+6 > -8, \end{cases}$ (d) $\begin{cases} x > 5, \\ x > 2.5, \\ x < 7. \end{cases}$

223. Solve the following two-sided inequalities:

(a) $-1 < 2x-3 < 7$, (b) $-10 \leq 3-x \leq 10$.

224. Solve the following inequalities: (a) $(2x-3) \times$

$(5-x) > 0$, (b) $\frac{-3}{2x+6} > 0$, (c) $\frac{2x-3}{3x+6} < 0$, (d) $\frac{4x+2}{2-5x} < 1$, (e) $\frac{3x-1}{2x+5} > 2$.

225. Solve the following inequalities: (a) $x^2 \geq 9$,

(b) $x^2 < 1$, (c) $x^2 > x$, (d) $x^3 > 1$, (e) $x^3 < 8$, (f) $x^4 > 1$,
 (g) $x^2 - x - 6 < 0$, (h) $2x^2 + x - 3 > 0$.

226. Find the domains of definition of the following

functions: (a) $y = \sqrt{5-x}$, (b) $y = \frac{\sqrt{-x}}{x+1}$, (c) $y = \sqrt{\frac{x-3}{2x-7}}$, (d) $y = \sqrt{\frac{4-x}{2x-5}}$, (e) $y = \sqrt{\frac{5-x}{x^2+4}}$.

Inequalities in Two Variables

227. Hatch on the coordinate plane the set of points which is specified by simultaneous inequalities in two variables and then find any two solutions for these inequalities:

- (a) $\begin{cases} y \geq x + 3, \\ y \leq -x + 3, \end{cases}$ (b) $\begin{cases} x^2 + y^2 = 1, \\ y \geq 0, \end{cases}$ (c) $\begin{cases} y \geq x^2, \\ y \leq 1, \end{cases}$
(d) $\begin{cases} y \leq -x^2 \\ y \geq -2. \end{cases}$

1.20. SEQUENCES

An *infinite sequence* is a function specified on the set of natural numbers.

If a function is specified on the set of the first n natural numbers, then it is called a *finite sequence*.

228. Find the first five terms of a sequence specified by the formula (a) $a_n = \frac{2}{n}$, (b) $b_n = 2^{n-1}$, (c) $c_n = 1 + (-1)^{n+1}$, (d) $x_n = \sqrt{n}$, (e) $x_n = 180^\circ (n - 1) + 30^\circ$.

229. A sequence is specified by the formula $x_n = n^2 - 2n$. Find (a) x_{k-1} , (b) x_{k+1} .

230. Can the sequence $y_n = n^2 - 15n + 30$ include (a) the number -20 , (b) the number 16 ? If it can, indicate the ordinal number of the term.

Arithmetic Progression

An *arithmetic progression* is a *number sequence* each term of which, beginning with the second, is equal to the sum of the preceding term and a constant.

The difference between any term of the progression, beginning with the second, and its predecessor is the *common difference* (or simply *difference*) d of the arithmetic progression:

$$d = a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1}.$$

231. 1. What must we know in order to specify an arithmetic progression?

2. What must the common difference of an arithmetic progression be for the progression to be an increasing sequence; a decreasing sequence?

3. How can a term of an arithmetic progression be expressed in terms of its predecessor and successor?

232. 1. Find a_1 and a_3 being given the second term a_2 of the arithmetic progression and its common difference d .

2. What conditions are needed for the three numbers a , b and c to be three consecutive terms of an arithmetic progression?

3. For what values of x do the numbers 1, x and $14 - x$, taken in this order, make an arithmetic progression?

The Formula for the n th Term of an Arithmetic Progression

$$a_n = a_1 + d(n - 1).$$

233. Find the 31st term of an arithmetic progression if $a_1 = -8.5$ and $d = -0.4$.

234. Is the number 146 a term of the arithmetic progression $-6, -2, \dots$? If it is, then indicate its ordinal number.

235. Given two terms $a_3 = 8.2$ and $a_{13} = 10.2$ of an arithmetic progression (a_n) , find the first term of the progression, its common difference and the number of the terms of the progression each of which is smaller than 20.

236. Given two numbers 8 and 0, insert between them three numbers which, together with the given numbers, will form an arithmetic progression.

Formulas for the Sum of the First n Terms of an Arithmetic Progression

$$S_n = \frac{(a_1 + a_n)n}{2} \tag{1}$$

or

$$S_n = \frac{2a_1 + d(n-1)}{2} n. \tag{2}$$

237. The number of terms of the arithmetic progression (a_n) is 16, and $a_1 + a_{16} = 40$. Indicate some more pairs of terms of the progression whose sum is also 40. Find the sum of all the terms of the progression.

238. Find the sum (a) of the first 100 natural numbers; (b) all odd numbers which are smaller than 100.

239. 1. How many terms of the progression 21, 18, 15, $\dots$, must we take for their sum to be zero?

2. How many terms of the progression 0, 4, 8, $\dots$ must we take for their sum to be greater than 112?

Geometric Progression

A *geometric progression* is a number sequence whose first term is nonzero and each term, beginning with the second, is equal to the product of its predecessor by a nonzero constant.

The common ratio of a geometric progression is the ratio of any of its terms to its predecessor. It is denoted by the letter q :

$$q = \frac{b_2}{b_1} = \frac{b_3}{b_2} = \dots = \frac{b_n}{b_{n-1}} = \dots$$

240. 1. What must we know in order to specify a geometric progression?

2. If we are given a geometric progression whose all terms are positive, how can we express any of its terms in terms of its predecessor and successor?

3. What is a geometric progression if $q > 0$, $q = 1$, $q < 0$?

241. 1. Find b_1 and b_3 being given b_2 and q .

2. Find b_2 and b_1 if $b_3 = 3\sqrt[3]{2}$ and $b_4 = 3$.

3. Find the unknown terms b_2 , b_4 and b_5 of the geometric progression 2, b_2 , 8, b_4 , b_5 .

4. Prove that for any $|a| \neq 1$ the value of the expressions $(a+1)^2$, a^2-1 and $(a-1)^2$ are three consecutive terms of a geometric progression.

The Formula for the n th Term of a Geometric Progression

$$b_n = b_1 q^{n-1}.$$

242. 1. Find the fifth term of a geometric progression if $b_1 = -0.001$ and $q = 10$.

2. Prove that if the sequence (c_n) is a geometric progression, then $c_2 c_{11} = c_6 c_8$.

3. Find the unknown terms of the finite geometric progression a_1 , 0.36, a_3 , a_4 , 360.

The Formula for the Sum of the First n Terms of a Geometric Progression

$$S_n = \frac{b_n q - b_1}{q - 1}, \quad q \neq 1, \quad (1)$$

or

$$S_n = \frac{b_1 (q^n - 1)}{q - 1}, \quad q \neq 1. \quad (2)$$

243. Given the seventh term, $b_7 = 1.28$, and the common ratio $q = 2$ of a geometric progression, find the first term of the progression and the sum of the first seven terms.

244. The sum of the first n terms of the geometric progression (a_n) can be calculated using the formula $S_n = 5(2^n - 1)$. Find S_6 , a_1 and q .

1.21. COMMON LOGARITHMS

The Concept of a Logarithm

The *common (decimal) logarithm* of the number a is the exponent indicating the power to which the number 10 must be raised in order to obtain a . The common logarithm is designated as $\log a$.

By definition, the equation $\log a = x$ means that $10^x = a$. Replacing x by $\log a$, we obtain

$$10^{\log a} = a.$$

This equation is known as the *fundamental logarithmic identity*. It is the definition of a common logarithm given in the form of a formula ($\log a$ is the exponent of the power to which the base 10 must be raised to obtain the number a).

245. 1. For what numbers do logarithms exist?

2. For what a is the identity $10^{\log a} = a$ valid?

246. Using the definition of a logarithm, verify the validity of the following equations: (a) $\log 100 = 2$,

(b) $\log 0.1 = -1$, (c) $\log \sqrt{10} = 1/2$, (d) $\log \sqrt[5]{0.1} = -\frac{1}{5}$, (e) $\log \frac{1}{\sqrt[3]{10}} = -\frac{1}{3}$.

247. Using the sign of a logarithm, replace the following equations by logarithmic equations: (a) $10^3 = 1000$, (b) $10^0 = 1$, (c) $10^{-3} = 0.001$, (d) $10^t = a$, (e) $10^{x+1} = b$.

248. Prove that if $10^a = 10^b$, then $a = b$.

249. Solve the following equations: (a) $10^x = 10$, (b) $10^x = 1$, (c) $10^x = 0.01$, (d) $10^x \div 4 = 0.025$, (e) $10^x = \sqrt[3]{10}$, (f) $10^x = \frac{1}{\sqrt[5]{100}}$.

250. Find the values of the following expressions: (a) $\log 10$, (b) $\log 10\,000$, (c) $\log 0.01$, (d) $\log 10 \sqrt{10}$, (e) $\log \sin^2 90^\circ$, (f) $\log (\cos 0^\circ + \cos 90^\circ)$.

251. Calculate (a) $\log \log 10$, (b) $\log \sqrt{10} + \sqrt{\log 10}$, (c) $(\log 0.1)^2 + \log 10 \log 100$, (d) $\frac{1 - (\log 0.01)^2}{1 + (\log 0.01)^2}$.

252. Which of the following expressions are meaningless: $\log 5^{-1}$, $\log 0$, $\log (-5)$, $\log 3^{-2}$, $\log \left(\frac{3}{4} - \frac{4}{5}\right)$, $-\log 0.3$, $\log (\sqrt{3} - 2)$?

253. Calculate the following expressions: (a) $10^{\log 3}$, (b) $10^{\log 2 + \log 5}$, (c) $10^{2 + \log 5}$, (d) $10^{\log 7 - 1}$, (e) $10^{1 - \log 5}$, (f) $10^{-2 \log 3}$, (g) $0.1^{-\log 4}$, (h) $5 \cdot 10^{-\log 5}$, (i) $10^{\log \log 100}$, (j) $5^{\log 100}$, (k) $0.25^{\log \sqrt{10}}$.

254. Solve the following equations: (a) $10^{\log x} = 1$, (b) $10^{\log |x|} = 2$, (c) $100^{\log x} = 4$.

255. Solve the following equations: (a) $\log x = 2$, (b) $\log x = 0$, (c) $\log (-x) = -1$, (d) $\log x = 0.1$, (e) $\log (x - 2) = 1$, (f) $\log (-x - 3) = -2$, (g) $\log (3 - x) = 0$, (h) $\log (x^2 - 6) = 1$, (i) $\log (x^2 - 3x) = 1$, (j) $\log (x^2 + 5x + 6.1) = -1$.

256. Solve the following equations: (a) $(\log x)^2 + \log x = 0$, (b) $(\log x)^2 - 3 \log x + 2 = 0$, (c) $2(\log x)^2 - \log x - 3 = 0$.

Formulas for Taking Logarithms

1. *The logarithm of the product of positive numbers is equal to the sum of logarithms of the factors:*

$$\log(ab) = \log a + \log b, \text{ where } a > 0 \text{ and } b > 0.$$

2. *The logarithm of the quotient resulting from the division of positive numbers is equal to the difference of the logarithms of the dividend and the divisor:*

$$\log \frac{a}{b} = \log a - \log b, \text{ where } a > 0 \text{ and } b > 0.$$

3. *The logarithm of the power with a positive base is equal to the exponent multiplied by the logarithm of the base:*

$$\log a^p = p \cdot \log a, \text{ where } a > 0;$$

in particular, $\log 10^p = p$.

257. Express (a) $\log 18$ in terms of $\log 2$ and $\log 3$, (b) $\log 75$ in terms of $\log 5$ and $\log 3$, (c) $\log \frac{\sqrt[3]{25}}{8}$ in terms of $\log 5$ and $\log 2$.

258. Given $\log 2 \approx 0.30$ and $\log 3 \approx 0.48$, find the approximate values of (a) $\log 60$, (b) $\log \frac{10}{3}$, (c) $\log \frac{1}{6}$, (d) $\log \sqrt[3]{12}$, (e) $\log 0.03$, (f) $\log (200 + 100)$, (g) $\log 200 + \log 100$, (h) $\frac{\log 10 + \log 90}{\log (10 + 90)}$.

259. Calculate (a) $\log (100a) - \log \frac{a}{10}$, where $a > 0$, (b) $\log \frac{x}{100} - \log (1000x)$, where $x > 0$.

260. Calculate (a) $\log \sqrt{0.1}$, (b) $\log \frac{1}{\sqrt[3]{10^2}}$, (c) $2 \log \sqrt[4]{10^3}$, (d) $\log 0.1 \sqrt{1000}$.

261. Find the values of the following expressions:

(a) $\frac{\log 8}{\log 2}$, (b) $\frac{\log 3}{\log 27}$, (c) $\frac{\log 25}{\log \sqrt[3]{5}}$, (d) $\frac{2 \log 49}{\log 7}$, (e) $\frac{\log 5 - 1}{\log 50}$, (f) $\frac{\log^2 49}{\log \sqrt{7}}$.

262. Given $\log 2 \approx 0.30$, find the approximate values of (a) $\log \sqrt[3]{0.8}$, (b) $\log \sqrt[4]{3.2}$.

263. Take the logarithms of the following expressions (letters denote positive numbers): (a) $x = \frac{a^2 b^3}{c \sqrt{a}}$,

(b) $x = 10a^4 \sqrt[3]{\frac{b^2}{c}}$, (c) $x = \frac{(a+3)^{2b^3}}{(a^2+2)^{2c^4}}$ (d) $x = \sqrt{\frac{(a+b)^6 c^3}{\sqrt{a}}}$.

264. Given $\log 2 \approx 0.30$, $\log 3 \approx 0.48$ and $\log 5 \approx 0.70$, find the approximate values of the roots of the following equations: (a) $2^x = 5$, (b) $3^x = 25$, (c) $2^x = 100$, (d) $0.1^x = 20$.

Taking Antilogarithms

To take antilogarithms is to find a number from its logarithm. This operation is an inverse of taking logarithms.

Formulas for taking antilogarithms

For $a > 0$ and $b > 0$ we have

$\log a + \log b = \log(ab)$, $\log a - \log b = \log \frac{a}{b}$, $k \log a = \log a^k$.

These are formulas for taking logarithms written in the reverse order.

265. Calculate the following expressions: (a) $\log 2 + \log 5$, (b) $\log 360 - \log 0.36$, (c) $\log 0.15 - \log 15$,
(d) $\frac{1}{2} \log 25 + \frac{1}{3} \log 64 - \frac{1}{5} \log 32$.

266. Prove that if the logarithms of two positive numbers are equal, then the numbers are also equal.

267. Find x if (a) $\log x = \log 2 + 3 \log 5$, (b) $\log x = 3 \log 2 + 2 \log 3$, (c) $\log x = \frac{1}{4} (\log 8 + \log 32)$,
(d) $\log x = 3 (\log 8 - \log 2)$, (e) $\log x = 2 \log (a - b) - \log (a^2 - b^2)$.

268. Solve the following equations: (a) $\log (2x - 5) = 2 \log 3 + \log 5$, (b) $\log (9 - 0.1x) = \log 6 - \log 2$,
(c) $\log (20 - 5x) = \log x + 1$, (d) $\log x = 2 - \log 4$,
(e) $\log (x + 6) = \log x + \log 2.5$, (f) $(x - 4) + \log (x + 5) = 1$.

269. Construct the graphs of the following functions:
(a) $y = 10^{\log(2x)}$, (b) $y = 10^{2 \log x}$, (c) $y = 10^{\log(x-2)}$,
(d) $y = 10^{\log x} - 2$, (e) $y = 10^{\log x - \log 2}$.

Using a Slide Rule and a Pocket Calculator for Computations

270. Use a slide rule and a calculator to calculate the following expressions. Then, using the calculator, round off the result leaving three significant digits in it.

You will thus verify the answer obtained. (a) $\frac{13.4 \cdot 3.78}{1.24}$,
(b) $\frac{34.4 \cdot 0.336}{4.64}$, (c) $\frac{4.15 \cdot 3.64}{16.7}$, (d) $\frac{1.9 \cdot 0.93}{0.15}$,
(e) $\frac{64.5 \cdot 0.432}{19.5}$.

271. Use a slide rule and a calculator to find (a) 2.5% of 358, (b) 43.5% of 192, (c) 18.6 is what per cent of 72.5?

272. Use a slide rule and a calculator to find the unknown term of the following proportions: (a) $\frac{x}{27.3} = \frac{4.25}{0.85}$, (b) $\frac{168}{x} = \frac{374}{0.735}$.

273. The length of a circle can be calculated using the formula $C = \pi d$. Use a slide rule and a calculator to find the diameter d if the length of the circle C is equal to (a) 4.21, (b) 14.2, (c) 260.

Elements of Geometry

2.22. FUNDAMENTAL CONCEPTS OF GEOMETRY

The fundamental concepts of geometry, which may not be defined, are a *point*, a *straight line*, a *plane*, a *distance* between two points, *mapping* of figures.

274. Define (a) a circumference, (b) a circle as a set of points.

275. What is (a) a line segment, (b) an open ray, a ray?

276. What is (a) a polygonal line, (b) a polygon?

277. What proposition is (a) an axiom, (b) a theorem?

Properties of Distances along a Straight Line

1°. *The distance between points A and B is positive if these points are distinct, and is equal to zero if the points coincide, i.e.*

$$|AB| > 0 \text{ if } A \neq B, \quad |AB| = 0 \text{ if } A = B.$$

2°. *Given any points A and B, the distance from A to B is equal to that from B to A: $|AB| = |BA|$.*

3°. *A point X lies between the points A and B if these three points are distinct and $|AX| + |XB| = |AB|$.*

4°. *If three points A, B and C do not belong to the same straight line, then $|AC| < |AB| + |BC|$.*

278. Given two distinct points A and B on a plane, find, on that plane, the set of points X such that the following conditions are fulfilled: (a) $|AX| \leq 2$ cm, (b) $|BX| = 2$ cm.

279. Do the points A, B and C belong to the same straight line if $|AC| = 4$ cm, $|CB| = 10$ cm and $|AB| = 6$ cm?

Theorem on the Length of a Broken Line

The length of a broken (polygonal) line exceeds the distance between its ends.

If $A_1A_2A_3A_4$ is a broken line, then $|A_1A_2| + |A_2A_3| + |A_3A_4| > |A_1A_4|$ (Fig. 9).

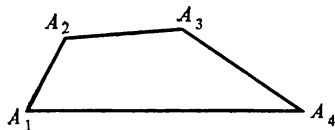


Fig. 9

280. 1. Prove that the length of the broken line ADC is smaller than that of the broken line ABC (Fig. 10).

2. Prove that the length of the broken line AMC is smaller than that of the broken line ABC (Fig. 11).

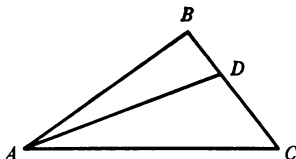


Fig. 10

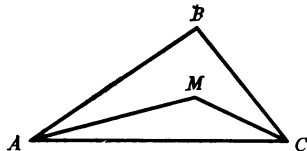


Fig. 11

281. Prove that the following inequalities hold true for the triangle ABC (formulate them),

$$|AC| > |AB| - |BC|, \quad |AB| > |AC| - |BC|, \\ |BC| > |AC| - |AB|.$$

282. 1. Two sides of an isosceles triangle are 3 cm and 6 cm long. What is the length of the third side?

2. What can the distance $|AC|$ be if the segments of the broken line ABC are 3 cm and 5 cm respectively?

Intersection and Union of Figures

An *intersection* of two or several figures is a figure consisting of all those and only those points which belong to each of the figures. The intersection of figures, like that of sets, is designated as $\cap$.

A *union* of two or several figures is a figure consisting of all those and only those points which belong to at least one of the figures. The union of figures, like that of sets, is designated as $\cup$.

The intersection of figures is shown in Fig. 12 by cross hatching and is outlined in dark black.

In Fig. 13 the union of figures is a figure whose boundary is outlined in dark black.

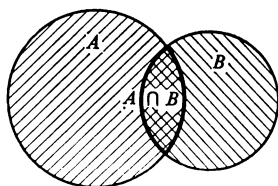


Fig. 12

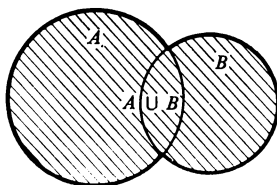


Fig. 13

283. 1. Show that the intersection of two circles may be an empty set, a point, or a circle.

2. Show that the intersection of two triangles may be an empty set, a set consisting of one point, a line segment, or a triangle.

3. Show that an isosceles triangle is the union of two right triangles.

Angles

An *angle* is a figure consisting of two different rays with a common origin and a part of a plane bounded by the rays. The rays are the *sides of the angle* and their common origin is the *vertex* of the angle.

284. 1. What is a straight angle, a right angle?

2. What two angles are known as adjacent angles; what is their sum equal to?

3. What two angles are known as vertical?

4. What is a bisector of an angle?

5. What two straight lines are said to be mutually perpendicular?

285. 1. What figure can be formed by the intersection of two acute angles with a common vertex? Make drawings of various cases.

2. What figure can be formed by the union of two angles of 30° and 60° which have a vertex and a side in common? (Make a drawing for each case.)

286. A straight angle is divided into three congruent parts. Prove that the bisector of the middle angle is perpendicular to the sides of the straight angle,

287. 1. Can both adjacent angles be acute, obtuse, right?

2. One of the adjacent angles is equal to α . Write a formula for the other adjacent angle β .

3. One of the adjacent angles is 90° larger than the other. Calculate each angle.

4. Calculate each of the adjacent angles if one of them is 50% larger than the other.

288. What is the value of the angle between the bisectors (a) of two adjacent angles, (b) of two vertical angles?

2.23. CONGRUENCE OF FIGURES AND THEIR DISPLACEMENT

Mapping of Figures

289. In Fig. 14 each point of the arc CD is associated with points of the arc AB (they lie on the same ray). (a) The mapping of what figure onto what is defined here?

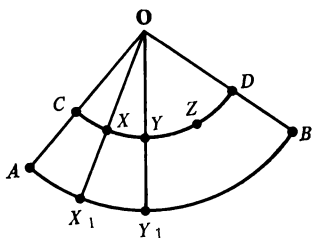


Fig. 14

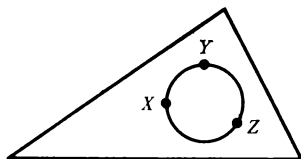


Fig. 15

(b) How can we construct a point corresponding to the point Z of the arc CD ? (c) Is this mapping invertible? What are the images of the points C , X , Y and D ?

290. A circle is inside a triangle (Fig. 15). (a) Define the mapping of the circle onto the contour of the triangle. (b) Construct the images of the points X , Y , Z . (c) Is this mapping invertible?

Congruent Figures

If a figure F can be mapped onto a figure F_1 so that the distance between any two points M and N of the figure F is equal to that between the corresponding points M_1 and N_1 of the figure F_1 , then we say that the figure F is congruent to the figure F_1 .

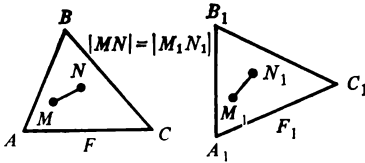


Fig. 16

Figure 16 shows congruent triangles ABC and $A_1B_1C_1$, the notation being $\triangle ABC \cong \triangle A_1B_1C_1$.

291. 1. Construct two congruent triangles ABC and $A_1B_1C_1$ in which $|AB| = |A_1B_1|$, $|BC| = |B_1C_1|$ and $|AC| = |A_1C_1|$. Take an arbitrary point M on the side AB and construct a point M_1 into which the point M will pass when the triangles are superimposed.

2. Solve the preceding problem under the condition that the point M is inside the triangle ABC .

Displacement. Rotation

A *displacement* is a mapping of a plane onto itself (meaning that under the mapping each point of the plane has an image on that plane) which retains the distances. Every figure being displaced is mapped onto a congruent figure.

Rotation of a Plane about a Point

The rotation of a plane about a centre O is a displacement of the plane under which (1) the point O is mapped onto itself; (2) the angle between the ray OX and the corresponding ray OX_1 (Fig. 17) has the same value α . The quantity α is called an *angle of rotation*.

The rotation angle is in the limits $0^\circ \leq \alpha \leq 180^\circ$. If we introduce the direction of rotation (positive, i.e. counterclockwise and negative, i.e. clockwise), then $-180^\circ \leq \alpha \leq 180^\circ$. The rotation about the centre O through the angle α is designated as R_O^α or simply R^α .

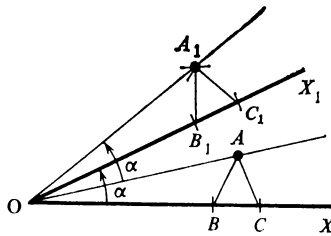


Fig. 17

292. Given a triangle ABC , construct the image of this triangle resulting from its rotation through 45° about the centre A .

293. Construct the image of a rhombus $ABCD$ resulting from its rotation which maps the segment AB onto the segment AD .

294. Indicate the centre and the angle of rotation for which a square passes into itself.

Composition of Rotations

The result of two successive rotations R_O^α and R_O^β about a common centre is known as a *composition of rotations* and is designated as $R_O^\beta \circ R_O^\alpha$ or $R^\beta \circ R^\alpha$. For any rotation angles α or β the equality $R^\beta \circ R^\alpha = R^\alpha \circ R^\beta = R^{\alpha+\beta}$ holds true.

295. Find all the values of the angle α for which the following equalities are satisfied: (a) $R^{60^\circ} = R^\alpha$; (b) $R^\alpha \circ R^{120^\circ} = R^{-\alpha}$.

Central Symmetry

The rotation through 180° about the centre O is known as a *symmetry with centre O* . In that case the centre of rotation is called the *centre of symmetry*.

The central symmetry is designated as Z_O , by definition $Z_O = R^{180^\circ}$.

Points A and A_1 , B and B_1 , C and C_1 shown in Fig. 18 are symmetric with respect to each other about the centre O . The triangle $A_1B_1C_1$ is symmetric with respect to the triangle ABC about the centre O .

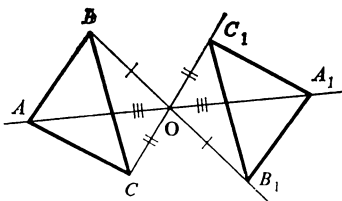


Fig. 18

296. What are the figures onto which the following figures are mapped under central symmetry (a) a straight line about any of its points, (b) a ray about its origin, (c) an angle with its vertex at the centre of symmetry, (d) a straight line about a centre which does not belong to it?

297. Construct a figure which is centrosymmetric with respect to a given circle taking as the centre of symmetry (a) a point lying on the circle, (b) a point lying inside the circle.

Axial Symmetry

Axial symmetry with an axis l is a displacement of a plane under which the points of the straight line l are mapped onto themselves and each point X of one half-plane with the boundary l is mapped onto a point X_1 of the other half-plane so that the segment $XX_1 \perp l$ and is bisected by the line l (Fig. 19).

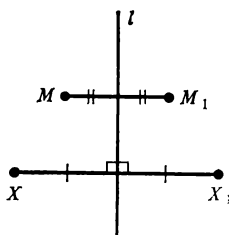


Fig. 19

If the symmetry about the l -axis maps the point X onto X_1 , then we write $S_l(X) = X_1$.

298. How will you construct a point M_1 if $M_1 = S_l(M)$?

299. Given (a) a line segment, (b) a ray, (c) a straight line, (d) a circumference and a circle, (e) an angle, (f) an equilateral triangle, (g) a square, find out how many axes of symmetry they possess.

300. 1. Construct a figure, which is symmetric with respect to a triangle ACB in which $\widehat{ACB} = 90^\circ$, about (AC) . What figure results from the union of the given and the constructed figure?

2. A point M is taken inside a right angle AOB and points $M_1 = S_{BO}(M)$ and $M_2 = S_{AO}(M)$ are constructed. Prove that the points M_1 , O and M_2 lie on the same straight line.

2.24. PERPENDICULAR AND PARALLEL STRAIGHT LINES

A Distance from a Point to a Straight Line

The distance from a point A to the foot of a perpendicular drawn through it to a straight line l is smaller than that from A to any other point on the line l . In brief, the perpendicular AO is smaller than the oblique line AB .

If $(AO) \perp l$ and (AB) is an oblique line to l (Fig. 20), then $|AO| < |AB|$.

301. What is the distance from a point to a straight line?

302. A point A is given inside a right angle. Consider-

ing different locations of the point A , find the shortest distances from A to the sides of the angle.

Theorem on Two Oblique Lines and a Perpendicular

1. If several oblique lines are drawn from the same point to a given straight line, then the oblique lines of the same length are associated with projections of the same length.

2. A longer oblique line is associated with a longer projection.

Thus, if $(OM) \perp a$ and $|OA| = |OB|$, then $|AM| = |MB|$ (Fig. 21a).

If $(OM) \perp a$ and $|OA| > |OB|$, then $|MA| > |MB|$ (Fig. 21b).

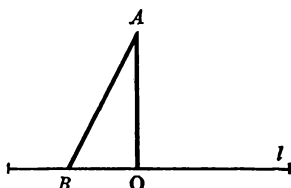


Fig. 20

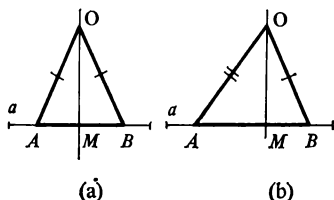


Fig. 21

303. What is (a) the projection of a point onto a straight line, (b) the projection of a line segment onto a straight line?

304. An altitude to the hypotenuse is drawn in a right triangle. Indicate (a) the projections of the legs onto the hypotenuse, (b) the projections of the legs onto the altitude.

Parallel Straight Lines

Two straight lines which lie in the same plane are said to be *parallel* if they have no points in common or coincide.

305. 1. Can we state that two straight lines are parallel if they do not intersect? Is this statement correct?

2. How can we measure the distance between two parallel lines?

306. 1. Formulate the parallel axiom.

2. Prove that if a perpendicular and an oblique line are drawn to the same straight line, then they intersect.

Directions

A *direction on a plane* is a set of rays which have the same *direction* as a certain ray on that plane.

An *angle between two directions* is the value of the angle between any two rays which have those directions and a common origin.

An *angle between the directions a and b* is the angle AOB (Fig. 22). The value of an angle does not depend on the origin of the rays which define the angle.

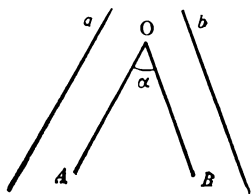


Fig 22

307. 1. What two rays lying on the same straight line are said to have the same direction and what two rays are said to have opposite directions?

2. What two parallel rays are said to have the same direction and what parallel rays are said to have opposite directions?

3. Given an angle and a point which belongs to it, construct an angle with the vertex at that point, whose sides (a) have the same direction as the sides of the given angle, (b) have directions opposite to those of the sides of the given angle.

308. Two straight lines are cut by a third line. Name (a) the corresponding angles, (b) interior alternate angles.

309. Two parallel straight lines are cut by a third line. (a) Write pairs of angles whose sides extend in the same direction. (b) Why are the corresponding angles congruent?

310. The angle ABC is equal to α . A point is taken in the interior of the angle and two straight lines are drawn through it, one parallel to $[BA)$ and the other parallel to $[BC)$. Determine the angles between the straight lines.

The Criteria of Parallelism of Straight Lines

1. If two straight lines are symmetric about a certain centre, then they are parallel.

If $b = Z_B(a)$, then $a \parallel b$ (Fig. 23).

2. If two straight lines are perpendicular to the same straight line, then they are parallel.

If $a \perp p$ and $b \perp p$, then $a \parallel b$ (Fig. 24).

3. If any two of the corresponding angles resulting from the intersection of two straight lines by a third line are congruent, then the two lines are parallel.

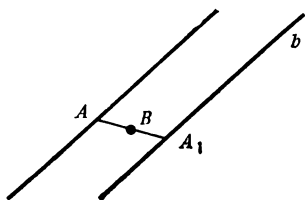


Fig. 23

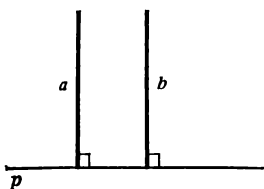


Fig. 24

If $\angle 1 \cong \angle 2$, then $a \parallel b$ (Fig. 25).

4. If any two of the interior alternate angles resulting from the intersection of two straight lines by a third line are congruent, then the lines are parallel.

If $\angle 3 \cong \angle 2$, then $a \parallel b$ (see Fig. 25).

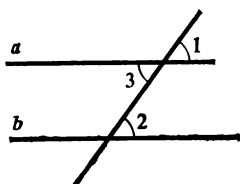


Fig. 25

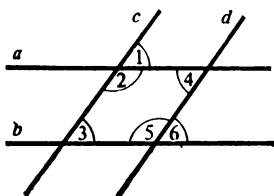


Fig. 26

311. 1. Using the properties of central symmetry, draw a straight line parallel to a given line through a point which does not belong to the given line.

2. All the angles of a quadrilateral are right angles. How are its opposite sides located?

312. Given the angles $\hat{2} = 132^\circ$, $\hat{3} = 49^\circ$, $\hat{4} = 48^\circ$ and $\hat{5} = 131^\circ$ (Fig. 26), name the parallel lines.

2.25. TRIANGLES

313. Given a triangle ABC , indicate (a) the sides opposite to the angles A , B and C , (b) the angles opposite to the sides AB , BC and AC , (c) the angles adjacent to the sides AB , BC and AC .

314. What is a bisector of a triangle, a median of a triangle, an altitude of a triangle?

Remark. We usually say “a bisector”, “a side”, “an angle” rather than “the length of a bisector”, “the length of a side”, “the value of an angle”, etc.

The Properties of an Isosceles Triangle

Theorems. 1. *The straight line which contains the bisector of the vertex angle of an isosceles triangle is the axis of symmetry of the triangle.*

2. *The bisector of the vertex angle of an isosceles triangle is at the same time its median and altitude.*

3. *The base angles of an isosceles triangle are congruent.*

If in the triangle ABC (Fig. 27) $|AB| = |BC|$ and the straight line l contains the bisector BM , then l is the axis of symmetry of the triangle.

$[BM] \perp [AC]$, $|AM| = |MC|$; $\angle A \cong \angle C$

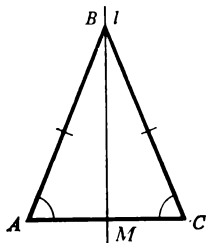


Fig. 27

315. 1. Which is the point of intersection of the altitudes of a right triangle?

2. Draw three altitudes in an obtuse triangle.

316. The perimeter of an isosceles triangle is $2p$ and the base is a . A lateral side of the triangle is at the same time a side of an equilateral triangle. Determine the perimeter of the second triangle.

The Criteria of the Congruence of Triangles

I. *If three sides of one triangle are congruent to three sides of another triangle, then the triangles are congruent.*

We shall abbreviate this criterion to “criterion SSS” (side, side, side).

II. *If two sides and the angle between them of one triangle are congruent to two sides and the angle between them of another triangle, then the triangles are congruent.*

We shall abbreviate this criterion to “SAS criterion” (side, angle, side).

III. *If a side and two angles adjacent to it of one triangle are congruent to a side and two angles adjacent to it of another triangle, then the triangles are congruent.*

We shall abbreviate this to “ASA” criterion (angle, side, angle).

Remark. Note that in this criterion we speak of the angles adjacent to a given side.

We can construct two triangles ABC and $A_1B_1C_1$ (Fig. 28) in which $[AC] \cong [A_1C_1]$ and two angles of one triangle are congruent to two angles of the other, but the triangles are not congruent.

Remember that every triangle has six principal elements: three sides and three angles. When solving problems, bear in mind that in congruent triangles congruent angles lie opposite congruent sides and conversely congruent sides lie opposite congruent angles.

317. Given triangles ABC and KLM (Fig. 29) in which $[AB] \cong [KL]$, $[BC] \cong [KM]$ and $\angle B \cong \angle K$. Which of the other elements of the triangles are congruent to one another?

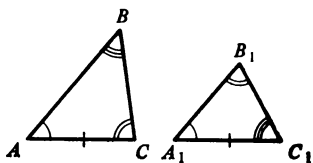


Fig. 28

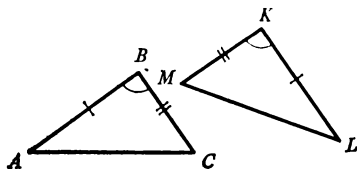


Fig. 29

318. How many principal elements must we have, and which in particular, to construct (a) a scalene triangle, (b) an isosceles triangle, (c) an equilateral triangle?

319. Prove that if we connect the endpoints of congruent chords with the centre, we obtain congruent triangles.

320. A point D is taken in the interior of an isosceles triangle ABC at the same distance from the endpoints of the base AC . The point D is connected with the vertices of the triangle. Prove that $\triangle ABD \cong \triangle BDC$.

321. Prove that line segments connecting the midpoint of the base of an isosceles triangle with the midpoints of the lateral sides are congruent.

322. In an isosceles triangle ABC points D and E divide the base AC into three congruent segments and the vertex B is connected to those points. Prove that $\angle ADB \cong \angle BEC$.

323. Prove that in an isosceles triangle (a) the medians drawn to the lateral sides are congruent, (b) the bisectors drawn to the lateral sides are congruent.

324. The bisector of the base angle is drawn in an isosceles nonequilateral triangle. Is it perpendicular to a lateral side?

325. In a scalene triangle ABC a perpendicular which cuts the sides BA and BC at points M and N is drawn to the bisector BD . Prove that the triangle MBN is isosceles.

The Criteria of Congruence of Right Triangles

Given two right-angled triangles, the right angle of one of which is congruent to the right angle of the other, and therefore the *SAS* and *ASA* criteria of congruence of the triangles can be formulated as follows:

1. *If the legs of one triangle are congruent to those of the other, then the triangles are congruent.*

2. *If a leg and an acute angle of one triangle are congruent to a*

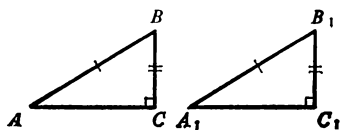


Fig. 30

leg and an acute angle of the other triangle, then the triangles are congruent.

The following criterion of congruence of right triangles requires proof

3. *If the hypotenuse and a leg of one triangle are congruent to the hypotenuse and a leg of the other, then the triangles are congruent.*

If in triangles ABC and $A_1B_1C_1$ we have $\hat{C} = \hat{C}_1 = 90^\circ$, $[AB] \cong [A_1B_1]$, $[BC] \cong [B_1C_1]$, then $\triangle ABC \cong \triangle A_1B_1C_1$ (Fig. 30).

326. In a triangle ABC the altitude BD bisects the side AC . Prove that the triangle ABC is isosceles.

327. Prove that in a triangle ABC the median AD ($D \in [BC]$) is at the same distance from the vertices B and C .

328. How many principal elements, and which in particular, must we have in order to construct a right triangle?

The Property of the Perpendicular Bisector of a Line Segment

Theorem. *If a point lies on the perpendicular bisector of a line segment, then it is equidistant from its endpoints.*

If (XO) is the perpendicular bisector to a segment AB , then $|XA| = |XB|$ (Fig. 31). And conversely, if $|XA| = |XB|$, then X lies on the perpendicular bisector of the segment AB .

329. How can you find the centre of a circle circumscribed about a triangle?

330. How can you construct the centre of a given circle if it is not marked in the drawing?

331. Two points A and B lie outside a straight line l at unequal distances from it. On the line l construct a point equidistant from A and B if A and B lie on different sides of l .

The Property of the Bisector of an Angle

Theorem. *If a point lies on the bisector of an angle, then it is equidistant from the sides of the angle.*

If $[OC)$ is the bisector of the angle AOB and $X \in [OC)$, then X is equidistant from $[OA)$ and $[OB)$ (Fig. 32).

And conversely, if a point X is equidistant from $[OA)$ and $[OB)$, which are the sides of the angle AOB , then $[OX)$ is the bisector of the angle AOB .

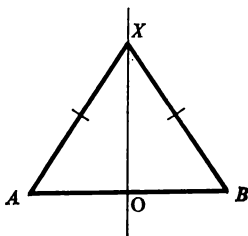


Fig. 31

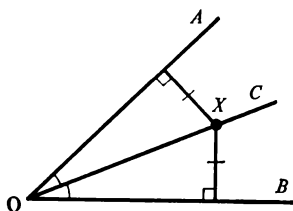


Fig. 32

332. How can you find the centre of a circle inscribed into a triangle?

333. Is every point of the diagonal AC of the rectangle $ABCD$ (different from a square) equidistant from its sides AB and AD ?

334. Points M and N lie outside the angle AOB . Construct a point which would be equidistant from the rays OA and OB and lie at the same distance from M and N .

The Sum of the Angles of a Triangle

Theorem. *The sum of the angles of a triangle is 180° .*

Corollary. *An exterior angle of a triangle is equal to the sum of two angles of the triangle which are not adjacent to that exterior angle.*

In the triangle ABC (Fig. 33) $\widehat{BCD} = \hat{A} + \hat{B}$.

335. 1. What is the sum of the acute angles of a right triangle?

2. What is each acute angle of a right isosceles triangle equal to?

3. What is each angle of an equilateral triangle equal to?

336. 1. In a triangle ABC $\hat{A} = \alpha$ and $\hat{B} = \beta$. Write a formula for the third angle γ .

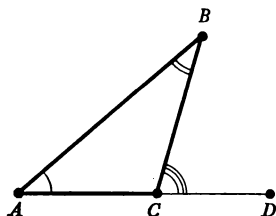


Fig. 33

2. An acute angle of a right triangle is α . Write a formula for the other acute angle β .

3. The vertex angle of an isosceles triangle is β . Write a formula for the base angle α .

337. In a triangle ABC , where $|AB| = |BC|$ and $\hat{B} = 36^\circ$, a bisector AE of the base angle A is drawn ($E \in [BC]$). Prove that $|BE| = |AE| = |AC|$.

338. What is the value of the obtuse angle between the bisectors of the angles of any right triangle?

339. 1. Why can a triangle not have two acute exterior angles?

2. Can an exterior angle of a triangle be smaller than its interior angle?

3. The vertex angle of an isosceles triangle is α . What is the value of the base exterior angle of that triangle?

340. 1. The base angle of an isosceles triangle is α . What is the value of the exterior vertex angle of that triangle?

2. Prove that the bisector of the exterior vertex angle of an isosceles triangle is parallel to the base.

341. 1. What property does the leg of a right triangle, which lies opposite the angle of 30° , possess?

2. In a right triangle a leg is equal to half the hypotenuse. Calculate the angles of the triangle.

3. A point M is taken on the side BC of an equilateral triangle ABC . Perpendiculars are drawn from M to the other two sides. Calculate the distances from the feet of the perpendiculars to the vertex A if $|BM| = 3$ cm and $|MC| = 4$ cm.

The Sum of the Interior and Exterior Angles of a Convex Polygon

Theorem. *The sum of the angles of a convex n -gon is $180^\circ (n - 2)$ or $2d (n - 2)$. The sum of the exterior angles of a convex polygon, taken one at each vertex, is 360° (4d).*

342. What is the sum of the angles (a) of a quadrilateral, (b) of a pentagon, (c) of a hexagon?

343. What is the interior angle (a) of a hexagon, (b) of an n -gon, whose all angles are equal?

344. A point M is taken inside the acute angle AOB and perpendiculars are drawn from it to the sides of the angle. Find the obtuse and acute angles between the perpendiculars if $\widehat{AOB} = \alpha$.

345. The sum of the interior angles of a polygon is 1260° . How many sides does the polygon possess? Into how many triangles can this polygon be partitioned by the diagonals drawn from one vertex?

346. Can a convex hexagon possess four right angles?

347. The exterior angle of a polygon whose all angles are equal is 72° . How many sides does the polygon possess?

Relations Between the Sides and the Angles of a Triangle

Theorem 1. *The larger angle lies opposite the larger side of a triangle.*

If in the triangle ABC (Fig. 34) $|BC| > |AB|$, then $\hat{A} > \hat{C}$.

Theorem 2. *The larger side lies opposite the larger angle of a triangle.*

If $\hat{A} > \hat{C}$ in the triangle ABC , then $|BC| > |AB|$ (Fig. 34).

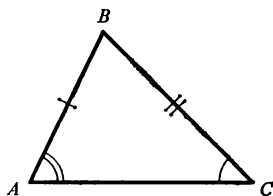


Fig. 34

348. 1. Indicate the largest side of the triangle ABC if $\hat{A} = 30^\circ$ and $\hat{B} = 72^\circ$.

2. The exterior base angle of an isosceles triangle is 126° . Which side of the triangle is the largest?

The Thales Theorem

If we lay off several congruent segments in succession on one side of an angle and through their endpoints draw parallel lines which cut the other side of the angle, then they will cut congruent segments on the other side of the angle.

If segments $[AB] \cong [BC] \cong [CD]$ are laid on the side OX of the angle XOY (Fig. 35) and $(AA_1) \parallel (BB_1) \parallel (CC_1) \parallel (DD_1)$, then $[A_1B_1] \cong [B_1C_1] \cong [C_1D_1]$.

349. In a triangle ABC the altitude BD divides the base AC into line segments 7 cm and 5 cm long. How long is the projection of the median AM ($M \in [BC]$) onto the straight line AC ?

A Median of a Triangle

A *median* of a triangle is a line segment which connects the midpoints of two sides of the triangle.

Theorem. A midline of a triangle is parallel to its third side and its length is equal to half the length of that side.

If in a triangle ABC (Fig. 36) $|AD| = |DB|$, $|BE| = |EC|$, then $[DE] \parallel [AC]$ and $|DE| = \frac{1}{2} |AC|$.

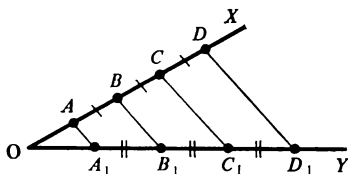


Fig. 35

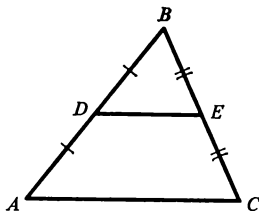


Fig. 36

350. A median is drawn in a triangle whose perimeter is $2p$. Find the perimeter of the triangle cut off by the median.

351. In an isosceles triangle whose perimeter is 30 cm a median 8 cm long is drawn between the lateral sides. Determine a lateral side of the triangle.

352. Can we state that the length of a median of a triangle is equal to the distance from the midpoint of one side of the triangle to its other side?

2.26. QUADRILATERALS

Parallelograms

A *parallelogram* is a quadrilateral with both pairs of opposite sides being parallel.

Theorem. The midpoint of a diagonal of a parallelogram is its centre of symmetry.

353. What are the properties of (a) the opposite sides, (b) the opposite angles, (c) the intersection point of the diagonals of a parallelogram?

354. What is the sum of the angles adjacent to one side of a parallelogram?

355. What are the necessary and sufficient conditions for a quadrilateral to be a parallelogram?

356. 1. Prove that a diagonal of a parallelogram divides it into two congruent triangles.

2. Prove that the vertices B and D of the parallelogram $ABCD$ are equidistant from the diagonal AC .

3. Prove that the segments of two parallel straight lines included between two other parallel straight lines are congruent.

357. How can you measure the distance between (a) the larger sides of a parallelogram, (b) the smaller sides of a parallelogram?

358. A diagonal divides a parallelogram into two isosceles right triangles. The larger side of the parallelogram is 8 cm. What is the length of the altitude drawn to that side?

359. In a parallelogram $ABCD$ altitudes 6 cm and 8 cm long are drawn from the vertex of the obtuse angle of 150° . Find the perimeter of the parallelogram.

360. 1. The bisector of an angle of a parallelogram divides one of its sides into segments a and b long. Express the perimeter of the parallelogram in terms of a and b .

2. The bisector of an angle of a parallelogram equal to 120° bisects its side, the bisector being 5 cm long. Find the perimeter of the parallelogram.

361. Prove that the diagonals of a parallelogram (different from a rhombus) do not bisect its angles.

Rectangles

A *rectangle* is a parallelogram each of whose four angles is a right angle.

Theorem. A perpendicular bisector of a rectangle is its axis of symmetry.

If in a rectangle $ABCD$ $[AM] \cong [MD]$ and $(MN) \perp [AD]$ (Fig. 37), then (MN) is the axis of symmetry of the rectangle $ABCD$.

362. Prove that (a) if one of the four angles of a parallelogram is a right angle, then the other three angles

are also right angles, (b) a quadrilateral whose three angles are right angles is a rectangle.

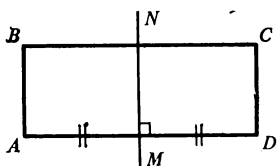


Fig. 37

363. 1. How many axes of symmetry does a rectangle possess?

2. What particular property do the diagonals of a rectangle possess (as distinct from those of a parallelogram)?

3. Prove that if the diagonals of a parallelogram are congruent, then the parallelogram is a rectangle.

4. Prove that the median of a right triangle drawn to the hypotenuse is equal to half the hypotenuse.

364. The smaller side of a rectangle is 2.5 cm, and the bisector of its angle divides the larger side of the rectangle into two segments one of which is twice as long as the other. Find the perimeter of the rectangle.

Rhombi

A rhombus is a parallelogram whose all sides are congruent.

Theorem. *The straight line which contains a diagonal of a rhombus is its axis of symmetry.*

365. What particular properties do the diagonals of a rhombus possess (as distinct from those of a parallelogram)?

366. Given a rhombus $ABCD$ and a point O which is the intersection point of its diagonals, find the points onto which the vertices of the rhombus will be mapped (a) in the case of symmetry about the straight line AC , (b) in the case of symmetry about the straight line BD , (c) in the case of symmetry about the point O ?

367. Prove that if the diagonals of a parallelogram are mutually perpendicular, then the parallelogram is a rhombus.

368. Is there a quadrilateral (besides a rhombus and a square) whose diagonals are mutually perpendicular?

369. Can a diagonal of a rhombus be perpendicular to its side?

370. The angle between a diagonal of a rhombus and its side is α . Express the angles of the rhombus in terms of α .

371. Calculate the angles of a rhombus if the altitude drawn from the vertex of an obtuse angle divides the opposite side into congruent segments.

372. Is the point of intersection of the diagonals of a rhombus at the same distance from (a) its vertices, (b) its sides?

Squares

A *square* is a rectangle whose all sides are congruent.

373. 1. Use the concept of a rhombus to define a square.

2. What particular properties do the diagonals of a square possess as compared to those of a rectangle and a rhombus (which are not squares)?

374. How, without measuring the angles of a quadrangular plot of land, can we make sure that it is a square?

375. Given a square $ABCD$ on whose diagonal BD , a segment BE , which is congruent to a side of the square, is laid off, and the point E is connected with the point C . Find the angles of the triangle CED .

376. A diagonal of a square whose side is 3 cm serves as a side of another square. Calculate the diagonals of the other square.

377. Points E and F divide the diagonal BD of a square $ABCD$ into three congruent segments, and the vertices C and A are connected with the points of division. Prove that the quadrilateral $AECF$ is a rhombus.

Trapezoids

A *trapezoid* is a quadrilateral two of whose sides are parallel (called bases) and the other two are nonparallel (called legs). The *median* of a trapezoid is a line segment which connects the mid-points of its nonparallel sides.

Theorem. *The median of a trapezoid is parallel to its bases and equal to half their sum.*

If in the trapezoid $ABCD$ shown in Fig. 38 $[AD] \parallel [BC]$, $|AM| = |MB|$, $|CN| = |ND|$, then $[MN] \parallel [AD]$ and $|MN| = \frac{|AD| + |BC|}{2}$.

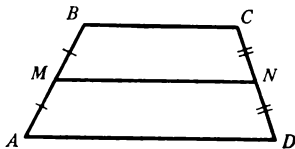


Fig. 38

378. 1. What trapezoid is isosceles, right-angled?

2. What line segment is the median of a trapezoid, its altitude?

379. Prove that in an isosceles trapezoid (a) the base angles are congruent, (b) the diagonals are congruent, (c) the diagonals are not bisected by their point of intersection.

380. 1. The length of one base of a trapezoid is a and that of its median is m . Write a formula for the length of the other base b .

2. The length of the median of an isosceles trapezoid is m and that of each nonparallel side is c . Write the formula for the perimeter P of the trapezoid.

381. 1. The median of a trapezoid 12 cm long is divided by its diagonal into two segments one of which is half of the other in length. Calculate the bases of the trapezoid.

2. The bases of a trapezoid are 8 cm and 12 cm long. Find the lengths of the segments into which one of the diagonals of the trapezoid divides the median.

382. The bases of a trapezoid are a and b in length. Find the distance between the midpoints of the diagonals.

383. In a right-angled trapezoid the length of the smaller base is a and that of the larger side is b . The acute angle is 60° . Find the length of the median of the trapezoid.

384. The midpoints of the sides of an isosceles trapezoid are connected by line segments in succession. Determine the kind of the resulting quadrilateral.

2.27. VECTORS

Translation

A *translation*, or *parallel displacement*, is a mapping of a plane onto itself under which each point X is mapped onto a point X_1 (Fig. 39) such that

- (1) the ray XX_1 has an assigned direction,
- (2) the segment XX_1 has an assigned length, i.e. all the points of the plane are displaced in the same direction by the same distance:

$$[XX_1] \parallel [YY_1]; \quad |XX_1| = |YY_1|.$$

Notation: if, under a translation, the point X goes into the point X_1 , then we write $T(X) = X_1$.

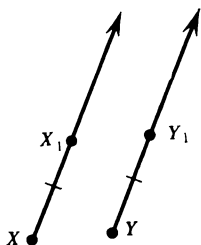


Fig. 39

385. 1. How can a translation be defined?

2. Formulate the properties of a translation.

386. We consider a translation. What is (a) the image of a straight line, (b) the image of a ray, (c) the image of a figure?

387. 1. What translation maps a circle onto a congruent circle?

2. Construct the image of a given rhombus $ABCD$ under the translation $T(A) = D$.

388. How can we find an angle if we cannot reach its vertex?

Vectors and the Ways of Defining Them

Translations are known as *vectors*. A vector that maps a point A onto a point B is designated as $\vec{AB}$ or $\mathbf{a}$ and is depicted as a directed segment (Fig. 40).

389. 1. What is a zero, or null, vector?

2. What is the length and the direction of the vector AB ?

390. What vectors are said (a) to have the same direction, (b) to have opposite directions, (c) to be collinear?

391. 1. How is a vector specified?

2. What vector is opposite to the vector $\vec{AB}$?

392. Given a parallelogram $ABCD$ (Fig. 41), name the vectors which are (a) equal, (b) opposite, (c) collinear.



Fig. 40

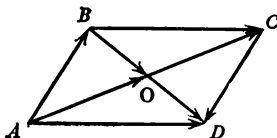


Fig. 41

393. A median MN is drawn in a trapezoid $ABCD$ ($[BC] \parallel [AD]$). Write the vectors which are collinear with the vector $\vec{MN}$ and defined by the pair of points formed from the vertices of the trapezoid.

394. Points E and F are the midpoints of the sides AB and BC of a parallelogram $ABCD$. Prove that the vectors $\vec{EF}$ and $\vec{CA}$ are collinear.

The Sum of Vectors

The composition of vectors $\mathbf{a}$ and $\mathbf{b}$ is customarily called the *sum of the vectors* $\mathbf{a}$ and $\mathbf{b}$. The notation is $\mathbf{a} + \mathbf{b} = \mathbf{c}$ (rather than $\mathbf{a} \cdot \mathbf{b} = \mathbf{c}$).

395. Given two vectors $\mathbf{a}$ and $\mathbf{b}$, how can you construct a vector representing their sum?

396. Prove that in an arbitrary quadrilateral $ABCD$ the equality $\vec{AB} + \vec{BC} = \vec{AD} + \vec{DC}$ holds true.

397. Assume that O is the point of intersection of the diagonals of a parallelogram. Find the sum of the vectors $\vec{AB}$ and $\vec{BC}$, $\vec{AB}$ and $\vec{CD}$, $\vec{OB}$ and $\vec{OD}$, $\vec{OC}$ and $\vec{OA}$.

The Difference of Vectors

The vector $\mathbf{c} = \mathbf{a} + (-\mathbf{b})$ is the difference of the vectors $\mathbf{a}$ and $\mathbf{b}$ which is designated as $\mathbf{a} - \mathbf{b}$.

398. Given two vectors $\mathbf{a}$ and $\mathbf{b}$, how can you construct the vector $\mathbf{c} = \mathbf{a} - \mathbf{b}$?

399. AC and BD are diagonals of a parallelogram $ABCD$. Express the vectors $\vec{AC}$ and $\vec{BD}$ in terms of the vectors $\vec{AB}$ and $\vec{AD}$.

400. Express the vectors $\vec{AB}$ and $\vec{AD}$ in terms of the vectors $\vec{OA}$ and $\vec{OB}$ if O is the point of intersection of the diagonals of the parallelogram $ABCD$.

Multiplying a Vector by a Scalar

The *product* of a vector $\mathbf{a}$ ($\mathbf{a} \neq 0$) by a scalar x ($x \neq 0$) is a vector whose length is equal to the product of the length of the vector $\mathbf{a}$ by $|x|$ and whose direction coincides with that of the vector $\mathbf{a}$, if $x > 0$, and is opposite to that of $\mathbf{a}$ if $x < 0$.

The product of the vector $\mathbf{a}$ by the scalar x is designated as $x\mathbf{a}$.

401. Under what conditions are the nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ collinear?

402. In a parallelogram $ABCD$ a point M is the midpoint of the side AB . Express $\vec{MC}$ in terms of $\vec{AB}$ and $\vec{BC}$.

403. Given a rhombus $ABCD$, express the vectors $\vec{AB}$ and $\vec{BC}$ in terms of the vectors $\vec{DB}$ and $\vec{CA}$.

Fundamental Laws of Vector Algebra

Addition Laws

1. $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$, associativity.
2. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$, commutativity.
3. $\mathbf{a} + \mathbf{0} = \mathbf{a}$, absorption of a zero vector.

Laws of Multiplication of a Vector by a Scalar

1. $(xy) \mathbf{a} = x (y\mathbf{a})$, associativity.
2. $x\mathbf{a} + y\mathbf{a} = (x + y) \mathbf{a}$, the first distributive law.
3. $x\mathbf{a} + x\mathbf{b} = x (\mathbf{a} + \mathbf{b})$, the second distributive law.
4. $0\mathbf{a} + x\mathbf{0} = \mathbf{0}$, absorption of zero and a zero vector.

404. AM is a median of a triangle ABC ($M \in [BC]$). Express the vector $\vec{AM}$ in terms of the vectors $\vec{AB} = \mathbf{a}$ and $\vec{AC} = \mathbf{b}$.

405. Given a parallelogram $ABCD$ (Fig. 42) in which $|OM| = \frac{1}{3}|OB|$, $|ON| = \frac{1}{3}|OC|$, express the vector $\vec{NM}$ in terms of the vector $\vec{AD} = \mathbf{a}$.

Vector Components

An arbitrary vector $\mathbf{c}$ (Fig. 43) can be represented as

$$\mathbf{c} = c_x \mathbf{i} + c_y \mathbf{j},$$

where $\mathbf{i}$ is a unit vector laid off from the origin O along the x -axis and $\mathbf{j}$ is a unit vector laid off along the y -axis; c_x is the projection

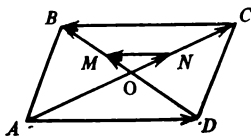


Fig. 42

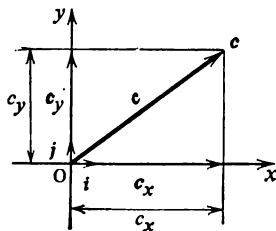


Fig. 43

of $|\mathbf{c}|$ onto the x -axis, and c_y is the projection of $|\mathbf{c}|$ onto the y -axis. The coefficients c_x and c_y of the resolution of the vector $\mathbf{c}$ into components with respect to the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are the *components of the vector $\mathbf{c}$* .

406. Given points $O(0, 0)$, $A(3, 2)$ and $B(4, 5)$ on the coordinate plane, resolve the vectors $\vec{OA}$, $\vec{OB}$, $\vec{OA} + \vec{OB}$ into components with respect to the unit vectors $\mathbf{i}$ and $\mathbf{j}$.

2.28. SIMILARITY AND HOMOTHETY OF GEOMETRIC FIGURES

Similar Figures

A figure F_1 is similar to a figure F if, there is a mapping of F onto F_1 under which the distances change in the same ratio $k > 0$.

If the figure F_1 is similar to the figure F , then we write $F_1 \sim F$, and if we wish to indicate the ratio of similitude then we write $F_1 \overset{h}{\sim} F$.

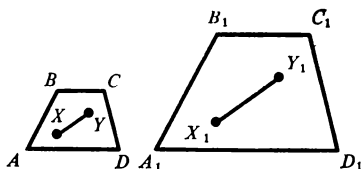


Fig. 44

If the ratio of similitude of the trapezoids $A_1B_1C_1D_1$ and $ABCD$ (Fig. 44) is 2, then the trapezoid $ABCD$ can be mapped onto $A_1B_1C_1D_1$ so that any two points X and Y of the first trapezoid will go into the points X_1 and Y_1 such that

$$|X_1Y_1| = 2|XY|.$$

407. 1. Can two congruent figures be similar?

2. Can two similar figures be congruent?

408. The plan of a land plot is given in two variants: the first plan is drawn to the scale of 1 : 100 and the second to the scale of 1 : 1000. What are the ratios of similitude in these plans?

Homothety

A *homothety* (or *similitude*) with centre O and ratio $k \neq 0$ is a mapping of a plane onto itself under which the image of an arbitrary point X is a point X_1 such that $\vec{OX}_1 = k\vec{OX}$. For example

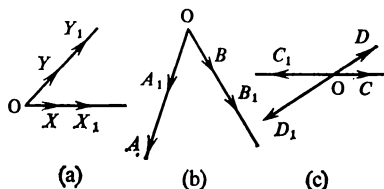


Fig. 45

(a) $\vec{OX}_1 = 2\vec{OX}$, $k = 2$, (b) $\vec{OA}_1 = \frac{1}{2}\vec{OA}$, $k = 1/2$, (c) $\vec{OC}_1 = -2\vec{OC}$, $k = -2$ (Fig. 45).

The homothety with centre O and ratio k is designated as H_O^h or simply H^k .

409. What is the image of each point X under the homothetic transformation with centre O if the ratio (a) $k = 1$, (b) $k = -1$.

410. What are the positions of the points X_1 and X about the centre of homothetic transformation if (a) $k > 0$, (b) $k < 0$?

411. 1. Why are homothetic figures similar?

2. Are similar figures always homothetic?

412. Construct a figure which would be homothetic to a given isosceles triangle ABC about the centre O with the homothetic ratio $k = -1.5$.

Proportional Line Segments

Line segments AB and CD are said to be *proportional* to the segments A_1B_1 and C_1D_1 if their lengths are proportional, i.e.

$$\frac{|AB|}{|AD|} = \frac{|A_1B_1|}{|C_1D_1|}.$$

Theorem. *Parallel straight lines which cut the sides of an angle intercept proportional segments on them.*

If $(AA_1) \parallel (BB_1)$ (Fig. 46), then $|OA| \div |OA_1| = |AB| \div |A_1B_1| = |OB| \div |OB_1|$.

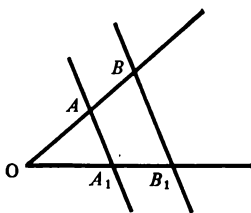


Fig. 46

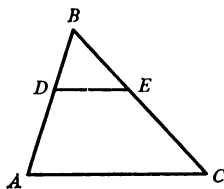


Fig. 47

Corollary. *A straight line which is parallel to a side of a triangle and cuts the other two sides off a triangle whose sides are proportional to the sides of the given triangle.*

If in the triangle ABC (Fig. 47) $[DE] \parallel [AC]$, then

$$\frac{|BA|}{|BD|} = \frac{|BC|}{|BE|} = \frac{|AC|}{|DE|}.$$

413. In the triangle ABC (Fig. 47) $[DE] \parallel [AC]$, $|BD| = 2$ cm, $|AD| = 3$ cm and $|BC| = 6.5$ cm. Calculate $|BE|$ and $|CE|$.

414. In the trapezoid $ABCD$ the leg AB is divided by a point M into segments $|AM| = 6.8$ cm and $|MB| = 6$ cm. A straight line MN is drawn through the point M parallel to the bases ($N \in [CD]$). Find $|CD|$ if $|CN| = 9$ cm.

Criteria of Similarity of Triangles

I. *If three sides of one triangle are proportional to three sides of another triangle, the triangles are similar.*

II. *If two sides of one triangle are proportional to two sides of another triangle and the angles between those sides are congruent, the triangles are similar.*

III. *If two angles of one triangle are congruent to two angles of another triangle, the triangles are similar.*

415. Can we state that the following figures are similar: (a) any two equilateral triangles, (b) any two right isosceles triangles, (c) isosceles triangles which have a congruent acute angle each?

416. 1. A triangle is observed with a lens whose magnifying power is $3X$. Will the angles of the triangle increase?

2. A plot of land is shaped as a triangle with sides 60, 70 and 78 m long. A plan is drawn of this plot of land on which the corresponding sides are 6, 7 and 7.5 cm long. Is the drawing correct?

417. Given a triangle ABC in which the medians AE ($E \in [BC]$) and BD ($D \in [AC]$) meet at a point O . Prove that the triangles AOB and DOE are similar. What is the ratio of similitude of these triangles?

418. Through a point D taken on the side BC of the triangle ABC draw, in the interior of the triangle, four distinct rays which cut off triangles similar to the given one.

419. In the triangle ABC $|AB| = 9$ cm, $|BC| = 15$ cm and $|AC| = 18$ cm. A straight line DE ($E \in [BC]$) is drawn from a point D taken on the side AC , the point being such that $\angle DEC \cong \angle A$. Calculate $|DC|$ and $|EC|$ if $|ED| = 6$ cm.

420. A common side of two similar isosceles triangles is 12 cm long. The base of the smaller triangle is 9 cm. Calculate the perimeters of the triangles.

421. Formulate the criteria of similarity of right triangles.

422. 1. The legs of one triangle are 1.8 cm and 0.8 cm and those of the other are 2.7 cm and 1.2 cm. Are the triangles similar?

2. The sides of a parallelogram are 18 cm and 12 cm long. The smaller altitude of the parallelogram is 9 cm. Find the larger altitude of the parallelogram.

423. Prove that the point of intersection of two medians separates a third of each median reckoning from the corresponding side.

424. Prove that in similar triangles the ratio of two respective (a) altitudes, (b) bisectors, (c) medians is equal to the ratio of two respective sides.

Geometric Means in a Right Triangle

A line segment x is known as a *geometric mean* (mean proportional) of the segments a and b if the equality $a : x = x : b$ holds true.

425. Read the following proportions: (a) $m : x = x : n$, (b) $a : p = q : a$. What is the repeated term equal to in these proportions if the other two terms are known?

Theorems. 1. A leg is a geometric mean of the hypotenuse and the projection of the leg onto the hypotenuse.

If in the triangle ABC (Fig. 48) CD is an altitude, $|CD| = h$, a_c and b_c are projections of the legs onto the hypotenuse, then

$$c : a = a : a_c \quad \text{or} \quad a^2 = c \cdot a_c,$$

$$c : b = b : b_c \quad \text{or} \quad b^2 = c \cdot b_c.$$

2. In a right triangle the altitude drawn from the vertex of the right angle is the geometric mean of the projections of the legs onto the hypotenuse:

$$a_c : h = h : b_c \quad \text{or} \quad h^2 = a_c \cdot b_c.$$

Pythagoras' Theorem. In a right triangle the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

In the right triangle ABC (Fig. 48) $c^2 = a^2 + b^2$.

426. 1. A leg of a right triangle is 3 cm and its projection onto the hypotenuse is 1.8 cm. Calculate the length of the hypotenuse and that of the other leg.

2. The projections of the legs of a right triangle onto the hypotenuse are 2 cm and 6 cm. Calculate the

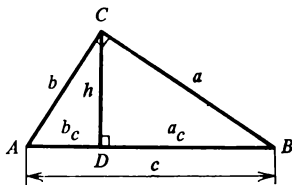


Fig. 48

lengths of the legs and that of the altitude drawn to the hypotenuse.

427. Proceeding from what theorem can we state that a triangle with sides of 3, 4 and 5 cm is right-angled?

428. A point inside a right angle is 6 cm and 8 cm distant from its sides. Find the distance from this point to the vertex.

429. The legs of a right triangle are 5 cm and 12 cm. Find the median drawn to the hypotenuse.

430. An iron bar 2 m long has been bent at its midpoint at right angles. Find the distance between its endpoints

431. The diagonals of a rhombus are 4.8 cm and 14 cm. Find its perimeter.

432. 1. A side of a square is a . Find the diagonal.

2. The diagonal of a square is d . Find its side.

433. 1. A side of an equilateral triangle is a . Find the altitude.

2. The altitudes of an equilateral triangle meet at a point O . Each of its sides is 2 cm. Find the distance from the point O to the sides and the vertices of the triangle.

434. 1. The bases of an isosceles trapezoid are 1 cm and 4 cm. Each of the nonparallel sides is 2.5 cm. Calculate the altitude of the trapezoid.

2. The bases of an isosceles trapezoid are 25 cm and 7 cm. The diagonal is 20 cm. Calculate the perimeter of the trapezoid.

435. The hypotenuse of a right triangle is 20 cm and one of the legs constitutes 75% of the other. Calculate the lengths of the legs.

Similar Polygons

436. Can we state that (a) the proportionality of the sides of two polygons (with more than three sides) implies the similarity of these polygons, (b) the congruence of the respective angles of the polygons implies their similarity?

437. What is the sufficient condition for the similarity of two polygons?

Ratio of the Perimeters of Similar Polygons

Theorem. *The ratio of the perimeters of similar polygons is equal to the ratio of similitude of those polygons.*

If the polygon $A_1B_1C_1D_1E_1$ is similar to the polygon $ABCDE$ (Fig. 49), P and P_1 are their respective perimeters and k is the ratio of similitude, then $P = kP_1$ or $P \div P_1 = k$.

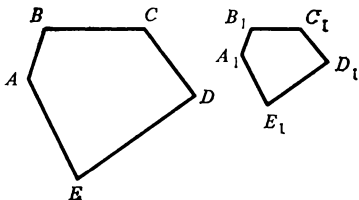


Fig. 49

438. 1. The largest sides of two similar polygons are related as $4/3$. The perimeter of the smaller polygon is 36 cm. Find the perimeter of the larger polygon.

2. The smaller sides of two similar polygons are 3 cm and 7 cm long. The perimeter of one polygon is 20 cm smaller than that of the other. Find the perimeter of each polygon.

2.29. CIRCLES

The Property of Central Angles

Theorem. *Two central angles of a circle are congruent if and only if the intercepted arcs are congruent.*

If in the circle (O, r) $\angle AOB \cong \angle A_1OB_1$ ($\widehat{AOB} = \widehat{A_1OB_1}$) (Fig. 50), then $\cap AB \cong \cap A_1B_1$ ($\widehat{AB} = \widehat{A_1B_1}$).

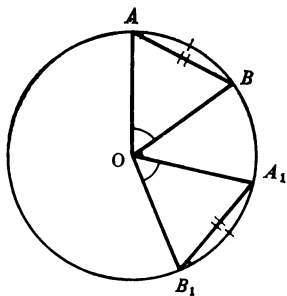


Fig. 50

And conversely, if in the circle (O, r) $\cap AB \cong \cap A_1B_1$ ($\widehat{AB} = \widehat{A_1B_1}$), then $\angle AOB \cong \angle A_1OB_1$ ($\widehat{AOB} = \widehat{A_1OB_1}$).

439. What are the values of the central angles if the intercepted arcs constitute (a) $1/3$, (b) $1/16$ of the circle?

440. Two points divide a circle into two arcs. What is the degree measure of each of the arcs if the degree measure of one of them is 90° larger than that of the other?

441. How can we bisect the arc of a circle?

442. The diameter of a circle is 12 cm. A chord drawn in that circle subtends an arc of 120° . Find the distance between the centre and the chord.

The Property of a Tangent to a Circle

A straight line which has only one point in common with a circle is known as a *tangent* to that circle.

Theorem. *If a straight line is perpendicular to a radius and passes through its endpoint lying on the circle, then it is a tangent to the circle.*

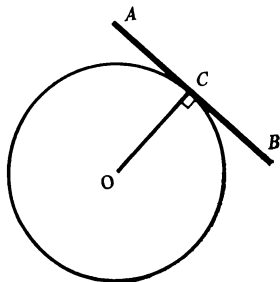


Fig. 51

If $[OC]$ is a radius of a circle (Fig. 51) (C is the point on the circle) and the straight line $(ACB) \perp [OC]$, then (ACB) is a tangent.

Inverse theorem. *If a straight line is a tangent to a circle, then the radius drawn to the point of tangency is perpendicular to the tangent.*

If the straight line (ACB) (Fig. 51) is a tangent (C is the point of tangency) and $[OC]$ is a radius, then $[OC] \perp (ACB)$.

443. A point A lies on a circle. How can we draw a tangent to the circle through this point?

444. What are the positions of the tangents to a circle which are drawn through the endpoints of a diameter?

445. The endpoints of a diameter are at the distances of 8 cm and 12 cm to a tangent to the circle. Calculate the length of the radius.

446. Two tangents are drawn to a circle from a point taken outside of the circle. Prove that the tangents are equal in length (the length of a tangent is the length of its segment from a given point to the point of tangency).

Inscribed Angles

A convex angle whose vertex is on the circle and whose side cut the circle is an *inscribed angle*.

Theorem. *An inscribed angle has half as many degrees as the intercepted arc.*

If $\angle ABC$ is an inscribed angle (Fig. 52), then

$$\widehat{ABC} = \frac{1}{2} \widehat{AC}, \text{ i.e. } \widehat{ABC} = \frac{1}{2} \widehat{AOC}.$$

447. 1. Prove that an inscribed angle resting on a diameter is a right angle.

2. Formulate two propositions converse of the preceding one. Are they true?

448. Under what condition is an inscribed angle (a) acute, (b) obtuse?

449. An inscribed angle is 30° smaller than the central angle subtended by the same arc. Find each of these angles.

450. Two angles are inscribed into the same circle and are subtended by the same arc. What can you say about the values of these angles?

451. Using the property of inscribed angles, prove that the arcs included between parallel chords are congruent.

452. A tangent and a chord are drawn from a point taken on a circle. Prove that the angle between the tangent and the chord is half the degree measure of the arc intercepted by this angle.

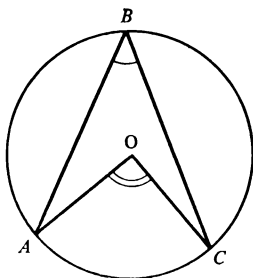


Fig. 52

2.30. INSCRIBED AND CIRCUMSCRIBED TRIANGLES

A triangle whose all vertices lie on a circle is said to be *inscribed* into that circle and the circle is said to be *circumscribed* about the triangle.

Theorem. *One and only one circle can be circumscribed about any triangle. The centre of the circle is the point of intersection of the perpendiculars drawn to the midpoints of the sides of the triangle.*

If O is the point of intersection of the perpendicular bisectors (Fig. 53), then it is the centre of the circumscribed circle with radii OA , OB and OC .

A triangle whose all sides touch a circle is said to be *circumscribed* about the circle and the circle is said to be *inscribed* in the triangle.

Theorem. *One and only one circle can be inscribed in a triangle. The centre of the circle is the point of intersection of the bisectors of the triangle.*

If O is the point of intersection of the bisectors of a triangle (Fig. 54), then it is the centre of the inscribed circle with radii OM , OE and OF .

453. When does the centre of the circle circumscribed about a triangle lie (a) inside the triangle, (b) outside of it, (c) on a side of the triangle?

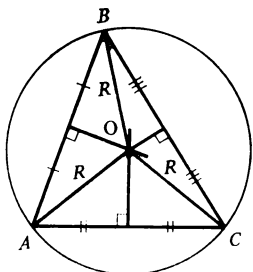


Fig. 53

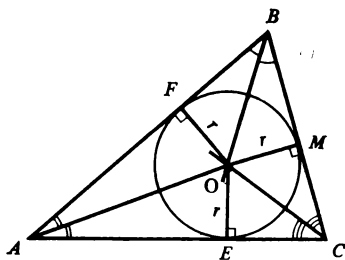


Fig. 54

454. How can an equilateral triangle be inscribed in a circle?

455. Can we state that the centre of the circle circumscribed about an equilateral triangle lies at the point of intersection of its bisectors?

456. The altitude of an equilateral triangle is h . Find the radius of the circumscribed circle.

457. The legs of a right triangle are 3 cm and 4 cm. Calculate (a) the diameter of the circle circumscribed about the triangle, (b) the median drawn from the vertex of the right angle.

458. One of the acute angles of a right triangle is 35° . Calculate the acute angle between the median drawn from the vertex of the right angle and the hypotenuse.

459. A lateral side of an isosceles triangle is a . The base angle is 30° . Find the diameter of the circumscribed angle.

460. A lateral side of an isosceles triangle is 10 cm and the altitude is 8 cm. Calculate the radius of the circumscribed circle.

461. Find whether the centres of the inscribed and the circumscribed circle coincide (a) in an isosceles triangle, (b) in an equilateral triangle, and if so why.

462. The radius of the circle inscribed into an equilateral triangle is r . Find the radius of the circumscribed circle and its altitude.

463. The legs of a right triangle are 9 cm and 12 cm. Calculate the radius of the inscribed circle.

2.31. TRIGONOMETRIC FUNCTIONS

The Sine, the Cosine and the Tangent of an Angle

We mark a point $P_0(1, 0)$ on a unit circle (Fig. 55). As a result of the rotation R_0^α the point P_0 is mapped onto a point P_α . Every angle α is associated with an abscissa x_α and an ordinate y_α of the point P_α . The ordinate y_α is the *sine* of the angle α , the abscissa x_α is the *cosine* of the angle α , and the ratio of $\sin \alpha$ to $\cos \alpha$ is the *tangent* of the angle α , i.e.

$$y_\alpha = \sin \alpha, \quad x_\alpha = \cos \alpha,$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha}.$$

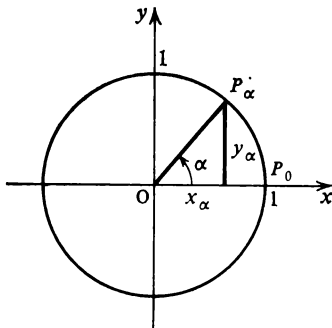


Fig. 55

464. Why are the sine, the cosine and the tangent of an angle its functions?

465. 1. What is the domain of definition and the range of a sine function and a cosine function?

2. Can a sine and a cosine of an angle be equal to $\sqrt{2}$, $10^{\log 2}$?

466. How do the values of the functions $\sin \alpha$ and $\cos \alpha$ change with the increase in the angle from 0 to 90° ?

467. Use a unit circle to find the values of the sine and cosine of the angles of (a) 90° , (b) 180° , (c) 270° , (d) 360° , (e) -90° .

468. Depict the graphs of the sine and cosine functions for the angles belonging to the interval $[-180^\circ, 180^\circ]$.

Some Identities for the Sine and Cosine Functions

$$\begin{aligned} \sin^2 \alpha + \cos^2 \alpha &= 1, \\ \sin(180^\circ - \alpha) &= \sin \alpha, \\ \cos(180^\circ - \alpha) &= -\cos \alpha, \\ \sin(-\alpha) &= -\sin \alpha, \\ \cos(-\alpha) &= \cos \alpha. \end{aligned}$$

469. Find the value of (a) $\cos \alpha$ if $\sin \alpha = 0.8$, $0^\circ < \alpha < 90^\circ$, (b) $\sin \alpha$ if $\cos \alpha = 1/2$, $0^\circ < \alpha < 90^\circ$.

470. Express the functions (a) $\sin 150^\circ$, (b) $\cos 150^\circ$, (c) $\sin 170^\circ$, and (d) $\cos 170^\circ$ in terms of the value of the trigonometric function of a positive angle smaller than 90° .

Relations Between the Sides and Angles of a Right Triangle

The *sine* of an acute angle of a right triangle (Fig. 56) is equal to the ratio of the opposite side to the hypotenuse.

The *cosine* of an acute angle of a right triangle is equal to the ratio of the adjacent side to the hypotenuse.

The *tangent* of an acute angle of a right triangle is equal to the ratio of the opposite side to the adjacent one, i.e.

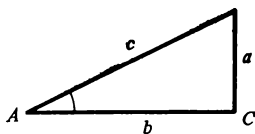


Fig. 56

$$\sin \hat{A} = \frac{a}{c}, \quad \cos \hat{A} = \frac{b}{c}, \quad \tan \hat{A} = \frac{a}{b}.$$

471. Use the formulas written above to express (a) a leg in terms of the hypotenuse and the trigonometric function of an acute angle, (b) the hypotenuse in terms of a leg and the trigonometric function of an acute angle.

472. Using the designations used in Fig. 56, write the values of $\sin \hat{B}$, $\cos \hat{B}$ and $\tan \hat{B}$.

473. 1. Write the values of the trigonometric functions of the angles α , β , x and y using the designations used in Fig. 57.

2. What trigonometric functions of the angles α , β , x and y express the ratios m/a , h/b , m/h and h/m ?

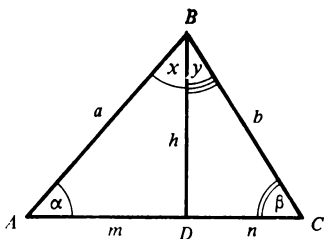


Fig. 57

474. Given the legs $a = 8$ cm and $b = 15$ cm of a right triangle (see Fig. 56), calculate the hypotenuse c , the sum of the sines of the acute angles A and B and the product of their cosines.

475. 1. Using a right triangle with an acute angle of 30° , find the values of the sine, the cosine and the tangent of the angles of 30° and 60° .

2. Using a right triangle with an angle of 45° , find the values of the sine, the cosine and the tangent of an angle of 45° .

476. Calculate $\sin^2 \alpha + \cos^2 \alpha$ if (a) $\alpha = 30^\circ$, (b) $\alpha = 45^\circ$, (c) $\alpha = 60^\circ$.

477. Compare the following values of the expressions:
(a) $\sin 2\alpha$ and $2 \sin \alpha$ for $\alpha = 30^\circ$, (b) $\sin (\alpha + \beta)$ and $\sin \alpha + \sin \beta$ for $\alpha = 30^\circ$ and $\beta = 60^\circ$.

478. Using the values of the trigonometric functions of angles of 30° , 45° and 60° , find (a) the altitude of an equilateral triangle being given its side a , as well as the radius of the inscribed circle, (b) the side of a square being given its diagonal d .

479. A railway embankment is 6 m wide at the top and 12 m wide at the bottom. Its lateral sides are at the angle of 35° to the horizontal. Calculate the height of the embankment.

2.32. AREAS OF POLYGONS

480. 1. What principal properties of areas are used to find the areas of polygons?

2. What are the units of areas?

3. What two figures are said to be of equal areas?

4. Are figures of equal areas always congruent?

481. A plot of land is 10 hectares in area. What does this amount to in square metres, in square kilometres?

The Area of a Rectangle

The area of a rectangle is equal to the product of its base by its altitude.

If in a rectangle a is the base, h is the altitude and S is the area, then $S = ah$.

482. What is the area of a square in which (a) the side is a , (b) the diagonal is d ?

483. Given two ropes 100 m each, one of which outlines a rectangle with a side of 30 m and the other outlines a square. Which area is larger, the area of the rectangle or that of the square?

484. A square is inscribed in a circle of radius R and a square is circumscribed about the same circle. Find the area of each square and the ratio of their areas.

485. The diagonal of one of the given squares is 3 cm and that of the other is 4 cm. Find the diagonal of the square whose area is equal to the sum of the areas of the given squares.

486. Will the area of a rectangle change if we increase the length of its base a by 50 per cent and decrease its altitude h by 50 per cent?

The Area of a Parallelogram

1. The area of a parallelogram is the product of the base and the altitude to that base (Fig. 58), i.e.

$$S = ah_1, \text{ or } S = bh_2.$$

2. The area of a parallelogram is equal to the product of its two adjacent sides by the sine of the angle between them (Fig. 58), i.e.

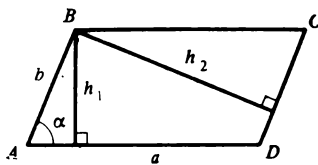


Fig. 58

$$S = ab \sin \alpha.$$

487. The sides of a parallelogram are 14 cm and 18 cm and its area is 126 cm^2 . Calculate the altitudes of the parallelogram and the

acute angle between its adjacent sides.

488. The sides of a parallelogram are 12 cm and 15 cm and one of the altitudes is 9 cm. Calculate the other altitude. Consider two cases.

489. Given a set of parallelograms, the sides of each of them being a and b , find which of these parallelograms has the greatest area.

490. Derive a formula which expresses the area S of a rhombus in terms of its diagonals d_1 and d_2 .

491. Calculate the area of a rhombus if (a) its diagonals are 5 cm and 8 cm long, (b) its side is 2.8 dm, and the acute angle is 30° , (c) the perimeter of the rhombus is P (express its area in terms of P and the function of the angle α between its adjacent sides).

The Area of a Triangle

The area of a triangle is equal to a half of the product of its base by its altitude.

If a is the base of a triangle and h_a is the altitude drawn to the side a (Fig. 59), then

$$S = \frac{1}{2} ah_a.$$

The area of a triangle is equal to a half of the product of two of its sides by the sine of the angle between them, i.e.

$$S = \frac{1}{2} ab \sin \gamma.$$

The area of a right triangle (Fig. 60) is equal to a half of the product of its legs, i.e.

$$S = \frac{1}{2} ab \quad \text{or} \quad S = \frac{1}{2} ch_c.$$

492. Write formulas which express (a) the area of a triangle in terms of its sides b and c and the altitudes

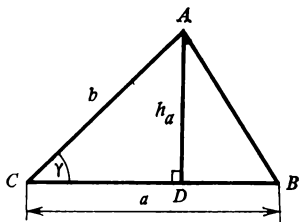


Fig. 59

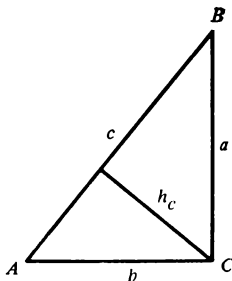


Fig. 60

drawn to those sides, (b) the altitudes of a triangle in terms of its sides and its area.

493. 1. A median is drawn in a triangle. Why do the two triangles into which the median divides the given triangle have equal areas?

2. M is the point of intersection of the medians in the triangle ABC . Prove that the area of the triangle AMD ($D \in [AC]$) constitutes $1/6$ of the area of the given triangle.

3. In the triangle ABC the median AM is perpendicular to the median BN . Find the area of the triangle ABC if $|AM| = 6$ cm and $|BN| = 8$ cm.

494. The area of a triangle is 12.5 cm^2 . Its altitude is congruent to its base. Calculate the base.

495. The base of an isosceles triangle is 12 cm. A lateral side exceeds the altitude by 2 cm. Calculate the area.

496. Calculate the area of an isosceles triangle whose lateral side is 4 cm and the vertex angle is (a) 30° , (b) 135° .

497. Find the area of an equilateral triangle if (a) its side is a , (b) its altitude is h .

498. Find the area of a right isosceles triangle if its hypotenuse is equal to c .

The Area of a Trapezoid

The area of a trapezoid is equal to the product of half the sum of its bases by its altitude.

If a and b are the bases of the trapezoid $ABCD$ and h is its altitude (Fig. 61), then

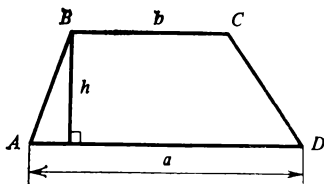


Fig. 61

$$S = \frac{a+b}{2} \cdot h.$$

499. How can the area of a trapezoid be expressed in terms of its median and altitude?

500. Find the area of an isosceles trapezoid if its smaller base is 12 cm, its

altitude is 10 cm and one of the nonparallel sides makes an angle of 45° with the base.

501. The smaller base of a trapezoid is 3 cm and one of the nonparallel sides is 4 cm and makes an angle of 30° with the larger base. The other angle at that base is 45° . Calculate the area of the trapezoid.

502. The bases of an isosceles trapezoid are 7 cm and 13 cm, and its area is 40 cm^2 . Calculate the perimeter of the trapezoid.

503. 1. Find the area of a trapezoid whose diagonals are mutually perpendicular and are m and n respectively.

2. The diagonals of an isosceles trapezoid are mutually perpendicular and its altitude is h . Find the area of the trapezoid.

The Ratio of the Areas of Similar Polygons

Theorem. *The ratio of the areas of similar polygons (Fig. 62) is equal to the square of the ratio of similitude.*

If the polygon $ABCDE$ is similar to the polygon $A_1B_1C_1D_1E_1$,

(Fig. 62) and $\frac{|AB|}{|A_1B_1|} = k$,

where k is the ratio of similitude, then

$$\frac{S_{ABCDE}}{S_{A_1B_1C_1D_1E_1}} = k^2.$$

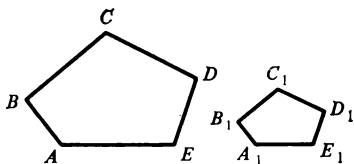


Fig. 62

504. How will the area of a polygon change if each of its sides increases three times and the angles remain unchanged?

505. The plan of a plot of land is drawn to the scale of $1 \div 100$. How many times is the area of the real plot greater than its area in the plan?

506. The median of a triangle divides it in two. How many times is (a) the area of the given triangle larger than that of the cut-off triangle, and (b) the area of the trapezoid obtained larger than the area of the cut-off triangle?

507. The ratio of the perimeters of two similar polygons is $2/3$. The area of the larger polygon is 90 cm^2 . Calculate the area of the smaller one.

2.33. METRIC RELATIONS IN A TRIANGLE

The Law of Cosines

The square of a side of a triangle (Fig. 63) is equal to the sum of squares of its other two sides minus the doubled product of these sides by the cosine of the angle between them.

If a , b and c are the sides of the triangle ABC (Fig. 63), then

$$a^2 = b^2 + c^2 - 2bc \cos \hat{A}.$$

508. What elements of a triangle can we calculate using the law of cosines?

509. The sides of a triangle are a and b . Find the third side c if the angle between the known sides is (a) 60° , (b) 45° , (c) 120° .

510. For what values of $\hat{A}$ are the sides a , b and c of a triangle related as (a) $a^2 < b^2 + c^2$, (b) $a^2 = b^2 + c^2$, (c) $a^2 > b^2 + c^2$?

511. The vertex A of the equilateral triangle ABC is connected with the point D which divides the side BC into segments $|BD| = 4 \text{ cm}$ and $|DC| = 6 \text{ cm}$. Calculate the length of the segment AD .

512. Given a triangle ABC in which $a = 7 \text{ m}$, $b = 8 \text{ cm}$ and $c = 5 \text{ cm}$, calculate $\hat{A}$.

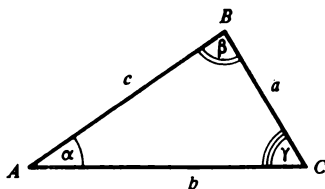


Fig. 63

The Law of Sines

The sides of a triangle are proportional to the sines of the opposite angles.

In the triangle ABC with sides a , b and c and angles α , β and γ (Fig. 63)

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}.$$

513. What elements of a triangle can we calculate using the law of sines?

514. Given a triangle ABC in which a side is a , $\hat{A} = \alpha$, $\hat{B} = \beta$. Find the sides b and c .

515. A diagonal of a parallelogram d long makes angles of 30° and 45° with the sides of the parallelogram. Find the sides of the parallelogram.

516. The bisector of the base angle of an isosceles triangle is 6 cm and cuts the opposite side at the angle of 135° (the angle faces the base). Calculate the base of the triangle.

2.34. REGULAR POLYGONS

A polygon in which all sides are congruent and all angles are congruent is said to be *regular*.

Theorem 1. *A circle can be circumscribed about every regular polygon.*

Theorem 2. *A circle can be inscribed in every regular polygon.*

517. How can we construct a regular n -gon using a circle?

518. 1. Can we say that the following polygons are regular: any rectangle, a rhombus, an isosceles triangle?

2. Can an inscribed polygon have (a) congruent angles but noncongruent sides, (b) congruent sides but noncongruent angles?

519. 1. What point is the centre of a regular polygon?

2. What is the apothem of a regular polygon?

520. 1. What is the degree measure of every interior angle of a regular (a) pentagon, (b) n -gon?

2. What is the degree measure of the central angle of (a) a regular hexagon, (b) a regular n -gon?

3. An apothem and a radius of the circumscribed circle are drawn to one of the sides of a regular n -gon. Find the degree measure of the angle between them.

521. The gauges for the angles of nuts have an angle of (a) 120° , (b) 135° . How many sides do the nuts have?

522. Can the parquet be made of blocks shaped as (a) regular hexagons, (b) regular pentagons?

Formulas for Calculating the Side and the Area of a Regular Polygon

If a is a side of a regular inscribed n -gon, R is the radius of the circle circumscribed about this n -gon, and $\frac{180^\circ}{n}$ is half the central angle (Fig. 64), then

$$a_n = 2R \sin \frac{180^\circ}{n}. \quad (1)$$

Assume that S is the area of a regular polygon, P is its perimeter and r is the radius of the inscribed circle. Then

$$S = \frac{1}{2} Pr. \quad (2)$$

If S_n is the area of a regular n -gon, R is the radius of the circumscribed circle, and $\frac{360^\circ}{n}$ is the central angle, then

$$S_n = \frac{1}{2} nR^2 \sin \frac{360^\circ}{n}. \quad (3)$$

523. 1. Using formula (1), prove that

$$a_6 = R, \quad a_4 = R\sqrt{2}, \\ a_3 = R\sqrt{3}.$$

2. Assuming the radius R of the circumscribed circle to be equal to 1 m, calculate the sides of the following regular polygons: a_8, a_{10}, a_{12} .

524. We have to saw a bar out of a log so that the section of the bar would be a square with a side of 18 cm. What must be the diameter of the log?

525. Calculate the area of a regular n -gon inscribed in a circle of radius $R = 1$ dm if (a) $n = 3$, (b) $n = 4$.

526. The sides of two regular polygons with the same number of sides are a and b . What is the ratio of the perimeters of these polygons and what is the ratio of their areas?

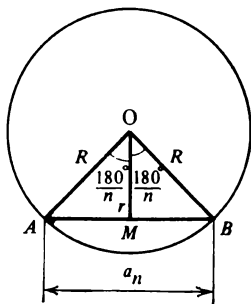


Fig. 64

2.35. THE LENGTH OF A CIRCUMFERENCE AND THE AREA OF A CIRCLE

Assume that C is the length of the circumference, S_c is the area of the circle, R is a radius and D is a diameter. Then we have

$$C = 2\pi R \quad \text{or} \quad C = \pi D,$$

$$S_c = \pi R^2 \quad \text{or} \quad S_c = \frac{\pi D^2}{4}.$$

527. How will the length of the circumference and the area of the circle change when the radius is increased two times?

528. The side of a square is 6 cm. Find the length of the circumference of the circle circumscribed about the square.

529. The side of a regular hexagon is 5 cm. Find the length of the circumference of the circle circumscribed about the hexagon.

530. In a right triangle the median drawn from the vertex of the right angle is 10 cm. Find the length of the circumference of the circle circumscribed about the triangle.

531. By how many centimetres is the length of the circumference of the circle with radius of 1 dm greater than the perimeter of the regular hexagon inscribed into the circle?

532. By how many centimetres will the length of the circumference of the circle increase if the radius of the circle is increased by 5 cm?

533. The side of a square is 4 cm long. Find the area of the circle inscribed in the square.

534. The side of a regular hexagon is 3 cm. Find the area of the circle inscribed in the hexagon.

535. What is larger, the area of a circle constructed on a segment 1 dm long as a diameter or the area of a semi-circle constructed on a segment 2 dm long as a diameter?

536. A square of the largest area has been cut out of a circle. What percentage of the circle was lost as trimmings?

537. Prove that the area S of a circle can be calculated from the formulas (C is the length of the circumference of the circle and d is its diameter) (a) $S = \pi \frac{d^2}{4}$, (b) $S = \frac{C}{2}r$, (c) $S = \frac{C^2}{4} \cdot \frac{1}{\pi}$.

538. Find the radius of the circle whose area (in sq. units) and length of the circumference (in the corresponding linear units) have the same numerical value.

539. Calculate the speed of a car if its drive wheel is 0.7 m in diameter and makes 500 revolutions per minute ($\pi \approx 22/7$).

540. We have to replace two water-supply pipes of the same diameter d by one pipe with the same capacity. Find the diameter D of the new pipe.

541. The area of a square is 1 m^2 . We have a circle with the same area. Prove that the perimeter of the square is larger than the length of the circumference of the circle.

2.36. ELEMENTARY CONSTRUCTIONS ON A PLANE

Carry out the following constructions with the aid of a pair of compasses and a ruler without using a drawing triangle or a protractor.

542. Construct an angle congruent to a given angle.

543. Bisect a given angle.

544. Bisect a given line segment.

545. Construct a point symmetric with respect to the given point X about the l -axis.

546. Construct a triangle ABC being given the three sides a , b and c .

547. Construct a triangle being given two sides b and c and the angle α between them.

548. Construct a triangle being given the side a and two angles β and γ adjacent to that side.

549. Construct a perpendicular to the midpoint of a line segment AB .

550. Construct a right triangle being given the hypotenuse and a leg.

551. Construct a tangent to a given circle which would pass through a given point.

552. Partition a given line segment into several congruent segments.

553. Construct a rectangle which would be of the same area as a given triangle and have a base in common with it.

554. Three line segments are a , b and c long. Construct a fourth segment which would be proportional to the given three segments, i.e. a segment of length x such that the proportion $a \div b = c \div x$ would hold true.

555. Construct a triangle ABC being given the angles α and β and the bisector l_C .

556. Construct a line segment x if $x = \sqrt{ab}$, where a and b are given segments.

557. Construct a line segment x if (a) $x = \sqrt{a^2 + b^2}$, (b) $x = \sqrt{a^2 - b^2}$ ($a > b$), (c) $x = a^2/b$, here a and b are the lengths of the given segments.

CHAPTER 3

Sample Examination Papers

(For Answers see p. 208)

Paper 1

1. Definition of a quadratic equation. Derive a formula for the roots of a quadratic equation.
2. Use the simplest technique to calculate
$$(0.56^3 - 0.56 \cdot 0.44^2) \cdot 8 \frac{1}{3}.$$
3. Calculate the area of a right triangle whose hypotenuse is 12 dm and an acute angle is 30° .

Paper 2

1. Formulate and prove Vieta's theorem.
2. For what values of the variable is the expression $\sqrt{8x-4}$ meaningful?
3. The side of an equilateral triangle is 2 dm. Calculate its altitude.

Paper 3

1. Derive a formula for factoring the quadratic trinomial $ax^2 + bx + c$, $a \neq 0$.
2. Use the simplest technique to calculate the expression $2.02^2 - 2 \cdot 2.02 \cdot 0.52 + 0.52^2$.
3. The hypotenuse of a right isosceles triangle is 6 cm. Find the legs and the altitude to the hypotenuse.

Paper 4

1. The function $y = kx$ and its graph.
2. Calculate the expression $\left(\frac{2}{3}\right)^{-1} + 8 \frac{1}{3} + (-3)^0$.
3. Find the sum of the vectors $\vec{AB}$, $\vec{BM}$ and $\vec{MN}$.

Paper 5

1. The function $y = k/x$ and its graph.
2. Solve the inequality $\frac{x}{x-1} > 0$.
3. Calculate the sum $\sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$.

Paper 6

1. Construct a triangle being given its three sides. Formulate the corresponding criterion of the congruence of triangles.
2. Reduce the fraction $\frac{5a^2 - 7a - 6}{3a - 6}$.
3. A side of a square is 8 cm. Calculate the length of the circumference and the area of the circle inscribed in the square.

Paper 7

1. Construct a triangle being given its two sides and the angle between them. Formulate the corresponding criterion of the congruence of triangles.
2. Find the number if 10 per cent of it constitute $1.3 \cdot 12.8 + 13 \cdot 1.72$.
3. The legs of a right triangle are 0.6 dm and 0.8 dm. Find the sines of the acute angles of the triangle.

Paper 8

1. Construct a triangle being given its side and two adjacent angles. Formulate the corresponding criterion of the congruence of triangles.
2. Prove that the expression $x(x - 2a) + a^2$ is equal to 4 for $x = a + 2$ and for $x = a - 2$.
3. Calculate the obtuse angle between the bisectors of two angles of an equilateral triangle.

Paper 9

1. Formulate and prove the theorem on the properties of an isosceles triangle.
2. Solve the equation $10x \div 3.5 = 0.4 \div 7$.
3. Find the area S_{rh} of a rhombus if its side is 5 cm and an acute angle is 60° .

Paper 10

1. Formulate and prove the theorem on a perpendicular and oblique line drawn from the same point to the same straight line.
2. Simplify the expression $5x^{3/5} \cdot x^{0.2} \cdot x^{1.2}$ and calculate its value for $x = 1.2$.

3. A chord of length $a = 16$ cm is drawn in a circle of radius $r = 10$ cm. Find the distance h between the chord and the centre of the circle.

Paper 11

1. Formulate and prove the theorem on the sum of the angles of a triangle.

2. Find the values of the function $y = x^2 - 6|x| + 5$ for $x = -1, -0.5, 0, 0.5$ and 1 .

3. Find the set of solutions of the inequality $\frac{1}{x} > 1$.

Paper 12

1. Formulate and prove the theorem on an exterior angle of a triangle.

2. Solve the equation $5(x - 2) + 9 = (1 + 2x)(2x - 1)$.

3. Construct a circle which would touch all the sides of a given triangle.

Paper 13

1. Formulate and prove the theorem on the sum of the interior angles of a convex polygon.

2. Find an arithmetic progression with a common difference $d = 1.5$ and find the sum of its first 15 terms.

3. The diagonal of a square $d = 10$ cm. Find a side of the square a and its area S .

Paper 14

1. Formulate and prove the theorem on the properties of a leg lying opposite an angle of 30° .

2. Prove that the inequality $n^2 \geq 8(n - 2)$ holds true for any n .

3. Draw the graph of the function $y = 6 - 2x$. Calculate the area of a right triangle formed by the axes of coordinates and the graph of this function.

Paper 15

1. Formulate and prove the theorem on the median of a triangle.

2. Solve the equation $2^{1/x} = 8$.

3. Draw the graph of the function $y = \frac{1}{2}x^2$. Using the graph find (a) the values of y if $x = 2.5$, (b) the values of x if $y = 6$.

Paper 16

1. Definition of a trapezoid. Formulate and prove the theorem on the median of a trapezoid.
2. Solve the equation $\log(x-3) = 2$. Verify the answer.
3. Without constructing a graph, find whether the points $A(-0.1, -0.005)$ and $B(-2, -50)$ belong to the graph of the function defined by the formula $y = 5x^3$.

Paper 17

1. Definition of a parallelogram. Formulate and prove the theorem on the properties of the sides and the angles of a parallelogram.
2. Find the set of solutions of the equation $\log |2x| = 1$.
3. Solve the equation $\left(3\frac{1}{3} - 1.5\right) \div a = 11$.

Paper 18

1. Formulate and prove the theorem on the diagonals of a parallelogram.
2. Solve the equation $(x+2)\sqrt{x-1} = 0$.
3. Construct the graph of the function $y = |x| + 1$.

Paper 19

1. Definition of a rectangle. Formulate and prove the theorem on the diagonals of a rectangle.
2. Solve the equation $|x-2| = 5$.
3. Calculate $\frac{4^{-6} \cdot 8^{-4}}{2^{-22}}$.

Paper 20

1. Definition of a rhombus. Formulate and prove the theorem on the diagonals of a rhombus.
2. Simplify the expression $\frac{a-b}{a^{1/2}-b^{1/2}} + \frac{b-a}{a^{1/2}+b^{1/2}}$.
3. What is an interior angle of a regular pentagon equal to?

Paper 21

1. The values of the sine and cosine of an angle of 30° .
2. Simplify the expression $2 \log 5 - \left(\log 3 - \frac{1}{8} + \log 8\right)$.

3. Construct the graph of the power function $y = x^n$ for $n = 1$ and $n = 2$.

Paper 22

1. The values of the sine and cosine of an angle of 60° .
2. Find the set of solutions of the inequality $x^2 > 100$.
3. Prove the congruence of the bisectors of an isosceles triangle drawn to its lateral sides.

Paper 23

1. The values of the sine, the cosine and the tangent of an angle of 45° .
2. Find the domain of definition of the function $y = \log |x|$.
3. Prove the congruence of the medians of an isosceles triangle drawn to its lateral sides.

Paper 24

1. Formulate and prove the Pythagorean theorem.
2. Factor the expression $(a - 8)^2 - 16$.
3. Solve the inequality $x^2 < 9$.

Paper 25

1. Derive a formula for the area of a parallelogram.
2. Calculate the expression $\left(9^{0.25} - \left(2^{1.5}\right)^{\frac{2}{3}}\right)\left(9^{0.25} + \left(2^{1.5}\right)^{\frac{2}{3}}\right)$.
3. An acute angle of one right triangle is 52° and that of the other is 38° . Are the triangles similar?

Paper 26

1. Derive a formula for the area of the triangle $S_{\text{tr}} = \frac{ah}{2}$.
2. Construct schematically the graphs of the function $y = kx$, where $k > 0$ and $k < 0$.
3. Solve the equation $\log(2x + 10) = 4 \log 2 + \log 3$.

Paper 27

1. Derive a formula for the area of the triangle $S_{\text{tr}} = \frac{1}{2} ab \sin \hat{C}$.
2. Solve the inequality $x^2(x + 1) > 0$.
3. Construct schematically the graphs of the function $y = k/x$, where $k > 0$ and $k < 0$.

Paper 28

1. Derive a formula for the area of a trapezoid.
2. Construct the graph of the function $y = 2x - 1$ and, using it, find the values of x for which $y = 0$, $y > 0$ and $y < 0$.
3. Find the first and the third term of the geometric progression (b_n) if $b_2 = -16$ and the common ratio $q = 1/2$.

Paper 29

1. Derive a formula for calculating a side of a regular n -gon $a = 2R \cdot \sin \frac{180^\circ}{n}$.
2. Reduce the fraction $\frac{\sqrt{x-4}}{x-16}$.
3. Find the set of solutions of the two-sided inequality $5.4 \leq 5x - 2 \leq 7.6$.

Paper 30

1. Using the formula $a_n = 2R \sin \frac{180^\circ}{n}$, derive formulas for calculating a_6 , a_4 and a_3 .
2. Solve the equation $x^4 - 29x^2 + 100 = 0$.
3. Use the coordinate plane to show the approximate positions of the graphs of the function $y = kx + l$ if (a) $k > 0$ and $l > 0$, (b) $k < 0$ and $l < 0$.

CHAPTER 4

Questions and Problems for an Oral Examination

(For Answers see p. 213)

2.37. ARITHMETIC AND ALGEBRA

EXPRESSIONS

Calculate (1-14) using the simplest techniques.

1. (a) $16 \cdot 125 \cdot 250$, (b) $64 \cdot 2.5 \cdot 12.5$.
2. (a) $4.85 - 2 \frac{4}{9} + 15 \frac{9}{20} \div 3$, (b) $215^2 - 214^2$.
3. (a) $3.7^2 + 2 \cdot 3.7 \cdot 6.3 + 6.3^2$, (b) $317^2 + 217^2 - 634 \cdot 217$.
4. $2.78 \cdot 36.1 + 36.1 \cdot 7.22 - 26.1 \cdot 5.64 - 26.1 \cdot 4.36$.
5. $12.8^3 - 12.8 \cdot 7.2^2 - 448 \cdot 3.2$.
6. $\left(-2 \frac{1}{5}\right) \left(-\frac{1}{5}\right) \left(-1 \frac{4}{11}\right) (-5) + (1 - 7.52 \times (-8.47)) \left(0.4 - \frac{2}{5}\right)$.
7. $3 \frac{11}{13} \cdot 0.26 + \frac{2.4 \cdot 2.73}{24 \cdot 0.273} + \frac{7.83 \div 31.6}{78.3 \div 316}$.
8. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \frac{1}{5 \cdot 6} + \frac{1}{6 \cdot 7}$.
9. (a) $\sqrt{3.13^2 - 3.12^2}$, (b) $\sqrt{52^2 + 39^2}$.
10. $\sqrt{(12.1^2 - 8.1^2) \div (0.25^2 + 0.25 \cdot 19.95)}$.
11. $2a^3b^2(5a + 7b)(4a - 8b)$ for $a = 2.5$, $b = 1.25$.
12. $\frac{(a^3 + 2a)^2}{a^4 + 4a + 4}$ for $a = 1 \frac{1}{3}$.
13. $\sqrt{a^3 - a^2b - ab^2 + b^2}$ for $a = 0.78$, $b = 0.22$.
14. $\sqrt{25a^2 + 16b^2 + 20ab}$ for $a = 0.4$, $b = -0.5$.

15. When a number a is divided by a certain number, the quotient is b and the remainder is r . Find the divisor.

16. Find two numbers if their sum is a and the difference between them is b .

17. Find two numbers if their sum is c and their ratio is p/q .

18. The set of values of x is $\{-28, -27, -26, \dots, 5, 6\}$. The set of values of y is $\{-1, -2, -3, \dots, -15, -16\}$. Find the least and the greatest value of $x + y$.

19. Indicate a number included between the numbers (a) 1.7 and 1.8, (b) $\frac{3}{8}$ and $\frac{1}{2}$, (c) $\frac{4}{7}$ and $\frac{5}{8}$, (d) -0.76 and -0.7 .

20. One worker can do a job in two hours and the other can do the same job in three hours. How many hours will the two workers need to do the job if they work together?

21. Half way through the journey a passenger began to look out of the window and continued to do so until the remainder of the journey was half as far as the distance covered while he was looking out of the window. How much of the journey had the passenger to go?

22. Prove that the sum of a two-digit number and the number written by the same digits in reverse order is divisible by 11.

23. Using the rule of calculating correct digits, calculate the area S of a rectangle with sides $a \approx 5.75$ dm and $h \approx 0.28$ dm.

PER CENT

24. Find (a) 50% of 17, (b) 25% of 36, (c) 12.5% of 24, (d) 75% of 48, (e) 50% of 40.

25. What is greater, 27% of 39 or 39% of 27?

26. 25% of a number constitute 0.25. Find the number.

27. On the first day the shop sold 50% of the available goods and on the second day, 25% of the remainder. What percentage of the available goods was not sold?

28. The length of a side of a square was increased by 50%. By how many per cent did the area of the resulting square increase?

29. The length of a rectangle was increased by 20% and the width was decreased by 20%. How did the area of the rectangle change?

30. A person deposited a roubles in the savings bank at the beginning of a year. A year later the sum increased by 3%. Another year later the resulting sum increased by another 3%. How much will the person have two years later?

POWERS AND ROOTS

31. Find the values of the following expressions:
 (a) $0.2^5 \cdot 5^6$, (b) $0.25^4 \cdot 8^3$, (c) $2.5^5 \cdot 8^4 \cdot 0.4^6 \cdot 0.125^4$,
 (d) $(-1)^{2n-1} (-1)^{1-n} (-1)^{n+2}$, (e) $2^0 + 2^{-1} + 2^{-2}$,
 (f) $2^{-1} + (-2)^{-1}$.

32. What is greater, (a) -3^2 or $(-3)^2$, (b) 0.5^{3+3} or $0.5^{3 \cdot 3}$, (c) 118^2 or $119 \cdot 117$, (d) 117^2 or $115 \cdot 119$?

33. Represent the numbers 2 and 5 as (a) square roots, (b) cubic roots.

34. Find two natural numbers which include the number (a) $\sqrt{10}$, (b) $\sqrt[3]{10}$, (c) $\sqrt[4]{20}$, (d) $\sqrt[3]{130}$ between them.

35. Find the value of (a) $\sqrt{a^2}$ if $a < 0$, (b) $\sqrt{(a-3)^2}$ if $a < 3$, (c) $\sqrt{(x-2)^2} + \sqrt{(x-4)^2}$ if $2 < x < 4$.

36. Take the factor outside of the root sign in the following expressions: (a) $\sqrt{25x^2}$, (b) $\sqrt{-36x^3}$.

37. Put the indicated factor under the root sign in the following expressions; (a) $b\sqrt{7}$ if $b > 0$, (b) $a\sqrt{6}$ if $a < 0$.

38. What is greater, (a) $\sqrt[3]{3}$ or $\sqrt{2}$, (b) $2\sqrt[3]{4}$ or $3\sqrt[3]{1.5}$?

39. Using the formula for the square of a binomial, extract a square root of the following binomials:

(a) $\sqrt{3+2\sqrt{2}}$, (b) $\sqrt{7-4\sqrt{3}}$.

40. Calculate (a) $\sqrt{3-\sqrt{5}} \cdot \sqrt{3+\sqrt{5}}$,
 (b) $\sqrt{5+2\sqrt{6}} \sqrt{5-2\sqrt{6}}$, (c) $(\sqrt{3}+\sqrt{2})^2 - (\sqrt{3}-\sqrt{2})^2$.

41. Reduce the following fractions: (a) $\frac{3}{\sqrt{3}}$,

$$(b) \frac{a^2-5}{\sqrt{5}-a}, \quad (c) \frac{x-a}{x^{1/2}-a^{1/2}}, \quad (d) \frac{a^{1/2}-b^{1/2}}{a^{1/4}-b^{1/4}},$$

$$(e) \frac{x+y+2x^{1/2}y^{1/2}}{x-y}.$$

42. Calculate (a) $y = 3x^{1/5} \cdot x^{0.6} \cdot x^{1/2}$ for $x = 10$,
 (e) $\sqrt[3]{\sqrt{8}} \cdot \sqrt[4]{4}$.

EQUATIONS AND INEQUALITIES

43. Find the values of x if (a) $|x| = 3$, (b) $|x| < 3$,
 (c) $|x| > 3$, (d) $|x-3| = 2$, (e) $|x-1| < 3$, (f) $|x+2| \leq 5$, (g) $|x-2| \geq 4$.

44. Solve the following equations: (a) $\frac{x-1}{x-1} = 1$,
 (b) $\frac{2x+3}{2x+3} = 1$, (c) $\sqrt{x^2} = 3$, (d) $\sqrt{y^2} = -4$, (e) $\sqrt{(x-2)^2} = 5$, (f) $\sqrt{x^2} = x$, (g) $\sqrt{y^2} = -y$, (h) $\frac{x^2}{x} = 0$, (i) $\frac{(x-5)(y-1)^2}{x-1} = 0$,
 (j) $\frac{(x-3)(x+2)}{x+2} = 1$, (k) $[x] = 2$, (l) $\{x\} = 0.2$.

45. Solve the following equations: (a) $(x+2)\sqrt{x} = 0$, (b) $(x+1)\sqrt{x-1} = 0$, (c) $(x-2)(x+4)\sqrt{x} = 0$,
 (d) $4x^2 + (x-1)^0 - 5 = 0$, (e) $(x^2+7)|x-3| = 0$, (f) $|x| = x^2$, (g) $x + \frac{1}{x} = 2\frac{1}{2}$, (h) $\sqrt{x} + \frac{1}{\sqrt{x}} = 3\frac{1}{3}$, (i) $1921x^2 - 3902x + 1981 = 0$, (j) $1917x^2 + 3896x + 1979 = 0$.

46. Solve the following equations: (a) $x^2 + y^2 = 0$,
 (b) $(x-3)^2 + (y-4)^2 = 0$, (c) $(x-5)^2 + (2x+5y)^2 + |2x+y-z| = 0$, (d) $x^3 + x^2 + x + 1 = 0$, (e) $x^3 - x = 0$.

47. At what points does the parabola $y = x^2 + x - 6$ cut the x -axis, the y -axis?

48. At what points does the graph of the function $y = (x-5)(x-1)(2x+1)(x+2)$ cut the x -axis, the y -axis?

49. Solve the equation (a) $S = \frac{1}{2}ah$ for a , (b) $S = \frac{1}{2}a^2 \sin \alpha$ for $\sin \alpha$, (c) $S = a^2h$ for a ($a > 0$), (d) $S =$

$(a_1 + a_n) n/2$ for a_n , (e) $S = \pi R^2$ for R ($R > 0$), (f) $V = \pi R^2 h/3$ for R ($R > 0$), (g) $S = v_0 + gt$ for v_0 and t , (h) $n = mc$ ($T - t$) for each of the variables m , c , T and t .

50. Solve the following inequalities: (a) $x^2 > 0$, (b) $x^2 < 0$, (c) $x^2 \geq 0$, (d) $x^2 > 16$, (e) $x^2 < 4$, (f) $(x-2)^2(x+1) > 0$, (g) $\frac{x-2}{-3} > 0$, (h) $\frac{2x-1}{\log 0.1} < 0$, (i) $\sqrt{x} > -1$, (j) $\sqrt{-x} > x$.

FUNCTIONS AND GRAPHS

51. Find the inverse functions of the following functions and construct their graphs in the same system of coordinates:

(a) $y = 2x - 3$, (b) $y = 4 - 2x$.

52. Construct the graphs of the following functions: (a) $y = x^0$, (b) $y = (x-2)^0$; (c) $y = 2x + (x+1)^0$, (d) $y = \sin^2 x + \cos^2 x$, (e) $y = |x| + 1$, (f) $y = x/|x|$, (g) $y = x|x|$, (h) $|y| = x + 1$.

53. Draw a quadrilateral whose vertices are the points of intersection of the straight lines $y = x + 2$, $y = x - 2$, $y = -x + 2$ and $y = -x - 2$. What kind of a quadrilateral is this?

54. Given that the graph of the function $y = ax + 4$ passes through the point $(-2, 5)$, find a and construct the graph.

55. The graph of the function $y = x^2$ was transferred (a) three units downwards, (b) four units to the right, (c) three units downwards and four units to the right. What will the equation of each of these graphs become?

56. Draw the graphs of the following functions and find their greatest and least values:

(a) $y = (x + 3)^2$, (b) $y = 7 - x^2$, (c) $y = x^2 + 2x - 3$.

57. Using graphical means, solve the equations (a) $x^2 = x + 2$, (b) $x^2 = |x|$, (c) $\sqrt{x} = 6 - x$.

58. Using graphical means, solve the following simultaneous equations:

(a) $\begin{cases} x^2 - y = 0, \\ 2x - y = 0, \end{cases}$ (b) $\begin{cases} xy - 6 = 0, \\ 2x - y = 0, \end{cases}$ (c) $\begin{cases} x^2 - 2y = 0, \\ 0.5x - y = 0. \end{cases}$

59. Solve the following simultaneous equations:

$$\begin{aligned} \text{(a)} \quad & \begin{cases} |x| = 2, \\ x + 2y = 4, \end{cases} \quad \text{(b)} \quad \begin{cases} 2x^2 - 3xy = 8, \\ 2x - 3y = 1, \end{cases} \quad \text{(c)} \quad \begin{cases} y(x - 2) = 0, \\ 2x + 3y = 13, \end{cases} \\ \text{(d)} \quad & \begin{cases} (x - y + 2)^2 + y = 8, \\ (x - y + 2)^2 + x = 7. \end{cases} \end{aligned}$$

60. On a coordinate plane find the set of points whose coordinates satisfy the conditions (a) $(x - y)(y + x^2) = 0$, (b) $(x - 2)(x^2 + y^2) = 0$, (c) $(x + y - 1)((x + y)^2 + y^2) = 0$.

PROGRESSIONS

61. 1. An arithmetic progression begins with $a, b, \dots$. Find its next two terms.

2. A geometric progression begins with $b, b^3, \dots$. Find its next two terms.

3. Given the second term $a_2 = m$ and the third term $a_3 = n$ of an arithmetic progression, find a_1 .

62. 1. The second term of a geometric progression $b_2 = m$ and the third term $b_3 = n$. Find the first term b_1 .

2. Given that $28 + 26 + \dots + x = 0$, find the number of terms on the left-hand side of the equation. What does the last term x equal to?

63. Find the 15th term of the arithmetic progression (a) $1 + \sqrt{3}, 1 + 3\sqrt{3}, 1 + 5\sqrt{3}, \dots$, (b) $\sqrt{2} - 1, \sqrt{2} - 4, \sqrt{2} - 7, \dots$.

64. Given the numbers $\sqrt{2}$ and $3\sqrt{2}$, insert between them a positive number such that the three resulting numbers be successive terms of a geometric progression.

65. A geometric progression is defined by the formula for the n th term: (a) $b_n = 5 \cdot 2^{n-1}$, (b) $b_n = 5\left(\frac{1}{2}\right)^{2n-1}$. Find b_{n-1} and b_{n+1} .

LOGARITHMS

66. Given $10^x = 3$, find (a) 10^{x+1} , (b) 10^{-x} , (c) 10^{x-1} , (d) 10^{2x} .

67. For what values of x do the following expressions have sense: (a) $\log(-x)$, (b) $\sqrt{-x}$, (c) $\log(1 - x)$, (d) $\sqrt{1 - x}$, (e) $\log \sqrt{x^2}$, (f) $\sqrt{x^2}$, (g) $\frac{1}{\log x}$, (h) $\frac{1}{\sqrt{x}}$?

68. Are the following expressions meaningful:
 (a) $\sqrt{\log 0.1}$, (b) $\log \sqrt{0.1}$, (c) $\log \cos 0^\circ$?
69. Calculate (a) $10^{\log 8}$, (b) $10^{\log 2} + 10^{\log 3}$,
 (c) $10^{\log 2 + \log 3}$, (d) $10^{\log 2 + 2}$, (e) $10^{\log 2} + 10^2$, (f) $10^{\log 3 - 1}$,
 (g) $10^{\log 3} + 10^{-1}$, (h) $10^{-\log 2}$.
70. Do the numbers $\log 0.01$, $\log 1$, $\log 100$, $\log 10\,000$ form an arithmetic progression?
71. Solve the following equations: (a) $\log^2 x = 1$,
 (b) $\log x^2 = 1$, (c) $\log^2 x^2 = 1$, (d) $10^{\log x^2} + x^2 = 8$.
72. Construct the graphs of the functions (a) $y = 10^{\log x}$,
 (b) $y = 10^{\log x^2}$.
73. Given that $\log a = 0.1$, calculate (a) $\log (10a)$,
 (b) $\log \frac{a}{10}$, (c) $\log \frac{10}{a}$, (d) $\log a^2$, (e) $\log^2 a$.

2.38. GEOMETRY

74. The distance from point A to point B is 10 km, and that from B to point C is 6 km. What are the greatest and the least distance from A to C ?
75. Draw two rays AB and CD , so that their intersection is (a) a line segment AC , (b) a ray CD .
76. Given three points A , B and C such that $|AB| + |BC| = |AC|$, $|AC| = a$. Find the distance between the midpoints of the segments AB and BC .
77. How many different line segments can be defined by four points taken pairwise?
78. Given a circle (O, r) , find the figure which is formed by the set of all points X of a plane for which (a) $|OX| \leq r$, (b) $|OX| < r$, (c) $0 < |OX| \leq r$.
79. The diameter of a circle is 5 cm larger than its radius. What is the diameter equal to?
80. A straight angle is divided into three equal parts. Prove that the bisector of the middle angle is perpendicular to the sides of the straight angle.
81. Can the difference between two adjacent angles be equal to 90° ?
82. Can the sum of two vertical angles be equal to 180° ?
83. Calculate the convex angle ABC if $\widehat{ABD} = 72^\circ$ and $\widehat{CBD} = 40^\circ$. Consider all cases.
84. Given a point A on the circumference of a circle, construct a chord whose endpoints are at the same distance from that point.

85. The coordinates of the points A and B of a straight line are equal to -2 and 5 respectively. What can be the coordinates of the point C of the line AB if it is known that the point C lies between the points A and B ; the point C does not lie between A and B ? What is the length of the segment AB ?

86. Prove that the line segment connecting the vertex of a triangle with any point of its base is smaller than half the perimeter of the triangle.

87. Draw two figures whose union is a circle and intersection is a triangle; whose union is a triangle and intersection is a circle.

88. How many axes of symmetry can (a) an angle, (b) an isosceles triangle, (c) an equilateral triangle possess?

89. Given a right triangle, construct a triangle which is symmetric with respect to it about a straight line containing the hypotenuse.

90. Given an isosceles triangle, construct a triangle which is symmetric with respect to it about the midpoint of a lateral side. What are the positions of the sides of the resulting quadrilateral?

91. In the quadrilateral $ABCD$ (Fig. 65) $|AB| = |BC|$ and $\widehat{BAD} = \widehat{BCD}$. Prove that $|AD| = |DC|$.

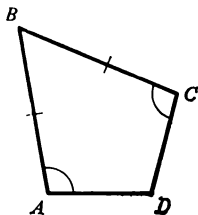


Fig. 65

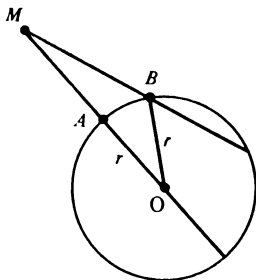


Fig. 66

92. 1. Why is the hypotenuse of a right triangle larger than its leg?

2. Can a lateral side of an isosceles triangle be equal to a half of the base?

93. A point X is taken in the interior of the triangle ABC . Prove that the sum of the distances from this point to the vertex of the triangle is larger than the sum of the distances from that point to the sides of the triangle.

94. Given an obtuse triangle ABC , construct the projections of its vertices and sides onto the straight lines AB , BC and AC .

95. We are given a circle (O, r) (Fig. 66). Points A and B lie on its circumference and a point M lies outside of it, the points M , A and O lying on the same straight line. Prove that $|MA| < |MB|$.

96. A point A lies on the circumference (O, r) . How can a tangent be drawn through this point and why is this tangent unique?

97. We are given congruent triangles ABC and DEF . It is known that $[AB] \cong [DE]$, $[AC] \cong [DF]$. Prove that the bisector BK is congruent to the bisector EL ($K \in [AC]$, $L \in [DF]$).

98. Is the following statement true: "In two triangles congruent sides lie opposite congruent angles"?

99. Why is at least one angle of a scalene triangle not larger than 60° ?

100. 1. Why is the exterior angle of a triangle larger than each interior angle not adjacent to that exterior angle?

2. Can an exterior angle of a triangle be smaller than its interior angle?

3. A point M is taken inside the triangle ABC and is connected to the vertices A and C . Prove that $\widehat{AMC} > \widehat{ABC}$.

101. Can the bisectors of two angles of a triangle be mutually perpendicular?

102. Prove that if a leg of a right triangle is equal to a half of the hypotenuse, then the angle lying opposite that leg is 30° .

103. An angle of a right triangle is 30° and the hypotenuse is 8 cm. Find the line segments into which the altitude drawn from the vertex of the right angle divides the hypotenuse.

104. Why does an acute angle always lie opposite the smaller side of a triangle?

105. The vertex angle of an isosceles triangle is α . The length of a lateral side exceeds that of the base. What is the range of variation of the angle α ?

106. The larger angle lies opposite the larger side of a triangle. Is there a direct proportionality here?

107. The legs of a right triangle are a and b . Perpendiculars are drawn from the midpoint of the hypotenuse to the legs. Find the perimeter of the resulting quadrilateral.

108. Two altitudes 6 cm long are drawn in an equilateral triangle. Find the projection of one of the altitudes onto the other.

109. Can the sum of the angles of a convex polygon be equal to a hundred right angles?

110. Why are the diagonals of a parallelogram, which is different from a rectangle, not congruent?

111. An angle of a parallelogram is 30° . What is the angle between its altitudes equal to?

112. The smaller side of a rectangle is a . What is its perimeter if the bisector of one of its angles bisects the larger side?

113. Is there a rectangle which the diagonals divide into four equilateral triangles?

114. Is the straight line which contains a diagonal of a rectangle its axis of symmetry?

115. Is there a rhombus in which (a) a diagonal is congruent to a side, (b) an altitude is congruent to a side, (c) a diagonal is equal to double its side?

116. Can a diagonal of an isosceles trapezoid be congruent to its leg?

117. Can one of the diagonals of a trapezoid be (a) the bisector of the angle at the smaller base, or (b) the bisector of the angle at the larger base?

118. Enumerate the common properties of a trapezoid and a parallelogram.

119. Does a trapezoid have (a) an axis of symmetry, (b) a centre of symmetry?

120. Can the median of a trapezoid pass through the point of intersection of its diagonals?

121. We are given two parallel line segments. Can we consider them to be homothetic? How many centres of homothety do they have?

122. The diagonals of a parallelogram divide it into four triangles. Find pairs of homothetic triangles among them. What is the homothetic ratio equal to?

123. A circle is circumscribed about an equilateral triangle and a circle is inscribed in that triangle. Where

does the centre of similitude of these circles lie? What is the homothetic ratio equal to?

124. Given $[AB] \perp [BC]$, $[BD] \perp [AC]$ (Fig. 67), find similar triangles and prove that they are similar.

125. In the triangle ABC (Fig. 68) a line segment BD

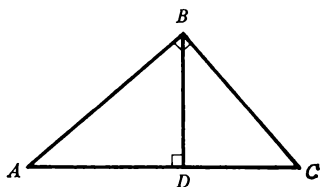


Fig. 67

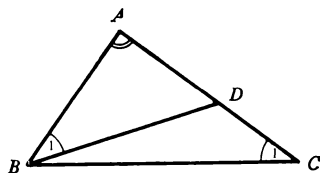


Fig. 68

is drawn such that $\widehat{ABD} = \widehat{BCA}$. Find a pair of similar triangles and write the proportionality of their corresponding sides.

126. A circle is circumscribed about the triangle ABC (Fig. 69). The bisector of the angle A cuts the side BC at a point D and the circle at a point E . Prove that the triangle ABE is similar to the triangle ADC .

127. What is the condition for the similarity of rhombi?

128. The legs of a right triangle are 6 cm and 8 cm. Find the median drawn to the hypotenuse.

129. An altitude 4 cm long is drawn from the vertex of the right angle of a right triangle. One of the resulting segments of the hypotenuse is 8 cm. Find the diameter of the circle circumscribed about the triangle.

130. Find the area of a right triangle if the radius of the circumscribed circle is 3 cm and the altitude drawn to the hypotenuse is 2 cm.

131. The base of an isosceles triangle is 24 cm and its lateral side is 15 cm. A student made an error in the calculations since he considered the area of the triangle to

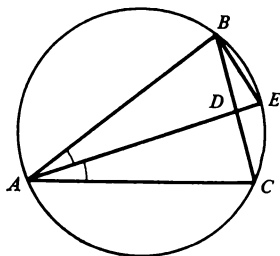


Fig. 69

be equal to half the product of the base by a lateral side. By how many per cent is the resulting area larger than the true one?

132. What is the simplest way of dividing any triangle into any number of triangles of equal areas?

133. An altitude BD ($D \in [AC]$) is drawn in the triangle ABC . A point M bisects that altitude. Find the area of the triangle AMC if the area of the given triangle is 1 dm^2 .

134. A median BD ($D \in [AC]$) is drawn in the triangle ABC . A point M bisects the median. Find the area of the triangle AMD if the area of the given triangle is S .

135. A lateral side of a triangle is divided in the ratio $1 : 3$ (reckoning from the vertex) and a straight line is drawn through the point of division parallel to the base. Find the ratio of the area of the cut-off triangle to the area of the given triangle and that of the area of the cut-off trapezoid to the area of the cut-off triangle.

136. Is there a regular polygon in which the interior angle and the central angle are congruent?

137. We are given a regular hexagon each of whose vertices is connected to the centre by a line segment. Indicate pairs of triangles which are symmetric about the larger diagonal and pairs of triangles symmetric about the centre of the triangle.

138. Three points divide a circle with the radius of 1 dm in the ratio $1 : 2 : 3$. Calculate the lengths of the chords which successively connect these points.

139. The legs of a right triangle are 3 cm and 4 cm . Find the lengths of the circumferences and the areas of the circles constructed on the hypotenuse and the legs (the hypotenuse and the legs serve as diameters).

140. The legs of a right triangle are 5 cm and 12 cm . Find the sine and cosine of the larger acute angle.

141. In the acute-angled triangle ABC $\hat{A} = 45^\circ$. The altitude CD divides the side AB into segments $|AD| = m$ and $|DB| = n$. Find (1) $\tan \hat{B}$, and (2) $\sin \hat{B}$.

142. In the acute-angled triangle ABC the altitude CD ($D \in [AB]$) makes angles α and β with the sides CA and CB . Calculate $|AD|$, $|BD|$, $|AC|$ and $|CB|$ if $|CD| = 1$.

143. An altitude CD is drawn to the hypotenuse AB of the right triangle ABC . Prove that (1) the ratio of similitude of the triangles ACD and BCD is $\tan \hat{B}$, (2) the ratio of similitude of the triangles ACD and ABC is $\sin \hat{B}$, and (3) $h = \frac{ab}{c}$, where a and b are the legs, c is the hypotenuse, and h is an altitude drawn to the hypotenuse.

CHAPTER 5

More Difficult Problems

(For Answers and Solutions see p. 227)

1. Use the simplest technique to calculate

$$(a) \frac{5932 \cdot 6001 - 69}{5932 + 60001 \cdot 5931} ; (b) \frac{254 \cdot 399 - 145}{399 \cdot 253 + 254}.$$

2. Prove that the sum of three successive powers of the number 2 is divisible by 7.

3. Prove that when the square of an odd number is divided by 8, the remainder is unity.

4. Prove that (a) $m^3 + 11m$, where $m \in N$, is divisible by 6. (b) $n^3 + 3n^2 + 8n$, where $n \in N$, is divisible by 6.

5. Prove that if $\frac{a}{b} = \frac{b}{c} = \frac{c}{a}$, then $a = b = c$.

6. What is the smallest number by which 2100 must be multiplied for the resulting product to be an exact cube of some number?

7. Calculate (a) $x^4 + y^4 - x^3y^2 - x^2y^3 + 16xy$ if $x + y = 2$, (b) $x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$ if $x = 11$.

8. Calculate the following sum: $\frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132}$.

9. Prove that the value of the expression $\sqrt{11+6\sqrt{2}} + \sqrt{11-6\sqrt{2}}$ is a natural number.

10. Simplify the following expressions: (a) $\sqrt{4+\sqrt{15}} + \sqrt{4-\sqrt{15}} - \sqrt{12-4\sqrt{5}}$, (b) $\sqrt{2+\sqrt{3}} \times \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}} \times \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$, (c) $(\sqrt[6]{9+4\sqrt{5}} + \sqrt[3]{2+\sqrt{5}}) \cdot \sqrt[3]{2-\sqrt{5}}$, (d) $(\sqrt[6]{7+4\sqrt{3}} + \sqrt[3]{2+\sqrt{3}}) \cdot \sqrt[3]{2-\sqrt{3}}$.

11. Remove the root signs in the denominator of the fraction $\frac{4+2\sqrt{3}}{\sqrt{6}-\sqrt{3}+\sqrt{2}-1}$.

12. Transform the following expressions:

(a) $\sqrt{x^2+6x+9} + \sqrt{x^2-2x+1} + \sqrt{x^2+14x+49}$,
 (b) $\sqrt{a+2\sqrt{a-1}} + \sqrt{a-2\sqrt{a-1}}$ for $1 \leq a \leq 2$.

13. Simplify the expression $\left(\frac{a\sqrt{a}+b\sqrt{b}}{\sqrt{a}+\sqrt{b}} - \sqrt{ab}\right) \times \left(\frac{\sqrt{a}+\sqrt{b}}{a-b}\right)^2$.

14. Find the value of the expression $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots + \frac{1}{\sqrt{99}+\sqrt{100}}$.

15. Prove the identity $a^3 + b^3 + c^3 = 3abc$ if it is known that $a + b + c = 0$.

16. Factor $a^3 + b^3 + c^3 - 3abc$.

17. Prove that the equality $a^2 + b^2 + c^2 = ac + bc + ab$ implies that $a = b = c$.

18. Solve the equations (a) $\frac{1}{x^2+2x} - \frac{1}{(x+1)^2} = \frac{1}{12}$,
 (b) $x(x+1)(x+2)(x+3) = 5040$, (c) $(x^2 - a^2)^2 = 4ax + 1$,
 (d) $\sqrt{x-1} + \sqrt{x} = \frac{1}{x^2}$.

19. Find x , y and z from the equation

$$x^2 + y^2 + 0.25z^2 - 2x + 2y - z + 3 = 0.$$

20. Find x and y from the equation

$$x^2 + y^2 + 10 = 2\sqrt{2}x + 4\sqrt{2}y.$$

21. (a). Solve the equation $xy = x + y$ in integers.

(b). Find several pairs of positive numbers the sum of which is equal to the product.

22. Prove that the equation $x^2 + y^2 = 1983$ has no solutions in integers.

23. Solve the equations (a) $|x - 2| + |x - 4| = 3$, (b) $(x + 1)(|x| - 1) = 1/2$.

24. Solve the equations (a) $x^2(x - 2)^2 + 2x^3 - 11x^2 + 20x - 12 = 0$, (b) $\frac{1}{4}x^2 + 1 - \frac{4}{x} = 0$, (c) $\frac{x^3}{c} + \frac{x^2}{b} + \frac{x}{a} = bc$, where $ab^2 + 1 = 0$, $abc \neq 0$.

25. Find (a) the domain of definition of the function $y = \frac{1}{x^3 - 5x^2 + 4x}$, (b) the domain of variation of the function $y = \frac{x - 3}{x - 2}$.

26. Solve the following simultaneous equations:

$$\begin{cases} |x| + 3y = 7, \\ 2x + |y - 1| = 3. \end{cases}$$

27. We have five locks and 5 keys to them. How many trials must we make in the least favourable cases to establish the correspondence between the locks and the keys?

28. The entrance fee to the stadium is 1 rouble and 50 kopecks. When the price of a ticket was lowered, the number of spectators increased by 50 per cent and the box-office returns increased by 25 per cent. How many kopecks cheaper did a ticket become?

29. The cost of twenty-five oranges in roubles is equal to the number of oranges that can be bought for a rouble. How many oranges can you buy if you have three roubles?

30. A bar of silver and zinc alloy with the mass of 3.5 kg contained 76 per cent of silver. It was alloyed with another bar and the resulting bar, whose mass was 10.5 kg, contained 84 per cent of silver. What percentage of silver did the second bar contain?

31. We have two alloys of gold and silver. In the first alloy the amounts of these metals are in the ratio $2/3$ and in the second they are in the ratio $3/7$. What amount of each alloy must be taken in order to obtain eight kilograms of a new alloy in which gold and silver would be in the ratio $5/11$?

32. An alloy consists of two metals, zinc and copper, which are in the ratio $1/2$, and another alloy consists of the same metals in the ratio $2/3$. How many parts of each alloy must be taken to obtain a third alloy which would contain the same metals in the ratio $5/8$?

33. One motor-cyclist left A for B and travelled at a speed of 60.9 km/h. Two hours later another motor-cyclist left A for B and travelled at a speed of 40.6 km/h. How long will it take the first motor-cyclist to cover the way from A to B if the second one arrived at B seven hours later than the first?

34. 1. A car travelled from A to B at a speed $v_1 = 60$ km/h and travelled back at a speed $v_2 = 40$ km/h. What is the average speed of the car?

2. Find the speed of the train being given that it needs five seconds to pass by a stationary observer and twenty seconds to pass by a station platform 420 m long. What is the length of the train?

35. Solve the simultaneous equations
$$\begin{cases} |x| + 3y = 7, \\ 2x + |y - 1| = 3. \end{cases}$$

36. Prove the inequalities (a) $3^{34} > 2^{51}$, (b) $38^{303} > 57^{202}$.

37. Prove the following inequalities (a) $a^2 + b^2 + c^2 \geq ab + bc + ac$, (b) $\frac{x^2}{1+x^4} \leq \frac{1}{2}$, (c) $a^2 + b^2 + c^2 + d^2 + e^2 \geq ab + ac + ad + ae$, (d) $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} > \sqrt{3}$; (e) $\frac{a^2+5}{\sqrt{a^2+4}} > 2$.

38. Prove that if $a > b > c > 0$ and a , b and c are three successive numbers of an arithmetic progression, then (a) $a^2 + c^2 > 2b^2$, and (b) $a^3 + c^3 > 2b^3$.

39. Prove that if the values of the angles of a convex pentagon form an arithmetic progression, then each angle is larger than 36° .

40. Find the domain of definition of the function

$$y = \sqrt{x} + \sqrt{3-x} + \sqrt{x^2-4x}.$$

41. Solve the inequality $x^2(2x-3)(x^2-2x+5) > 0$.

42. Solve the inequalities (a) $x^2 - 3|x| + 2 < 0$,

(b) $|x+2| > |x|$, and (c) $\left| \frac{x-2}{x+1} \right| > 1$.

43. Find the least value of the function $y = \frac{1}{\sqrt{1-x}} + \frac{1}{\sqrt{1+x}}$.

44. Construct the graph of the function defined by the formula (a) $y = x^2 - 2|x|$ and then use graphical means to solve the inequalities $x^2 - 2|x| > 0$ and $x^2 - 2|x| < 0$, (b) $y = x^2 - 3|x| + 2$ and then use it to solve the inequality 42 (a).

45. Construct the graphs of the functions (a) $y = \sin^2 \log x + \cos^2 \log x$, (b) $y = \frac{x \sqrt{\frac{x^4-2x^2+1}{4x^2}+1}}{x^2+1}$, (c) $y = \sqrt{x+2\sqrt{x-1}} + \sqrt{x-2\sqrt{x-1}}$, where $x \in [1, 2]$.

46. Construct the set of points of a plane whose coordinates satisfy the equations (a) $|x| + |y| = 2$, (b) $x^2 + y^2 = 0.25^{1/x}$, (c) $x^2y^2 - x^2 - y^2 + 1 = 0$, and (d) $x^2 + y^2 - 2xy - x^3y - xy^3 = 0$.

47. Construct the set of points of a plane whose coordinates satisfy the following simultaneous inequalities $\begin{cases} |x| - y \geq 1, \\ 3x + y < 3. \end{cases}$

48. Solve the equations (a) $\log^2 x^2 = \log 1000$, (b) $\log x^2 + \log(-x) = 3$.

49. Find the values of x for which the following equations hold true: (a) $\sqrt{x} = x$, $\sqrt{x} > x$, $\sqrt{x} < x$, (b) $\log \sqrt{x} = \sqrt{\log x}$, $\log \sqrt{x} > \sqrt{\log x}$, $\log \sqrt{x} < \sqrt{\log x}$.

50. In the triangle ABC , where $|AC| = 2$ m, an altitude BD is drawn ($D \in [AC]$), $|BD| = \sqrt{2}$ m, with $|AD| = |BC|$. Find the length of the side AB .

51. In the triangle ABC $|AB| = 6$ m, $|AD| = 4$ m and $|BC| = 4$ m. Find the altitude $|CD|$ ($D \in [AB]$) and the area of the triangle.

52. A point D is the midpoint of the hypotenuse AB of a right triangle ABC . The perpendicular bisector of $[AB]$ cuts the leg AC at a point M and the extension of the leg BC at a point N . Find $|AB|$ if $|DM| = a$ and $|MN| = b$.

53. In the trapezoid $ABCD$ the base AD is 2 m long, and the base BC and each of the nonparallel sides AB and CD are 1 m long. Find the length of the diagonal of the trapezoid.

54. In the rectangle $ABCD$ $|AD| = a$ and $|AB| = b$ ($a > b$). A point M is taken on the side AD such that $\angle BMA \cong \angle CMB$. Find $|MD|$.

55. A point M is taken on the chord AB of a circle with centre at O ($|AM| \neq |MB|$). A circle drawn through the points A , M and O cuts the first circle at the points A and C . Prove that $|MB| = |MC|$.

56. There is an internal tangency between two circles at a point A . The line segment AB is a diameter of the larger circle. The chord BD of the larger circle touches the smaller circle at a point C . Prove that AC is the bisector of the angle BAD .

57. A common external tangent BC is drawn to two circles O and O_1 which touch each other externally at a point A (B and C are the points of tangency). Prove that $\widehat{BAC} = 90^\circ$.

58. In the trapezoid $ABCD$ the nonparallel sides AB and CD are 8 cm and 10 cm respectively and the base BC is 2 cm. The bisector of the angle ADC passes through the midpoint of the side AB . Find the area of the trapezoid.

59. In the trapezoid $ABCD$ the nonparallel sides AB and CD are perpendicular. Prove that the sum of the squares of the diagonals of such a trapezoid is equal to the sum of the squares of its bases.

60. The bisector CD of the triangle ABC ($D \in [AB]$) cuts the circumscribed circle at a point E . Prove that $|AC| \cdot |BC| = |CD| \cdot |CE|$.

61. Prove that the sum of the distances from any point taken on a side of an equilateral triangle to its other two sides is a constant quantity.

62. The centre of the circle inscribed in a right-angled trapezoid is 8 cm and 6 cm apart from the endpoints of a leg. Find the altitude of the trapezoid and its perimeter.

63. The base AD , the diagonal AC and the leg AB of the trapezoid $ABCD$ are 2 cm each. The leg CD is 1 cm. Find the length of the diagonal BD .

64. 1. The bisectors AD and CE ($D \in [BC]$, $E \in [AB]$) which meet at a point O are drawn in the triangle ABC whose angle $\hat{B}$ is 60° . Prove that $|OD| = |OE|$.

2. The bisectors AD and CE of the triangle ABC meet at a point O . The points B , D , O and E lie on the same circle. Prove that $\hat{B} = 60^\circ$.

65. AC and BD are the diagonals of the trapezoid $ABCD$. A perpendicular BE ($E \in [AC]$) is drawn from the vertex B to the diagonal AC , $|BE| = 1$ m, and $|AC| = 2$ m. Find the area of the triangle BCD .

66. The altitudes AF and BK , which meet at a point O , are drawn in the triangle ABC . Prove that the vertex C lies on the circle which passes through the points F , O and K .

67. 1. Prove that in the triangle ABC , where a and b are the sides, the following relations hold true:

$$(a) \ S \leq \frac{a^2 + b^2}{4} \text{ and } (b) \ S \leq \frac{(a+b)^2}{8}.$$

68. Prove that if the areas of an equilateral triangle and a square are equal, then the perimeter of the triangle is larger than that of the square.

69. The sides of the triangle ABC are a , b and c long. It is known that $c^3 = a^3 + b^3$. Is the angle C acute, right, or obtuse?

70. Find the angles of the triangle if its area S can be expressed in terms of the lengths of its sides a and b by the formula $S = \frac{1}{4}(a^2 + b^2)$.

71. The vertices of a regular triangle with a side a lie on three parallel straight lines, which are p and q apart ($p \neq q$). Prove that $a > 2\sqrt{pq}$.

72. Prove that in any triangle the radius R of the circumscribed circle can be calculated from the formula $R = \frac{abc}{4S}$, where a , b and c are the sides of the triangle and S is its area.

73. Being given several triangles two of whose sides are a and b long, prove that the triangle whose given sides are perpendicular has the greatest area.

74. Prove that in any triangle a smaller altitude corresponds to the greatest side.

75. In the trapezoid $ABCD$ the leg CD is equal to a and the distance from the midpoint of AB to CD is b . Find the area of the trapezoid.

76. Prove that the perpendicular drawn from the midpoint of the segment which connects the feet of two altitudes of the triangle bisects the third side.

77. 1. Find the area of the triangle ABC in which $\hat{C} = 90^\circ$, $|AB| = c$ and $\hat{A} = 15^\circ$.

2. An acute angle of a right triangle is 15° . Prove that the product of its legs is equal to the square of a half of the hypotenuse.

78. In the rectangle $ABCD$ $|AB| \div |BC| = 1 \div 2$. From a point M taken on the side BC the sides AB and AD can be seen at the same angle. Find the angles.

79. The diagonals partition a trapezoid into four triangles. Find the area of the trapezoid if the areas of the triangles adjacent to the bases of the trapezoid are S_1 and S_2 .

80. Altitudes BD and AE ($D \in [AC]$, $E \in [BC]$) are drawn in the triangle ABC . Given that $\hat{A} = \alpha$ and $\hat{B} = \beta$, find the angles of the triangle DEC .

81. We are given a square and a rectangle, the diagonals of the square being equal to those of the rectangle. The area of which figures, a square or a rectangle, is larger?

82. Calculate the angles of an isosceles triangle if the bisector of the base angle cuts off a triangle which is similar to the given triangle.

83. A median m_c is drawn from the vertex of the triangle ABC , where $|AB| = c$. Establish the relationship between m_c and c if the triangle ABC is right-angled, acute-angled, obtuse-angled.

84. Two parallel chords are drawn in a circle of radius r , the distances from the chords to the centre of the circle being d and $2d$.

1. Express the lengths of chords a and b in terms of d and r .

2. For what ratio of d and r are the lengths of the chords proportional to their distances from the centre?

85. Prove that if the sides of a right triangle form an arithmetic progression, then the common difference of

the progression is equal to the radius of the circle inscribed in the triangle.

86. Prove that the radius of a circle inscribed in a triangle is smaller than a half of each of its altitudes.

87. Prove that in a right triangle $a + b < c + h_c$, where a and b are legs, c is the hypotenuse, and h_c is an altitude drawn to the hypotenuse.

88. The lengths of the sides of the triangle ABC are a , b and c . It is given that $c^3 = a^3 + b^3$. Is the angle C acute? right? obtuse?

89. A point M taken inside an angle of 60° is 2 cm and 11 cm apart from the sides of the angle. Find the distance from the point M to the vertex of the angle.

90. The distance between the feet of two altitudes of a rhombus drawn from one vertex is half the larger diagonal of the rhombus. Find the angles of the rhombus.

91. An isosceles trapezoid with an acute angle α is circumscribed about a circle, the angle between a base and a diagonal of the trapezoid being β . Prove that $\sin \alpha = \tan \beta$.

Answers, Hints and Solutions

CHAPTER 1

SECTION 1.1

1. (a) The *intersection of the sets* A and B is a new set containing those and only those elements which belong both to the set A and to the set B . The intersection of the sets A and B is designated as $A \cap B$. Examples:

$$\begin{aligned}\{a, b, c, d\} \cap \{c, d, e, f\} &= \{c, d\}, \{1, 2\} \cap \{3, 4\} = \emptyset, \\ \{5, 6, 7\} \cap \{5, 6, 7\} &= \{5, 6, 7\}.\end{aligned}$$

The *intersection of two figures* is a figure consisting of all those and only those points which belong to each of the figures.

(b) The *union of two sets* A and B is a set consisting of all elements which enter into at least one of the given sets. The union of the sets A and B is designated as $A \cup B$. For example, $A = \{a, b, c, d, e, f\}$, $B = \{a, g, h, f\}$, $A \cup B = \{a, b, c, d, e, f, g, h\}$. The *union of two figures* is a figure consisting of all those and only those points which belong to at least one of the given figures. For example, the union of two adjacent angles is a straight angle.

2. $E = \{1, 2, 3, 4, 6, 12\}$, $F = \{1, 2, 3, 6, 9, 18\}$, $E \cap F = \{1, 2, 3, 6\}$, $E \cup F = \{1, 2, 3, 4, 6, 9, 12, 18\}$. 3. $A = \{0, 1, 2, 3, 4\}$, $B = \{-1, 0, 1, 2\}$, $A \cup B = \{-1, 0, 1, 2, 3, 4\}$.

SECTION 1.2

5. The numbers used to count objects are known as *natural numbers*. The set $N = \{1, 2, 3, \dots\}$ of all natural numbers is infinite and is known as a *natural scale*.

6. If $a + b = c$, then a and b are *summands*, and c , as well as $a + b$, is a *sum*.

If $a - b = c$, then a is a *minuend*, b is a *subtrahend*, and c , as well as $a - b$, is the *difference*.

If $a \cdot b = c$, then a is a *multiplicand*, b is a *multiplier*, c , as well as $a \cdot b$, is a *product*; a and b are also called *factors*.

If $a \div b = c$, when a is a *dividend*, b is a *divisor*, and c , as well as $a \div b$, is a *quotient*.

7. (a) To find the unknown summand, we must subtract the known summand from the sum, i.e. if $x + a = b$, then $x = b - a$, (b) to find the unknown minuend, we must add the subtrahend to the difference, i.e. if $y - c = b$, then $y = b + c$, (c) to find the unknown subtrahend, we must subtract the difference from the minuend, i.e. if $a - z = b$, then $z = a - b$, (d) to find the unknown multiplier, we must divide the product by the known multiplier, i.e. if $ax = b$, then $x = b \div a$, (e) to find the unknown dividend, we must multiply the quotient by the divisor, i.e. if $y \div a = b$, then $y = ab$, (f) to find the unknown divisor, we must divide the dividend by the quotient, i.e. if $a \div z = b$, then $z = a \div b$.

8. (a) The last operation on the left-hand side of the expression is division. The unknown is a component of the dividend. It is equal to the divisor multiplied by the quotient, i.e. $(2x - 18) 15 + 120 = -4.45$. Now we have $(2x - 18) 15 = -180 - 120$, $(2x - 18) 15 = -300$, $2x - 18 = (-300) \div 15$, $2x = -20 + 18$, $x = -1$, (b) $112 - (96 - 3x) 4 = -208 \div 2$, $(96 - 3x) 4 = 112 + 104$, $96 - 3x = 216 \div 4$, $3x = 96 - 54$, $3x = 42$, $x = 14$.

9. (a) 4266, (b) 82, (c) 0, (d) the expression is meaningless.

10. (a) $(1282 + 1318) + (247 + 353) + 800 = 2600 + 600 + 800 = 4000$, (b) $(25 \cdot 4) \cdot (8 \cdot 125) \cdot 13 = 1\,300\,000$, (c) $25 \cdot 3 \cdot 25 \cdot 4 \cdot 4 = (25 \cdot 4) \cdot (25 \cdot 4) \cdot 3 = 30\,000$, (d) $(161 + 139) \cdot 27 = 8100$.

11. (a) $(200 + 40 + 8) \cdot 5 = 200 \cdot 5 + 40 \cdot 5 + 8 \cdot 5 = 1240$, or $248 \div 2 \cdot 10 = 1240$, (b) $244 \div 4 \cdot 100 = 6100$, (c) $72 \div 4 \cdot 1000 = 18\,000$, (d) $35 \cdot (10 - 1) = 350 - 35 = 315$, (e) $34 \cdot (10 + 1) = 374$, (f) $17 \cdot (100 + 1) = 1717$.

12. A *prime number* is a number which has only two divisors, unity and the number itself. Thus the set of prime numbers of the first ten numbers is $\{2, 3, 5, 7\}$. A *composite number* is a number which has more than two divisors.

13. Numbers which have no other natural divisors in common besides unity are said to be *coprime numbers*. For example, 3 and 8, 9 and 25 are coprime numbers.

14. The numbers 678, 1980 and 719 820 are divisible by 2, the numbers 678 ($6 + 7 + 8 = 21$), 1980 and 719 820 are divisible by 3, the numbers 1980, 65 425 and 719 820 are divisible by 5, the numbers 1980 ($1 + 9 + 8 + 0 = 18$) and 719 820 are divisible by 9.

15. The *least common multiple* of two or several numbers is the least of the natural numbers which is divisible by all given numbers. For example, the least common multiple of the numbers 9 and 12 is 36. This number can be obtained by successively multiplying the larger of these numbers, 12 by 1, 2 and 3 until the resulting product (36) becomes divisible by 9.

16. The least common divisor of several natural numbers is unity, the greatest common multiple of natural numbers does not exist. Indeed, however large a common multiple may be, we get a new common multiple larger than the preceding one by multiplying it, say, by 2.

SECTION 1.3

17. Two fractions a/b and c/d are equal if $ad = bc$. For example, $2/4 = 3/6$ (since $2 \cdot 6 = 3 \cdot 4$), $-1/2 = -5/10$ (here $(-1) \cdot (10) = 2(-5)$).

18. If $|a| < |b|$.

19. A *mixed number* is a sum of an integer and a proper fraction; it is written without the summation sign, for instance $3\frac{1}{2} = 3 + \frac{1}{2}$, $4\frac{5}{6} = 4 + \frac{5}{6}$.

20. If we multiply or divide the numerator and the denominator of a fraction by the same nonzero number we obtain a fraction equal to the initial fraction, i.e. $\frac{a}{b} = \frac{am}{bm}$ if $m \neq 0$. Reduction of fractions is based on this property, i.e. $\frac{5}{10} = \frac{1}{2}$, $\frac{-10}{100} = \frac{-1}{10}$, as well as reduction of fractions to a common denominator, i.e. $\frac{2}{3} = \frac{8}{12}$, $\frac{3}{4} = \frac{9}{12}$.

21. Two numbers whose product is unity are known as *mutually inverse*. For example, 5 and $1/5$, a/b and b/a are numbers inverse to each other.

22. To add fractions with different denominators, we must first reduce them to a common denominator (better to the least common denominator) and then add the resulting fractions with the same denominator. For example, $\frac{7}{10} + \frac{2}{15} = \frac{21}{30} + \frac{4}{30} = \frac{25}{30} = \frac{5}{6}$.

23. A product of fractions is equal to a fraction whose numerator is equal to the product of their numerators and the denominator is equal to the product of their denominators.

Remarks. 1. We must first reduce fractions wherever possible, and then multiply them, for example, $\frac{4}{9} \cdot \frac{18}{5} = \frac{4 \cdot 18}{9 \cdot 5} = \frac{8}{5} = 1\frac{3}{5}$.

2. When multiplying an integer by a fraction, it is sufficient to represent the integer as a fraction with the denominator equal to unity, for example, $\frac{7}{9} \cdot (-6) = \frac{7}{9} \cdot \frac{-6}{1} = \frac{7 \cdot (-6)}{9 \cdot 1} = \frac{-14}{3} = -4\frac{2}{3}$.

24. To divide a fraction by a fraction, we must multiply the dividend by the inverse of the divisor, for example, $\frac{9}{5} \div \frac{8}{15} = \frac{9}{5} \cdot \frac{15}{8} = \frac{9 \cdot 15}{5 \cdot 8} = \frac{27}{8} = 3\frac{3}{8}$, $\frac{7}{12} \div 5 = \frac{7}{12} \cdot \frac{1}{5} = \frac{7}{60}$, $(-3) \div \frac{2}{3} = -3 \cdot \frac{3}{2} = -4.5$.]

25. (a) $\frac{25}{9} + \left(14\frac{5}{6} - 11\frac{3}{6}\right) \cdot \frac{4}{45} = \frac{25}{9} + 3\frac{1}{3} \cdot \frac{4}{45} = \frac{25}{9} + \frac{10 \cdot 4}{3 \cdot 45} = \frac{25}{9} + \frac{8}{27} = \frac{83}{27} = 3\frac{2}{27}$, (b) $\left(25\frac{1}{3} - \frac{135 \cdot 2}{27}\right) \cdot \frac{3}{16} = \left(25\frac{1}{3} - 10\right) \cdot \frac{3}{16} = 15\frac{1}{3} \cdot \frac{3}{16} = \frac{46 \cdot 3}{3 \cdot 16} = \frac{23}{8} = 2\frac{7}{8}$, (c) the numerator of the fraction is $3\frac{4}{5}$, the denominator is $6\frac{1}{2}$, $3\frac{4}{5} \div 6\frac{1}{2} = \frac{38}{65}$, (d) (1) $2\frac{3}{5} \cdot 5 = 2 \cdot 5 + \frac{3}{5} \cdot 5 = 13$, (2) $3\frac{2}{5} \cdot 5 = 3 \cdot 5 + \frac{2}{5} \cdot 5 = 17$, (3) $13 + 17 = 30$, (4) $30 - 15\frac{4}{15} = 14\frac{11}{15}$, (5) $14\frac{11}{15} \cdot 15 = 14 \cdot 15 + \frac{11}{15} \cdot 15 = 221$, (6) $\frac{5}{9} \cdot \frac{3}{5} = \frac{1}{3}$, (7) $1 - \frac{1}{3} = \frac{2}{3}$, (8) $\frac{2}{3} \div 8 = \frac{2 \cdot 1}{3 \cdot 8} = \frac{1}{12}$, (9) $221 - \frac{1}{12} = 220\frac{11}{12}$.

26. To add decimal fractions, we must (1) make the number of decimal places in the summands equal, (2) write the summands under one another so that the decimal points are under one another, (3) add the resulting numbers as we add natural numbers, (4) place the decimal point in the resulting number under the decimal point in the summands. For example, the notation for the operations $12.05 + 0.708 + 3.2$ and $7.6 - 4.99$ has the form

$$\begin{array}{r} 12.050 \\ + 0.708 \\ \hline 3.200 \\ \hline 15.958 \end{array} \qquad \begin{array}{r} 7.60 \\ - 4.99 \\ \hline 2.61 \end{array}$$

27. To multiply one decimal fraction by another, we must carry out the multiplication without paying attention to the decimal points and then, in the product obtained, separate from the right by a point as many digits as there are digits after the decimal point in both multipliers taken together. For example,

$$\begin{array}{r} \times 3.48 \\ 0.5 \\ \hline 1.740 = 1.74 \end{array} \qquad \begin{array}{r} \times 0.265 \\ 0.04 \\ \hline 0.01060 = 0.0106. \end{array}$$

Remark. We must not delete the zeros in the product before we put the decimal point. If the resulting product contains less digits than the number of digits to be separated by a point, then we write several zeros in front of the fraction.

28. To multiply a decimal fraction by 10, 100, 1000, . . . , 10^n , we must transfer the decimal point to the right by 1, 2, 3, . . . , n digits. For example, $0.136 \cdot 10 = 1.36$, $2.75 \cdot 1000 = 2.750 \cdot 1000 = 2750$.

29. When we divide a decimal fraction by a natural number, we must put a point in the quotient when the division of the integer

part ends, without paying attention to the decimal point, in the initial fraction. For example, we have

$$\begin{array}{r} 30.4 \\ 8 \overline{) 243.2} \\ \underline{24} \\ 3 \\ \underline{32} \\ 32 \\ \underline{0} \end{array}$$

30. To divide a decimal fraction by 10^n , we must transfer the decimal points in this fraction to the left by 1, 2, 3, . . . , n digits. For example, $9.46 \div 100 = 0.0946$ and $0.0946 \div 100 = 0.00946$.

31. To divide a number by a decimal fraction, we must transfer the decimal point in the dividend and the divisor to the right by as many digits as there are in the divisor and then carry out the division by the natural number. For example,

$$\begin{array}{r} 0.574 \\ 85 \overline{) 48.79} \\ \underline{487} \\ 425 \\ \underline{629} \\ 595 \\ \underline{340} \\ 340 \\ \underline{0} \end{array}$$

Warning! Sometimes, instead of transferring the decimal point to the right in the divisor, students erroneously make this operation in the dividend and thus complicate the calculations.

32. This can be done in two ways, either by dividing the numerator by the denominator or by multiplying the denominator and the numerator of the fraction by a number which turns the denominator into 10, 100, 1000 and so on. We have

$$\begin{array}{r} 0.75 \\ 4 \overline{) 3} \\ \underline{0} \\ 30 \\ \underline{28} \\ 20 \\ \underline{20} \\ 0 \end{array} \quad \frac{3}{4} = \frac{3}{2 \cdot 2} = \frac{3 \cdot 5 \cdot 5}{(2 \cdot 5) \cdot (2 \cdot 5)} = \frac{75}{100} = 0.75;$$

$$\frac{5}{8} = 5 \div 8 = 0.625 \quad \text{or} \quad \frac{5}{8} = \frac{5}{2 \cdot 2 \cdot 2} = \frac{5 \cdot 5 \cdot 5}{(2 \cdot 5) \cdot (2 \cdot 5) \cdot (2 \cdot 5)} = \frac{625}{1000} = 0.625,$$

$$1 \frac{4}{25} = 1 \frac{4}{5 \cdot 5} = 1 \frac{4 \cdot 2 \cdot 2}{(5 \cdot 2) \cdot (5 \cdot 2)} = 1 \frac{16}{100} = 1.16,$$

$$\frac{6}{125} = \frac{6}{5 \cdot 5 \cdot 5} = \frac{6 \cdot 2 \cdot 2 \cdot 2}{(5 \cdot 2) \cdot (5 \cdot 2) \cdot (5 \cdot 2)} = \frac{48}{1000} = 0.048.$$

33. We can write the fractions $\frac{3}{50}$, $\frac{9}{18} = \frac{1}{2}$, $\frac{12}{15} = \frac{4}{5}$, $1\frac{3}{25}$, $\frac{7}{16}$ as terminating decimal fractions since the factorization of the denominator of the fraction into prime factors contains only 2 and 5. The fractions $\frac{5}{48}$ and $\frac{15}{54} = \frac{5}{18}$ cannot be written as terminating decimal fractions since the factorization of the denominator includes, in addition to 2 and 5, some other prime factors which cannot be reduced to any of the denominators 10, 100, 1000 and so on.

34. (a) It is more convenient here to make calculations in ordinary fractions, i.e. $\frac{5}{32} \div 0.25 = \frac{5}{32} \div \frac{1}{4} = \frac{5}{32} \cdot \frac{4}{1} = \frac{5}{8}$,

(b) it is more convenient to make calculations in decimal fractions, i.e. $14.56 \div 0.14 = 1456 \div 14 = 104$, (c) we multiply the numerator and denominator of the first fraction by 6 and those of the second fraction by 10, i.e.

$$\frac{1\frac{1}{3} \cdot 6}{3\frac{1}{2} \cdot 6} \cdot \frac{3.9 \cdot 10}{4.2 \cdot 10} = \frac{8}{21} + \frac{39}{42} = \frac{16}{42} + \frac{39}{42} = \frac{55}{42} = 1\frac{13}{42},$$

(d) it is simpler to calculate this expression in decimal fractions, i.e. (1) $4\frac{4}{25} + 2.59 = 4.16 + 2.59 = 6.75$, (2) $\frac{3}{4} - 0.15 = 0.75 - 0.15 = 0.6$, (3) $0.6 \div 4 = 0.15$, (4) $6.75 \div 0.15 = 675 \div 15 = 45$, (e) we calculate in ordinary fractions, i.e. (1) $\frac{7}{24} \div 0.125 = \frac{7}{24} \div \frac{1}{8} = 2\frac{1}{3}$, (2) $2\frac{1}{3} + 3.5 = 2\frac{1}{3} + 3\frac{1}{2} = 5\frac{5}{6}$, (3) $\frac{2}{3} - 0.25 = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$, (4) $5\frac{5}{6} \div \frac{5}{12} = \frac{35}{6} \cdot \frac{12}{5} = 14$, (f) we calculate the numerator in decimal fractions and the denominator in ordinary fractions, i.e.

$$\begin{aligned} \frac{1.05 + 0.16}{\left(7\frac{1}{2} - 3\frac{1}{3}\right) \cdot \frac{3}{5}} &= \frac{1.21}{4\frac{1}{6} \cdot \frac{3}{5}} = \frac{1.21}{\frac{25}{6} \cdot \frac{3}{5}} = \frac{1.21}{2.5} \\ &= \frac{1.21 \cdot 4}{2.5 \cdot 4} = 0.484, \end{aligned}$$

(g) besides an ordinary solution, i.e. "operation by operation", we can use here the following technique: using the distributive law, we multiply each expression in brackets in the numerator by $\frac{2}{3}$ and divide that in the denominator by $2\frac{2}{3}$, or multiply it by $\frac{3}{8}$, i.e.

$$\frac{4.5 \cdot \frac{5}{3} \cdot \frac{2}{3} - 6.75 \cdot \frac{2}{3}}{\frac{2}{3} \cdot \frac{3}{8} + \frac{10}{3} \cdot \frac{3}{10} \cdot \frac{3}{8}} = \frac{5 - 4.5}{\frac{1}{4} + \frac{3}{8}} = 0.8,$$

$$(h) \frac{1 \frac{1}{12} - \frac{53}{24}}{0.415 - 0.54} + \frac{\frac{34}{18} - 1 \frac{2}{3} + \frac{7}{9}}{0.45 + 1.8} = \frac{-\frac{27}{24}}{-0.125} + \frac{\left(\frac{17}{9} - 1 \frac{6}{9} + \frac{7}{9}\right) \cdot 4}{2.25 \cdot 4} = \frac{9}{8} \div \frac{1}{8} + \frac{4}{9} = 9 + \frac{4}{9} = 9 \frac{4}{9}, \quad (i) \quad 0,$$

(j) the expression is meaningless.

35. (a) $[25.3] = 25$, $\{25.3\} = 25.3 - 25 = 0.3$, (b) $[0.25] = 0$, $\{0.25\} = 0.25$, (c) $[-6.3] = -7$, $\{-6.3\} = 0.7$.

Warning! Sometimes students erroneously assume that -6 is the integer part and 0.3 is the fractional part. This is not so since $-6 > -6.3$ and by definition this must be the greatest number not exceeding -6.3 . In order to make no mistakes, it is convenient to write the number -6.3 as $-6.3 = -6 - 0.3 = -6 - 1 + 1 - 0.3 = -7 + 0.7$, (d) $[-10.5] = -11$, $\{-10.5\} = 0.5$,

$$\left[-\frac{5}{6}\right] = -1, \quad \left\{-\frac{5}{6}\right\} = \frac{1}{6}.$$

36. $y(-0.1) = -1 - 0.9 = -1.9$, $y(-1.1) = -2 - 0.9 = -2.9$, $y(-2.1) = -3 - 0.9 = -3.9$.

SECTION 1.4

$$37. 50\%; 25\%. 38.1. \frac{10}{100} = 0.1, \quad \frac{75}{100} = \frac{3}{4}, \quad \frac{0.1}{100} = 0.001, \quad 0.005,$$

$$\frac{33 \frac{1}{3}}{100} = \frac{1}{3}, \quad \frac{16 \frac{2}{3}}{100} = \frac{1}{6}, \quad \frac{p}{100}. \quad 2. \text{ To express several per cent}$$

as a fraction, we must divide the given percentage by 100, for example, 12.5 per cent constitute $\frac{12.5}{100} = \frac{1}{8}$.

39. (a) To find one per cent of a number, it is sufficient to divide it by 100, for instance 1% of 287 is 2.87, (b) to find 50 per cent of a number, it is sufficient to divide the given number by 2, for instance 50% of 1268 is 634, (c) divide the given number by 4, (d) divide the given number by 8, (e) multiply the given number by $3/4$, (f) divide the given number by 3.

40. (a) We must add 40% of a or $0.4a$, to the number a , i.e. $a + 0.4a = 1.4a$, (b) $a - 0.1a = 0.9a$.

41. If we assume the output in 1983 to be 100 per cent, then in 1985 the output was 128 per cent, which is 115 200 000 roubles of 90 000 000 roubles since $90 \div 100 \cdot 128 = 90 \cdot 1.28 = 115.2$.

42. *The 1st technique.* Assume that x is the unknown number, 42% of this number are equal to 12.6, 1% of x is 42 times as small, or $12.6 \div 42$, and the whole unknown number assumed to be 100% is 100 times as large, i.e. $x = (12.6 \div 42) \cdot 100 = 30$. *The 2nd technique.* Using the hypothesis, we derive an equation. Since 1% of x is $\frac{x}{100}$, it follows that 42% are $\frac{x}{100} \cdot 42$, i.e. $0.42x$, and, therefore $0.42x = 12.6$, $x = 12.6 \div 0.42 = 30$. *The 3rd technique.* We form a proportion

$$\begin{array}{l} 12.6 - 42\% \\ x - 100\%, \text{ whence } x \cdot 42 = 12.6 \cdot 100, x = 30. \end{array}$$

43. By the hypothesis, 106 per cent of the daily quota x is 159, i.e. $1.06x = 159$, whence $x = 150$.

44. We first find 40% of 33 and get 13.2. Then we derive an equation, taking into account that 10% of the unknown number x is 13.2. We have $0.1x = 13.2$, whence $x = 132$.

45. (a) 1%, (b) 10%, (c) 200%, (d) 74%.

46. *The 1st technique.* Assume that the number of the members of the Young Communist League is $x\%$. By the hypothesis, $x\%$ of 60 is 42, i.e. $60 \cdot \frac{x}{100} = 42$, $0.6x = 42$, whence $x = 42 \div 0.6 = 70$. Thus the members of the Young Communist League constitute 70% of the whole number of the workers. *The 2nd technique.*

$$\begin{array}{l} 60 - 100\% \\ 42 - x\%, \quad 60x = 42 \cdot 100, \quad x = \frac{42 \cdot 100}{60} = 70. \end{array}$$

47. *The 1st technique.* Since $x\%$ of 240 is 288, or $\frac{240 \cdot x}{100} = 288$,

it follows that $x = \frac{288 \cdot 100}{240} = 120$. The productivity increased by 20% since $120 - 100 = 20$. *The 2nd technique.*

$$\begin{array}{l} 240 - 100\% \\ 288 - x\%, \quad 240x = 288 \cdot 100, \quad x = 120. \end{array}$$

48. (a) 10% of 50% is $1/10$ of 50%, i.e. 5%, (b) 6.25%.

49. By 16.6%. *Hint.* If in the first year the output was 100%, then in the second year it was 106% and in the third year it was 110% of 106% = $106\% + 10\%$ of $106\% = 106\% + 1/10$ of $106\% = 116.6\%$.

50. *The 1st technique.* If we assume the capacity of the tank to be 100%, then 95% is the capacity filled with petrol. We designate this capacity as x and then

$$0.95x = 38, \quad x = 40.$$

This means that to fill up the tank we must add 2 tons of petrol.

The 2nd technique. We can solve this problem using the proportion, i.e.

$$\begin{array}{l} 38 - 95\% \\ x - 5\%, \quad 95x = 38 \cdot 5, \quad x = 2. \end{array}$$

51. Ten litres of the 30% solution contain $\frac{10 \cdot 30}{100} = 3$ litres acid anhydride. If we add 5 litres of water, we get 15 litres of solution. Consequently, the percent concentration of the resulting solution is 20% since $3 \cdot 100 \div 15 = 20$.

52. 1. By 100% since the second number is twice as large as the first. 2. Yes, it would. Assume that the initial price of the vegetables was a roubles. Then, after the first reduction, the price of the vegetables was $0.9a$ roubles, and after the second reduction, it was 0.9 of $0.9a$ roubles, i.e. $0.81a$ roubles, or 81% of the initial price. If we reduced the initial price by 20% instantaneously, it would be 80% ($100\% - 20\% = 80\%$) of the initial price. Consequently, the vegetables would be 1% cheaper.

SECTION 1.5

53. 1. Assume that the proportion $a : b = c : d$ is true. We designate its left-hand side as k , i.e. $a : b = k$ and $c : d = k$, whence $a = bk$ and $c = dk$. We seek the product of the extremes a and d : $ad = (bk)d = bdk$, and the product of the means b and c : $bc = b(dk) = bdk$. We see that $ad = bc$ since both products are equal to the same value bdk . 2. $b = c$. 3. $c/d = m/n$.

54. We can form several proportions of this kind: (a) if 25 and 3 are extreme and 15 and 5 are means, then $25 : 15 = 5 : 3$, (b) $2 : 3 = b : a$, (c) $a : m = n : b$, (d) $x : a = b : x$.

55. (a) We solve the problem orally: 19 is twice as large as 9.5, consequently, 13 is twice as large as x , hence $x = 6.5$,

$$(b) 7.5x = 20.8 \cdot \frac{50}{13}, \quad x = \frac{20.8 \cdot 50}{7.5 \cdot 13} = \frac{208 \cdot 50}{75 \cdot 13} = \frac{16 \cdot 2}{3 \cdot 1} = 10 \frac{2}{3},$$

$$(c) 7(x+1) = 8 \cdot 3.75, \quad x+1 = \frac{30}{7}, \quad x = 4 \frac{2}{7} - 1 = 3 \frac{2}{7},$$

$$(d) 2.6 \div 1.3 = x \div 5, \quad x = 5 \cdot 2 = 10.$$

SECTION 1.6

56. 1. A *coordinate axis* is a straight line on which a reference point (or origin), a unit segment and a direction are chosen. 2. The coordinate of a point is the number which locates a position of a point on a straight line. 4. Two numbers whose sum is zero are said to be opposite numbers. Thus the numbers 3 and -3 , -7.2 and 7.2 , 0 and 0 are opposite numbers. Two opposite numbers are represented on the coordinate axis by points symmetric about the reference point O .

57. A *repeating, or periodic, fraction* is an infinite decimal fraction in which, beginning with a certain place, one digit or a block of digits repeats indefinitely, for instance, 0.666 , . . . This fraction can be written as $0.\overline{6}$ (and is read as "zero point and 6 is the period"). The fraction $3.5\overline{13}13$. . . can be written as $3.5\overline{1\bar{3}}$ (it is read as "three point five ~~tens~~ and 13 is the period").

The repeating digit or block of digits in the notation for a decimal fraction is called a *period*. If a period begins immediately after the decimal point, the fraction is said to be *purely periodic*, for instance $5.\overline{23} = 5.2323 \dots$. If there are other digits at the decimal places between the point and the period, then the fraction is said to be *mixed periodic*, for instance, $2.5\overline{18} = 2.51818 \dots$.

58. The absolute value of a positive number is equal to that number, the absolute value of a zero is zero, the absolute value of a negative number is equal to the opposite number, i.e.

$$|a| = \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

59. (a) $x = 6$ and $x = -6$, the set of solutions of this equation can be written as $\{-6, 6\}$, (b) $\{-8.5, 8.5\}$, (c) $\{0\}$, (d) $\emptyset$.

(e) *The 1st technique.* The distance between the point with coordinate x and the point with coordinate 3 is 2. We lay off two units from the second point to both sides along the coordinate axis and get $x = 5$ and $x = 1$.

The 2nd technique. Since $x - 3 = 2$ and $x - 3 = -2$, it follows that $x = 5$ and $x = 1$, $\{1, 5\}$, (f) $\{-5, -1\}$. *Hint.* The distance from the point with coordinate -3 is 2, i.e. $|x - (-3)| = 2$.

60. (a) $\{-2, -1, 0, 1, 2\}$, (b) $\{-3, -2, -1, 0, 1, 2, 3\}$.

61. (a) -0.5 , (b) 46.5 , (c) 46.5 , (d) 0.5 .

SECTION 1.7

62. When we round off a decimal fraction to some decimal place, we delete all digits following that place.

If the first digit to be deleted is 5 or greater than 5, then the last remaining digit is increased by unity. If the first digit to be deleted is smaller than 5, then the preceding digit remains unchanged. For example, if we round off 6.419 to unity, we get $6.419 \approx 6$ (rounding down). If we round off 36.874 to the nearest tenth, we get $36.874 \approx 36.9$ (rounding up). If we round off 5675 with an accuracy not exceeding 10, we get $5675 \approx 5680$.

63. 1. An *error of approximation* is the difference between the exact and the approximate value of a quantity. If we know, for instance, that the exact value of a quantity $x = 6.34$, the approximation by defect is $a = 6.3$ and the approximation by excess is $b = 6.4$, then the error of the approximation by defect is $x - a = 6.34 - 6.3 = 0.04$ and the error of the approximation by excess is $x - b = 6.34 - 6.4 = -0.06$. The error of approximation by defect is positive and that by excess is negative.

2. The *absolute error of approximation* is the absolute value of the difference between the exact and the approximate value of a quantity. In the example cited above the absolute error is $|x - a| = |6.34 - 6.3| = 0.04$ (by defect), $|x - b| = |6.34 - 6.4| = 0.06$ (by excess). The approximation by defect is the better of the two since it differs less from x ($0.04 < 0.06$).

3. The *relative error of approximation* is the ratio of the absolute approximation error to the modulus of the approximate value of the quantity. For instance in the example cited above the relative

error $\frac{|x-a|}{a} = \frac{|6.34-6.3|}{6.3} = \frac{0.04}{6.3} \approx 0.006$. The relative error is usually given in per cent, in our example it is 0.6%.

64. (a) $4.36 \approx 4.4$, $x = 4.36$, $a = 4.4$. The absolute error $|x - a| = |4.36 - 4.4| = 0.04$, the relative error

$$\frac{|x-a|}{|a|} = \frac{0.04}{4.4} = \frac{1}{110} \approx 1\%, \quad (b) \quad 0.825 \approx 0.8, \quad |x-a| = 0.025,$$

$$a = 0.8, \quad \frac{|x-a|}{|a|} = \frac{0.025}{0.8} = \frac{1}{32} \approx 3\%.$$

65. (a) $16.8 - 0.5 \leq x \leq 16.8 + 0.5$, $16.3 \leq x \leq 17.3$,
(b) $0.25 \leq x \leq 0.55$.

66. If the absolute error of approximation does not exceed h , i.e. if $|x - a| \leq h$.

67. (a) We seek the absolute error of approximation:

$$\left| \frac{3}{11} - \frac{27}{100} \right| = \left| \frac{300 - 297}{1100} \right| = \frac{3}{1100};$$

$\frac{3}{1100} < \frac{1}{100}$ since $\frac{3}{1100} < \frac{11}{1100}$, and therefore the inequality

$$\left| \frac{3}{11} - 0.27 \right| \leq 0.01 \text{ holds true, (b) } \left| \frac{2}{3} - \frac{6}{10} \right| = \left| \frac{1}{15} \right| < 0.1.$$

68. Approximate values of numbers are written so that all digits in the notation are *true*, i.e. the absolute error of approximation does not exceed unity in the last remaining decimal place. For instance, the notation $x \approx 7.6$ means that the absolute error does not exceed 0.1, or $x = 7.6 \pm 0.1$. The notation $x \approx 7.63$ means that the absolute error does not exceed 0.01.

69. (a) 0.01, (b) 0.001, (c) 0.1, (d) 0.01, (e) 1.

70. In the first case the measurement is accurate to 1 mm, in the second case to 0.1 mm, and in the third case to 0.01 mm.

71. Assume that P is the perimeter of the rectangle, S is its area, and $S = xy$. Then we have

$$\begin{array}{rcl} 6.3 \leq x \leq 6.5 & & 6.3 \cdot 2.3 \leq xy \leq 6.5 \cdot 2.5 \\ 2.3 \leq y \leq 2.5 & & 14.49 \leq xy \leq 16.25 \\ \hline 8.6 \leq x+y \leq 9 & & 14 \leq xy \leq 16 \\ 17.2 \leq 2(x+y) \leq 18 & & S = 15 \pm 1 \\ P = 17.6 \pm 0.4 & & \end{array}$$

(when we seek S , we take the lower bound to be equal to 14 and the upper bound to 16).

72. In the number 2.6 there is one decimal place and two significant digits, in 26 there are no decimal places and two significant digits, in 0.26 there are two decimal places and two significant digits, in 0.206 there are three decimal places and three significant digits, in 2.060 there are three decimal places and four significant digits, in 0.0026 there are four decimal places and two significant digits.

73. (a) We isolate the least exact term, it is 1.3. It has the least number of true decimal digits. We round off the second and the third

term to the nearest hundredth (one decimal digit more than in the isolated term): $12.341 \approx 12.34$, $0.4764 \approx 0.48$. Next we find the sum $1.3 + 12.34 + 0.48 = 14.12$ and round off the result to the nearest tenth, i.e. leave as many decimal digits as there were in the least exact term, and then $x + y + z \approx 14.1$, (b) we isolate the factor with the least number of *significant digits*. It is 0.64. We round off 12.26 so that it has one significant digit more (one reserve digit), $12.26 \approx 12.3$, and then $12.3 \cdot 0.64 = 7.872$. We round off the result to two significant digits (we leave as many significant digits as there were in the factor 0.64), i.e. $7.872 \approx 7.9$; the final result is $ab \approx 7.9$, (c) $a \div b \approx 3.6 \div 1.39 \approx 2.59 \approx 2.6$.

SECTION 1.8

74.2. The *domain of definition of an expression* is the set of values of the variables for which the expression has sense. For example, the expression $x^2 + 1$ has sense for any value of x . The expression $\frac{3a}{a-3}$ has sense only when the denominator is nonzero, and therefore the domain of this expression is the set of all numbers except 3. The domain of the expression $\frac{x}{2x-y}$ is the set of pairs (x, y) in which $y \neq 2x$. 2. (a) $x \neq 5$, (b) $a \neq -2$, (c) $x \neq y$.

75. (a) $a^0 = 1$ if $a \neq 0$, or every number (except zero) in degree zero is equal to unity. For example, $5^0 = 1$, $\left(\frac{2}{7}\right)^0 = 1$. The expression 0^0 is meaningless, (b) $a^{-n} = 1/a^n$, where $a \neq 0$, $n \in \mathbb{Z}$ and $n > 0$. For example, $10^{-3} = 1/10^3 = 0.001$, $(-2)^{-2} = 1/(-2)^2 = 1/4$, $\left(-\frac{3}{4}\right)^{-1} = 1/\left(-\frac{3}{4}\right)^1 = -\frac{4}{3}$. The notation 0^{-n} is meaningless.

76. 1. We prove that $a^3 a^2 = a^{3+2} = a^5$. Indeed, $a^3 a^2 = (a \cdot a \cdot a) \cdot (a \cdot a) = aaaaa = a^5$. 2. (a) When we multiply powers with the same bases, we leave the base of the power unchanged and add up the exponents, (b) when we divide powers with the same bases, we leave the base unchanged and subtract the exponent of the divisor from that of the dividend, (c) when we raise a power to a power, we leave the base unchanged and multiply the exponents, (d) the n th power of a product is equal to the product of the n th powers of the factors, (e) the n th power of a fraction is equal to the n th power of the numerator divided by the n th power of the denominator.

77. (a) $4 \cdot 3 - (-1)^3 \cdot 2^2 = 16$, (b) $(4 \cdot 3 - (-1)^3 \cdot 2)^2 = 196$, (c) $(4 \cdot 3 - (-1)^3) \cdot 2^2 = 52$, (d) 1024 , (e) $4(3 - (-1)^3 \cdot 4) = 28$, (f) $4(3 - (-1)^3 \cdot 2)^2 = 100$, (g) $4(3 - (-1)^3)^2 \cdot 2^2 = 64$.

78. (a)
$$\frac{0.5 \cdot 4^2 + (0.5 \cdot 4)^2}{5(4 + (-1))^2 - (2 \cdot 4 + (-1))^2} = \frac{0.5 \cdot 16 + 2^2}{5 \cdot 3^2 - 7^2} = \frac{12}{-4} = -3$$
, (b) it is convenient here to replace -0.25 by $-1/4$, and then
$$\frac{2 \cdot \left(-\frac{1}{4}\right)^2 + \left(2 \cdot \left(-\frac{1}{4}\right)\right)^2 + 2^2 \cdot \left(-\frac{1}{4}\right)}{\left(1 - 2 \cdot \left(-\frac{1}{4}\right)\right)^2 \div \left(2 + 4^3 \left(-\frac{1}{4}\right)^2\right)} = \frac{\frac{1}{8} + \frac{1}{4} - 1}{\frac{9}{4} : \frac{9}{4}} =$$

$$-\frac{5}{8} = -0.625, \text{ (c) } \frac{3 \cdot \left(-\frac{1}{2} + \frac{2}{3}\right)^2 + 3 \cdot \left(-\frac{1}{2} - \left(\frac{2}{4}\right)^2\right)}{\left(-\frac{1}{2}\right) \cdot \left(-\frac{2}{3}\right)^2 - \left(\left(-\frac{1}{2}\right) \cdot \left(-\frac{2}{3}\right)\right)^2} =$$

$$\frac{3 \cdot \left(\frac{1}{6}\right)^2 + 3 \cdot \left(-\frac{17}{18}\right)}{-\frac{2}{9} - \frac{1}{9}} = \frac{\frac{1}{12} - \frac{17}{6}}{-\frac{1}{3}} = \frac{-\frac{33}{12}}{-\frac{1}{3}} = 8.25,$$

$$\text{(d) } |-3 - 2.5| + |3.5 - (-3)| - |-3 + 2| = 5.5 + 6.5 - 1 = 11, \text{ (e) } 10 \cdot (-0.4)^2 - 5 \cdot |-0.4| + 1 = 1.6 - 2 + 1 = 0.6.$$

79. 1. (a) To obtain the number $1 \leq a < 10$, we put a point between 7 and 6 in the notation 0.00764 and delete zeros to the left of the digit 7. We get a decimal fraction 7.64, which is the given number increased 10^3 times. To leave the initial number unchanged, we must divide it by 10^3 . Then we get $0.00764 = 7.64 : 10^3 = 7.64 \cdot 10^{-3}$. The order of the number is $n = -3$, (b) $0.02 = 2 : 100 = 2 \cdot 10^{-2}$, (c) $17\,000 = 1.7 \cdot 10^4$, (d) $\approx 0.333 \cdot 10^{-3} = 3.33 \cdot 10^{-4}$. 2. (a) $(4.8 \cdot 1.5) \cdot 10^{-4+8} = 7.2 \cdot 10^4$, (b) $(2.4 \cdot 12.5) \cdot 10^2 = 30 \cdot 10^2 = 3 \cdot 10^3$.

80. (a) The expressions are identically equal on the set of all numbers, except zero, (b) the expressions are identically equal for $x \neq -y$.

81. (a) It is sufficient to give one value of x for which the values of the expressions are not equal, say, for $x = 1$ we get $1^2 + 1 \neq 1^3$, (b) for $a = 1$ we have $(1 + 5)^2 \neq 1^2 + 25$, (c) for $x = 1$ we get $|-x|^3 = |-1|^3 = 1$, $a - x^3 = -1^3 = -1$.

SECTION 1.9

82. 1. The *degree of a monomial*, which contains several variables (letters), is the sum of the exponents in the degrees of those variables. For example, the monomial $6a^2b^3$ is of the fifth degree since the sum of the exponents in the variables is five. The degree of the monomial $3ab$ is two. Every number is a monomial. The degree of such a monomial is zero. For example, the number 5 can be represented as $5a^0$, or $5a^0b^0$. The degree of every monomial of this kind is zero.

The *degree of a polynomial* is the degree of its leading term. For example, the degree of the polynomial $5a^3 - 2a^2 + 4a - 1$ is 3. 2. A monomial is said to be *standard* if its numerical factor (coefficient) occupies the first place and the product of similar variables is represented by a power. For example, $6x^3y^2$, $-7c^2$ are standard monomials.

A *polynomial has a standard form* if all its monomials are written in standard form and it has no like terms. For example, the polynomial $6a^4 + 4a^3 - 0.2a^2 + 9a - 2$ has a standard form.

83. (a) We apply the rule for removing brackets stating that if the brackets are preceded by the minus sign, we may omit them and change the signs of all terms in brackets. If the brackets are preceded by the plus sign, then we may omit them and retain the

signs of all the terms in brackets. We thus obtain $-2a^2 + 3b + 4b + 6a^2 - 1 = 4a^2 + 7b - 1$, (b) $0.25x^2 + 1.75x - 3$.

84. (a) $6.3x^2y - 2.4xy + 2y^3 + 3.7x^2y - 6.6xy - 2y^3 = 10x^2y - 9xy$; for $x = 1$ and $y = 5$ this expression is equal to $10 \cdot 1^2 \cdot 5 - 9 \cdot 1 \cdot 5 = 50 - 45 = 5$, (b) $5a^2 - 3ab + a^2 - 5ab - 6a^2 = 2ab$; for $a = 0.4$ and $b = -0.5$ the expression is equal to -0.4 , (c) $9x^3 - 8x^3 + 1 - 2x^3 = -x^3 + 1$, $-(-0.5)^3 + 1 = 0.125 + 1 = 1.125$.

85. (a) $5(x+2) \cdot (x-1) \cdot (-2x-1) - 5^2 = 25$, (b) $0.2 \cdot 3(x-3) \cdot (-x-5) \times 4(x-1) \cdot (-x-2) = 0.2 \cdot 3^2 \cdot 4 = 7.2$, (c) $120 \cdot 5^2 = 3\ 000$.

86. (a) $5x^2 - 20xy - 10y^2 + 20xy = 5x^2 - 10y^2$, $5 \cdot (-0.2)^2 - 10 \cdot (-0.5)^2 = 0.2 - 2.5 = -2.3$, (b) $a^2 + 2a - b^2 - b$, $\left(-\frac{1}{3}\right)^2 + 2\left(-\frac{1}{3}\right) - \left(\frac{2}{3}\right)^2 - \left(-\frac{2}{3}\right) = \frac{1}{9} - \frac{2}{3} - \frac{4}{9} + \frac{2}{3} = \frac{1}{9} - \frac{4}{9} = -\frac{1}{3}$, (c) $mn - 2n^2$, $0.4 \cdot (-0.6) - 2 \cdot (-0.6)^2 = -0.24 - 0.72 = -0.96$.

87. $(2a - 3b)(4a + 0.5b) - a(8a - 11b) = 8a^2 + ab - 12ab - 1.5b^2 - 8a^2 + 11ab = -1.5b^2$; $b^2 \geq 0$ and $-1.5b^2 \leq 0$ for any b .

SECTION 1.10

88. (a) $a^2 - 0.01$, (b) $2.25x^4 - 9y^6$, (c) $(a^2/4 - 9b^2)(a^2/4 + 9b^2) = a^2/16 - 81b^2$, (d) $a^{2x} + 2 + a^{-2x}$, (e) $(3^m - 3^{m-1})^2 = (3^m + (-3^{m-1}))^2 = (3^m)^2 + 2 \cdot 3^m \cdot (-3^{m-1}) + (-3^{m-1})^2 = 3^{2m} - 2 \cdot 3^{m+m-1} + 3^{2m-2} = 3^{2m} - 2 \cdot 3^{2m-1} + 3^{2m-2}$, (f) $((a+b) \times (a-b))^2 = (a^2 - b^2)^2 = a^4 - 2a^2b^2 + b^4$, (g) $((2a+1)^3 + (2a-1)^3) = ((2a)^3 + 3(2a)^2 \cdot 1 + 3 \cdot (2a) \cdot 1^2 + 1^3) + ((2a)^3 + 3 \cdot (2a)^2 \cdot (-1) + 3(2a) \cdot (-1)^2 + (-1)^3) = 16a^3 + 12a$, (h) $x^2 - 1$.

89. (a) $99 \cdot 101 = (100 - 1) \cdot (100 + 1) = 100^2 - 1^2 = 9999$, (b) $20.1 \cdot 19.9 = (20 + 0.1) \cdot (20 - 0.1) = 399.99$, (c) 899.75 , (d) $(30 + 1)^2 = 30^2 + 2 \cdot 30 \cdot 1 + 1^2 = 961$, (e) $(100 - 2)^2 = 9604$, (f) $(10 - 0.1)^2 = 98.01$.

90. (a) $(4a^2 - 12a + 9) - (4a^2 + 6a) = -18a + 9 = 9(-2a + 1)$, $9(-2 \cdot 2.5 + 1) = -36$, (b) $(9x^2 - 6x + 1) - (9x^2 + 4.5x - 2x - 1) = 9x^2 - 6x + 1 - 9x^2 - 4.5x + 2x + 1 = -8.5x + 2$, $-8.5 \cdot 0.2 + 2 = 0.3$.

91. The expression is equal to $24n$, it is divisible by 24 for any natural n .

SECTION 1.11

92. (a) We put the common factor $6a^2b$ before the brackets. To find out what must be written in brackets, we must divide each term of the given polynomial by $6a^2b$, i.e. $12a^3b \div 6a^2b = 2a$, $18a^2b^2 \div 6a^2b = 3b$, $6a^2b \div 6a^2b = 1$. We thus have $12a^3b - 18a^2b^2 + 6a^2b = 6a^2b(2a - 3b + 1)$. Warning! Sometimes, when solving such problems, students forget to write the last term, i.e. unity, in brackets, (b) $5x^4(3x^2 - 2x + 1)$, (c) we first put the last

two terms in brackets, and then the binomial $(x - 2y)$ is the common factor. When factoring out, the common factor must also be put in brackets and each term of this given polynomial is divided by $(x - 2y)$. And then $3a(x - 2y) + 5b(x - 2y) + (x - 2y) = (x - 2y)(3a + 5b + 1)$, (d) $5x(2a - 3) - 3y(2a - 3) - (2a - 3) = (2a - 3)(5x - 3y - 1)$.

93. 1. We divide each term of the given polynomial by 5 and obtain $5 \left(a + \frac{3}{5}b - \frac{2}{5}c \right)$. 2. $ax^2 + bx + c = a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right)$.
3. $9x - 3y + 2 = \frac{1}{2}(18x - 6y + 4)$.

94. (a) We represent the given polynomial as the sum of two binomials, $(ax + ay) + (bx + by)$, put a before the brackets in the first binomial and b in the second binomial and obtain $a(x + y) + b(x + y)$. Then we factor out the common factor $(x + y)$ and then we have $ax + ay + bx + by = (x + y)(a + b)$. We could factor this polynomial grouping its terms in another way: $(ax + bx) + (ay + by) = x(a + b) + y(a + b) = (a + b) \times (x + y)$, (b) $ac + bc + a + b = c(a + b) + 1 \cdot (a + b) = (a + b)(c + 1)$, (c) $a^3 + a^2b - ab^2 - b^3 = a^2(a + b) - b^2(a + b) = (a + b)(a^2 - b^2) = (a + b)^2(a - b)$.

95. (a) $(0.2a)^2 - 3^2 = (0.2a - 3)(0.2a + 3)$, (b) $\left(\frac{1}{3}x - \frac{3}{2} \right) \times \left(\frac{1}{3}x + \frac{3}{2} \right)$, (c) $(a + 4 - 5)(a + 4 + 5) = (a - 1)(a + 9)$, (d) $x^2 \times (3x - 1) - 9(3x - 1) = (3x - 1)(x^2 - 9) = (3x - 1)(x - 3)(x + 3)$, (e) $(2a - a + 1)(2a + a - 1) = (a + 1)(3a - 1)$, (f) $(a + b^2) - (a + c^2) = (a + b - a - c)(a + b + a + c) = (b - c)(2a + b + c)$, (g) $x^3 + 3^3 = (x + 3)(x^2 - 3x + 9)$, (h) $a^3 - 0.5^3 = (a - 0.5)(a^2 + 0.5a + 0.25)$.

96. (a) For $a = -\frac{3}{4}$ the value of the expression $(4a + 3)^2$ is zero, (b) $((x + 6) + (x - 1))^2 = (2x + 5)^2$, for $x = -0.5$ this expression is equal to 16.

97. (a) $(113 - 112) \cdot (113 + 112) = 225$, (b) 0.454, (c) $2\frac{1}{3}$, (d) $(26 + 14)^2 = 1600$, (e) 400.

SECTION 1.12

98. (a) For $x = 0$ and for $x = -3$, (b) for $a = 1/2$ and for $a = -1/2$.

99. (a) $\frac{a-2}{2-a} = -\frac{a-2}{a-2} = -1$, (b) $\frac{(a-b)^2}{b^2-a^2} = \frac{(b-a)^2}{(b-a)(a+b)} = \frac{b-a}{a+b}$, (c) $\frac{1}{1-x}$, (d) $\frac{m(m^2-1)}{(m-1)^2} = \frac{m(m+1)}{m-1}$,

$$(e) \frac{a^2 - (b^2 + c^2 + 2bc)}{(a^2 + b^2 - 2ab) - c^2} = \frac{a^2 - (b+c)^2}{(a+b)^2 - c^2} = \frac{(a-b-c)(a+b+c)}{(a+b-c)(a+b+c)} =$$

$$= \frac{a-b-c}{a+b-c}, (f) \frac{(ab-a)-(b-1)}{(b-bc)-(1-c)} = \frac{(a-1)(b-1)}{(1-c)(b-1)} = \frac{a-1}{1-c}.$$

100. (a) $\frac{(x-y)^2}{y(x-y)} = \frac{x-y}{y}$, for $x = -0.5$ and $y = -1.5$ we get

$$-2/3, (b) \frac{(a+b)^2 - c^2}{a+b+c} = \frac{(a+b+c)(a+b-c)}{a+b+c} = a+b-c, 0.25 +$$

$$0.5 - (-0.5) = 1.25.$$

101. (a) $\frac{1}{x(x-3)} + \frac{1}{3(3-x)} = \frac{1}{x(x-3)} - \frac{1}{3(x-3)} =$

$$\frac{3-x}{3x(x-3)} = -\frac{x-3}{3x(x-3)} = -\frac{1}{3x}, -\frac{1}{3\left(-1\frac{1}{3}\right)} = 0.25,$$

(b) $\frac{2^{12}}{a(4-a^2)} + \frac{-a^{12}}{2(4-a^2)} = \frac{4-a^2}{2a(4-a^2)} = \frac{1}{2a}$, for $a = -0.1$ this

expression is equal to -5 , (c) -11 . *Hint.* $\frac{x-x^3}{x^2-1} = -x$, $\frac{x^2}{0.2} =$

$$5x^2, \frac{x+x \div 0.1}{x} = \frac{x(1+1 \div 0.1)}{x} = 11.$$

102. (a) $\frac{5a^{1a-b}}{a+b} - \frac{2a-3b^{1a+b}}{a-b} - \frac{2a^2-6ab+2b^{211}}{a^2-b^2} =$

$$\frac{5a(a-b) - (2a-3b)(a+b) - 1 \cdot (2a^2-6ab+2b^2)}{a^2-b^2} = \frac{a^2+2ab+b^2}{a^2-b^2} =$$

$$\frac{a+b}{a-b}, (b) (1) \frac{xy}{x^2-y^2} - \frac{y}{2x-2y} = \frac{xy^2}{(x-y)(x+y)} - \frac{y^{1x+y}}{2(x-y)} =$$

$$\frac{2xy-y(x+y)}{2(x-y)(x+y)} = \frac{xy-y^2}{2(x-y)(x+y)} = \frac{y}{2(x+y)}, (2) \frac{y}{2(x+y)} \div$$

$$\frac{2y}{x^2-y^2} = \frac{y(x^2-y^2)}{2(x+y)2y} = \frac{x-y}{4}, (3) \frac{x-y}{4} - \frac{x-5y}{4} = y, (c) \text{ it is}$$

convenient here to use the distributive law of multiplication;

taking into account that $(b-a)^2 = (a-b)^2$, we get $\frac{a}{(b-a)^2} \times$

$$\frac{(b-a)}{a} = \frac{a}{b^2-a^2} \cdot \frac{(b-a)^2}{a} + \frac{2b}{a+b} = 1 - \frac{b-a}{a+b} + \frac{2b}{a+b} = 1 +$$

$$\frac{a-b+2b}{a+b} = 2, (d) \left(\frac{2x}{x+1} + \frac{2}{x-1} - \frac{4x}{x^2-1} \right) \div \frac{2x-2}{x+1} =$$

$$\frac{2x(x-1)+2(x+1)-4x}{(x-1)(x+1)} : \frac{2(x-1)}{x+1} = \frac{(2x^2-4x+2)(x+1)}{(x-1)(x+1) \cdot 2(x-1)} =$$

$$\begin{aligned}
\frac{2(x-1)^2(x+1)}{2(x-1)^2(x+1)} &= 1, \text{ (e) } 0, \left(\frac{x}{6(x+y)} + \frac{y}{6x} \right) \frac{x^2-y^2}{x^2y^2} - \frac{1}{6y^2} + \\
\frac{y}{6x^2} &= \frac{(x^2+xy+y^2)(x^2-y^2)}{6x(x+y) \cdot x^2y^2} - \frac{1}{6y^2} + \frac{y}{6x^3} = \frac{(x^2+xy+y^2)(x-y)}{6x^3y^2} - \\
\frac{1}{6y^2} + \frac{y}{6x^3} &= \frac{x^3-y^3}{6x^3y^2} - \frac{1}{6y^2} + \frac{y}{6x^3} = \frac{x^3}{6x^3y^2} - \frac{y^3}{6x^3y^2} - \frac{1}{6y^2} + \\
\frac{y}{6x^3} &= 0, \text{ (f) } \frac{(a-1)^2 - (a^2+a+1)}{(a-1)(a^2+a+1)} \cdot \frac{a^2+a+1}{a} = \frac{-3a}{a(a-1)} = \\
\frac{3}{1-a}, \text{ (g) } (1) \frac{5x+9}{(5x-1)^2} - \frac{10x}{25x^2-1} &= \frac{(5x+9)(5x-1) - 10x(5x+1)}{(5x+1)^2(5x-1)} = \\
\frac{-(5x-3)^2}{(5x+1)^2(5x-1)}, (2) \frac{20x}{1-5x} + \frac{(10x+2)^2}{(5x-3)^2} \cdot \frac{(5x-3)^2}{(5x+1)^2(5x-1)} &= \\
\frac{20x}{1-5x} + \frac{(2(5x+1))^2(5x-3)^2}{(5x-3)^2(5x+1)^2(5x-1)} &= \frac{20x}{1-5x} + \frac{4}{5x-1} = \frac{20x-4}{1-5x} = \\
\frac{4(5x-1)}{1-5x} &= -4, \text{ (h) if } x > 1, \text{ then } |x-1| = x-1 \text{ and the given} \\
\text{expression is equal to } \frac{x-1+x}{2x-1} - \frac{x-1}{x-1} &= 1-1=0. \text{ If } x < 1 \text{ but} \\
x \neq 1/2 \text{ (for } x=1/2 \text{ the denominator in the first fraction} & \\
\text{vanishes), then } |x-1| = 1-x, \text{ and then } \frac{1-x+x}{2x-1} - \frac{1-x}{x-1} &= \\
\frac{1}{2x-1} + 1 = \frac{1+2x-1}{2x-1} = \frac{2x}{2x-1}, \text{ (i) if } a > 3, \text{ then } |a-3| = & \\
a-3, |a| = a \text{ and we have } \frac{a(a-3)}{a} = a-3, \text{ if } 0 < a < 3, \text{ then} & \\
|a-3| = 3-a, |a| = a \text{ and we have } \frac{a(3-a)}{a} = 3-a, \text{ if } a < 0, & \\
\text{then } |a-3| = 3-a, |a| = -a \text{ and we have } \frac{a(3-a)}{-a} = a-3. &
\end{aligned}$$

SECTION 1.13

103. 1. For $a < 0$ the expression $\sqrt[3]{a}$ is meaningless since there is no number whose square is equal to a negative number. For $a = 0$ the expression $\sqrt[3]{a}$ is meaningful since $\sqrt[3]{0} = 0$. For $a > 0$ the expression $\sqrt[3]{a}$ is meaningful. Thus the expression $\sqrt[3]{a}$ has sense for any $a \geq 0$. 2. For $a = 1, 4, 9, \dots, \frac{1}{9}, \frac{4}{9}, \frac{16}{25}$ the expression $\sqrt[3]{a}$ is a rational number, it is expressed by an infinite repeating decimal fraction; for $a = 2, 3, 5$, and so on $\sqrt[3]{a}$ is an irrational number, it is expressed by an infinite nonrepeating decimal fraction. The approximate values of these fractions can be found by means of the graph of the function $y = \sqrt[3]{x}$ or taken from the requisite tables.

The number $\pi = 3.14159 \dots$ belongs to irrational numbers. Rational and irrational numbers form a set of real numbers which is designated as $\mathbb{R}$.

104. By the definition of an arithmetic square root, for $a \geq 0$ $\sqrt{a}$ is a number whose square is a , and this means that $(\sqrt{a})^2 = a$ on the set of nonnegative numbers.

105. No, the equality holds true only for $a \geq 0$. For any $a < 0$ the equality $\sqrt{a^2} = -a$ holds true.

Warning! It is sometimes erroneously assumed that $-a$ is a negative number whereas for $a < 0$ the number $-a$ is positive. For instance, if $a = -3$, then $\sqrt{a^2} = \sqrt{(-3)^2} = 3$. If the sign of the number a is unknown, then we write $\sqrt{a^2} = |a|$.

106. (a) For $a \in (-\infty, 0]$, (b) for $x = 0$, (c) for $a \in [1.5, \infty)$, (d) for $x \in (-\infty, -3]$.

$$107. \frac{a^2 - 5}{a - \sqrt{5}} = \frac{(a - \sqrt{5})(a + \sqrt{5})}{a - \sqrt{5}} = a + \sqrt{5}.$$

108. If n is an odd number, then $\sqrt[n]{a}$ has sense for any a . For instance, $\sqrt[3]{27} = 3$, $\sqrt[3]{-8} = -2$. If n is an even number, then $\sqrt[n]{a}$ has sense only for $a \geq 0$. For instance, $\sqrt[4]{16}$ has sense since it is equal both to 2 (an arithmetic root) and to -2 . $\sqrt[4]{0} = 0$, $\sqrt[4]{-16}$ is meaningless.

109. $(\sqrt[n]{a})^n = a$ by the definition of the n th root of the number a . This equality is valid only when $\sqrt[n]{a}$ has sense. For instance $(\sqrt[3]{-27})^3 = -27$. The equality $(\sqrt[4]{-81})^4 = -81$ does not hold true since $\sqrt[4]{-81}$ is meaningless.

110. (a) $x \leq 0$, (b) for any values of x , (c) $x \leq 0$, (d) for any values of x .

111. (a) $\sqrt{(\sqrt{5}-2)^2} = |\sqrt{5}-2| = \sqrt{5}-2$, since $\sqrt{5} > 2$, (b) $\sqrt{(1-\sqrt{2})^2} = |1-\sqrt{2}| = \sqrt{2}-1$, since $1 < \sqrt{2}$, (c) $\sqrt{7}-\sqrt{6}$.

112. $\sqrt{(2a-1)^2} + 3a = |2a-1| + 3a = |2 \cdot 0.2 - 1| + 3 \cdot 0.2 = |-0.6| + 0.6 = 1.2$.

114. (a) $\sqrt[3]{\sqrt{30} + \sqrt{3}} \cdot \sqrt[3]{\sqrt{30} - \sqrt{3}} = \sqrt[3]{(\sqrt{30})^2 - (\sqrt{3})^2} = \sqrt[3]{27} = 3$, (b) $\sqrt[5]{(3\sqrt{2})^2 - (\sqrt{17})^2} = \sqrt[5]{1} = 1$, (c) $\sqrt{(\sqrt{5})^2 - 2\sqrt{5} \cdot 1 + 1^2} \cdot \sqrt[4]{6 + 2\sqrt{5}} = \sqrt[4]{5 - 2\sqrt{5} + 1} \times \sqrt[4]{6 + 2\sqrt{5}} = \sqrt[4]{6^2 - (2\sqrt{5})^2} = 2$.

115. (a) $\sqrt{(3 \cdot 7)(4 \cdot 7)(3 \cdot 4)} = \sqrt{3^2 \cdot 4^2 \cdot 7^2} = 3 \cdot 4 \cdot 7 = 84$, (b) $\sqrt[3]{(2^4)(2^2 \cdot 3^2)(3^4)} = \sqrt[3]{2^6 \cdot 3^6} = 2^2 \cdot 3^2 = 36$.

$$(c) \sqrt[3]{(313-312) \cdot (313+312)} \cdot \sqrt[3]{(13-12) \cdot (13+12)} = \sqrt[3]{1 \cdot 625 \cdot 1 \cdot 25} = \sqrt[3]{5^4 \cdot 5^2} = \sqrt[3]{5^6} = 25, \quad (d) \sqrt[3]{27 \cdot 8} - \sqrt[4]{8 \cdot \sqrt[3]{8}} = \sqrt[3]{27} \cdot \sqrt[3]{8} - \sqrt[4]{2^3 \cdot 2} = 6 - 2 = 4, \quad (e) \sqrt[5]{81 \cdot \sqrt[4]{81}} - \sqrt[4]{16 \cdot 81} = \sqrt[5]{3^4 \cdot 3} - \sqrt[4]{2^4 \cdot \sqrt[4]{3^4}} = 3 - 6 = -3.$$

116. (a) $\sqrt{-5y^3} = \sqrt{-5y^2 y} = |y| \sqrt{-5y} = -y \sqrt{-5y}$ since $|y| = -y$ for $y < 0$, (b) $3\sqrt[3]{a}$, (c) $\sqrt[4]{3^4 x^4 x^2} = 3|x| \sqrt[4]{x^2} = -3x \sqrt{-x}$.

117. (a) $x \sqrt{2} = -(-x) \sqrt{2} = -\sqrt{2(-x)^2} = -\sqrt{2x^2}$, (b) the expression is meaningful if $a < 0$, and therefore $a \sqrt{-3/a} = -\sqrt{a^2(-3/a)} = -\sqrt{-3a}$.

118. (a) $\sqrt[4]{16} \sqrt[4]{a^4} \sqrt[4]{b} = 2|a| \sqrt[4]{b}$, for $a = -0.5$ and $b = 1$ this expression is equal to 1, (b) $\sqrt[6]{10a^6 a^3 b^6} = |ab| \sqrt[6]{10a^3}$, for $a = 2$ and $b = -2$ the expression is equal to $4\sqrt[6]{80}$.

119. (a) $\sqrt{2} = \sqrt[6]{2^3} = \sqrt[6]{8}$, $\sqrt[3]{3} = \sqrt[6]{3^2} = \sqrt[6]{9}$, $\sqrt[6]{8} < \sqrt[6]{9}$, and therefore $\sqrt{2} < \sqrt[3]{3}$, (b) $\sqrt[4]{4} = \sqrt[4]{2^2} = \sqrt{2}$, (c) $2\sqrt[3]{2} = \sqrt[3]{2^3} \sqrt[3]{2} = \sqrt[3]{16} = \sqrt[6]{16^2} = \sqrt[6]{256}$, $\sqrt[3]{7 \sqrt{3}} = \sqrt[3]{\sqrt{49} \sqrt{3}} = \sqrt[3]{\sqrt{147}} = \sqrt[6]{147}$, $\sqrt[6]{256} > \sqrt[6]{147}$, and therefore $2\sqrt{2} > \sqrt[3]{7 \sqrt{3}}$, (d) we assume that $\sqrt{14} > \sqrt{3} + 2$ and square both sides of this positive inequality, and then we have $14 > 3 + 4\sqrt{3} + 4$, or $7 > 4\sqrt{3}$; we again square both sides of this inequality and get a true inequality $49 > 48$; consequently $\sqrt{14} > \sqrt{3} + 2$.

120. (a) $\frac{2 + \sqrt{2}}{\sqrt{6} + \sqrt{3}} = \frac{(\sqrt{2})^2 + \sqrt{2}}{\sqrt{3}\sqrt{2} + \sqrt{3}} = \frac{\sqrt{2}(\sqrt{2} + 1)}{\sqrt{3}(\sqrt{2} + 1)} = \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{2}\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{\sqrt{6}}{3} = \frac{1}{3} \sqrt{6}$, (b) $\frac{x - \sqrt{xy}}{y - \sqrt{xy}} = \frac{(\sqrt{x})^2 - \sqrt{x}\sqrt{y}}{(\sqrt{y})^2 - \sqrt{x}\sqrt{y}} = \frac{\sqrt{x}(\sqrt{x} - \sqrt{y})}{\sqrt{y}(\sqrt{y} - \sqrt{x})} = -\sqrt{\frac{x}{y}} = -\frac{1}{y} \sqrt{xy}$.

121. (a) $\frac{15}{\sqrt{3}} = \frac{15\sqrt{3}}{\sqrt{3}\sqrt{3}} = 5\sqrt{3}$, (b) $\frac{2}{\sqrt[3]{4}} = \frac{2\sqrt[3]{2}}{\sqrt[3]{4}\sqrt[3]{2}} = \frac{2\sqrt[3]{2}}{\sqrt[3]{8}} = \sqrt[3]{2}$, (c) we multiply the numerator and denominator of the fraction by $\sqrt{3} + \sqrt{2}$, and then in the denominator we get a

rational number $\frac{2}{\sqrt{3} - \sqrt{2}} = \frac{2(\sqrt{3} + \sqrt{2})}{(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2})} = \frac{2(\sqrt{3} + \sqrt{2})}{(\sqrt{3})^2 - (\sqrt{2})^2} = 2(\sqrt{3} + \sqrt{2}) \approx 2(1.73 + 1.41) \approx 6.28$,

(d) $2(\sqrt{10} + \sqrt{3})$, (e) $(3\sqrt{2} + 4)/2$.

$$\begin{aligned}
 122. \quad (a) \quad & \frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} = \frac{1 \cdot (\sqrt{2}+1) - 1 \cdot (\sqrt{2}-1)}{(\sqrt{2})^2 - 1^2} = 2, \\
 (b) \quad & 8\sqrt{5}, \quad (c) \quad (16-8\sqrt{3}+3)(19+8\sqrt{3}) - (9+4\sqrt{2}) \times \\
 & (1-4\sqrt{2}+8) = (19^2 - (8\sqrt{3})^2) - (9^2 - (4\sqrt{2})^2) = (361 - 192) - \\
 & (81 - 32) = 169 - 49 = 120, \quad (d) \quad \frac{(2-6\sqrt{2}+9) \cdot (11+6\sqrt{2})}{2(9+4\sqrt{2}) \cdot (2-8\sqrt{2}+16)} = \\
 & \frac{11^2 - (6\sqrt{2})^2}{2 \cdot 2 \cdot (9^2 - (4\sqrt{2})^2)} = \frac{121-72}{4 \cdot (81-32)} = \frac{49}{4 \cdot 49} = 0.25.
 \end{aligned}$$

SECTION 1.14

$$\begin{aligned}
 123. \quad (a) \quad & (10^3)^{\frac{1}{3}} \cdot ((0.1)^4)^{\frac{1}{4}} + ((0.3)^3)^{\frac{1}{3}} \cdot 1 - \left(\frac{13}{10}\right)^1 = 10 \cdot 0.1 + \\
 & 0.3^1 - 1.3 = 0, \quad (b) \quad \left(\frac{64}{27}\right)^{-\frac{2}{3}} \div \left(\frac{100}{9}\right)^{-0.5} + (2.5^2)^{0.5} \div \frac{1}{(-4)^1} = \\
 & \left(\left(\frac{4}{3}\right)^3\right)^{-\frac{2}{3}} \div \left(\left(\frac{10}{3}\right)^2\right)^{-0.5} + 2.5 \cdot (-4) = \left(\frac{4}{3}\right)^{-2} \div \left(\frac{10}{3}\right)^{-1} + \\
 & 2.5 \cdot (-4) = \frac{9}{16} \div \frac{3}{10} - 10 = 1 \frac{7}{8} - 10 = -8 \frac{1}{8} = -8.125. \\
 124. \quad (a) \quad & a^{\frac{5}{6} + \frac{2}{3} - \frac{1}{6}} = a^{\frac{4}{3}}, \quad (b) \quad a^{\frac{4}{9} \cdot (-3) + \frac{1}{6} \cdot (-2)} = a^{-\frac{4}{3} - \frac{1}{3}} = \\
 & a^{-\frac{5}{3}}.
 \end{aligned}$$

$$\begin{aligned}
 125. \quad (a) \quad & \left(2a^{\frac{1}{2}} - 5\right)^2 + 20a^{\frac{1}{2}} = 4a + 25, \text{ for } a=1, 2 \text{ we get } 29.8, \\
 (b) \quad & 45.
 \end{aligned}$$

$$126. \quad (a) \quad \frac{a-9a^{\frac{1}{2}}}{\frac{3}{a^{\frac{1}{4}}} - 3a^{\frac{1}{2}}} = \frac{a^{\frac{1}{2}}(a^{\frac{1}{2}} - 9)}{a^{\frac{1}{2}}(a^{\frac{1}{4}} - 3)} = \frac{(a^{\frac{1}{4}})^2 - 3^2}{a^{\frac{1}{4}} - 3} = a^{\frac{1}{4}} + 3,$$

$$(b) \quad \frac{x-36x^{\frac{1}{3}}}{\frac{2}{x^{\frac{2}{3}}} - 6x^{\frac{1}{3}}} = \frac{x^{\frac{1}{3}}(x^{\frac{2}{3}} - 36)}{x^{\frac{1}{3}}(x^{\frac{1}{3}} - 6)} = \frac{(x^{\frac{1}{3}})^2 - 6^2}{x^{\frac{1}{3}} - 6} = x^{\frac{1}{3}} + 6.$$

$$127. \quad (a) \quad \left(\frac{b^{\frac{1}{2}}}{a^{\frac{1}{2}} \left(a^{\frac{1}{2}} + b^{\frac{1}{2}} \right)} - \frac{b^{\frac{1}{2}}}{a^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right)} \right) \cdot \frac{b-a}{2a^{\frac{1}{2}}b^{\frac{1}{2}}} =$$

$$\begin{aligned}
& \frac{b^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right) - b^{\frac{1}{2}} \left(a^{\frac{1}{2}} + b^{\frac{1}{2}} \right)}{a^{\frac{1}{2}} \left(a^{\frac{1}{2}} + b^{\frac{1}{2}} \right) \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right)} \cdot \frac{b-a}{2a^{\frac{1}{2}} b^{\frac{1}{2}}} = \frac{-2b(b-a)}{a^{\frac{1}{2}} (a-b) 2a^{\frac{1}{2}} b^{\frac{1}{2}}} = \\
& \frac{b^{\frac{1}{2}}}{a} = \frac{\sqrt{b}}{a}, \quad (b) \left(\frac{a-b}{a^{\frac{1}{2}} \left(a^{\frac{1}{4}} + b^{\frac{1}{4}} \right)} - \frac{a^{\frac{1}{2}} - b^{\frac{1}{2}}}{a^{\frac{1}{4}} + b^{\frac{1}{4}}} \right) \cdot \frac{1}{b^{-\frac{1}{2}} \left(a^{\frac{1}{4}} - b^{\frac{1}{4}} \right)} = \\
& \frac{(a-b) - a^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right)}{a^{\frac{1}{2}} \left(a^{\frac{1}{4}} + b^{\frac{1}{4}} \right) \cdot b^{-\frac{1}{2}} \left(a^{\frac{1}{4}} - b^{\frac{1}{4}} \right)} = \frac{a-b-a+a^{\frac{1}{2}} b^{\frac{1}{2}}}{a^{\frac{1}{2}} b^{-\frac{1}{2}} \left(a^{\frac{1}{4}} - b^{\frac{1}{4}} \right)} = \\
& \frac{b^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right)}{a^{\frac{1}{2}} b^{-\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}} \right)} = \frac{b}{a^{\frac{1}{2}}} = \frac{b}{\sqrt{a}} = \frac{b \sqrt{a}}{a}, \quad (c) \frac{\left(\frac{1}{a^{\frac{1}{3}}} \right)^3 - \left(\frac{1}{b^{\frac{1}{3}}} \right)^3}{\frac{1}{a^{\frac{1}{3}}} - \frac{1}{b^{\frac{1}{3}}}} = \\
& \frac{\left(\frac{1}{a^{\frac{1}{3}}} \right)^3 + \left(\frac{1}{b^{\frac{1}{3}}} \right)^3}{\frac{1}{a^{\frac{1}{3}}} + \frac{1}{b^{\frac{1}{3}}}} = \frac{\left(\frac{1}{a^{\frac{1}{3}}} - \frac{1}{b^{\frac{1}{3}}} \right) \left(a^{\frac{2}{3}} + a^{\frac{1}{3}} b^{\frac{1}{3}} + b^{\frac{2}{3}} \right)}{\frac{1}{a^{\frac{1}{3}}} - \frac{1}{b^{\frac{1}{3}}}} = \\
& \frac{\left(\frac{1}{a^{\frac{1}{3}}} + \frac{1}{b^{\frac{1}{3}}} \right) \left(a^{\frac{2}{3}} - a^{\frac{1}{3}} b^{\frac{1}{3}} + b^{\frac{2}{3}} \right)}{\frac{1}{a^{\frac{1}{3}}} + \frac{1}{b^{\frac{1}{3}}}} = 2a^{\frac{1}{3}} b^{\frac{1}{3}}, \quad (d) \frac{\left(\frac{1}{x^{\frac{1}{3}}} \right)^3 - \left(\frac{1}{y^{\frac{1}{3}}} \right)^3}{\frac{2}{x^{\frac{2}{3}} + x^{\frac{1}{3}} y^{\frac{1}{3}} + y^{\frac{2}{3}}}} = \\
& \frac{\left(\frac{1}{x^{\frac{1}{3}}} \right)^3 + \left(\frac{1}{y^{\frac{1}{3}}} \right)^3}{\frac{2}{\frac{2}{3} - x^{\frac{1}{3}} y^{\frac{1}{3}} + y^{\frac{2}{3}}}} = \left(x^{\frac{1}{3}} - y^{\frac{1}{3}} \right) - \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right) = -2y^{\frac{1}{3}}.
\end{aligned}$$

SECTION 1.15

128. We can define a function in the following three ways:

The 1st technique. By enumerating all pairs of numbers for the functions whose domain is a finite set, say, $\{(0, 0), (1, 1), (2, 4), (3, 9), (4, 16), (5, 25)\}$. We can also *tabulate* this set of pairs, one column being the domain of the function $\{0, 1, 2, 3, 4, 5\}$ and the other the range of the function $\{0, 1, 4, 9, 16, 25\}$. The tabulated values of functions are often used in practice (tables of squares, of square roots, cubes, etc.).

The 2nd technique. It is convenient to define the numerical values of functions by *formulas* in two variables. For example, the sum (S) of the angles of a convex polygon is expressed by the formula $S = 180^\circ (n - 2)$ ($n \in \mathbb{N}$ and $n \geq 3$) according as the number n of its sides. This method of defining a function is known as an *analytical method*.

The 3rd technique. A function is often defined by a *graph*, say, the graph of the temperature of a patient or the graph of the growth of a child with his age.

129. 1. To find the domain of a function, we must project its graph onto the x -axis, if we project the graph onto the y -axis, we get the range of the function. Figure 70a shows a function whose

domain is the segment $[-2, 5]$ and the range is the segment $[3, 9]$.

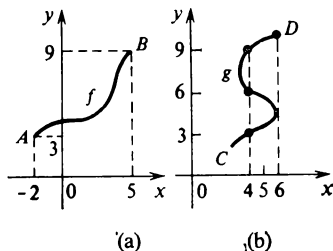


Fig. 70

2. Not every one. If every abscissa is associated with a single ordinate, then the correspondence defined by the graph is a function. Fig. 70a depicts the correspondence f whose graph, i.e. the curve AB is a function. Fig. 70b shows the correspondence g whose graph, i.e. the curve CD , is not a function since there are points on the graph for which one and the same abscissa is associated with

different ordinates. For example, points with the abscissa $x = 4$ have three different ordinates, i.e. $y_1 = 3$, $y_2 = 6$, $y_3 = 9$, and points with the abscissa $x = 6$ have two different ordinates, $y_1 = 4$, $y_2 = 10$.

130. The number a is a *root* (zero) of the function $f(x)$ if the value of the function is zero for $x = a$, i.e. if $f(a) = 0$. To find the roots of the function $f(x)$, we must solve the equation $f(x) = 0$. For example, the root of the function $f(x) = 3x - 6$ is a root of the equation $3x - 6 = 0$, $x = 2$. The roots of the equation $f(x) = x^2 - 1$ are $x_1 = 1$ and $x_2 = -1$. On the graph the *root of the function* is the point of intersection (or a point of tangency) of the abscissa axis and the graph of the function (at that point $y = 0$).

131. The function f is said to be *increasing* on the number set X if the larger value of the function corresponds to any larger value of the argument belonging to the set X , i.e. if the condition $x_2 > x_1$, $x_1 \in X$, $x_2 \in X$ implies $f(x_2) > f(x_1)$. For example, the function $f(x) = x^2$ increases on the interval $[0, 5]$. A *decreasing function* is a function on the set X such that the condition $x_2 > x_1$, $x_1 \in X$, $x_2 \in X$ implies $f(x_2) < f(x_1)$. For example, the function $f(x) = x^2$ decreases on the interval $[-3, 0]$. The function which increases or decreases on the given set is known as a *monotonic function* on that set. A point which moves along the graph of an *increasing* function from left to right rises *upwards* and a point which moves along the graph of a *decreasing* function from left to right descends *downwards*.

132. (a) Assume $x_2 > x_1$, i.e. $x_2 - x_1 > 0$, and then $f(x_2) - f(x_1) = (2x_2 - 3) - (2x_1 - 3) = 2(x_2 - x_1) > 0$, or $f(x_2) > f(x_1)$. Hence the given function increases, (b) assume $x_2 > x_1$, or $x_2 - x_1 > 0$, and then $f(x_2) - f(x_1) = x_2^3 - x_1^3 = (x_2 + x_1) \times (x_2 - x_1)$, but $x_2 + x_1 < 0$ ($x_2 < 0$ and $x_1 < 0$ by the hypothesis), $x_2 - x_1 > 0$, and therefore $f(x_2) - f(x_1) < 0$ and $f(x_2) < f(x_1)$.

133. 1. The function f with the domain X and the range Y is said to be *invertible* if the inverse relation g between the set Y

and the set X is a function, i.e. every element of the set Y is associated with no more than one element of the set X . In this case the function g is the *inverse* of the function f . For example, the function $y = 2x$ on the set $X = \{-1, 0, 1, 2\}$ is invertible since the range of this function is $Y = \{-2, 0, 2, 4\}$ and every value of the set Y is associated with only one element of the set X , which can be seen from the pairs of the inverse relation: $(-2, -1), (0, 0), (2, 1), (4, 2)$. However, the function $y = x^2$ is not invertible on this set since the set of its values is $Y = \{1, 0, 1, 4\}$; the pairs of the inverse relation $(1, -1), (0, 0), (1, 1), (4, 2)$ show that one and the same element of the set Y ($y = 1$) is associated with two distinct elements of the set X ($x = -1$ and $x = 1$). However, on the intervals of monotonicity of the function it has an inverse. An invertible function and its inverse are said to be *functions inverse to each other*.

2. The graphs of the functions, which are inverse to each other, are symmetric about the straight line $y = x$, i.e. the bisector of the first and the third quadrant.

134. $B = \{-7, -5, -3, -1, 1\}$. The inverse relation is $(-7, -2), (-5, -1), (-3, 0), (-1, 1), (1, 2)$, the function f is invertible.

135. If the function $y = f(x)$ is invertible, then to find the formula defining the inverse, we must express the variable x appearing in this formula in terms of y and change x to y and y to x . For example, to define the inverse of the function $y = 3x$, we express x in terms of y and get $x = y/3$, and substitute y for x and x for y in the resulting equation. Thus the inverse of the given function is defined by the formula $y = x/3$. The graphs of two functions inverse to each other $y = 3x$ and $y = x/3$ are symmetric about the straight line $y = x$ (Fig. 71).

136. (a) We express x in terms of y and get $x = 2y + 6$. The inverse of the given function is $y = 2x + 6$, (b) $y = -3x + 3$.

137. $D = \frac{C}{\pi} = \frac{1}{\pi} C$. D and C

are in direct proportion, the proportionality factor is $1/\pi$.

138. 1. To construct a straight line, it is sufficient to know the positions of two of its points, and therefore the graph of direct proportionality can be constructed using any two of its points. It is convenient to take the origin as one of them and the point $(1, k)$ as the other.

139. (a) 45° , (b) 135° .

140. 1. The set of all nonzero numbers: $(-\infty, 0) \cup (0, +\infty)$.

2. If $k > 0$, then the branches of the hyperbola lie in the first and the third quadrant and if $k < 0$, then they lie in the second and the fourth quadrant.

141. (a) S and h are in direct proportion, (b) S and a are in direct proportion, (c) $a = S/h$, a and h are in inverse proportion.

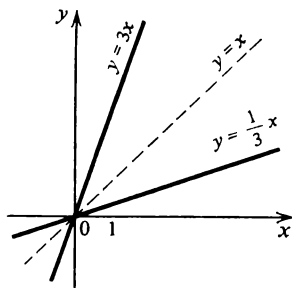


Fig. 71

142. (a) A hyperbola and a straight line which passes through the origin meet at points (2, 6) and (-2, -6), (b) (3, -2), (-3, 2).

143. Substituting the coordinates of the given point for x and y , we get $5 = k(-1) + 3$, $k = -2$, $y = -2x + 3$; since $k < 0$, the function decreases. The root of this function is a number for which $y = 0$, or $-2x + 3 = 0$, hence the root of the function is $x = 1.5$.

144. (a) To find the point of intersection of the graph of the function and the y -axis, we must assume that $x = 0$, and then $y = -2$. We get the point of intersection with the x -axis when we find the root of the function, i.e. if we assume that $y = 0$; then

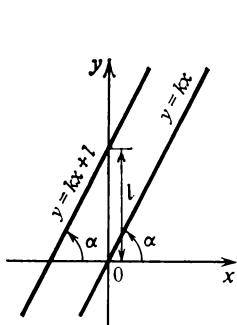


Fig. 72

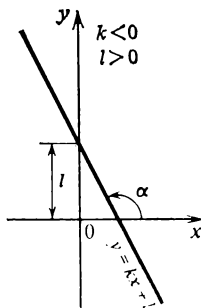


Fig. 73

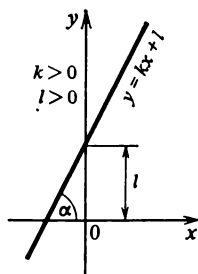


Fig. 74

$x - 2 = 0$, whence $x = 2$. Thus the graph of this function cuts the coordinate axes at the points $A(0, -2)$ and $B(2, 0)$ (b) $(0, -3)$, $(6, 0)$, (c) $(0, -6)$, $(-2.5, 0)$.

145. For the same value of k and different l the graphs of the function are parallel to the graph of the direct proportionality $y = kx$ (Fig. 72). If $k > 0$, then the angle of inclination of the graph is acute, and if $k < 0$, then the angle is obtuse (Fig. 73).

146. (a) The graph is shown in Fig. 74, (b) see Fig. 73, (c) see Fig. 75, (d) see Fig. 76, (e) see Fig. 77.

147. (a) Each of these graphs lies in the upper half-plane, passes through the origin and is symmetric about the y -axis. The graph

of the function $y = \frac{1}{2}x^2$ is a more gently sloping curve than the

curve $y = x^2$ since the points with similar abscissas have ordinates half as large; for instance, for $x = 3$ we have $y_1 = 9$ for the first graph and $y_2 = 4.5$ for the second graph (Fig. 78), (b) the graph of the function $y = 2x^2$ is a steeper curve than that of $y = x^2$ (Fig. 79), (c) every point of the first graph is associated with a point of the second graph which is symmetric about the x -axis; for instance, for $x = 1$ we have $y_1 = 2$ for the first graph and $y_2 = -2$ for the second graph (Fig. 80). This means that the second graph is symmetric with respect to the first one about the x -axis, the first graph lying in the upper half-plane and the second in the lower half-plane. Both graphs are symmetric about the y -axis and pass through the origin.

148. For $a > 0$ the function $y = ax^2$ increases if $X = [0, +\infty)$, for $a < 0$ the function increases if $X = (-\infty, 0]$.

149. (a) $-9 = a \cdot 3^2$, $a = -1$, $y = -x^2$, (b) $3 = a(-1)^2$, $a = 3$, $y = 3x^2$.

150. (a) We can consider the numbers $\sqrt[3]{0.6}$ and $\sqrt[3]{2/3}$ to be the values of the increasing function $y = \sqrt[3]{x}$. The inequality

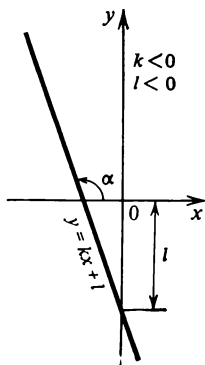


Fig. 75

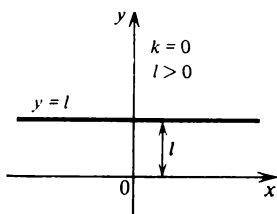


Fig. 76

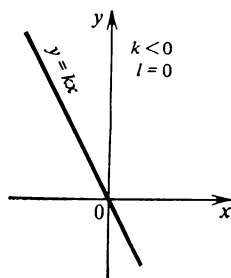


Fig. 77

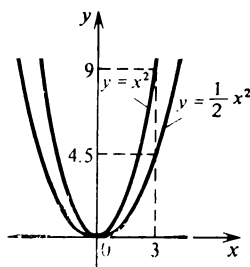


Fig. 78

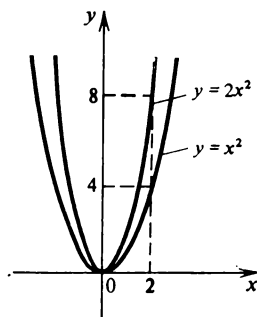


Fig. 79

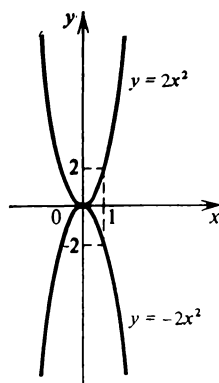


Fig. 80

$0.6 < 2/3$ ($\frac{2}{3} = 0.666 \dots$) holds true and, consequently, the

inequality $\sqrt[3]{0.6} < \sqrt[3]{2/3}$ also holds true, (b) $\sqrt[3]{\frac{5}{7}} > \sqrt[3]{0.7}$,

(c) $\sqrt[8]{0.88} < \sqrt[8]{1}$, (d) $\sqrt[10]{(6.2-6.1)(6.2+6.1)} = \sqrt[10]{1.23} > \sqrt[10]{1.22}$,

SECTION 1.16

151. (1) A *root of an equation* in one variable is the value of the variable which reduces the equation to a true equality.

2. To *solve an equation* means to find the set of its roots.

3. The set of roots of an equation in one variable can consist of

(a) one element (for instance, the set of roots of the equation $\frac{1}{2}x = -1$ consists of one element -2), (b) several elements (for instance the set of roots of the equation $x^2 = 1$ consists of two elements, 1 and -1), (c) an infinite number of elements (for instance, in the equation $\sqrt{x^2} = -x$, any nonpositive number $x \in (-\infty, 0]$ is its root), (d) the set of roots may be empty (the equation $x^2 = -4$, for instance, has an empty set of roots).

152. If $a \neq 0$, then the equation $ax = b$ has a unique root $x = b/a$. If $a = 0$ and $b = 0$, then the equation assumes the form $0x = 0$. The set of roots of such an equation is the set of all real numbers. If $a = 0$ and $b \neq 0$, the equation assumes the form $0x = b$. This equation has no roots since its left-hand side is zero and the right-hand side is nonzero.

153. (a) This equation can be reduced to the equation $x^2 + 12x = 0$. We put x before the brackets, i.e. $x(x + 12) = 0$. The product of two numbers is zero if and only if at least one of the factors is zero, i.e. $x = 0$ or $x + 12 = 0$. Therefore the set of roots of the given equation is $\{-12, 0\}$. (b) We multiply both sides of the equation by 6, i.e. by the least common multiple of the numbers 2 and 6, and obtain

$$(5x - 1) \cdot 3 - (x + 5) = 6, \quad 15x - 3 - x - 5 = 6;$$

from this we find that the root of the equation is 1. (c) $\{51\}$, (d) $\{17\}$, (e) $\emptyset$. *Hint.* The equation can be reduced to $0x = 109$, (f) $\{0.754\}$. *Hint.* Multiply the numerator and denominator of each fraction

by 100. (g) $\left\{ \frac{1 + \sqrt{3}}{2}, \sqrt{3} - 1 \right\}$.

154.(a) To use the condition for the equality of a fraction to zero ($a/b = 0$ only if $a = 0$ and $b \neq 0$), we transfer $7/8$ to the left-hand side of the equation, i.e.

$$\frac{6x - 37}{2(x - 8)} - \frac{2(5x - 39)}{3(x - 8)} - \frac{7}{8} = 0.$$

Then we transform the left-hand side of the equation into a fraction, i.e.

$$\frac{12(6x - 37) - 2 \cdot 8(5x - 39) - 7 \cdot 3(x - 8)}{24(x - 8)} = 0, \quad \frac{-29x + 348}{24(x - 8)} = 0.$$

This fraction turns into zero for those and only those values of x for which $-29x + 348 = 0$ and $24(x - 8) \neq 0$, i.e. the given equation is equivalent to the simultaneous equations

$$\begin{cases} -29x + 348 = 0, \\ 24(x - 8) \neq 0. \end{cases}$$

The equation $-29x + 348 = 0$ has a root 12, for this value of x we have $24(x - 8) \neq 0$, and this means that the number 12 is the root of the given equation. *Answer:* $\{12\}$,

$$(b) \frac{x(x-4)-5-1(x-4)}{(x-1)^2} = 0, \quad \frac{x^2-4x-5-x^2+8x-16}{(x-1)^2} = 0,$$

$$\frac{4x-21}{(x-4)^2} = 0, \quad \begin{cases} 4x-21=0, \\ (x-4)^2 \neq 0. \end{cases} \quad \text{Answer: } \{5.25\},$$

$$(c) \frac{30+(x-1)(x+7)-(x^2-49)}{x^2-49} = 0, \quad \begin{cases} 6x+72=0, \\ x^2-49 \neq 0. \end{cases} \quad \text{Answer: } \{-12\},$$

(d) we can solve this equation using the condition for the equality of a fraction to zero, but we can also solve it using the condition for the equality of two fractions with the same denominator, i.e.

$\frac{a}{b} = \frac{c}{b}$, whence $a = c$ and $b \neq 0$. Reducing the fractions to a common denominator, we obtain

$$\frac{4(x+2)}{x^2-4} - \frac{5(x-2)}{x^2-4} = \frac{12}{x^2-4}.$$

We represent the left-hand side of the equation as a fraction, i.e.,

$$\frac{4(x+2)-5(x-2)}{x^2-4} = \frac{12}{x^2-4},$$

since the fractions have the same denominator, their values are only equal for the values of x for which their numerators are equal and the denominators are nonzero. Consequently we have simultaneous equations

$$\begin{cases} 4(x+2)-5(x-2)=12, \\ x^2-4 \neq 0, \end{cases}$$

solving which we get the answer: $\{6\}$,

$$(e) \frac{(x+3)^2-(12x-15)-(x-3)^2}{x^2-9} = 0, \quad \frac{15}{x^2-9} = 0. \quad \text{Since the nu-}$$

merator is nonzero, the equation has no roots. Answer: $\emptyset$.

155. We designate the required distance as x km. The passenger train covers this distance in $x/60$ h and the goods train in $x/48$ h. By the hypothesis, the time the passenger train needs is less by 36 min = 36/60 h = 3/5 h than the time the goods train needs,

and therefore we have an equation $\frac{x}{48} - \frac{x}{60} = \frac{3}{5}$, from which we

find that $x = 144$ km. *Verification.* The passenger train will travel 144 km in 2.4 h ($144 : 60 = 2.4$), and the goods train in 3 h ($144 : 48 = 3$). Hence the passenger train needs 0.6 h less to travel 144 km ($3 - 2.4 = 0.6$). Answer. The distance between stations A and B is 144 km.

It is sometimes convenient to tabulate the dependence between the unknown and the given variable before setting up an equation.

For instance, the solution of this problem can be written as follows:

	Distance, km	Speed, km/h	Time, h
Passenger train	x	60	$\frac{x}{60}$
Goods train	x	48	$\frac{x}{48}$

156. 60 km/h, 80 km/h. *Hint.* If the speed of the first car is x km/h, then we get an equation $3.5x - 2.5(x + 20) = 10$.

157. 4 h. *Hint.* If the time needed now to machine one part is x h, then we arrive at an equation $30x = 24(x + 1)$.

158. 500 kg, 600 kg, 300 kg. *Hint.* If x kg were sold on the first day, then we have an equation $x + (x + 100) + \frac{3}{5}x = 1400$.

159. Assume that we have to take x litres of the first solution, then we need $(100 - x)$ litres of the second solution. The first solution of nitric acid contains 30 per cent of x , i.e. $0.3x$ litres and the second solution contains 55 per cent of $(100 - x)$ litres, i.e. $0.55(100 - x)$ litres. By the hypothesis, the resulting solution must contain 50 litres of nitric acid (50 per cent of 100 litres is 50 litres). We get

$$0.3x + 0.55(100 - x) = 50, \quad x = 20 \text{ litres.}$$

Verification. Twenty litres of the first solution contain $20 \cdot 0.3 = 6$ litres of nitric acid and 80 litres of the second solution contain $(100 - 20 = 80) \cdot 0.55 = 44$ litres of nitric acid. The total amount of nitric acid in the two solutions is $6 + 44 = 50$ l, and this corresponds to the hypothesis. *Answer:* 20 l and 80 l.

160. 1.25 l. *Hint.* Assume that we have to add x litres of the 80% solution, and then $3.5 + 0.8x = 0.72(5 + x)$.

161. Problems of this kind are known as problems involving *concerted action*. We assume that the time the first excavator needs to do the whole job is x h, then the second excavator will need $2x$ h to do the whole job.

Warning! Sometimes students erroneously assume that the first excavator needs x h and the second $2x$ h, forgetting that "faster" means spending less time. We form the following table:

	Time needed to do the whole job, h	Work done in 1 h	Time spent, h	Work done in 12 h
Excavator I	x	$\frac{1}{x}$	12	$\frac{12}{x}$
Excavator II	$2x$	$\frac{1}{2x}$	12	$\frac{12}{2x}$

Since as a result of the concerted action the whole job, taken to be unity, was done, we have $\frac{12}{x} + \frac{12}{2x} = 1$, whence $x = 18$. *Answer:* 18 h and 36 h.

162. 1. If $a = 0$, then the equation is not quadratic but of the first degree. 2. An equation of the form $ax^2 + bx = 0$ can be solved by factoring its left-hand side, i.e. $x(ax + b) = 0$. Its roots are $x_1 = 0$ and $x_2 = -b/a$. This equation can be solved by graphical means, writing it in the form $ax^2 = -bx$. The abscissas of the points of intersection of the graphs $y = ax^2$ and $y = -bx$ are the roots of the equations. The equation $ax^2 + c = 0$ has two opposite roots if a and c have opposite signs: $x_1 = \sqrt{-c/a}$ and $x_2 = -\sqrt{-c/a}$. The equation $ax^2 = 0$ has two equal roots $x_1 = x_2 = 0$.

163. (a) $\{0, 1\}$, (b) $\{-3, 3\}$, (c) $\{0\}$.

164. (a) The parabola $y = x^2$ and the straight line $y = 2x$ meet at two points whose abscissas are $x_1 = 0$ and $x_2 = 2$. They are the roots of the equation, (b) $\{-2, 2\}$.

165. 1. $R = \sqrt{S/\pi}$, $t = \sqrt{2S/g}$. 2. $t = 2v/g$.

166. (a) $D = 2^2 - 4m^2$; the equation has one root if $D = 0$; $4 - 4m^2 = 0$, hence $m = -1$ or $m = 1$, (b) $m = -30$ or $m = 30$, (c) $m = 9$.

167. (a) $a = 1$, $b = 1$, $c = -6$; $D = 1^2 - 4 \cdot 1 \cdot (-6) = 25$; $x = \frac{-1 \pm \sqrt{25}}{2}$, whence $x_1 = -3$ and $x_2 = 2$. (b) The equation

can be reduced to the form $3x^2 + 11x + 10 = 0$; $D = 11^2 - 4 \cdot 3 \cdot 10 = 1$; $x_1 = -2$ and $x_2 = -5/3$. (c) $b = -4\sqrt{3}$ is an even number. We use formula (2) since $\frac{D}{4} = \left(\frac{b}{2}\right)^2 - ac = (-2\sqrt{3})^2 - 1 \cdot 9 = 3$, we have $x = \frac{2\sqrt{3} \pm \sqrt{3}}{1}$; $x_1 = \sqrt{3}$,

$x_2 = 3\sqrt{3}$, (d) $\left\{-\frac{3}{2}, \sqrt{2}\right\}$. *Hint.* $D = (3 - 2\sqrt{2})^2 - 4 \cdot 2 \cdot (-3\sqrt{2}) = 9 - 12\sqrt{2} + 8 + 24\sqrt{2} = 9 + 12\sqrt{2} + 8 = (3 + 2\sqrt{2})^2$, (e) $\{-5\}$. *Hint.* The equation can be reduced to the system

$$\begin{cases} x^2 + 2x - 15 = 0, \\ x(x - 3)(x + 3) \neq 0. \end{cases}$$

Solving the first equation, we get $x_1 = -5$ and $x_2 = 3$. We reject the second root since only the number -5 satisfies the hypothesis $x(x - 3)(x + 3) \neq 0$, (f) $\{1\}$. *Hint.* We reject the second root $x = -1/2$ of the resulting equation.

168. (a) This is an equation of the form $ax^4 + bx^2 + c = 0$. Equations of this kind are known as *biquadratic*. We designate x^2 as z and get

$$4z^2 - 13z + 9 = 0; \quad z = 1 \quad \text{or} \quad z = 9/4.$$

The union of the roots of the equations $x^2 = 1$ and $x^2 = 9/4$ is the set of roots of the given equation; the roots of the first equation are -1 and 1 , and those of the second equation are $-3/2$ and $3/2$. Thus the set of roots of the given equation is $\{-3/2, -1, 1, 3/2\}$, (b) we

designate $\sqrt{x}$ as z , and then $x = z^2$ and $z^2 + z = 12$, $z^2 + z - 12 = 0$; $z = 3$ or $z = -4$, i.e. $\sqrt{x} = 3$ or $\sqrt{x} = -4$. The last equation has no roots ($\sqrt{x} \geq 0$). Solving the first equation, we get $x = 9$. *Answer:* {9}, (c) we introduce the designation $x^2 + 2 = z$, and then $x^2 + 3 = z + 1$, and the given equation can be reduced to the equation

$$\frac{2}{z} + \frac{3}{z+1} = 2, \quad \frac{2(z+1) + 3z - 2z(z+1)}{z(z+1)} = 0,$$

whose roots are $z = 2$ or $z = -1/2$, whence

$$x^2 + 2 = 2 \quad \text{or} \quad x^2 + 2 = -1/2.$$

The last equation has no roots. The equation $x^2 + 2 = 2$ has a unique root $x = 0$. *Answer:* {0}, (d) if we designate $\frac{x^2+1}{x}$ as z ,

we have $\frac{x}{x^2+1} = \frac{1}{z}$, and then the given equation assumes the form $z + 1/z = 2.9$. The numbers 2.5 and 0.4 are solutions of this equation, hence

$$\frac{x^2+1}{x} = 2.5 \quad \text{or} \quad \frac{x^2+1}{x} = 0.4.$$

The second equation has no roots, the set of roots of the first equation and, hence, of the given equation is $\{1/2, 2\}$, (e) we assume $x - 2 = a$, and then we have

$$2a^2 - 3a + 1 = 0; \quad a_1 = 1/2 \quad \text{and} \quad a_2 = 1,$$

whence $x - 2 = 1/2$ or $x - 2 = 1$. The set of solutions of the given equation is the union of the sets of solutions of the last two equations: $\{2.5, 3\}$, (f) we assume that $x^2 = a$, and then $2x^2 + 7 =$

$2a + 7$, $a - \frac{50}{2a+7} = 7$. From this we get

$$2a^2 - 7a - 99 = 0; \quad a = 9 \quad \text{or} \quad a = -11/2.$$

Solving the equation $x^2 = 9$, we find the set of roots of the given equation $\{-3, 3\}$. The equation $x^2 = -11/2$ has no roots, (g) we designate $x^2 - 4x + 10 = a$, and then we have $x^2 - 4x + 6 = a - 4$ and

$$21/a = a - 4, \quad a^2 - 4a - 21 = 0; \quad a = -3 \quad \text{or} \quad a = 7.$$

The set of roots of the given equation is the union of the sets of roots of the equations $x^2 - 4x + 10 = -3$ and $x^2 - 4x + 10 = 7$. The first equation has no roots since its discriminant $D = 16 - 52 < 0$. We can find the set of roots of the given equation by solving the equation $x^2 - 4x + 3 = 0$. It is $\{1, 3\}$.

169. (a) We write the given equation in the form $x^2 = x + 2$. Then we plot a parabola $y = x^2$ and a straight line $y = x + 2$. The abscissas of their intersection points are -1 and 2 . Consequently, the roots of the given equation are $x_1 = -1$ and $x_2 = 2$.

Remark. It is a good practice to employ a template in order to avoid plotting the parabola $y = x^2$ using its points each time, (b) we reduce the given quadratic equation to the form $x^2 + 0.5x - 1.5 = 0$. Then we draw the graphs of the functions $y = x^2$ and $y = -0.5x + 1.5$. The abscissas of their intersection points are -1.5 and 1 . Consequently the roots of the given equation are $x_1 = -1.5$

and $x_2 = 1$, (c) the parabola $y = x^2$ and the straight line $y = -4x - 4$ have only one point in common, $(-2, 4)$. The equation has a unique root $x = -2$, (d) the parabola $y = x^2$ and the straight line $y = 2x - 5$ have no points in common. The equation has no roots.

170. Yes, it is. It can be formulated as follows: the numbers m and n are roots of the quadratic equation $x^2 - (m+n)x + mn = 0$.

171. (a) $D/4 = 4 + 2.25 = 6.25 > 0$; $x_1 + x_2 = -16/9$, $x_1 x_2 = -4/9$, (b) $x_1 + x_2 = 0$, $x_1 x_2 = -20/3$, (c) $x_1 + x_2 = 48$, $x_1 x_2 = 0$.

172. $x_1 = 1$; $x_1 x_2 = -9$, whence $x_2 = -9$; $b = -(x_1 + x_2) = -(1 - 9) = 8$.

173. (a) According to the inverse of the Vieta theorem, we have an equation $x^2 - (-0.6 + 1.6)x + (-0.6) \cdot 1.6 = 0$, whence we find that $x^2 - x - 0.96 = 0$, (b) $x^2 - (-\sqrt{3} + \sqrt{3})x + (-\sqrt{3})\sqrt{3} = 0$, $x^2 - 3 = 0$, (c) $x^2 - (5 - 2\sqrt{6} + 5 + 2\sqrt{6})x + (5 - 2\sqrt{6})(5 + 2\sqrt{6}) = 0$, $x^2 - 10x + 1 = 0$, (d) $ax^2 - 2kx + c = 0$.

174. (a) $x_1 = m$, $x_2 = n$, (b) $x_1 = \sqrt{3}$, $x_2 = \sqrt{5}$.

175. 1. We must equate the trinomial to zero and solve the quadratic equation $ax^2 + bx + c = 0$. 2. A quadratic trinomial, as well as a quadratic equation, may have two distinct roots, one root (two coinciding roots), and may have no roots. This depends on the discriminant $D = b^2 - 4ac$ of the quadratic trinomial which is also a discriminant of the corresponding quadratic equation.

176. (a) $x^2 + px + q = (x - x_1)(x - x_2)$, where x_1 and x_2 are the roots of the trinomial, (b) in this case the trinomial has two equal roots and then $ax^2 + bx + c = a(x - x_1)(x - x_1) = a(x - x_1)^2$.

177. (a) $D = (-3)^2 - 4 \cdot 5 \cdot (-2) = 49$, $49 > 0$. The trinomial has two distinct roots $x_1 = -0.4$ and $x_2 = 1$. Consequently, $5x^2 - 3x - 2 = 5(x - (-0.4))(x - 1) = 5(x + 0.4)(x - 1) = (5x + 2) \times (x - 1)$, (b) $12x^2 - 11x + 2 = 12\left(x - \frac{1}{4}\right)\left(x - \frac{2}{3}\right) = 4\left(x - \frac{1}{4}\right) \times 3\left(x - \frac{2}{3}\right) = (4x - 1)(3x - 2)$.

178. (a) The roots of the trinomial in the numerator are $a_1 = -4$, $a_2 = 3$, therefore $\frac{a^2 + a - 12}{9 - a^2} = \frac{(a + 4)(a - 3)}{(3 - a)(3 + a)} = -\frac{(a + 4)(a - 3)}{(a - 3)(a + 3)} = -\frac{a + 4}{a + 3}$, (b) $\frac{(a + 3)^2}{-(a^2 + a - 6)} = \frac{(a + 3)^2}{-(a - 2)(a + 3)} = \frac{a + 3}{2 - a}$, (c) $\frac{6a^2 - 7a - 3}{2a^2 - a - 3} = \frac{6\left(a + \frac{1}{3}\right)\left(a - \frac{3}{2}\right)}{2\left(a - \frac{3}{2}\right)(a + 1)} = \frac{3a + 1}{a + 1}$, (d) $\frac{5x^{n+2} + 4x^{n+1} - 9x^n}{x^{n+2} - x^n} = \frac{x^n(5x^2 + 4x - 9)}{x^n(x^2 - 1)} = \frac{5(x - 1)\left(x + \frac{9}{5}\right)}{(x - 1)(x + 1)} = \frac{5x + 9}{x + 1}$.

179. (a) 0. *Hint.* First factor the denominators of the fractions and cancel like terms, (b) $\left(\frac{a^2-6a+5}{a^2-7a+10} - \frac{a^2-6a+8}{a^2-5a+4} \right) \times (a^2-3a+2) = \left(\frac{(a-5)(a-1)}{(a-5)(a-2)} - \frac{(a-2)(a-4)}{(a-1)(a-4)} \right) (a-2)(a-1) = \frac{a-1}{a-2} (a-2)(a-1) - \frac{a-2}{a-1} (a-2)(a-1) = (a-1)^2 - (a-2)^2 = 2a-3$. This expression becomes meaningless for $a=1$, $a=2$, $a=4$ and $a=5$.

$$180. (a) \frac{x}{x+2} - \frac{3}{x+3} - \frac{5}{(x+2)(x+3)} = 0, \\ \frac{x(x+3) - 3(x+2) + 5}{(x+2)(x+3)} = 0; \text{ hence}$$

$$\begin{cases} x^2 + 3x - 3x - 6 + 5 = 0, \\ (x+2)(x+3) \neq 0, \end{cases} \quad \text{or} \quad \begin{cases} x^2 - 1 = 0, \\ (x+2)(x+3) \neq 0; \end{cases}$$

$x_1 = -1$, $x_2 = 1$. Answer: $\{-1, 1\}$, (b) $\{0\}$.

181. (a) Here $x \neq 1$; reducing the fraction, we get $y = x + 2$. The graph is a straight line in which the point with the abscissa $x = 1$ is "punched" (Fig. 81), (b) see Fig. 82.

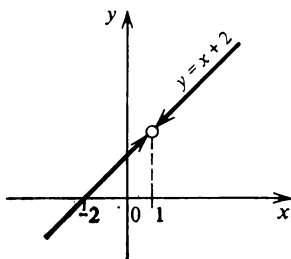


Fig. 81

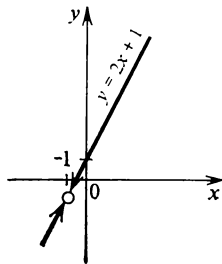


Fig. 82

182. (a) $y = x^2 - 8x + 10 = (x^2 - 2 \cdot 4x + 16) - 16 + 10 = (x - 4)^2 - 6$. The parabola is convex downwards ($a > 0$), its vertex is at the point $O_1(4, -6)$, the straight line $x = 4$ is the axis of symmetry. For $x = 0$ we have $y = 10$. Consequently the parabola cuts the y -axis at the point $(0, 10)$. To find the points at which the parabola cuts the x -axis, we set $y = 0$, and then $x_1 = 4 - \sqrt{6} \approx 1.5$ and $x_2 \approx 4 + 2.5 \approx 6.25$ (Fig. 83).

Remark. We could find the coordinates of the vertex of the parabola from the formulas $x = -\frac{b}{2a}$, $y = -\frac{D}{4a}$, and then $x = -\frac{-8}{2} = 4$, $y = \frac{64 - 4 \cdot 1 \cdot 10}{4} = -6$, or $y_{x=4} = 4^2 - 8 \cdot 4 + 10 = -6$,

$$(b) y = x^2 - 5x + 4 = -(x^2 + 5x - 4) = -\left(x^2 + 2 \cdot \frac{5}{2}x + \left(\frac{5}{2}\right)^2 - \right.$$

$\left(\frac{5}{2}\right)^2 - 4 = -\left(\left(x + \frac{5}{2}\right)^2 - 10\frac{1}{4}\right) = -(x + 2.5)^2 + 10.25$. The parabola is convex upwards, the coordinates of the vertex are $(-2.5, 10.25)$, the parabola cuts the y -axis at the point $(0, 4)$. To find the points of intersection of the parabola and the x -axis,

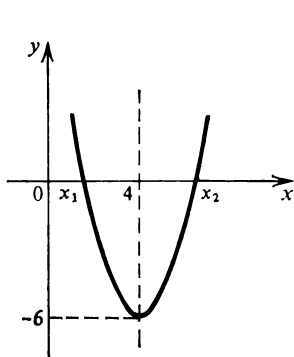


Fig. 83

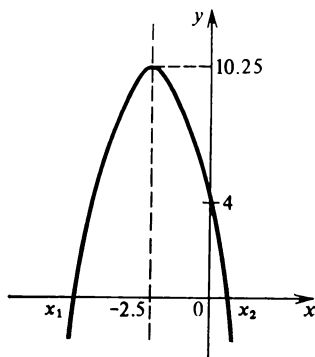


Fig. 84

we set $y=0$ and solve the equation $-x^2 - 5x + 4 = 0$ or $x^2 + 5x - 4 = 0$; we have $x_1 = \frac{-5 - \sqrt{41}}{2} \approx -5.7$, $x_2 = \frac{-5 + \sqrt{41}}{2} \approx 0.7$ (Fig. 84).

183. (a) $y = (x + 1.5)^2 - 6.25$ (Fig. 85). The function decreases if $x < -1.5$ and increases if $x > -1.5$. The values of this quadratic function are positive for all x belonging to the union

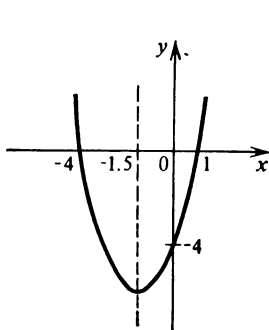


Fig. 85

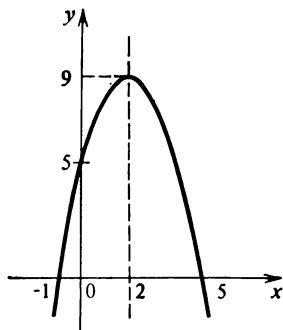


Fig. 86

of the intervals $(-\infty, -4) \cup (1, \infty)$ and negative for all x belonging to the interval $(-4, 1)$, (b) $y = -(x - 2)^2 + 9$. The function decreases if $x \in (2, \infty)$ and increases if $x \in (-\infty, 2)$. The values

of the function are positive if $x \in (-1, 5)$, $y < 0$ if $x \in (-\infty, 1) \cup (5, \infty)$ (Fig. 86), (c) $2(x^2 + 2x) - 6 = 2(x^2 + 2x + 1 - 1) - 6 = 2(x + 1)^2 - 8$. The parabola is congruent to the parabola $y = 2x^2$, the straight line $x = -1$ is the axis of symmetry. The coordinates of the vertex are $(-1, -8)$, the roots are $x_1 = -3$ and $x_2 = 1$. The function decreases if $x \in (-\infty, -1)$ and increases if $x \in (-1, \infty)$. The values of the function are $y > 0$ if $x \in (-\infty, -3) \cup (1, \infty)$, $y < 0$ if $x \in (-3, 1)$.

184. Assume that there are x rows in the cinema and hence x seats in each row, and this means that there are x^2 seats in the cinema. If we double the number of rows and reduce the number of seats in a row by 10, then the number of seats in the cinema will be $2x(x - 10)$ and we get an equation $2x(x - 10) = x^2 + 300$ whose roots are -10 and 30 . By the hypothesis $x > 0$, and, consequently, the problem has a unique solution. *Answer*: 30 rows.

185. 6 cm. *Hint*. If we designate the length of each leg as x , we get an equation $0.5 \cdot 2x(x - 2) = 0.5x^2 + 6$.

186. Designating the initial speed of the train as x , we get an equation $\frac{36}{x} - \frac{36}{x+4} = \frac{1}{10}$ ($6 \text{ min} = \frac{1}{10} \text{ h}$), from which we find that $x_1 = -40$ and $x_2 = 36$. By the hypothesis $x > 0$, and, consequently, the initial speed of the train is 36 km/h.

187. 12 min. *Hint*. Assume that the normal speed of the ship is x km/h. Then we arrive at an equation $\frac{36}{x-3} + \frac{36}{x+3} = 5$.

Solving this equation, we find that the average speed at which the ship must cover the whole distance is 15 km/h and it must cover the whole distance in $72/15$ h, i.e. in 4 h 48 min. Actually it needed 5 h.

188. 20 centners, 420 hectares. *Hint*. If the harvest was x centners from a hectare, we arrive at an equation $\frac{8400}{x} - \frac{8400}{x+1} = 20$,

$$\text{or } \frac{420}{x} - \frac{420}{x+1} = 1.$$

189. 10 h, 15 h. *Hint*. Assuming the time in hours the second crane needs to do the whole job alone to be x , we get an equation $\frac{6}{x} + \frac{6}{x+5} = 1$ (see the explanation of the solution of Problem 161).

190. 11 cm and 3 cm. *Hint*. If we assume the radius of one circle to be r cm, then the radius of the other circle is $(14 - r)$ cm, and the areas are πr^2 and $\pi(14 - r)^2$ respectively. Hence we get an equation $\pi r^2 + \pi(14 - r)^2 = 130\pi$.

191. Assume that initially there were a people in the settlement and the yearly increment of the population was x per cent. Then in a year the population in the settlement will be

$$a + a \frac{x}{100} = a \left(1 + \frac{x}{100} \right), \text{ in two years it will be } a \left(1 + \frac{x}{100} \right) + a \left(1 + \frac{x}{100} \right) \frac{x}{100} = a \left(1 + \frac{x}{100} \right) \left(1 + \frac{x}{100} \right) = a \left(1 + \frac{x}{100} \right)^2.$$

By the hypothesis, in two years the population in the settlement will be $3a$, and hence we get an equation $a \left(1 + \frac{x}{100}\right)^2 = 3a$, or $1 + \frac{x}{100} = \sqrt{3}$, whence $x = 100(\sqrt{3} - 1) \approx 73\%$, i.e. the average yearly increment in the population is approximately 73%.

SECTION 1.18

192. 1. The graph of a linear equation of the form $ax + by = c$, where at least one of the coefficients, a or b , is nonzero, is a straight line. This follows from the fact that for $b \neq 0$ the equation can be reduced to the form $y = -\frac{a}{b}x + \frac{c}{b}$ (here $k = -a/b$, $l = c/b$). If $a = 0$, $b \neq 0$ and $c \neq 0$, then $y = c/b$ and the graph is a straight line parallel to the x -axis. If $c = 0$ and $b \neq 0$, then the linear equation assumes the form $y = -\frac{a}{b}x$ and the graph is a straight line passing through the origin. If $b = 0$ and the coefficient $a \neq 0$, then the linear equation assumes the form $ax + 0y = c$, and hence $x = c/a$, and the graph is a straight line parallel to the y -axis. If $a = 0$, $b = 0$ and $c = 0$, then we get an equation $0x + 0y = 0$, whose solution is any pair of numbers and the graph is the whole plane. 2. A solution of an equation in two variables is an ordered pair of numbers which turns the equation in two variables into a true equality. For example, the pair of numbers $(-1, 2)$ is a solution of the equation $2x + 3y = 4$; giving an arbitrary value to the variable x , say, $x = 0$, we find the corresponding value of y : $y = 4/3$. Therefore a linear equation in two variables has an infinite number of solutions.

193. 1. A solution of simultaneous equations in two variables is a pair of values of the variables which turns each equation into a true equality. For example, the pair of values of the variables $x = 2$ and $y = -1$ is a solution of the simultaneous equations $3x - 2y = 8$ and $5x + y = 9$. 2. To solve simultaneous equations is to find the set of their solutions.

194. (a) In each equation we express y in terms of x :

$$\begin{aligned}y &= -2x + 4, \\y &= 0.5x + 3.5.\end{aligned}$$

The slopes of the straight lines, which are the graphs of the simultaneous equations, are different, the slope of the first graph is equal to -2 and that of the second is 0.5 , and therefore the straight lines intersect. The coordinates of their intersection point, $(-1, 3)$ (Fig. 87), are a solution of the equations, the solution being unique since two straight lines intersect at one point, (b) the straight lines $y = 2x - 3$ and $y = -2.5x + 1.5$ meet at a point $(1, -1)$, and this is a solution of the simultaneous equations (Fig. 88).

195. (a) The straight lines $y = -\frac{2}{3}x + 2$ and $y = -\frac{2}{3}x + \frac{10}{3}$

have the same slope (Fig. 89) and hence they are parallel. In addition, they cut the y -axis at different points $(0, 2)$ and $(0, \frac{10}{3})$. This means that they have no points in common, i.e. the simul-

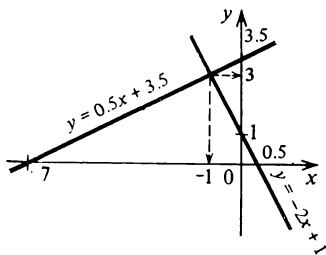


Fig. 87

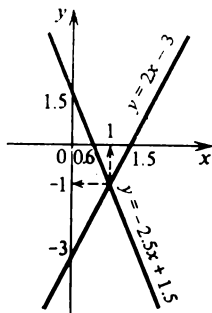


Fig. 88

taneous equations have no solutions. We can verify this by multiplying all the terms of the given equation by 2:

$$\begin{cases} 2x + 3y = 6, \\ 2x + 3y = 10. \end{cases}$$

The left-hand sides of the resulting equations are similar and the right-hand sides are different, and this is impossible for the same values of x and of y .
Answer: $\emptyset$.

196. (a) The graphs of the equations coincide since multiplying the second equation by 2, we get simultaneous equations

$$\begin{cases} 2x - 6y = 3, \\ 2x - 6y = 3, \end{cases}$$

which are similar, and, as is known, one equation in two variables has an infinite number of solutions.

197. We reduce each of the simultaneous equations to the form $y = kx + l$ and then

$$\begin{cases} y = 2x - 4, \\ y = ax - c. \end{cases}$$

The equations have a unique solution if the angular coefficients are different, i.e. if $a \neq 2$; they have no solutions if the angular coefficients are equal and the constant terms are different, i.e. if $a = 2$ and $c \neq 4$; in this case the graphs of the equations are parallel; the equations have an infinite number of solutions if the graphs of the equations coincide, i.e. if $a = 2$ and $c = 4$.

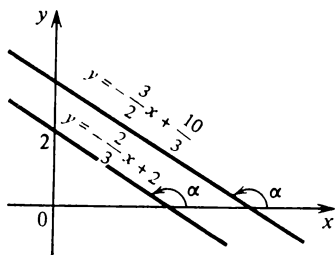


Fig. 89

198. (a) For the coefficients in y to become opposite numbers, we find the least common multiple of the numbers 27 and 18 (it is 54) and multiply both sides of each equation by the corresponding complementary factor, i.e.

$$\begin{cases} 16x - 27y = 20, \\ 5x + 18y = 41.5, \end{cases} \begin{vmatrix} 2 \\ 3 \end{vmatrix} \quad \text{then} \quad \begin{cases} 32x - 54y = 40, \\ 15x + 54y = 124.5. \end{cases}$$

The term-by-term addition leads to an equation $47x = 164.5$. Hence we get $x = 164.5 : 47 = 3.5$. Substituting the value $x = 3.5$ into the second equation, we get $5 \cdot 3.5 + 18y = 41.5$, $y = 1 \frac{1}{3}$. *Answer:* $(3.5, 1 \frac{1}{3})$. (b) In the first equation we express the variable x in terms of y and get simultaneous equations

$$\begin{cases} x = 5y/8, \\ 4(5y/8) + 0.25y = 11. \end{cases}$$

Solving the second equation, we find that $y = 4$ and obtain $x = 2.5$ from the first equation. *Answer:* $\{(2.5, 4)\}$.

Remark. This technique of solution is known as the *method of substitution*. It is convenient when the coefficient in one of the variables of simultaneous linear equations is 1 or -1 , and also when one of the simultaneous equations is in the form of a proportion. In addition, this method is always used when solving two simultaneous equations, one of which is of the second degree

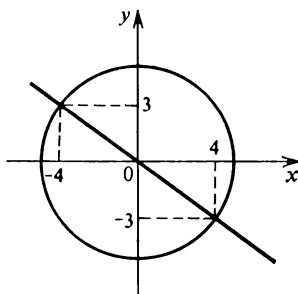


Fig. 90

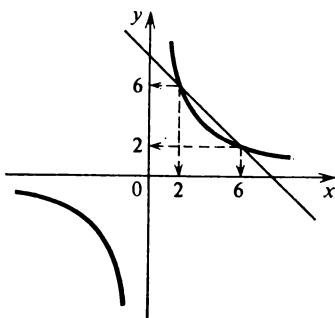


Fig. 91

and the other of the first degree. (c) $\{(2, -2)\}$, (d) $\{(5.5 \cdot 10^{10}, 4.5 \cdot 10^9)\}$. *Hint.* To find the variable x , do a term-by-term addition, and to find y , do a term-by-term subtraction.

199. Substituting successively the coordinates of each point into the equation for the straight line, we get simultaneous equations

$$\begin{cases} -3 = 2k + l, \\ 4.5 = -k + l. \end{cases}$$

Solving them, we find that $k = -2.5$ and $l = 2$. The equation of the straight line has the form $y = -2.5x + 2$.

200. (a) A circle with centre at the origin and a straight line $y = -\frac{3}{4}x$ (Fig. 90), which passes through the origin, meet at

two points $(-4, 3)$ and $(4, -3)$. *Answer:* $\{(-4, 3), (4, -3)\}$, (b) a hyperbola $y = 12/x$ (Fig. 91) and a straight line $x + y = 8$ meet at two points $(2, 6)$ and $(6, 2)$, (c) $\{(-3, 9), (2, 4)\}$, (d) $\{(2, -1), (-1, 2)\}$, (e) $\{(1, 6)\}$.

201. (a) Expressing in the first equation the variable x in terms of y , we get $x = 4 - y$. Then we replace in the second equation the variable x by the expression $4 - y$:

$$\begin{cases} x = 4 - y, \\ (4 - y)^2 - 2y^2 - (4 - y)y + 5(4 - y) + 15 = 0. \end{cases}$$

Solving the second equation, we find that $y_1 = -7.5$ and $y_2 = 3$. Substituting these values of y into the equation $x = 4 - y$, we get the values of x : $x_1 = 11.5$ and $x_2 = 1$. *Answer:* $\{(11.5, -7.5), (1, 3)\}$, (b) $\{(-2, -2.5), (5, 1)\}$, (c) $\{(5, -2), (-7, 2.8)\}$.

202. (a) $\{(-1, 6), (0, 5)\}$. *Hint.* The given system of simultaneous equations reduces to two systems:

$$\begin{cases} x = 0, \\ x^3 + y = 5, \end{cases} \quad \begin{cases} x = -1, \\ x^3 + y = 5, \end{cases}$$

(b) $\{(1, 1), (3, -3)\}$, (c) the given system reduces to the system

$$\begin{cases} (3x - 1)(y - 1) + (x + 1) = y + 2, \\ x = 4y - 7, \\ (x + 1)(y - 1) \neq 0, \end{cases} \quad \text{or} \quad \begin{cases} 3xy - 2y - 2x = 0, \\ x = 4y - 7, \\ (x + 1)(y - 1) \neq 0. \end{cases}$$

Answer: $\{(1, 2), (-4 \frac{2}{3}, \frac{7}{12})\}$.

203. 6/25. *Hint.* If we designate the numerator of the fraction as x and the denominator as y , then we arrive at a system consisting of first-degree equations $(x - 1)/y = 1/5$ and $x/(y - 1) = 1/4$.

204. 24, 42. *Hint.* 1. If x is the tens digit and y is the units digit, then we obtain two simultaneous equations $x + y = 6$ and $\frac{10x + y}{10y + x} = \frac{4}{7}$. 2. This problem can also be solved by means of one equation in one variable. If we designate the tens digit as x , then the units digit will be $6 - x$, and we arrive at an equation $\frac{10x + (6 - x)}{10(6 - x) + x} = \frac{4}{7}$.

Comparing the two techniques of solving the problem, we see that when we introduce only one variable, it is more difficult to express the required quantities in terms of this variable and set up an equation.

205. 50, 100. *Hint.* If x is the amount of the 800-mm pipes and y is the amount of 1200-mm pipes and we take into account that 800 mm = 0.8 m and 1200 mm = 1.2 m, then we have two simultaneous equations $x + y = 150$ and $0.8x + 1.2y = 160$.

206. 16, 24. *Hint.* If we designate as x and y the number of machine-tools the first and the second mill had to produce, respectively, then we obtain

$$\begin{cases} x + y = 360, \\ 1.1x + 1.12y = 400; \end{cases} \quad x = 160, \quad y = 200.$$

207. 5 and 3.

208. 6 cm and 4 cm.

209. 5 cm and 12 cm. *Hint.* The sum of the lengths of the legs is 17 cm ($30 - 13 = 17$). Derive simultaneous equations using the Pythagoras theorem.

210. 6 cm and 8 cm. If the lengths of the legs are x and y , then we arrive at simultaneous equations $x + y = 14$ and $\frac{1}{2}xy = 24$.

Remark. Sometimes students erroneously assume that the equation $\frac{1}{2}xy = 24$ is of the first degree. But it is not so since the sum of the exponents of the variables in this equation is two.

211. Assume that the length of the plot is x m and its width is y m, with $x > y$. Then we have two simultaneous equations $xy = 600$ and $x + y = 50$. Solving them, we find that $x = 30$ m and $y = 20$ m.

212. Assume that x is the tens digit and y is the units digit in the required number. Then the two-digit number has the form $10x + y$, the sum of its digits is $x + y$ and their product is xy . Bearing in mind that the dividend is equal to the product of the divisor by the quotient plus the remainder, we obtain simultaneous equations

$$\begin{cases} 10x + y = 6(x + y) + 2, \\ 10x + y = 5xy + 2, \end{cases} \quad \text{or} \quad \begin{cases} 4x - 5y = 2, \\ 10x + y - 5xy = 2. \end{cases}$$

Solving them, we find that $x = 3$ and $y = 2$. *Answer:* 32.

SECTION 1.19

213. (a) We form the difference between the left-hand and right-hand sides of the inequality and transform it, i.e. $a^2 + 2 - 2(2a - 1) = a^2 + 2 - 4a + 2 = (a - 2)^2$. For $a = 2$ this expression is zero and for $a \neq 2$ we have $(a - 2)^2 > 0$, and hence the inequality $a^2 + 2 \geq 2(2a - 1)$ holds true for any a , (b) we have $2(p + 1)^2 + 3p - p(p + 5) = 2p^2 + 4p + 2 + 3p - p^2 - 5p = p^2 + 2p + 2 = (p + 1)^2 + 1 > 0$, i.e. the inequality holds true for any p , (c) *Hint.* The inequality $9 - (x - y)^2 \leq 0$ holds true for any x and y .

214. 1. $\left(a + \frac{1}{a}\right) - 2 = \frac{a^2 + 1 - 2a}{a} = \frac{(a - 1)^2}{a}$. Since $\frac{(a - 1)^2}{a} \geq 0$

for $a > 0$ (we have an equality only for $a = 1$), we have $a + \frac{1}{a} \geq 2$ for $a > 0$. This can be formulated as follows: *the sum of two positive numbers, inverse of each other, is not smaller than two.* 2. We

have $\left(b + \frac{1}{b}\right) - (-2) = b + \frac{1}{b} + 2 = \frac{(b + 1)^2}{b}$. Since $(b + 1)^2 \geq$

0 for any b , and $b < 0$ by the hypothesis, we have $\frac{(b + 1)^2}{b} < 0$,

and hence $b + \frac{1}{b} \leq -2$ for $b < 0$ (we have an equality only for

$b = -1$). Thus the sum of two negative numbers, inverse of each other, is no larger than -2 . 3. We transform the inequality as follows:

$$\frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2} = \frac{(\sqrt{a})^2 + (\sqrt{b})^2 - 2\sqrt{ab}}{2} = \frac{(\sqrt{a} - \sqrt{b})^2}{2}.$$

Since $\frac{(\sqrt{a} - \sqrt{b})^2}{2} \geq 0$ (we have an equality

only for $a=b$), it follows that $\frac{a+b}{2} \geq \sqrt{ab}$. This can be formulated as follows: the arithmetic mean of two nonnegative numbers is larger than or equal to their geometric mean.

215. (a) Adding the inequalities $3a > 9$ and $2.5b > 10$ term-by-term, we get $3a + 2.5b > 19$, (b) multiplying these inequalities term-by-term, we get $ab > 12$, whence $2ab > 24$, (c) since $a^2 > 3^2$ and $b^2 > 4^2$, it follows that $a^2 + b^2 > 3^2 + 4^2$, $a^2 + b^2 > 25$.

216. (a) $7 < 5a < 8$

$$\frac{1 < 0.2b < 1.2}{8 < 5a + 0.2b < 9.2}.$$

$$217. \quad \frac{1}{3} > \frac{1}{a} > \frac{1}{4} \quad \text{or} \quad \frac{1}{4} < \frac{1}{a} < \frac{1}{3},$$

$$\frac{1}{6} < \frac{1}{b} < \frac{1}{5}$$

$$\frac{\frac{1}{4} + \frac{1}{6} < \frac{1}{a} + \frac{1}{b} < \frac{1}{3} + \frac{1}{5}}$$

whence $\frac{5}{12} < \frac{1}{a} + \frac{1}{b} < \frac{8}{15}.$

218. We have $8.5 \cdot 2.5 \leq ab \leq 8.6 \cdot 2.6$, or $21.25 \leq ab \leq 22.36$, or rounding off to two significant digits (there are two of them in each factor), we get $21 \leq ab \leq 22$, i.e. $21 \text{ cm}^2 \leq S \leq 22 \text{ cm}^2$.

219. To solve an inequality means to find the set of its solutions.

220. (a) We multiply both sides of the given inequality by the positive number 6; i.e.

$$7x + 5 - 3(3x - 9) > 8 \cdot 6, \quad 7x + 5 - 9x + 27 > 48.$$

We transfer the known term 32 from the left-hand side to the right-hand side with the opposite sign (see the corollary of Property 3 on p. 44), and then

$$7x - 9x > 48 - 32, \quad -2x > 16.$$

Dividing both sides of the inequality by -2 and changing the sign of the inequality to the opposite, we get an inequality $x < -8$, which is equivalent to the given one. Answer: $(-\infty, -8)$, (b) the set of solutions is an open number ray from -5.5 to $+\infty$: $(-3.5, +\infty)$, (c) $\emptyset$. Hint. The inequality reduces to $15x - 9 >$

$15x + 5$, or $0 \cdot x > 14$. The set of solutions of the inequality obtained and, hence, of the given one, is empty. We can arrive at the same conclusion in another way. Dividing each term of the numerator of the first fraction by 10 and that of the second fraction by 6, we obtain $\frac{x}{2} - \frac{3}{10} > \frac{x}{2} + \frac{1}{6}$.

221. (a) $\{1, 2, 3\}$, (b) $\{1, 2, 3, 4, 5\}$.

222. (a) Replacing each inequality of the system by the elementary inequality, equivalent to it, we get a system equivalent to the given one: $\begin{cases} x > -2, \\ x < 3. \end{cases}$ The set of solutions of the first inequality is the number interval $(-2, +\infty)$ and that of the second inequality is the number interval $(-\infty, 3)$.

We can find the intersection of the sets of solutions using the coordinate axis (Fig. 92): $[-2, +\infty) \cap (-\infty, 3) = (-2, 3)$.



Fig. 92



Fig. 93

The set of solutions of the given simultaneous equations is the number interval $(-2, 3)$ (in Fig. 92 this interval is double hatched), (b) $(2.5, +\infty)$ (Fig. 93), (c) $\emptyset$ (Fig. 94), (d) $(5, 7)$.

223. (a) *The 1st technique.* The given two-sided inequality can

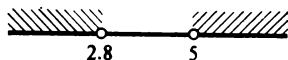


Fig. 94

be written in the form of a system of simultaneous inequalities $\begin{cases} 2x - 3 > -1, \\ 2x - 3 < 7. \end{cases}$ The set of solutions of this system is the number interval $(1, 5)$.

The 2nd technique. We add 3 to each side of the inequality and get

$$\begin{cases} -1 + 3 < 2x - 3 + 3 < 7 + 3, \\ 2 < 2x < 10, \quad 1 < x < 5, \end{cases}$$

(b) $[-7, 13]$.

224. (a) The product of two factors is positive only when both factors are either positive or negative, i.e. when the factors are of the same sign. This means that the given inequality is equivalent to the set of systems

$$\begin{cases} 2x - 3 > 0, \\ 5 - x > 0 \end{cases} \quad \text{or} \quad \begin{cases} 2x - 3 < 0, \\ 5 - x < 0. \end{cases}$$

The set of solutions of the first system is an interval $(1.5, 5)$ and the set of solutions of the second system is an empty set. *Answer:*

$(1.5, 5) \cup \emptyset = (1.5, 5)$, (b) $(-\infty, -3)$. *Hint.* Since the numerator is smaller than zero ($-3 < 0$), the given inequality is equivalent to the inequality $2x + 6 < 0$, (c) the value of the fraction is negative only when the numerator and the denominator have values of different signs, and therefore

$$\begin{cases} 2x-3 > 0, \\ 3x+6 < 0 \end{cases} \quad \text{or} \quad \begin{cases} 2x-3 < 0, \\ 3x+6 > 0. \end{cases}$$

The set of solutions of the first system is an empty set and the set of solutions of the second system is a number interval $(-2, 1.5)$. The set of solutions of the given inequality is $\emptyset \cup (-2, 1.5) = (-2, 1.5)$, (d) we replace the given inequality by an equivalent one whose right-hand side is zero by transferring unity with the opposite sign to the left-hand side of the inequality. Then we get in succession

$$\frac{4x+2}{2-5x} - 1 < 0, \quad \frac{4x+2-2+5x}{2-5x} < 0, \quad \frac{9x}{2-5x} < 0.$$

Hence we have

$$\begin{cases} 9x < 0, \\ 2-5x > 0 \end{cases} \quad \text{or} \quad \begin{cases} 9x > 0, \\ 2-5x < 0. \end{cases}$$

The set of solutions of the first system is $(-\infty, 0)$ and that of the second is $(0.4, +\infty)$. *Answer:* $(-\infty, 0) \cup (0.4, +\infty)$.

Warning! Sometimes students try to solve this inequality by deleting the denominator and get only one open number ray $(-\infty, 0)$. But it cannot be done since by doing this, we multiply both sides of the inequality by the expression $2 - 5x$, which can be either positive or negative. In the last case the sign of the inequality must be changed to the opposite;

$$(e) \quad \frac{3x-1}{2x+5} - 2 > 0; \quad \frac{x-11}{2x+5} > 0. \quad \text{Answer: } (-\infty, -2.5) \cup (11, +\infty).$$

225. (a) *The 1st technique.* $x^2 - 9 \geq 0$, $(x-3)(x+3) \geq 0$. The value of the product of two factors is nonnegative if and only

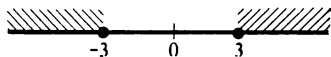


Fig. 95

if the factors are nonnegative or nonpositive. This means that the given inequality holds true for the values of x for which

$$\begin{cases} x-3 \geq 0, \\ x+3 \geq 0 \end{cases} \quad \text{or} \quad \begin{cases} x-3 \leq 0, \\ x+3 \leq 0. \end{cases}$$

Answer: $(-\infty, -3) \cup (3, +\infty)$.

The 2nd technique. Since both sides of the inequality are nonnegative, the given inequality is equivalent to the inequality $|x| \geq 3$. This problem can be formulated thus: find on the coordinate axis the set of points whose distance from the point with the zero coordinate is three or exceeds three. We get the same answer (Fig. 95).

This inequality can also be solved by graphical means. We construct the graphs of the functions $y = x^2$ and $y = 9$ in the same system of coordinates. The inequality $x^2 \geq 9$ holds true for the values of x for which the graph of the function $y = x^2$ lies not lower than that of the function $y = 9$ (Fig. 96). For clarity

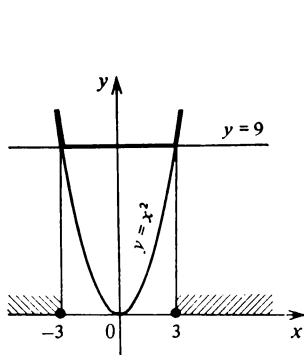


Fig. 96

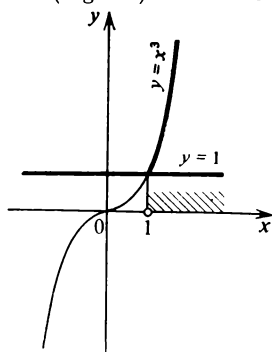


Fig. 97

we project the indicated part of the graph of the function $y = x^2$ onto the x -axis, (b) $(-1, 1)$, (c) $(-\infty, 0) \cup (1, +\infty)$, (d) $(1, +\infty)$. *Hint.* $x^3 - 1 > 0$, $(x - 1)(x^2 + x + 1) > 0$, but $x^2 + x + 1 = x^2 + x + \frac{1}{4} - \frac{1}{4} + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$ for any x . There is another technique of establishing this fact. The discriminant of the quadratic trinomial $x^2 + x + 1$ is smaller than zero, i.e. $D = 1 - 4 = -3 < 0$. The coefficient a is larger than zero i.e. $a = 1 > 0$. Consequently, $x^2 + x + 1 > 0$ for any x (the graph of this trinomial is a parabola lying above the x -axis). The graphical solution of the inequality is given in Fig. 97, (e) $(-\infty, 2)$, (f) $(-\infty, -1) \cup (1, +\infty)$. *Hint.* $x^4 - 1 = (x - 1)(x + 1) \times$

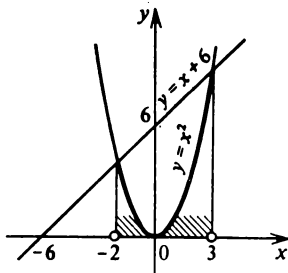


Fig. 98

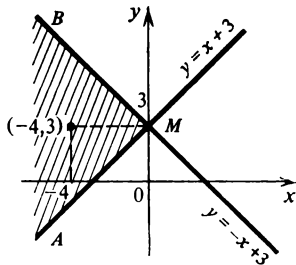


Fig. 99

$(x^2 + 1)$, but $x^2 + 1 > 0$ for any value of x , (g) $(-2, 3)$. *Hint.* Factoring the quadratic trinomial, we get $(x + 2)(x - 3) < 0$, or, writing the given inequality in the form $x^2 < x + 6$, we project onto the x -axis the part of the parabola $y = x^2$ which lies below the straight line $y = x + 6$ (Fig. 98), (h) $(-\infty, -1.5) \cup (1, +\infty)$.

226. (a) The domain of the function $y = \sqrt{x}$ is $(0, +\infty)$, and therefore $5 - x \geq 0$ for the given function, whence we have $x \leq 5$. Thus the domain of the given function is the number ray $(-\infty, 5)$, (b) we have a system $-x \geq 0, x + 1 \neq 0$. *Answer:* $(-\infty, -1) \cup (-1, 0)$, (c) $(-\infty, 3] \cup (3.5, +\infty)$, (d) $(2.5, 4]$, (e) $(-\infty, 5]$.

227. (a) The required set is the angle AMB (Fig. 99) formed by the intersection of the straight lines $y = x + 3$ and $y = -x + 3$. One of the solutions is $x = -4$ and $y = 3$, (b) the required set is a semicircle bounded by a circumference, which has its centre at the origin and a unit radius, and the abscissa axis. One of the solutions is $x = -0.5$ and $y = 0.6$ (Fig. 100), (c) a part of the plane bounded by the parabola $y = x^2$ and the straight line $y = 1$ (the boundaries inclusive).

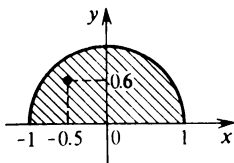


Fig. 100

228. (a) $2, 1, \frac{2}{3}, \frac{1}{2}, \dots$, (b) $1, 2, 4,$

$8, 16$, (c) $2, 0, 2, 0, 2, \dots$, (d) $1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \dots$, (a) $30^\circ, 210^\circ, 390^\circ, 570^\circ, 750^\circ$.

229. (a) $x_{k-1} = (k-1)^2 - 2(k-1) = k^2 - 4k + 3$, (b) $x_{k+1} = (k+1)^2 - 2(k+1) = k^2 - 1$.

230. (a) We must find the natural values of the variable n for which the expression $n^2 - 15n + 30$ is equal to -20 . Solving the equation $n^2 - 15n + 30 = -20$, we get the roots 5 and 10. Hence the number -20 is encountered in the sequence twice: the fifth and tenth terms are -20 , (b) the first and fourteenth terms are 14.

231. 1. To define an arithmetic progression, it is sufficient to know its first term a_1 and the common difference d .

2. If $d > 0$, then the arithmetic progression is an *increasing sequence* and if $d < 0$, then it is a *decreasing sequence*.

3. Any term of an arithmetic progression, beginning with the second, is an arithmetic mean of the predecessor and successor, i.e. $a_n = (a_{n-1} + a_{n+1})/2$. And conversely, if a sequence is such that its any term, beginning with the second, is an arithmetic mean of the preceding and the succeeding term, then the sequence is an arithmetic progression.

232. 1. $a_1 = a_2 - d, a_3 = a_2 + d$. 2. If $b - a = c - b$, then $b = \frac{a+c}{2}$. 3. The equality $x = \frac{1+(14-x)}{2}$ must hold true, whence $x = 5$. An arithmetic progression is 1, 5, 9.

233. $a_{31} = -8.5 + (-0.4)(31-1) = -8.5 - 12 = -20.5$.

234. From the formula for any term of an arithmetic progression, $a_n = -6 + 4(n-1) = -6 + 4n - 4 = 4n - 10$. Solving the equation $4n - 10 = 146$, we find that its root is the number 39 and therefore the number 146 is the 39th term of the progression.

235. Since $a_3 = a_1 + 2d, a_{13} = a_1 + 12d$, we have a system of simultaneous equations $a_1 + 2d = 8.2$ and $a_1 + 12d = 10.2$. From this system we find that $a_1 = 7.8$ and $d = 0.2$. Solving the inequality $7.8 + 0.2(n-1) < 20$, we find that $n < 62$. Thus each of the first 61 terms of this progression is smaller than 20.

236. The arithmetic progression will contain five terms (two given terms and three inserted terms); $a_1 = 8$, $a_5 = 0$ and $n = 5$. We find the common difference d of the progression using the formula for the n th term: $0 = 8 + 4d$, $d = -2$. We get an arithmetic progression 8, 6, 4, 2, 0.

237. For instance, $a_2 + a_{15} = (a_1 + d) + (a_{16} - d) = 40$;
 $S_{16} = \frac{(a_1 + a_{16}) 16}{2}$, or 320.

238. (a) We have an arithmetic progression in which $a_1 = 1$, $a_{100} = 100$, $n = 100$; $S_{100} = \frac{(1 + 100) 100}{2} = 5050$, (b) the first odd number is 1, the last is 99, and there are exactly 50 odd numbers smaller than 100. This means that $S_{50} = \frac{(1 + 99) \cdot 50}{2} = 2500$.

239. 1.15. *Hint.* $a_1 = 21$, $d = -3$, $S_n = 0$. We derive an equation $0 = \frac{(2 \cdot 21 + (-3)(n-1)) n}{2}$ (see formula (2)). Now we have

$$\frac{(42 - 3n + 3) n}{2} = 0, \quad (45 - 3n) n = 0; \quad n_1 = 0, \quad n_2 = 15.$$

The first root must be rejected. 2. Since $a_1 = 0$, $d = -4$ and $S_n > 112$, the problem reduces to the solution of the inequality

$$\frac{(0 + 4(n-1)) n}{2} > 112, \text{ from which we find that } n^2 - n - 56 > 0,$$

or $(n - 8)(n + 7) > 0$, whence $n > 8$, or $n < -7$. Since $n \in \mathbb{N}$, the values $n = 9, 10, 11, \dots$ satisfy the hypothesis.

240. 1. To define a geometric progression, it is sufficient to know its first term b_1 and the common ratio q . 2. *Any term of a geometric progression, beginning with the second, is equal to the arithmetic square root of the product of the preceding and the succeeding term*, i.e. $b_n = \sqrt{b_{n-1} b_{n+1}}$. The converse is also true: if a sequence is such that its any term, beginning with the second, is equal to the arithmetic square root of the product of the preceding and the succeeding term, then the progression is geometric. 3. If $q > 0$ and $q \neq 1$, then the progression is either increasing or decreasing. For instance, if $b_1 = 1$ and $q = 2$, then the progression 1, 2, 4, ... is increasing, now if $b_1 = -5$ and $q = 2$, then the progression $-10, -20, -40, \dots$ is decreasing. If $q = 1$, then all the terms of the progression are equal. If $q < 0$, then the progression is neither increasing nor decreasing, for instance, if $b_1 = 2$ and $q = -3$, then the progression is 2, -6 , 18, -54 , 162, ...

241. 1. $b_1 = b_2 : q$, $b_3 = b_2 q$. 2. Since $q = b_4 : b_3$, and $b_2 = b_3 : q$, it follows that $q = 1/\sqrt{2}$ and $b_2 = 6$, $b_1 = 6\sqrt{2}$. 3. $b_2 = 4$, $b_4 = 16$, $b_5 = 32$. 4. *Hint.* We must prove that $\frac{a^2 - 1}{(a + 1)^2} = \frac{(a - 1)^2}{a^2 - 1}$ for any $|a| \neq 1$.

242. 1. $b_5 = (-0.001) \cdot 10^4 = -10$. 2. *Hint.* $c_1 q \cdot c_1 q^{10} = c_1 q^4 \cdot c_1 q^7$. 3. From the condition that $a_2 = 0.36$ and $a_5 = 360$ we derive two simultaneous equations

$$\begin{cases} a_1 q = 0.36, \\ a_1 q^4 = 360. \end{cases}$$

From this we obtain $q^3 = 1000$, $q = 10$; $a_1 = 0.036$, $a_3 = 3.6$, $a_4 = 36$.

243. We have $l_7 = b_1 q^6$, $b_1 = b_7 / q^6$, or $b_1 = 0.02$; $S_7 = \frac{b_7 q - b_1}{q - 1}$, or $S_7 = \frac{1.28 \cdot 2 - 0.02}{2 - 1} = 2.54$.

244. $S_3 = 315$, $a_1 = S_1 = 5$, $S_2 = 5 \cdot (2^2 - 1) = 15$, $a_2 = 10$, $q = 2$.

SECTION 1.21

245. 1. Logarithms exist only for positive numbers; negative numbers and zero have no logarithms. For instance, the expressions $\log(-3)$, $\log 0$, $\log(1 - \sqrt{2})$ are meaningless since if we assume that $\log(-3)$ is equal to the real number x , then the number 10^x must equal the number -3 , and this is impossible since $10^x > 0$ for any x . 2. It is valid for $a > 0$.

246. (a) $10^2 = 100$, (b) $10^{-1} = 0.1$, (c) $10^{1/2} = \sqrt{10}$, (d) $10^{-\frac{1}{5}} = \frac{1}{10^{1/5}} = \left(\frac{1}{10}\right)^{1/5} = \sqrt[5]{0.1}$, or $10^{-1/5} = (10^{-1})^{1/5} = \sqrt[5]{0.1}$, (e) $10^{-\frac{1}{3}} = \frac{1}{10^{1/3}} = \sqrt[3]{\frac{1}{10}}$.

247. (a) $\log 1000 = 3$, (b) $\log 1 = 0$, (c) $\log 0.001 = -3$, (d) $\log a = l$, (e) $\log b = x + 1$.

248. Dividing both sides of the given equation by $10^b \neq 0$, we get $10^a \div 10^b = 10^b \div 10^b$, or $10^{a-b} = 1$, whence $a - b = 0$, i.e. $a = b$.

249. (a) $10^x = 10^1$, $x = 1$, (b) $10^x = 10^0$, $x = 0$, (c) $10^x = 10^{-2}$, $x = -2$, (d) $10^x = 4 \cdot 0.025$, $10^x = 0.1$, $10^x = 10^{-1}$, $x = -1$, (e) $10^x = 10^{1/3}$, $x = 1/3$, (f) $10^x = \frac{1}{\sqrt[5]{10^2}}$; $10^x = 10^{-\frac{2}{5}}$; $x = -0.4$.

250. (a) 1, (b) 4, (c) -2 , (d) $\log(10 \cdot 10^{1/2}) = \log 10^{3/2} = 3/2$, (e) $\log 1^2 = \log 1 = 0$, (f) $\log(1+0) = \log 1 = 0$.

251. (a) $\log \log 10 = \log 1 = 0$, (b) $\log \sqrt{10} + \sqrt{\log 10} = \log 10^{1/2} + \sqrt{1} = 1/2 + 1 = 1.5$, (c) $(\log 0.1)^2 + \log 10 \log 100 = (-1)^2 + 1 \cdot 2 = 3$. (d) $\frac{1 - (\log 0.01)^2}{1 + (\log 0.01)^2} = \frac{1 - (-2)^2}{1 + (-2)^2} = \frac{1 - 4}{1 + 4} = -0.6$.

252. The following expressions are meaningless: $\log 0$, $\log(-5)$, $\log\left(\frac{3}{4} - \frac{4}{5}\right)$, $\log(\sqrt{3} - 2)$.

253. (a) From the principal logarithmic identity ($10^{\log a} = a$, $a > 0$), we have $10^{\log 3} = 3$, (b) $10^{\log 2} + 10^{\log 5} = 2 + 5 = 7$,

(c) $10^{2+\log 5} = 10^2 \cdot 10^{\log 5} = 100 \cdot 5 = 500$, (d) $10^{\log 7 - 1} = 10^{\log 7} \times 10^{-1} = 7 \cdot 10^{-1} = 0.7$, (e) $10^1 \cdot 10^{-\log 5} = 10 \div 10^{\log 5} = 10 \div 5 = 2$, (f) using the identity $a^{mn} = (a^m)^n$, we have $10^{-2 \log 3} = (10^{\log 3})^{-2} = 3^{-2} = 1/9$, (g) $0.1^{-\log 4} = (10^{-1})^{-\log 4} = 10^{\log 4} = 4$, (h) $5 (10^{\log 5})^{-1} = 5 \cdot \frac{1}{5} = 1$, (i) $10^{\log \log 100} = 10^{\log 2} = 2$, (j) $5^{\log 100} = 5^2 = 25$, (k) $0.25^{\log \sqrt{10}} = 0.25^{\log 10^{1/2}} = 0.25^{1/2} = \sqrt{0.25} = 0.5$.

254. (a) $x = 1$, (b) $|x| = 2$, whence $x = 2$ or $x = -2$, (c) $(10^2)^{\log x} = 4$, $(10^{\log x})^2 = 4$; $x^2 = 4$ whence $x = 2$; the value $x = -2$ is not suitable since $\log(-2)$ is meaningless.

255. (a) By the definition of a logarithm, $x = 10^2 = 100$, (b) $x = 10^0 = 1$, (c) $-x = 10^{-1}$, whence $x = -0.1$, (d) $x = 10^{0.1} = \sqrt[10]{10}$, (e) $x - 2 = 10$, $x = 12$, (f) $-x - 3 = 0.01$, $x = -3.01$, (g) $3 - x = 1$, $x = 2$, (h) $x^2 - 6 = 10$, $x_1 = -4$, $x_2 = 4$, (i) $x^2 - 3x = 10$; $x_1 = -2$, $x_2 = 5$, (j) $x^2 + 5x + 6.1 = 0.1$, $x_1 = -3$, $x_2 = -2$.

256. (a) We designate $\log x$ as a and then we have $(\log x)^2 = a^2$, and we can write the given equation as $a^2 + a = 0$, or $a(a + 1) = 0$, whence we have $a = -1$ or $a = 0$, i.e. $\log x = -1$ or $\log x = 0$; consequently, $x_1 = 0.1$ and $x_2 = 1$, (b) $\log x = 1$ or $\log x = 2$; $x_1 = 10$ and $x_2 = 100$, (c) $\log x = -1$ or $\log x = 3/2$, $x_1 = 0.1$, $x_2 = 10^{3/2} = \sqrt{10^3} = 10 \sqrt{10}$.

257. (a) $\log 18 = \log (9 \cdot 2) = \log 3^2 + \log 2 = 2 \log 3 + \log 2$, (b) $\log 75 = \log (5^2 \cdot 3) = 2 \log 5 + \log 3$; (c) $\log \frac{\sqrt[3]{25}}{8} = \log 5^{\frac{2}{3}} - \log 2^3 = \frac{2}{3} \log 5 - 3 \log 2$.

258. (a) $\log 60 = \log (2 \cdot 3 \cdot 10) = \log 2 + \log 3 + \log 10 \approx 0.30 + 0.48 + 1 \approx 1.78$, (b) $\log \frac{10}{3} = \log 10 - \log 3 \approx 1 - 0.48 \approx 0.52$, (c) $\log \frac{1}{6} = \log 1 - \log 6 = 0 - (\log 2 + \log 3) \approx -(0.30 + 0.48) \approx -0.78$, (d) $\log \sqrt[3]{12} = \log 12^{\frac{1}{3}} = \log (2^2 \cdot 3)^{\frac{1}{3}} = \frac{1}{3} (2 \log 2 + \log 3) \approx \frac{1}{3} (2 \cdot 0.30 + 0.48) \approx 0.36$, (e) $\log 0.03 = \log 3 - \log 100 \approx 0.48 - 2 \approx -1.52$; (f) $\log (200 + 100) = \log 300 = \log 3 + \log 100 \approx 0.48 + 2 \approx 2.48$, (g) $\log 200 + \log 100 = \log (2 \cdot 100) + 2 = \log 2 + 4 \approx 0.30 + 4 \approx 4.30$, (h) $\frac{\log 10 + \log 90}{\log (10 + 90)} = \frac{1 + 2 \log 3 + 1}{2} = \frac{2 + 2 \log 3}{2} = 1 + \log 3 \approx 1.48$.

$$259. (a) \log(100a) - \log \frac{a}{10} = \log 100 + \log a - (\log a - \log 10) = 2 + \log a - \log a + 1 = 3, (b) \log \frac{x}{100} - \log(1000x) = \log x - \log 100 - (\log 1000 + \log x) = \log x - 2 - 3 - \log x = -5.$$

$$260. (a) \log \sqrt[4]{0.1} = \log 0.1^{1/2} = \frac{1}{2} \log 0.1 = \frac{1}{2} \cdot (-1) = -0.5, (b) \log \frac{1}{\sqrt[3]{10^2}} = \log 10^{-2/3} = -\frac{2}{3} \log 10 = -\frac{2}{3}; (c) 2 \log \sqrt[4]{10^3} = 2 \log 10^{3/4} = 2 \cdot \frac{3}{4} \log 10 = 1.5, (d) \log(0.1 \sqrt[4]{1000}) = \log 0.1 + \log 10^{3/2} = -1 + \frac{3}{2} = 0.5.$$

$$261. (a) \frac{\log 8}{\log 2} = \frac{\log 2^3}{\log 2} = \frac{3 \log 2}{\log 2} = 3.$$

Warning! Sometimes students erroneously replace $\frac{\log 8}{\log 2}$ by $\log \frac{8}{2}$ and get $\log 8 - \log 2 = 2 \log 2$. One must remember that in the general case

$$\frac{\log a}{\log b} \neq \log a - \log b, \quad a > 0, \quad b > 0, \quad b \neq 1.$$

$$262. (a) \log \sqrt[3]{0.8} = \log \left(\frac{8}{10} \right)^{1/3} = \log \left(\frac{2^3}{10} \right)^{1/3} = \log \frac{2}{10^{1/3}} = \log 2 - \frac{1}{3} \log 10 \approx 0.30 - 0.33 \approx -0.03, (b) \log \sqrt[4]{3.2} = \log \left(\frac{2^5}{10} \right)^{1/4} = \log \frac{2^{5/4}}{10^{1/4}} = 1.25 \log 2 - 0.25 \log 10 = 1.25 \cdot 0.30 - 0.25 \approx 0.375 - 0.25 \approx 0.13.$$

$$263. (a) x = \frac{a^2 b^3}{\frac{1}{cd^2}} = a^2 b^3 c^{-1} d^{-1/2}; \log x = \log a^2 + \log b^3 + \log c^{-1} + \log d^{-1/2} = 2 \log a + 3 \log b - \log c - \frac{1}{2} \log d, (b) x = 10 a^4 b^{2/3} c^{-1/3}; \log x = \log 10 + \log a^4 + \log b^{2/3} + \log c^{-1/3} = 1 + 4 \log a + \frac{2}{3} \log b - \frac{1}{3} \log c, (c) \log x = \log (a+3)^2 + \log b^3 - \log (a^2+2)^2 - \log c^4 = 2 \log (a+3) + 3 \log b - 2 \log (a^2+2) - 4 \log c.$$

Warning! Sometimes students erroneously replace $\log(a+3)$ by $\log a + \log 3$. One must remember that in the general case

the logarithm of the sum is not equal to the sum of logarithms, i.e. $\log(a+b) \neq \log a + \log b$. (d) $\log x = 2.5 \log(a+b) + 1.5 \log c - 0.25 \log d$.

264. (a) If $a > 0$, $b > 0$ and $a = b$, then $\log a = \log b$. Using this property of logarithms, we take logarithms of both sides of the given equation: $\log 2^x = \log 5$, $x \log 2 = \log 5$, whence $x = \frac{\log 5}{\log 2} \approx \frac{0.70}{0.30} \approx 2.3$, (b) $x \log 3 = 2 \log 5$, $x = \frac{2 \log 5}{\log 3} \approx \frac{2 \cdot 0.70}{0.48} \approx 2.9$, (c) $x = \frac{\log 100}{\log 2} \approx \frac{2}{0.30} \approx 6.7$, (d) $x = \frac{\log 20}{\log 0.1} = \frac{\log 2 + \log 10}{-1} \approx -(0.30 + 1) = -1.30$.

265. (a) By formulas for taking antilogarithms, we have $\log 2 + \log 5 = \log(2 \cdot 5) = 1$, (b) $\log 360 - \log 0.36 = \log(360 \div 0.36) = \log 1000 = 3$, (c) $\log 0.15 - \log 15 = \log 0.01 = -2$, (d) $\frac{1}{2} \log 25 + \frac{1}{3} \log 64 - \frac{1}{5} \log 32 = \log(5^2)^{1/2} + \log(2^6)^{1/3} - \log(2^5)^{1/5} = \log 5 + \log 4 - \log 2 = \log(5 \cdot 4 \div 2) = \log 10 = 1$.

266. Assume $a > 0$, $b > 0$ and $\log a = \log b$. We shall prove that $a = b$. By the hypothesis, $\log a - \log b = 0$, or, taking antilogarithms, we have $\log \frac{a}{b} = 0$, whence we get, according to the definition of logarithms, $10^0 = a/b$, or $a/b = 1$ and $a = b$. The converse is also true: if $a = b$, then $\log a = \log b$.

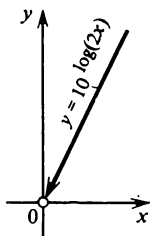


Fig. 101

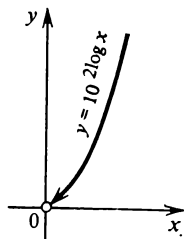


Fig. 102

267. (a) $\log x = \log 2 + \log 5^3 = \log(2 \cdot 5^3) = \log 250$, $x = 250$, (b) $\log x = \log 2^3 + \log 3^2 = \log(2^3 \cdot 3^2) = \log 72$, $x = 72$, (c) $\log x = \frac{1}{4} \log(8 \cdot 32) = \log(2^3 \cdot 2^5)^{1/4} = \log(2^8)^{1/4} = \log 2^2 = \log 4$, $x = 4$, (d) $\log x = \log \left(\frac{8}{2}\right)^3 = \log 64$, $x = 64$, (e) if $a > b > 0$, $a \neq b$, then $\log x = \log \frac{(a-b)^2}{a^2 - b^2} = \log \frac{a-b}{a+b}$, $x = \frac{a-b}{a+b}$.

268. (a) $\log(2x - 5) = \log(3^2 \cdot 5)$, $2x - 5 = 45$; $x = 25$, (b) $9 - 0.1x = 3$, $0.1x = 6$, $x = 60$, (c) we replace 1 by $\log 10$ and then $\log(20 - 5x) = \log x + \log 10$, $\log(20 - 5x) = \log 10x$,

$20 - 5x = 10x$, $x = 1\frac{1}{3}$, (d) we replace 2 by $2 \log 10 = \log 100$ and then $\log x = \log 100 - \log 4$, $x = 25$, (e) $\log (x + 6) = \log 2.5x$, $x = 4$, (f) $\log (x - 4)(x + 5) = \log 10$, $x^2 + x - 30 = 0$, whence $x_1 = -6$ and $x_2 = 5$; we must reject the first root since

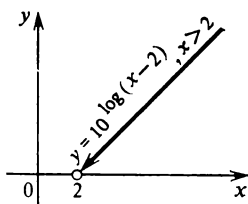


Fig. 103

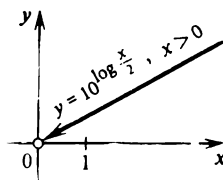


Fig. 104

for $x = -6$ the expressions $\log (x - 4)$ and $\log (x + 5)$ are meaningless.

269. (a) A straight line $y = 2x$, where $x > 0$ (Fig. 101), (b) $y = 10^{2 \log x} = 10^{\log x^2}$. The graph is a parabola where $x > 0$ (Fig. 102), (c) the graph is a straight line $y = x - 2$, where $x > 2$ (Fig. 103), (d) the straight line $y = x - 2$, where $x > 0$, (e) a straight line $y = \frac{1}{2}x$, where $x > 0$ (Fig. 104).

270. (a) 40.8, (b) 2.49, (c) 0.905, (d) 11.8, (e) 1.43.

271. (a) $3.58 \cdot 2.5 \approx 8.96$, (b) $43.5 \cdot 1.92 \approx 83.7$, (c) $(18.6 \div 72.5) 110\% \approx 25.7\%$.

272. (a) $x = \frac{27.3 \cdot 4.25}{0.85} \approx 135$; (b) $x = \frac{168 \cdot 0.735}{374} \approx 0.339$.

273. (a) $d = \frac{C}{\pi}$, therefore $\frac{4.21}{\pi} \approx 1.34$, (b) $d \approx 4.52$, (c) $d \approx 82.8$.

CHAPTER 2

SECTION 2.22

274. A *geometric figure* is any set of points. For instance, a line segment, an angle, a triangle, a circumference, a circle are all plane figures, whereas a sphere, a cube, a rectangular parallelepiped are solid figures.

(a) A *circumference* is the set of all points of the plane, which are at a given distance from a given point, (b) a *circle* is the set of all points of the plane the distance from each of which to a given point of that plane is not larger (is equal to or smaller) than the given distance.

275. (a) A *line segment* is a set which consists of two different points and all the points lying between them. It is designated as $[AB]$, the points A and B being its endpoints. The *length of a segment* is the distance between its endpoints and is designated as

$|AB|$, (b) if we take an arbitrary point O on the straight line AB , then the point divides the line into three subsets. The subsets A and B are called open rays and the subset C consists of one point O . The union of an open ray and its origin, point O , is a *ray*. A ray which begins at point O and contains a point A is designated as (OA) .

276. (a) A *broken (polygonal) line* is the union of the segments, $A_1A_2, A_2A_3, A_3A_4, \dots$ (Fig. 105) such that the end of each segment (except for the last one) is the beginning of the next one and the adjacent segments do not lie on the same straight line. The sum of the lengths of all the segments of the broken line is its *length*. (b) A *polygon* is the union of a closed polygonal line and its interior domain. The elements of a polygon are its sides, vertices and angles (interior angles). The *diagonal of a polygon* is a line segment which connects its two nonadjacent vertices.

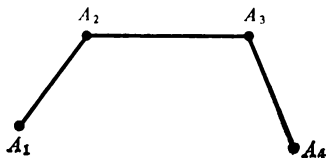


Fig. 105

277. (a) An *axiom* is a proposition which is accepted without a proof. For example, one and only one straight line passes through any two distinct points (*the axiom of a straight line*). If two points of a straight line belong to a plane, then all the points of that straight line belong to the plane, (b) a *theorem* is a proposition whose validity must be proved.

278. (a) A circle with centre at the point A and the radius of 2 cm, (b) a circumference with centre at the point B and the radius of 2 cm.

279. The points A, B and C belong to the same straight line since there holds an equality $|CA| + |AB| = |CB|$.

280. 1. By the property of distance for three points which do not lie on the same straight line, we have an inequality $|AD| < |AB| + |BD|$. We add $|DC|$ to both sides of the inequality and get

$$|AD| + |DC| < |AB| + |BD| + |DC|,$$

or

$$|AD| + |DC| < |AB| + |BC|.$$

2. We have $|AM| + |MD| + |DC| < |AB| + |BC|$ (see the preceding problem). We replace the sum $|MD| + |DC|$, by a smaller quantity $|MC|$ and get $|AM| + |MC| < |AB| + |BC|$.

281. On the basis of the property of a straight line we have (Fig. 106)

$$|AB| < |BC| + |AC|, \quad |BC| < |AC| + |AB|, \quad |AC| < |AB| + |BC|.$$

If we subtract $|BC|$ from both sides of the first inequality, $|AC|$ from both sides of the second and $|AB|$ from those of the third then we obtain

$$|AB| - |BC| < |AC|, \quad |BC| - |AC| < |AB|, \\ |AC| - |AB| < |BC|.$$

We unite these inequalities with the initial ones and get

$$\begin{aligned} |AB| - |BC| &< |AC| < |AB| + |BC|, \\ |BC| - |AC| &< |AB| < |BC| + |AC|, \\ |AC| - |AB| &< |BC| < |AC| + |AB|. \end{aligned}$$

We can formulate this as follows: *the length of each side of a triangle is smaller than the sum of the lengths of the other two sides but not larger than their difference.*

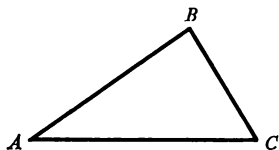


Fig. 106

282. 1. There are two possibilities here: the length of the third side is either 3 cm or 6 cm, but a triangle with the sides of 3, 3 and 6 cm is impossible. The only possible triangle has sides of 3, 6 and 6 cm since $6 < 6 + 3$. Thus the side 3 cm long is the base of the triangle and each lateral side is 6 cm long.

2. We have $|AB| - |BC| < |AC| < |AB| + |BC|$ (see 281). Hence we have $2 \text{ cm} < |AC| < 8 \text{ cm}$.

284. 1. A *straight angle* is an angle whose sides are opposite rays. A *right angle* is half a straight angle.

2. *Adjacent angles* are two angles whose union is a straight angle and the intersection is a ray. In Fig. 107 the angles ABC and

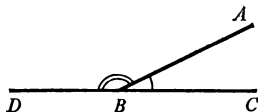


Fig. 107

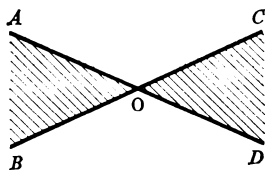


Fig. 108

ABD are adjacent. The sum of two adjacent angles is 180° .

3. *Vertical angles* are two angles which are smaller than a straight angle if the sides of one of them are rays, which are opposite to the sides of the other angle. In Fig. 108 the angles AOB and COD are vertical, the ray OC is opposite to the ray OB , and the ray OD is opposite to the ray OA . The vertical angles are *congruent* and

are therefore equal, i.e. $\widehat{AOB} = \widehat{COD}$, $\widehat{AOC} = \widehat{BOD}$.

4. The *bisector of an angle* is a ray which begins at the vertex of the angle and divides it in two.

5. Two straight lines are said to be *mutually perpendicular* if their intersection yields right angles. If the straight line AB is perpendicular to the straight line CD , then we write $(AB) \perp (CD)$.

285. 1. A point, a ray, and an angle.

2. A right angle or an angle of 60° .

287. 2. $\beta = 180^\circ - \alpha$.

3. If we designate the value of the smaller angle as x , then the larger angle is $x + 90^\circ$, and we have an equation $x + (x + 90) =$

180. Solving it, we find that the smaller angle is 45° and the larger one is 135° .

4. 108° and 72° . *Hint.* If we designate the value of the smaller angle as x , then the larger angle is $1.5x$ (150% of $x = 1.5x$).

288. (a) If the value of one of the adjacent angles is α and that of the other is β , then $\alpha + \beta = 180^\circ$. The angle between the bisectors of the angles is $\frac{\alpha}{2} + \frac{\beta}{2} = \frac{\alpha + \beta}{2} = 90^\circ$, (b) 180° .

SECTION 2.23

289. (a) The mapping of the arc CD onto the arc AB , (b) construct a ray $[OZ]$ and then $\{Z_1\} = [OZ] \cap AB$, (c) the mapping is invertible since under this mapping any two different points of the arc CD have different images.

The points C, X, Y and D have the images $X \rightarrow X_1, Y \rightarrow Y_1$ and $D \rightarrow B$.

290. (a) To define a mapping of a circle onto the contour of a triangle, we can take any point O inside the circle and draw rays beginning at that point. They cut the circle and the contour of the triangle, (b) the points X, Y and Z of the circle are mapped, respectively, into the points of intersection of the rays OX, OY, OZ and the contour of the triangle, (c) the mapping is invertible.

291. (a) Lay off $|A_1M_1| = |AM|$ on the side A_1B_1 , (b) draw two circles of radii AM and CM from the vertices A_1 and C_1 . These circles meet at two points, the point which lies inside the triangle $A_1B_1C_1$ is the required point (Fig. 109).

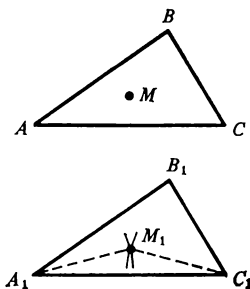


Fig. 109

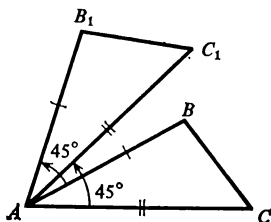


Fig. 110

292. We construct a ray $[AB_1]$ so that $\widehat{BAB_1} = 45^\circ$. We lay off $[AB_1] \cong [AB]$ on $[AB_1]$ and then $B_1 = R_A^{45^\circ}(B)$. Similarly we construct $C_1 = R_A^{45^\circ}(C)$. The triangle AB_1C_1 is the required one (Fig. 110).

293. *Hint.* The centre of rotation is the point A . The rotation angle is $\widehat{BAD}$.

294. The square is mapped onto itself upon rotations about the intersection points of the diagonals through $90^\circ, 180^\circ$ and 270° both clockwise and counterclockwise.

295. (a) $\alpha = 60^\circ + 360^\circ n$, where $n \in \mathbb{Z}$, (b) $\alpha + 120^\circ = -\alpha + 360^\circ n$, whence we have $\alpha = -60^\circ + 180^\circ n$, where $n \in \mathbb{Z}$.

296. (a) The straight line is mapped onto itself, (b) onto the opposite ray with the same origin, (c) onto the angle which is vertical with respect to the given one, (d) onto the parallel straight line.

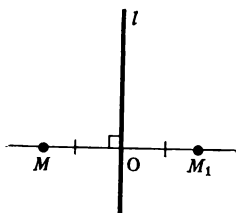


Fig. 111

297. (a) We construct a point symmetric with respect to the centre of the given circle and draw a circle of the same radius from the point obtained, (b) two cases are possible here: (1) if the centre of symmetry coincides with the centre of the given circle, then we get the same circle, (2) if the centre of symmetry does not coincide with that of the circle, we get a circle which cuts the first circle and is congruent to it.

298. We must draw a straight line through the point M at right angles to the l -axis and mark a point M_1 on the ray MO ($O \in l$) so that $|OM_1| = |OM|$ (Fig. 111).

299. (a) Two, the straight line which contains the segment and the straight line perpendicular to that segment and passing through its middle point, (b) one, the straight line on which the ray lies, (c) an infinite number: the straight line itself and any straight

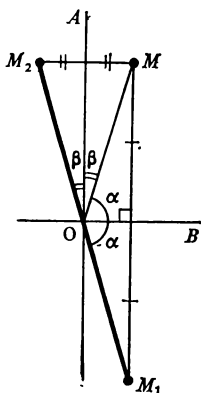


Fig. 112

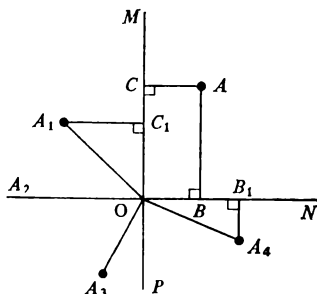


Fig. 113

line perpendicular to the given line, (d) an infinite number: the straight lines which pass through the centre of the circle, (e) one bisector of the angle, (f) three: the bisector of each angle, (g) four: two of them contain the diagonals and the other two contain the perpendicular bisectors of the adjacent sides.

300. 1. An isosceles triangle. 2. *Hint.* $\angle M_1OM_2 = 2\alpha + 2\beta = 2(\alpha + \beta) = 2 \cdot 90^\circ = 180^\circ$, and therefore the angle $\angle M_1OM_2$ is a straight angle (Fig. 112).

SECTION 2.24

301. It is the smallest of the distances from the point to various points of the straight line or the distance from the point to the foot of the perpendicular drawn through the point to the straight line.

302. (1) If the point A is in the interior of an angle, say, MON (Fig. 113), then $|AB|$ and $|AC|$ are the distances from that point to the sides of the angle ($[AB] \perp [ON]$, $[AC] \perp [OM]$).

(2) If the point A_1 is in the interior of the angle MOA_2 , then

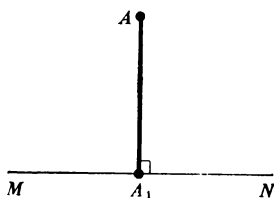


Fig. 114

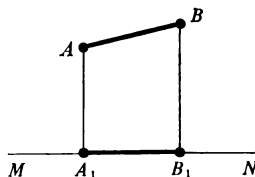


Fig. 115

the distance from A_1 to $[OM]$ is $|A_1C_1|$ ($[A_1C_1] \perp [OM]$), and the distance from A_1 to $[ON]$ is $|A_1O|$; if the point A_3 lies in the interior of the angle A_2OP , then $|A_3O|$ is the distance from the point A_3 to $[ON]$ and to $[OM]$.

(4) If the point A_4 lies in the interior of the angle NOP , then $|A_4B_1|$ ($[A_4B_1] \perp [ON]$) is the distance from the point A_4 to $[ON]$ and $|A_4O|$ is the distance from the point A_4 to $[OM]$.

303. (a) The projection of the point A onto the straight line MN is the foot A_1 of the perpendicular drawn from the point A to the straight line MN (Fig. 114).

(b) The projection of the line segment AB onto the straight

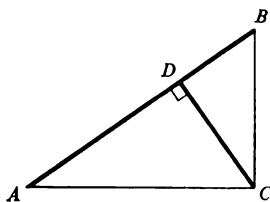


Fig. 116

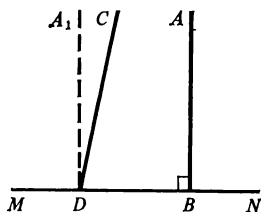


Fig. 117

line MN is the segment A_1B_1 ; the endpoints of this segment are the projections of the endpoints of the given segment (Fig. 115).

304. Assume that we are given a right triangle ABC (Fig. 116). The projection of the leg BC onto the hypotenuse AB is the segment BD ; the projection of the leg AC onto the hypotenuse is the segment AD ; the projection of the legs BC and AC onto CD is the segment CD .

305. No, we cannot. For instance, although the semaphore at the railway bed and the rail on the same bed do not meet, we cannot say that they are parallel since they lie in different planes.

306. 1. Not more than one straight line can be drawn through a given point parallel to a given straight line.

2. Let $(AB) \perp MN$ and (CD) be an oblique line (Fig. 117). We assume that (AB) does not cut (CD) , i.e. $(AB) \parallel (CD)$. We draw

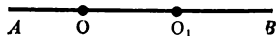


Fig. 118

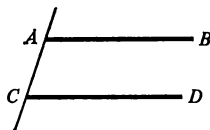


Fig. 119

a perpendicular to MN through the point D , i.e. $(DA_1) \perp MN$, and then $(DA_1) \parallel BA$, i.e. two straight lines DA_1 and DC are drawn parallel to (AB) through the point D , and this contradicts the parallel axiom.

307. 1. Figure 118 shows two rays OB and O_1B which have the same direction and lie on the same straight line AB . One of them contains the other one. The rays O_1B and OA have opposite directions, $[O_1B)$ is not contained in $[OA)$.

2. In Fig. 119 the rays (AB) and (CD) are of the same direction. The straight line AC divides the plane into two half-planes: $[AB)$ and (CD) lie in the same half-plane. In Fig. 120 $[AB)$ and (CD) have opposite directions; they lie in different half-planes.

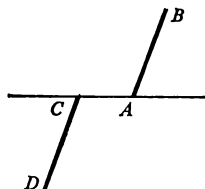


Fig. 120

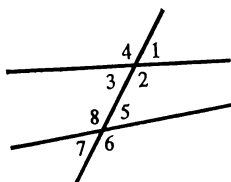


Fig. 121

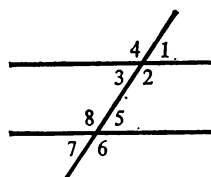


Fig. 122

308. Angles 1 and 5, 2 and 6, 4 and 8, 3 and 7 are the corresponding angles (Fig. 121). Angles 3 and 5, 2 and 8 are interior alternate angles.

309. (a) The sides of the angles 1 and 5, 2 and 6, 3 and 7, 4 and 8 extend in the same direction (Fig. 122), (b) the corresponding angles at the parallel straight lines are congruent since their sides extend in the same direction.

310. α and $180^\circ - \alpha$.

311. 2. The opposite sides are pairwise parallel (according to the criterion of parallelism of straight lines).

312. $\hat{1} = 180^\circ - 132^\circ = 48^\circ$, $\hat{1} \neq \hat{3}$, and therefore $a \neq b$, $\hat{3} = \hat{6} = 49^\circ$, and therefore $c \parallel d$,

SECTION 2.25

313. (a) BC is the side opposite to the angle A , (b) the angle C is opposite to the side AB , (c) the angles A and B lie at the side AB .

314. The bisector of a triangle is a segment of the bisector of the angle of the triangle between its vertex and the opposite side. The median of a triangle is a segment which connects a vertex of the triangle with the midpoint of the opposite side. The altitude of a triangle is the segment of the perpendicular drawn from a vertex of the triangle to the opposite side.

315. 1. At the vertex of the right angle. See Fig. 123.

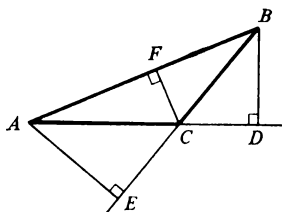


Fig. 123

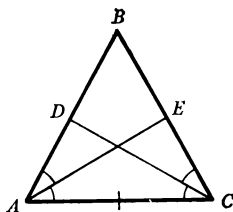


Fig. 124

316. $3(p - a/2)$.

317. We have $\triangle ABC \cong \triangle KLM$, and therefore $[AC] \cong [ML]$, $\angle A \cong \angle L$, $\angle C \cong \angle M$.

318. (a) Three principal elements, at least one of which must be linear, (b) two principal elements, at least one of which is linear, (c) we must be given only one side of the triangle.

319. The constructed triangles are congruent according to the SSS criterion.

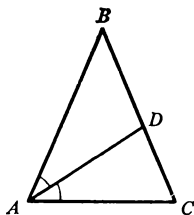


Fig. 125

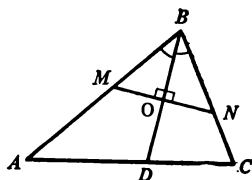


Fig. 126

321. These segments lie opposite congruent angles in congruent triangles.

322. These angles lie in the congruent triangles ABD and BEC opposite the congruent sides AB and BC .

323. (b) *Hint.* $\triangle ADC \cong \triangle AEC$ according to the ASA criterion (Fig. 124).

324. No, it is not perpendicular. If we assume that $[AD] \perp [BC]$ (Fig. 125), then $\triangle ADC \cong \triangle ADB$ (according to the ASA criterion, and then $|AB| = |AC|$, and this is incorrect since the triangle ABC is nonequilateral.

325. We have $[BD] \cap [MN] = 0$, $\triangle MBO \cong \triangle BON$ (the ASA criterion) and hence $|MB| = |BN|$ (Fig. 126).

326. This follows from the fact that $\triangle ABD \cong \triangle DBC$ (two legs are equal).

327. *Hint.* The distances BF and CE are equal since the triangles BFD and CDE are congruent (Fig. 127).

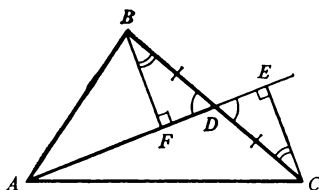


Fig. 127

328. Besides a right angle we must have two elements, at least one of which is linear.

329. The centre of such a circle is the point of intersection of the perpendicular bisectors of the sides of the triangle.

330. We must choose three arbitrary points, A , B and C , on the circumference of this circle.

The point of intersection of the perpendicular bisectors of the chords AB and BC is the centre of the circle.

331. The required point X is the intersection point of the perpendicular bisector of $[AB]$ and the straight line l (Fig. 128).

332. The centre of the circle inscribed into the given triangle ABC must be equidistant from the sides AB and AC as well as from the sides AC and BC ; this means that the centre belongs to

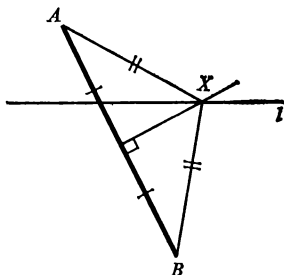


Fig. 128

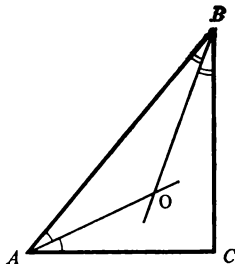


Fig. 129

the bisector of the angle A and to the bisector of the angle C , and the point of intersection of these bisectors is the centre of the inscribed circle.

333. No, it is not, since the diagonal of such a rectangle is not a bisector of its angle.

334. The solution is the point of intersection of the bisector of the angle AOB and the perpendicular bisector of the segment MN .

335. 1. 90° . 2. 45° . 3. 60° .

336. 1. $\gamma = 180^\circ - (\alpha + \beta)$. 2. $\beta = 90^\circ - \alpha$. 3. $\alpha = 90^\circ - \beta/2$.

337. *Hint.* Calculate the angles of the triangles ABE and AEC .

338. Let O be the point of intersection of the bisectors BO and

AO (Fig. 129). We have $\widehat{BAC} + \widehat{ABC} = 90^\circ$, $\frac{1}{2}\widehat{BAC} + \frac{1}{2}\widehat{ABC} = \widehat{BAO} + \widehat{ABO} = 45^\circ$, whence $\widehat{AOB} = 135^\circ$.

339. 2. It can, if that exterior angle is adjacent to the interior obtuse angle of the triangle.

3. $90^\circ + \alpha/2$.

340. 1. 2α .

2. This follows from the congruence of the base angle of the triangle and the angle between the bisector of the exterior angle and a lateral side of the triangle.

341. 1. The leg of a right triangle which lies opposite the angle of 30° is equal to a half of the hypotenuse.

2. 30° , 60° and 90° .

3. 5 cm and 5.5 cm. *Hint.* Use the property of a leg which lies opposite the angle of 30° , and then the perpendicular divides one of the sides of the triangle into segments 1.5 cm and 5.5 cm long and the other side into segments 2 cm and 5 cm long.

342. (a) $180^\circ(4-2) = 360^\circ$; (b) $180^\circ(5-2) = 540^\circ$, (c) $180^\circ(6-2) = 720^\circ$.

343. (a) $\frac{180^\circ(6-2)}{6} = 120^\circ$, (b) $\frac{180^\circ(n-2)}{n}$.

344. $180^\circ - \alpha$, α .

345. Solving the equation $180^\circ(n-2) = 1260^\circ$, we find that the number of sides of the polygon is $n = 9$. The number of triangles is smaller by two, i.e. 7.

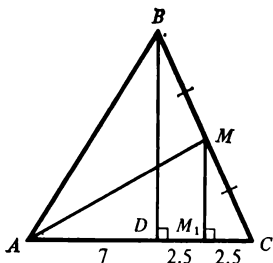


Fig. 130

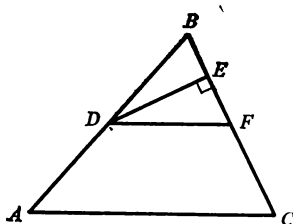


Fig. 131

346. No, it cannot, since in that case the sum of the exterior angles of this hexagon would be larger than 360° , and this is impossible.

347. The sum of all the exterior angles of the polygon is 360° . There are $360 \div 72 = 5$ angles and, hence, as many sides.

348. (a) AB , (b) the base, since it is opposite the larger angle equal to 72° .

349. We have $|BM| = |MC|$ (Fig. 130), and therefore $|M1C| = |DM1|$ (by the Thales theorem). The projection of the median AM : $|AM1| = |AD| + |DM1|$, $|AM1| = 7 \text{ cm} + 2.5 \text{ cm} = 9.5 \text{ cm}$.

350. p (the length of each side of the cut-off triangle is half the corresponding side of the given triangle).

351. 7 cm.

352. No, we cannot. To find this distance, we must, say, draw a perpendicular DE from the midpoint of $[AB]$ to $[BC]$ (Fig. 131), the length of DE is smaller than that of the median DF .

SECTION 2.26

353. (a) The opposite sides of a parallelogram are pairwise congruent, (b) the opposite angles of a parallelogram are pairwise congruent, (c) the diagonals of a parallelogram are bisected by the point of their intersection.

354. The sum of the angles at the same side of a parallelogram is 180° (2d).

355. 1. For a rectangle to be a parallelogram, it is necessary and sufficient that its opposite sides be pairwise congruent. This must be understood as follows: if a given rectangle is a parallelogram, then its opposite sides are pairwise congruent, and conversely, if the opposite sides of a rectangle are pairwise congruent, then the rectangle is a parallelogram.

2. For a rectangle to be a parallelogram, it is necessary and sufficient that its two opposite sides be congruent and parallel.

356. 2. This follows from the congruence of the right triangles ABE and CDF (Fig. 132).

3. This is because these segments are opposite sides of a parallelogram.

357. (a) We must draw a perpendicular from an arbitrary

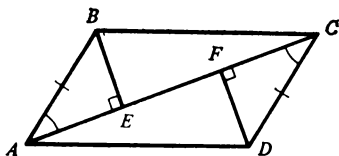


Fig. 132

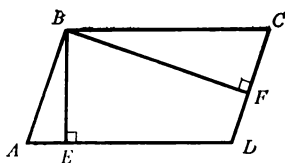


Fig. 133

point of the larger side (the best choice is a vertex of the parallelogram) to the side parallel to it, (b) we must draw a perpendicular from an arbitrary point of the smaller side of the parallelogram (the best choice is a vertex of the parallelogram) to the side parallel to it (or to its extension). In Fig. 133 $|BE|$ is the distance between the larger sides of the parallelogram, $|BF|$ is the distance between its smaller sides.

358. 4 cm.

359. 56 cm. *Hint.* Prove that the acute angle of the given parallelogram is 30° and then use the property of the leg opposite the angle of 30° (Fig. 134).

360. 1. $2(2a + b)$ or $2(2b + a)$. *Hint.* Prove that two angles of the triangle cut off by the bisector are congruent.

2. 30 cm. *Hint.* Prove that the indicated bisector cuts off an equilateral triangle.

361. If we assume that in the parallelogram shown in Fig. 135 $\angle 1 \cong \angle 2$, then $\angle 1 \cong \angle 3$ since they are interior alternate angles at parallel lines (whose sides extend in the same direction) and, consequently, $\angle 2 \cong \angle 3$. Then $|BC| \cong |AB|$, and hence $ABCD$ is a rhombus, and this contradicts the hypothesis.

362. (a) This follows from the fact that the opposite angles of a parallelogram are congruent and the sum of its angles at the

same side is 180° , (b) *Hint*. Two perpendiculars drawn to the same straight line are parallel.

363. 1. Two. Perpendicular bisectors of its two adjacent sides.
2. The diagonals of a rectangle are congruent.

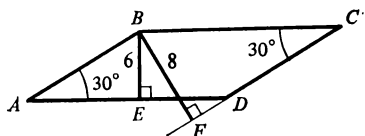


Fig. 134

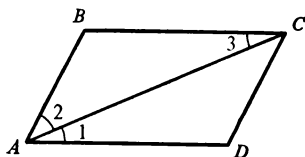


Fig. 135

3. The triangles ABD and ACD (Fig. 136) are congruent (SSS criterion), and therefore $\angle BAD \cong \angle ADC$, but the sum of these angles is 180° and therefore each of them is a right angle.

4. *Hint*. Assuming the midpoint of the hypotenuse to be the centre of symmetry, complete the given right triangle to get a rectangle.

364. 20 cm, 12.5 cm.

365. The diagonals of a rhombus are mutually perpendicular and bisect its angles.

367. This follows from congruence of all the four right triangles into which the diagonals divide the parallelogram.

368. Yes, there is. The diagonals of such a quadrilateral are not bisected at the point of their intersection (Fig. 137).

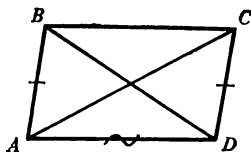


Fig. 136

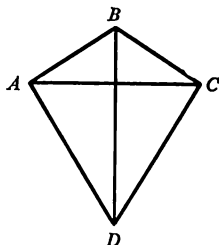


Fig. 137

369. No, it cannot. If we assumed that a diagonal of a rhombus is perpendicular to its side, then the length of a leg (diagonal) would be equal to the length of the hypotenuse (a side of the rhombus).

370. 2α , $180^\circ - 2\alpha$.

371. 60° and 120° . *Hint*. If a leg of a right triangle is half the hypotenuse, then the angle opposite that leg is 30° .

372. (a) No, it is not, since the diagonals of a nonrectangular rhombus are not congruent, (b) yes, it is, it follows from the property of points lying on the bisector of an angle.

373. 1. A square is a rhombus whose angles are right.

2. As distinct from the diagonals of a rectangle, those of a square are mutually perpendicular and bisect its angles. As distinct from the diagonals of a rhombus, those of a square are congruent.

374. We must measure the diagonals and sides of the quadrilateral, and if the diagonals are congruent, as well as all its sides, then the land plot is a square.

375. $22^\circ 30'$, 45° and $112^\circ 30'$. *Hint.* First calculate the angles of the isosceles triangle BCE .

376. 6 cm (Fig. 138).

377. *Hint.* The triangles ABE , BEC , CFD and AFD (Fig. 139)

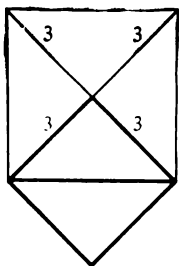


Fig. 138

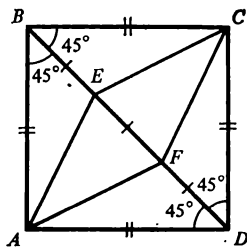


Fig. 139

are congruent (SSS criterion) and hence $|EC| = |EA| = |CF| = |AF|$.

378. A trapezoid is *isosceles* if its nonparallel sides are congruent. A trapezoid is said to be *right-angled* if one of its angles is a right angle.

379. This follows (a) from the fact that the perpendicular drawn from the midpoint of the base of an isosceles trapezoid is its axis of symmetry, (b) from the congruence of the triangles ABD and ACD (Fig. 140), (c) if we assume that $|AO| = |OC|$

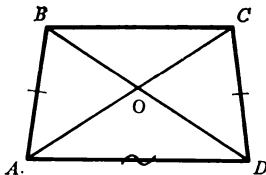


Fig. 140

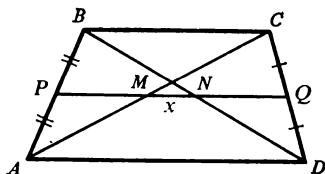


Fig. 141

and $|BO| = |OD|$, then the triangles AOD and BOC are congruent and $BC \cong AD$, and this is impossible for a trapezoid.

380. 1. $\frac{a+b}{2} = m$, whence $b = 2m - a$.

2. $P = 2m + 2c = 2(m + c)$.

381. 1. 8 cm and 16 cm. *Hint.* If we designate the length of the smaller segment of the median as x , then the length of the larger segment is $2x$; solving the equation $x + 2x = 12$, we find

the lengths of the segments, 4 cm and 8 cm. We can find the lengths of the bases from the property of the median of a triangle.

2. 4 cm and 6 cm.

382. We extend the required distance MN until it intersects the nonparallel sides of the trapezoid at points P and Q (Fig. 141).

Then MQ is the median of the triangle ACD , NQ is the median of the triangle BCD , and $|MN| = |MQ| - |NQ|$;

$$|MN| = \frac{a}{2} - \frac{b}{2} = \frac{a-b}{2}.$$

383. $a + b/4$. *Hint.* Draw an altitude of the trapezoid from the vertex of the obtuse angle.

384. *Hint.* We have $[ME] \parallel [FN]$ (Fig. 142), since each of these segments is paral-

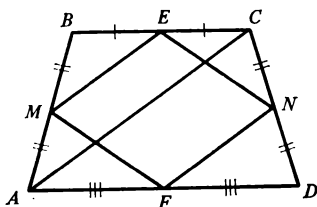


Fig. 142

lel to the diagonal AC ; $|ME| = \frac{1}{2} |AC|$ and $|FN| = \frac{1}{2} |AC|$; consequently, $|ME| \cong |FN|$ and the quadrilateral $MENC$ is a parallelogram. In addition, since $\triangle AMF \cong \triangle FND$, it follows that $[MF] \cong [FN]$, and therefore, the quadrilateral $MENF$ is a rhombus.

SECTION 2.27

385. 1. To define a parallel displacement, it is sufficient to define a point on the plane and its image.

2. A parallel displacement maps each figure onto a congruent figure.

386. (a) Under a parallel displacement the image of a straight line is a straight line parallel to it (Fig. 143), (b) the image of a ray is a ray of the same direction (Fig. 144), (c) the image of a figure is a congruent figure (Fig. 145).

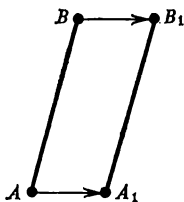


Fig. 143

387. 1. A parallel displacement defined by the centres of the circles.

388. We draw straight lines parallel to the sides of the given angle through any point taken in the interior of that angle. The angle between these straight lines is congruent to the given angle.

389. 1. A *zero vector* is a parallel displacement to a zero distance (an identity projection of a plane), 0, $\vec{AA}$ and $\vec{BB}$ are different notations for a zero vector.

2. The distance $|AB|$ is the *length of the vector* $\vec{AB} = a$ and is designated as $|\vec{AB}|$. The direction defined by the ray AB is the *direction of the vector* $\vec{AB}$. A zero vector has no direction.

390. (a) The vectors $\vec{AB}$ and $\vec{CD}$ are of the same direction if the rays AB and CD are of the same direction, (b) the vectors $\vec{AB}$ and $\vec{CD}$ are of the opposite directions if the rays AB and CD have opposite directions, (c) nonzero vectors are collinear if their directions coincide or are opposite.

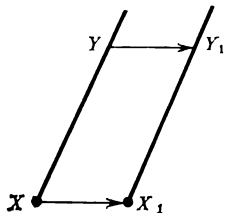


Fig. 144

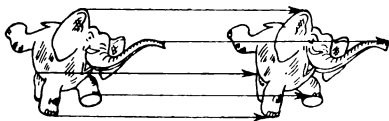


Fig. 145

391. 1. To define a nonzero vector, it is sufficient to indicate its length and direction.

2. The vector opposite to the vector $\vec{AB}$ is $\vec{BA}$; the vector $\vec{BA}$ "shifts" all the points of the plane in the same direction, which is opposite to the direction of the vector $\vec{AB}$, with $|\vec{AB}| = |\vec{BA}|$.

392. (a) $\vec{AD} = \vec{BC}$, $\vec{BO} = \vec{OD}$, $\vec{AO} = \vec{OC}$, (b) $\vec{AB}$ and $\vec{CD}$, (c) $\vec{BD}$ and $\vec{OD}$, $\vec{BO}$ and $\vec{OD}$, $\vec{AD}$ and $\vec{BC}$, $\vec{AB}$ and $\vec{CD}$.

393. $\vec{AD}$, $\vec{DA}$, $\vec{BC}$ and $\vec{CB}$.

394. Since $[EF]$ is the median of the triangle ABC (Fig. 146), it follows that $[EF] \parallel [AC]$. Consequently, the vectors $\vec{EF}$ and $\vec{CA}$ are collinear.

395. We lay off a vector $\vec{AB} = \mathbf{a}$ from an arbitrary point A (Fig. 147), then a vector $\vec{BC} = \mathbf{b}$ from the point B . The composition

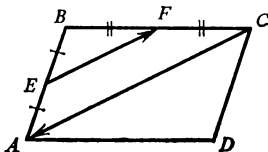


Fig. 146

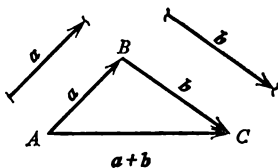


Fig. 147

of the vectors $\mathbf{a}$ and $\mathbf{b}$ maps the point A onto C . Consequently, $\vec{a} + \vec{b} = \vec{AC}$. Replacing $\mathbf{a}$ by $\vec{AB}$ and $\mathbf{b}$ by $\vec{BC}$, we get a rule for finding the sum of vectors, $\vec{AB} + \vec{BC} = \vec{AC}$. This is a rule of triangle.

To add these two vectors together, according to the "parallelogram rule", the vectors $\mathbf{a}$ and $\mathbf{b}$ are laid off from the same point A and a parallelogram with sides AB and AD is constructed. Then $\vec{AC} = \mathbf{a} + \mathbf{b}$ (Fig. 148).

396. $\vec{AB} + \vec{BC} = \vec{AC}$, $\vec{AD} + \vec{DC} = \vec{AC}$, and therefore $\vec{AB} + \vec{BC} = \vec{AD} + \vec{DC}$ (Fig. 149).

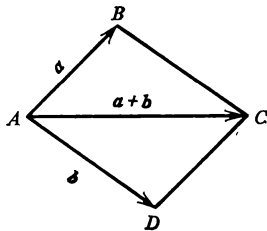


Fig. 148

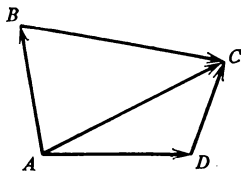


Fig. 149

397. $\vec{AB} + \vec{BC} = \vec{AC}$, $\vec{AB} + \vec{CD} = \mathbf{0}$, $\vec{OB} + \vec{OD} = \mathbf{0}$, $\vec{OC} + \vec{OA} = \mathbf{0}$, $\vec{OB} + \vec{OD} + \vec{OC} + \vec{OA} = \mathbf{0}$.

398. *The 1st technique.* We must first construct the vector $\mathbf{c} = \mathbf{a} + (-\mathbf{b})$. We lay off a vector $\mathbf{a}$ ($\vec{OA} = \mathbf{a}$) from an arbitrary point O (Fig. 150a) and a vector $-\mathbf{b}$ ($\vec{AB} = -\mathbf{b}$) from the point A , and then $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = \vec{OA} + \vec{AB} = \vec{OB}$.

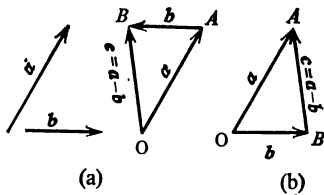


Fig. 150

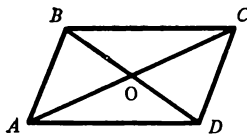


Fig. 151

The 2nd technique. We lay off the vectors $\mathbf{a}$ and $\mathbf{b}$ from the same point O (Fig. 150b) and then we have $\mathbf{a} - \mathbf{b} = \vec{OA} - \vec{OB} = \vec{BA}$.

399. $\vec{AC} = \vec{AB} + \vec{AD}$, $\vec{BD} = \vec{AD} - \vec{AB}$.

400. $\vec{AB} = \vec{OB} - \vec{OA}$, $\vec{AD} = \vec{OD} - \vec{OA} = -\vec{OB} - \vec{OA}$ (Fig. 151).

401. Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are collinear if and only if there is a number x such that $\mathbf{b} = x\mathbf{a}$.

402. $\vec{MC} = \frac{1}{2} \vec{AB} + \vec{BC}$.

403. $\vec{AB} = \frac{1}{2} \vec{DB} - \frac{1}{2} \vec{CA}$, $\vec{BC} = -\frac{1}{2} \vec{CA} - \frac{1}{2} \vec{DB}$ (Fig. 152).

404. *The 1st technique.* We complete the triangle ABC to obtain a parallelogram (Fig. 153). According to the "parallelogram rule" $\vec{AE} = \vec{AB} + \vec{AC}$; since $\vec{AM} = \frac{1}{2} \vec{AE}$, we have $\vec{AM} = \frac{1}{2} \times (\vec{AB} + \vec{AC}) = \frac{1}{2} (\mathbf{a} + \mathbf{b})$.

The 2nd technique. According to the "triangle rule" $\vec{AM} = \vec{AB} + \vec{BM}$ and $\vec{AM} = \vec{AC} + \vec{CM}$. Adding these equations term-by-

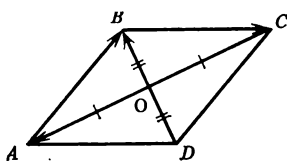


Fig. 152

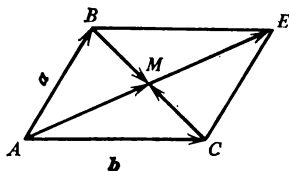


Fig. 153

term, we get $2\vec{AM} = \vec{AB} + \vec{BM} + \vec{AC} + \vec{CM}$, but $\vec{BM} + \vec{CM} = \mathbf{0}$ being two opposite vectors, and hence $\vec{AM} = \frac{1}{2} (\vec{AB} + \vec{AC})$.

$$405. \vec{NM} = \vec{OM} - \vec{ON} = \frac{1}{3} \vec{OB} - \frac{1}{3} \vec{OC} = \frac{1}{3} (\vec{OB} - \vec{OC}) = \frac{1}{3} \vec{CB} = -\frac{1}{3} \vec{AD} = -\frac{1}{3} \mathbf{a}.$$

406. $\vec{OA} = 3\mathbf{i} + 2\mathbf{j}$, $\vec{OB} = 4\mathbf{i} + 5\mathbf{j}$; each coordinate of the sum of the vectors is equal to the sum of the corresponding coordinates: $\vec{OA} + \vec{OB} = (3\mathbf{i} + 2\mathbf{j}) + (4\mathbf{i} + 5\mathbf{j}) = 7\mathbf{i} + 7\mathbf{j}$.

SECTION 2.28

407. 1. Yes, they can. Their ratio of similitude is unity.

2. No, they cannot. Figures are congruent only when their ratio of similitude is unity.

408. The plan of the first plot of land is similar to that of the second, $k = 10$.

409. (a) Under such a homothetic transformation each point is mapped onto itself, (b) by the definition of homothety, in this case $\vec{OX}_1 = -\vec{OX}$, i.e. the point X is mapped onto the point X_1 which is centrosymmetric to the point X .

410. (a) For $k > 0$ the points X_1 and X lie on a straight line on the same side of the centre of homothety (the vectors $\vec{OX}_1$ and $\vec{OX}$ have the same direction), (b) for $k < 0$ the points X_1 and X lie on different sides of the centre of homothety.

411. 1. *Hint.* Under homothetic transformation with, say, a ratio k , the distance between the points changes in the same ratio $|k|$.

2. No, not always. Similar figures are only then homothetic when their positions are such that the points O (the centre of homothety), X and the corresponding point X_1 lie on the same ray.

412. We construct $\vec{CA_1} = -1.5\vec{CA}$, $\vec{CB_1} = -1.5\vec{CB}$ (Fig. 154).

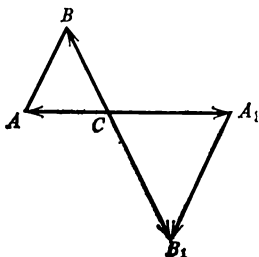


Fig. 154

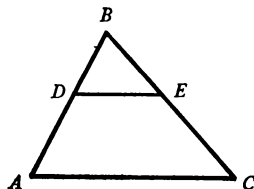


Fig. 155

We can also obtain the vertex B_1 by drawing $[A_1B_1] \parallel [AB]$ so that they intersect the ray BC .

413. By the parallel-proportion theorem we have $\frac{|BD|}{|DA|} = \frac{|BE|}{|EC|}$ (Fig. 155). We designate $|BE|$ as x and then $|EC| = 6.5 - x$. Solving the proportion $\frac{2}{3} = \frac{x}{6.5 - x}$, we get $x = 2.6$ cm, $|BE| = 2.6$ cm and $|EC| = 3.9$ cm.

414. 10.2 cm.

415. (a) Yes, we can. The triangles are similar since their sides are proportional, or due to the congruence of the angles of

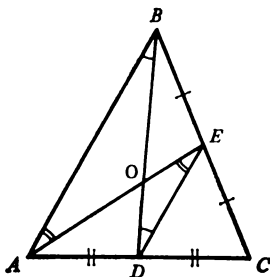


Fig. 156

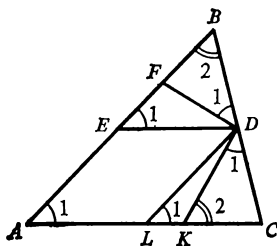


Fig. 157

these triangles (each of them is 60°). (b) Yes, we can. The angles of the triangles are 90° , 45° and 45° . (c) They are similar only if the base angles or vertex angles are congruent.

416. 1. No, they will not. When each side of the triangle is increased three times, a triangle results, which is similar to the given one, and therefore the angles of the triangle we see through the lens are respectively congruent to those of the given triangle.

2. No, it is not. The length of the third side on the plan must be 7.8 cm.

417. In the triangle ABC (Fig. 156) $|DE|$ is a median, and therefore $|DE| \parallel |AB|$. The triangles AOB and DOE are similar (two angles are equal). By the property of the median of a triangle, $|AB| : |DE| = 2$, and therefore the ratio of similitude is $k = 2$.

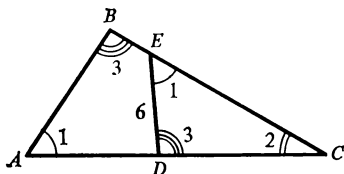


Fig. 158

418. The rays $[DE)$, $[DK)$, $[DL)$ and $[DF)$ cut off triangles DBE , DCK , DLC and DBF , which are similar to the triangle ABC (two angles are congruent) (Fig. 157). Congruent angles are labelled by the same figures.

419. We have $\triangle ABC \sim \triangle DEC$ (Fig. 158) since $\angle DEC \cong \angle A$ ($\angle C$ is common). Hence follows the proportionality of the corresponding sides. (Sides which lie opposite equal angles in similar triangles are called *corresponding sides*). To indicate these sides, we label the congruent angles of the triangle by the same figures:

Angles	$\triangle ABC$	$\triangle DEC$
Opposite $\angle 1$	BC	DC
Opposite $\angle 2$	AB	ED
Opposite $\angle 3$	AC	EC

We first write the sides of one triangle, i.e.

$$|AB| = |BC| = |AC|$$

Under each of these sides we write the corresponding side of the triangle DEC :

$$\frac{|AB|}{|ED|} = \frac{|BC|}{|DC|} = \frac{|AC|}{|EC|}.$$

We take the proportion of the first ratios $\frac{|AB|}{|ED|} = \frac{|BC|}{|DC|}$.

(If we took the last two ratios, there would be two unknown terms in the proportion.) Substituting the given numbers, we obtain $\frac{9}{6} = \frac{15}{|DC|}$, whence we have $|DC| = 10$ cm. Now we form a proportion of the first and the third ratio, i.e. $\frac{|AB|}{ED} = \frac{|AC|}{|EC|}$ and find that $|EC| = 12$ cm.

420. 44 cm and 33 cm. *Hint.* The base of one triangle must coincide with a lateral side of the other (Fig. 159). Otherwise the triangles would be congruent.

421. Since right angles are always congruent, right triangles are similar if (1) the legs of one of the right triangles are proportional to the legs of the other, or (2) an acute angle of one triangle is congruent to an acute angle of the other.

422. 1. Yes, they are.

2. 13.5 cm.

423. Assume that in the triangle ABC (Fig. 160) AM and CN are medians which meet at a point O , then $[NM]$ is a median

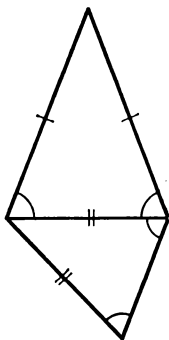


Fig. 159

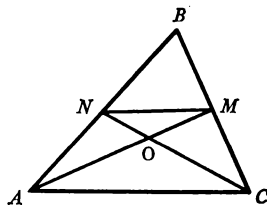


Fig. 160

of the triangle ABC and therefore $[MN] \parallel [AC]$. The triangles ONM and AOC are similar (two angles are equal), whence we have $\frac{|OM|}{|OA|} = \frac{|ON|}{|OC|} = \frac{|NM|}{|AC|}$, but $\frac{|NM|}{|AC|} = \frac{1}{2}$ (in accordance with the property of the median of a triangle) and therefore $\frac{|OM|}{|OA|} = \frac{1}{2}$ and $\frac{|ON|}{|OC|} = \frac{1}{2}$; consequently, $\frac{|OM|}{|AM|} = \frac{1}{3}$ and $\frac{|ON|}{|CN|} = \frac{1}{3}$, i.e. $|OM| = |AM|/3$ and $|ON| = |CN|/3$.

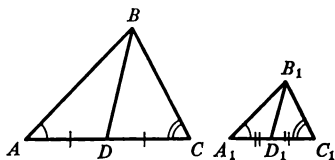


Fig. 161

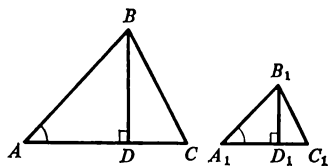


Fig. 162

424. (a) This follows from the similarity of the right triangles ABD and $A_1B_1D_1$ (Fig. 161); $\angle A \cong \angle A_1$ since $\triangle ABC \sim \triangle A_1B_1C_1$ and $\widehat{ADB} = \widehat{A_1D_1B_1} = 90^\circ$. (c) *Hint*. It follows from the

similarity of the triangles ABC and $A_1B_1C_1$ (Fig. 162) that

$$\frac{|AB|}{|A_1B_1|} = \frac{|AC|}{|A_1C_1|} = \frac{\frac{1}{2}|AC|}{\frac{1}{2}|A_1C_1|} = \frac{|AD|}{|A_1D_1|}$$

and

$$\angle A \cong \angle A_1;$$

consequently, $\triangle ABD \sim \triangle A_1B_1D_1$ (two sides of one triangle are proportional to two sides of the other and the angles between them are congruent).

425. (a) Besides the usual reading (m is related to x as x is related to n), this proportion can be read as follows: x is the mean proportional (geometric mean) of the numbers m and n , i.e. $x = \sqrt{mn}$. (b) a is the mean proportional (geometric mean) of the numbers p and q , i.e. $a = \sqrt{pq}$.

426. 1. By the hypothesis, $a = 3$ cm, $a_c = 1.8$ cm. From the equation $a^2 = ca_c$ we find that $c = a^2 \div a_c = 9 \div 1.8 = 5$. We find the leg using the Pythagorean theorem: $c^2 = a^2 + b^2$, whence $b^2 = c^2 - a^2$, $b^2 = 25 - 9 = 16$, $b = 4$ cm.

2. $a^2 = 8 \cdot 2 = 16$, $a = 4$; $b^2 = 8 \cdot 6 = 48$, $b = 4\sqrt{3}$; $h^2 = 6 \cdot 2 = 12$, $h = 2\sqrt{3}$.

427. Using the converse of the Pythagorean theorem, we have: if the square of the length of one side of a triangle is equal to the sum of the squares of the lengths of the other two sides, then the triangle is right-angled.

428. 10 cm.

429. 6.5 cm. *Hint.* Extend the median, lay off a segment equal to the median in length, and then complete the given triangle to get a rectangle. The length of the median is equal to a half of the hypotenuse.

430. $\sqrt{2}$ m ≈ 1.4 m.

431. 29.6 cm. *Hint.* Do not forget that the diagonals of a rhombus are mutually perpendicular and are bisected by the point of their intersection.

432. 1. $a\sqrt{2} \approx 1.4a$.

2. If we denote the side of the square by x , then we get an equation $2x^2 = d^2$, whence $x = \frac{1}{2}d\sqrt{2}$.

433. 1. $a\sqrt{3}/2$.

2. The distances to the sides are $\sqrt{3}/3 \approx 0.58$ cm, the distances to the vertices are $\frac{2}{3}\sqrt{3}$ cm ≈ 1.2 cm each.

434. 1. 2 cm. *Hint.* Draw two altitudes from the vertices of the obtuse angles.

2. 62 cm.

435. We denote the larger leg by x , and then the smaller leg is $0.75x$. Using the Pythagorean theorem, we get an equation $x^2 + (0.75x)^2 = 20^2$, whence $x = 16$, 75% of 16 is 12; hence the smaller leg is 12 cm.

436. (a) No, we cannot. For instance, a square is not similar to a rhombus although their sides are proportional. (b) No, we

cannot. For instance, a square is not similar to a rectangle although their angles are congruent.

437. If the sides of one polygon are respectively proportional to the sides of the other polygon and the corresponding angles of the polygons are congruent, then the polygons are similar.

438. 1. $k = 4/3$, $P = kP_1 = 36 \cdot (4/3)$ cm = 48 cm.

2. $k = 3/7$, $P \div (P + 20) = 3/7$; from this we find that the perimeter of the smaller polygon is 15 cm and that of the larger one is 35 cm.

SECTION 2.29

439. (a) 120° , (b) $22^\circ 30'$.

440. Assume that the degree measure of the smaller arc is x° , then that of the larger arc is $(x + 90)^\circ$. Solving the equation $x + (x + 90) = 360$, we find that the angular measure of the smaller arc is 135° and that of the larger one is 225° .

441. Connect the endpoints of the arc by a chord and then draw a perpendicular to the chord from the centre of the circle. The

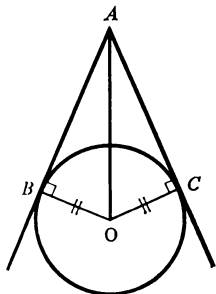


Fig. 163

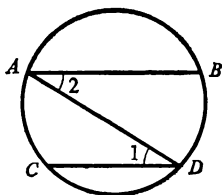


Fig. 164

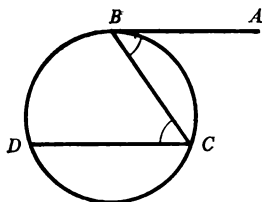


Fig. 165

point of intersection of the perpendicular and the arc bisects the arc.

442. 3 cm. *Hint.* Connect the centre with the endpoints of the chord and then use the property of the leg lying opposite an angle of 30° .

443. We must draw a perpendicular to the radius OA through the point A .

444. They are parallel.

445. 1 dm. *Hint.* The required radius is the median of a right-angled trapezoid.

446. The equality of the lengths of the tangents AB and AC (Fig. 163) follows from the congruence of the right triangles AOB and AOC .

447. 1. It follows from the fact that it is subtended by an arc whose degree measure is 180° .

2. The inverse propositions are true, i.e. if the inscribed angle is a right angle, then it rests on the diameter. If a right angle rests on a diameter, then it is inscribed into the circle.

448. If it is subtended by an arc whose degree measure is (a) smaller than 180° , (b) larger than 180° .

449. If we denote the inscribed angle by x , then the central angle is $(x + 30)$. Using the property of an inscribed angle, we have $2x = x + 30$. Hence the inscribed angle is 30° and the central angle is 60° .

450. If the vertices of the inscribed angles lie on the same side of the chord which connects the endpoints of the arc, then the angles are congruent, and if they lie on different sides of the chord, then the sum of the values of these angles is 180° .

451. Assume that the chords AB and CD are parallel (Fig. 164). We draw a chord AD , and then $\hat{1} = \hat{2}$ (as the interior alternate angles at parallel lines), but $\hat{1} = \frac{1}{2} \widehat{AC}$, $\hat{2} = \frac{1}{2} \widehat{BD}$; consequently, $\widehat{AC} \cong \widehat{BD}$.

452. *Hint.* Draw the chord CD parallel to the tangent BA (Fig. 165).

SECTION 2.30

453. If the triangle is (a) acute-angled, (b) obtuse-angled, (c) right-angled; the centre of the circumscribed circle is the mid-point of its hypotenuse.

454. We can draw, say, a diameter AB (Fig. 166), then draw

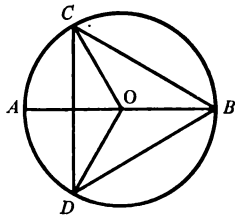


Fig. 166

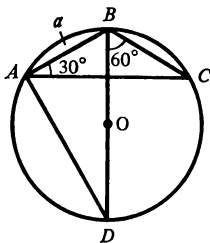


Fig. 167

a perpendicular bisector of the radius OA until it intersects the circle at points C and D . Then the resulting triangle BCD will be equilateral ($\widehat{MCO} = 30^\circ$, $\widehat{AOC} = 60^\circ$, $\widehat{COD} = 120^\circ$, $\widehat{CBD} = 60^\circ$).

455. We can, since in an equilateral triangle each bisector is at the same time a perpendicular bisector of its side.

456. $\frac{2}{3}h$.

457. (a) 5 cm, (b) 2.5 cm. *Hint.* The hypotenuse of this triangle coincides with the diameter of the circumscribed circle and the median coincides with its radius.

458. 70° . *Hint.* Circumscribe a circle about this triangle.

459. 2a. *Hint.* Draw a diameter BD and a chord AD (Fig. 167).

460. 6.25 cm. *Hint.* Let the perpendicular bisector of the lateral side BC (Fig. 168) cut the altitude of the triangle at a point O , then the required radius OB can be found from the similarity of the triangles DBC and OBM .

461. (a) No, they do not, since the perpendicular bisector of a lateral side of an isosceles nonequilateral triangle does not coincide with the bisector of the angle drawn to that side, (b) they coincide, since in an equilateral triangle a perpendicular bisector of each side is at the same time the bisector of the opposite angle.

462. $2r$, $3r$.

463. 3 cm. *Hint.* Assume that M , N and P are the points of tangency of the inscribed circle and the sides of the triangle

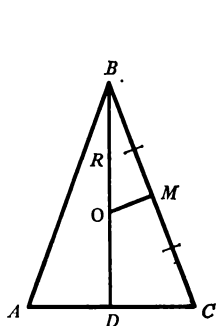


Fig. 168

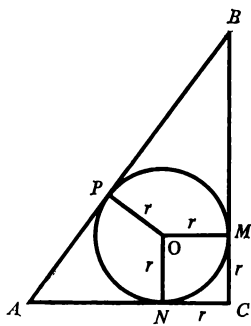


Fig. 169

(Fig. 169). We denote the radius of the inscribed circle by r , and then $|AN| = |AP| = 9 - r$, $|BM| = |BP| = 12 - r$ (the tangents are equal in length, see 446). The length of the hypotenuse AB is 15 cm. But $|AB| = |AP| + |PB|$ and we get an equation $15 = (9 - r) + (12 - r)$, whence $r = 3$ cm.

SECTION 2.31

464. The sine is a function since each angle of rotation α is associated with one and only one point P_α , and consequently one definite number y_α ; the cosine and the tangent are functions for the same reason.

465. 1. The *domain of definition* of the sine and cosine functions is the whole set of angular values, and the *range* of these functions is the interval $[-1, 1]$ since the ordinates of the points P_α can assume the values from -1 to 1 .

2. No, they cannot, since $\sqrt{2} > 1$ and $10^{\log 2} = 2 > 1$.

466. The values of the function $\sin \alpha$ increase from 0 to 1, and the values of the function $\cos \alpha$ decrease from 1 to 0.

467. (a) $\sin 90^\circ = 1$, $\cos 90^\circ = 0$, (b) $\sin 180^\circ = 0$, $\cos 180^\circ = -1$, (c) $\sin 270^\circ = -1$, $\cos 270^\circ = 0$, (d) $\sin 360^\circ = 0$, $\cos 360^\circ = 1$, (e) $\sin (-90^\circ) = -1$, $\cos (-90^\circ) = 0$.

469. (a) Since $\sin^2 \alpha + \cos^2 \alpha = 1$, it follows that $\cos^2 \alpha = 1 - \sin^2 \alpha$, or $\cos^2 \alpha = 1 - 0.8^2 = 0.36$, $\cos \alpha = 0.6$, (b) $\sin^2 \alpha =$

$1 - \cos^2 \alpha$, $\sin^2 \alpha = 1 - (1/2)^2 = 3/4$, $\sin \alpha = \sqrt{3/4} = \sqrt{3}/2$.

470. (a) $\sin 150^\circ = \sin (180^\circ - 30^\circ) = \sin 30^\circ$ (we have applied the formula $\sin (180^\circ - \alpha) = \sin \alpha$), (b) $\cos 150^\circ = \cos (180^\circ - 30^\circ) = -\cos 30^\circ$ (we have applied the formula $\cos (180^\circ - \alpha) = -\cos \alpha$), (c) $\sin 170^\circ = \sin 10^\circ$, (d) $\cos 170^\circ = -\cos 10^\circ$.

471. (a) $a = c \cdot \sin \hat{A}$, $b = c \cdot \cos \hat{A}$, (b) $c = \frac{a}{\sin \hat{A}}$, $c = \frac{b}{\cos \hat{A}}$.

472. $\sin \hat{B} = b/c$; $\cos \hat{B} = a/c$; $\tan \hat{B} = b/a$.

473. 1. $\sin \alpha = h/a$, $\cos \alpha = m/a$, $\sin x = m/a$ and so on.

2. $m/a = \cos \alpha$, $h/b = \sin \beta$, $m/h = \tan x$, $h/n = \tan \beta$.

474. $c = 17$ cm; $\sin \hat{A} + \sin \hat{B} = \frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$, or $1 \frac{6}{17}$;

$\cos \hat{A} \cdot \cos \hat{B} = \frac{b}{c} \cdot \frac{a}{c} = \frac{ab}{c^2}$, or $\frac{120}{289}$.

475. 1. Note that if $\hat{A} = 30^\circ$, then $\hat{B} = 60^\circ$ (Fig. 170). We de-

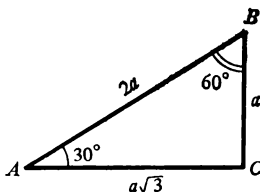


Fig. 170

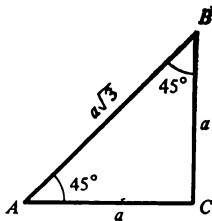


Fig. 171

note $|BC|$ by a , and then $|AB| = 2a$ (in accordance with the property of a leg lying opposite an angle of 30°). Now we have

$|AC| = \sqrt{(2a)^2 - a^2} = \sqrt{3a^2} = a\sqrt{3}$, and therefore $\sin 30^\circ =$

$\frac{a}{2a} = \frac{1}{2}$, $\cos 30^\circ = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}$, $\tan 30^\circ = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}} =$

$\frac{1 \cdot \sqrt{3}}{(\sqrt{3})^2} = \frac{\sqrt{3}}{3}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\tan 60^\circ = \sqrt{3}$.

2. If $\hat{A} = 45^\circ$, then $\hat{B} = 45^\circ$ (Fig. 171). We denote $|BC|$ and

$|AC|$ by a , and then $|AB| = \sqrt{2a^2} = a\sqrt{2}$; $\sin 45^\circ = \frac{a}{a\sqrt{2}} =$

$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$, $\cos 45^\circ = \frac{\sqrt{2}}{2}$, $\tan 45^\circ = 1$.

$$476. (a) \sin^2 30^\circ + \cos^2 30^\circ = \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} + \frac{3}{4} = 1,$$

$$(b) \sin^2 45^\circ + \cos^2 45^\circ = \left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 = 1.$$

$$477. (a) \sin (2 \cdot 30^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2} \approx \frac{1.73}{2} \approx 0.865, \text{ and}$$

$$2 \sin 30^\circ = 2 \cdot \frac{1}{2} = 1, \text{ i.e. } 2 \sin 30^\circ > \sin (2 \cdot 30^\circ), \quad (b) \sin (30^\circ +$$

$$60^\circ) = \sin 90^\circ = 1 \text{ and } \sin 30^\circ + \sin 60^\circ = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2} \approx$$

$$\frac{1 + 1.73}{2} \approx 1.365, \text{ i.e. } \sin 30^\circ + \sin 60^\circ > \sin (30^\circ + 60^\circ).$$

478. (a) We draw an altitude BD in the equilateral triangle ABC . We have $\frac{|BD|}{|AB|} = \sin 60^\circ$, or $\frac{|BD|}{a} = \frac{\sqrt{3}}{2}$, whence

$$|BD| = \frac{a\sqrt{3}}{2}, \quad \frac{|OD|}{|DC|} = \tan 30^\circ \text{ (from the right triangle$$

$$ODC), \text{ whence } |OD| = \frac{a\sqrt{3}}{6}, \quad (b) \frac{d\sqrt{2}}{2}.$$

479. 2. 1 m. *Hint.* The section of the embankment is an isosceles trapezoid, the height of the embankment is a leg of a right triangle, whose other leg is 3 m, and the adjacent acute angle is 35° .

SECTION 2.32

480. 1. Congruent polygons have equal areas; if a polygon consists of nonoverlapping polygons, then its area is equal to the sum of the areas of those polygons.

2. The unit area is the area of the square whose side is of unit length. For example, a square metre is the area of the square whose side is one metre.

3. The figures of the same area. For example, a rectangle with the sides of 4 and 9 cm and a square with the side of 6 cm, (d) no, not always. For example, a square with the side of 4 cm and a rectangle with the sides of 2 and 8 cm have the same area but they are not congruent.

$$481. 10 \text{ hectares} = 10 \cdot 10\,000 \text{ m}^2 = 100\,000 \text{ m}^2 = 0.1 \text{ km}^2.$$

$$482. (a) S = a^2. (b) \text{ according to the Pythagorean theorem, } a^2 + a^2 = d^2, \text{ or } 2a^2 = d^2, \text{ whence } a^2 = d^2/2, \text{ i.e. } S = d^2/2.$$

We can obtain the same result if we assume a square to be a rhombus with equal diagonals (see 490).

483. The area of the rectangle is $20 \cdot 30 = 600 \text{ (m}^2\text{)}$, that of the square is $25^2 = 625 \text{ (m}^2\text{)}$,

484. If we designate the side of the inscribed square as a (Fig. 172), then $a^2 = R^2 + R^2 = 2R^2$ (according to the Pythagorean theorem), and this is the area of the inscribed square, i.e. $S_{\text{insc}} = 2R^2$. We seek the area of the circumscribed square: $S_{\text{circ}} = (2R)^2 = 4R^2$. The ratio of the areas of these squares is two.

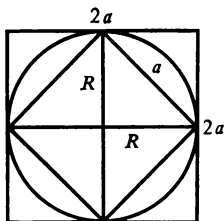


Fig. 172

485. We designate the diagonal of the required square as d . Using the formula $S = \frac{1}{2} d^2$ (see 482), we arrive at

an equation $\frac{1}{2} d^2 = \frac{1}{2} 3^2 + \frac{1}{2} 4^2$, whence $d = 5$ cm.

486. Yes, it will change. The area of the given rectangle is ah . The increased base is $1.5a$, the decreased altitude is $0.5h$, and the new area of the rectangle is $1.5a \cdot 0.5h = 0.75ah$. This means that the area of the resulting rectangle will decrease by 25%.

487. Let a and b be the sides of the parallelogram and S its area. Then we can find one of the altitudes from the formula $h_1 = \frac{S}{a}$ and the other from the formula $h_2 = \frac{S}{b} = \frac{126}{14} = 9$ (cm), $\frac{126}{18} = 7$ (cm). The angle is 30° .

488. *The first case.* Assume that the altitude is drawn to the larger side. Then the area of the parallelogram is 135 cm^2 ($9 \cdot 15 = 135$), and the other altitude is 11.75 cm ($135 \div 12$). *The second case.* If the altitude is drawn to the smaller side, then the area of the parallelogram is 108 cm^2 and the other altitude is 7.2 cm.

489. The parallelogram in which the angle between the adjacent sides is a right angle is the largest since the expression $ab \sin \alpha$ is the largest if $\sin \alpha = 1$ and, hence, $\alpha = 90^\circ$.

490. The area of the rhombus is equal to half the product of the diagonals, i.e. $S = \frac{1}{2} d_1 d_2$.

$$491. (a) S = \frac{1}{2} \cdot 5 \cdot 8 = 20 \text{ cm}^2, (b) S = 2.8^2 \sin 30^\circ \approx 3.9 \text{ dm}^2,$$

$$(c) S = \frac{1}{16} P^2 \sin \alpha.$$

$$492. (a) S = \frac{1}{2} b h_b, S = \frac{1}{2} c h_c, (b) h_a = \frac{S}{\frac{1}{2} a} = \frac{2S}{a},$$

$$h_b = \frac{2S}{b}; h_c = \frac{2S}{c}.$$

493. 1. The triangle ABD and DBC (Fig. 173) are of the same area since they have congruent bases AD and DC and a common altitude BE .

2. The area of the triangle ADM (Fig. 173) is $1/3$ of that of the triangle ABD since $|DM| = \frac{1}{3} |BD|$ (in accordance with the property of the medians of a triangle), and the altitude AF is their common altitude. Consequently, $S_{AMD} = \frac{1}{3} S_{ABD} = \frac{1}{3} \cdot \frac{1}{2} S_{ABC} = \frac{1}{6} S_{ABC}$.

3. The medians AM and BN (Fig. 174) meet at a point O ; $|OM| = \frac{1}{3} |AM|$, and therefore $|AO| = 4$ cm. The median AM

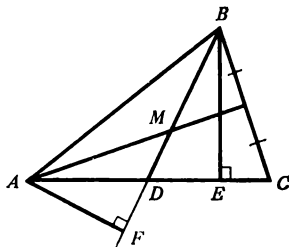


Fig. 173

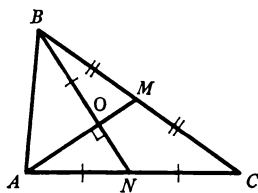


Fig. 174

is perpendicular to the median BN ; assuming BN to be the base of the triangle ABN , and AO to be its altitude, we find that the area of the triangle ABN is 16 cm^2 since $S_{ABN} = \frac{1}{2} |BN| \times |AO| = \frac{1}{2} 8 \cdot 4$. The area of the triangle ABC is twice the area of the triangle ABN (see 493, 1), i.e. 32 cm^2 .

494. 5 cm. *Hint.* If we denote the base of the triangle by x , then its altitude will also be x , whence $0.5x^2 = 12.5$.

495. 48 cm^2 . *Hint.* If we denote the altitude of the triangle by x , then its lateral side will be $(x + 2)$.

496. (a) From the formula $S = \frac{1}{2} \cdot 4 \cdot 4 \cdot \sin 30^\circ$ we find that $S = 4 \text{ cm}^2$, (b) $S = \frac{1}{2} \cdot 4 \cdot 4 \cdot \sin 135^\circ$, $\sin 135^\circ = \sin (180^\circ - 45^\circ) = \sin 45^\circ = \frac{\sqrt{2}}{2}$, whence $S = 4\sqrt{2} \text{ cm}^2 \approx 5.7 \text{ cm}^2$.

497. (a) From the formula $S = \frac{1}{2} \cdot a \cdot a \cdot \sin 60^\circ$ we have $S = \frac{1}{4} a^2 \sqrt{3}$, (b) $a = \frac{h}{\sin 60^\circ}$, $S = \frac{1}{2} \frac{h^2}{\sin 60^\circ} = \frac{h^2 \sqrt{3}}{3}$.

498. We designate the leg of this triangle as a , and then $a = c \cdot \sin 45^\circ$, $S = \frac{1}{2} a^2 = \frac{1}{2} c^2 \sin^2 45^\circ = \frac{1}{2} c^2 \left(\frac{\sqrt{2}}{2} \right)^2 = \frac{1}{4} c^2$,

or in another notation, $2a^2 = c^2$, $a^2 = \frac{c^2}{2}$, $S = \frac{1}{2} a^2 = \frac{1}{4} c^2$.

499. The area of a trapezoid is equal to the product of its median by the altitude.

500. 2.2 dm^2 . *Hint.* The projection of a leg onto the larger base is 10 cm.

501. $(8 + 2\sqrt{3}) \text{ cm}^2 \approx 11.5 \text{ cm}^2$. *Hint.* The altitude of the trapezoid is 2 cm, the projection of one leg is $2\sqrt{3} \text{ cm}$ and that of the other is 2 cm.

502. 30 cm. *Hint.* The median of this trapezoid is 10 m since $(7 + 13) \div 2 = 10$, $h = 4 \text{ m}$ since $40 \div 10 = 4$. The projection of a leg onto the larger base is 3 cm.

503. $1. mn/2$. *Hint.* The area of the trapezoid $ABCD$ (Fig. 175)

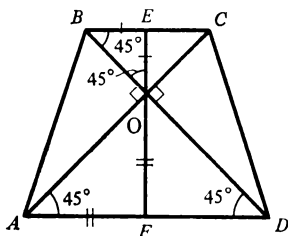


Fig. 175

is equal to the sum of the areas of the triangles ABC and BCD , whose common base is BC , and the altitudes are AO and OC ,

$$\text{or } S = \frac{1}{2} |BC| \cdot |AO| + \frac{1}{2} |BC| \cdot |CO|.$$

2. h^2 . *Hint.* The median of this trapezoid is equal to $\frac{|AD| + |BC|}{2} = \frac{|AD| +$

$$\frac{|BC|}{2} = |AF| + |BE| \text{ (Fig. 175).}$$

The altitude of this trapezoid is equal to the median since $|OF| = |AF|$, and $|OE| = |BE|$.

504. $k = 3$, the area of the polygon increases $3^2 = 9$ times.

505. $k = 100$, the area of the real plot of land is $100^2 = 10\,000$ times larger than in the plan.

506. (a) The area of the given triangle is four times the area of the cut-off triangle (the ratio of similitude is two), (b) the arc of the trapezoid is thrice that of the cut-off triangle.

507. Assume that P and P_1 , S and S_1 are the perimeters and the areas of the similar triangles respectively. Then $P : P_1 = k$, $S : S_1 = k^2$, where k is the ratio of similitude, $k = \frac{P}{P_1} = \frac{2}{3}$, $k^2 = \frac{4}{9}$, $S : S_1 = 4 : 9$, $S : 90 = 4 : 9$, whence $S = 40 \text{ cm}^2$.

SECTION 2.33

508. We can calculate the length of one side of the triangle if we know the other two sides and the angle between them; if we know the length of each side of the triangle, we can calculate each angle using the formulas

$$\cos \hat{A} = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos \hat{B} = \frac{a^2 + c^2 - b^2}{2ac}, \quad \cos \hat{C} = \frac{a^2 + b^2 - c^2}{2ab}.$$

509. (a) $c^2 = a^2 + b^2 - 2ab \cos 60^\circ = a^2 + b^2 - 2ab \cdot \frac{1}{2} = a^2 + b^2 - ab$, $c = \sqrt{a^2 + b^2 - ab}$, (b) $c = \sqrt{a^2 + b^2 - ab \sqrt{2}}$, (c) $c = \sqrt{a^2 + b^2 - 2ab \cos 120^\circ} = \sqrt{a^2 + b^2 + ab}$ ($\cos 120^\circ = \cos (180^\circ - 60^\circ) = -\cos 60^\circ = -1/2$).

510. (a) If $\hat{A} < 90^\circ$, since in that case $\cos \hat{A} = \frac{b^2 + c^2 - a^2}{2bc} > 0$;
 (b) if $\hat{A} = 90^\circ$, since in that case $\cos \hat{A} = \frac{b^2 + c^2 - a^2}{2bc} = 0$;
 (c) if $\hat{A} > 90^\circ$, since in that case $\cos \hat{A} = \frac{b^2 + c^2 - a^2}{2bc} < 0$.

511. Using the cosine law, we get from the triangle ABD (Fig. 176) $|AD|^2 = |AB|^2 + |BD|^2 - 2|AB| \cdot |BD| \cdot \cos 60^\circ$, i.e. $|AD|^2 = 10^2 + 4^2 - 2 \cdot 10 \cdot 4 \times \cos 60^\circ = 76$, $|AD| = \sqrt{76} \approx 8.8$ (cm).

512. From the formula $\cos \hat{A} = \frac{b^2 + c^2 - a^2}{2bc}$ we find that $\cos \hat{A} = 1/2$, $\hat{A} = 60^\circ$.

513. Given a side and two angles of a triangle, we can calculate its other elements.

514. $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta}$, whence

$b = \frac{a \sin \beta}{\sin \alpha}$, $\frac{a}{\sin \alpha} = \frac{c}{\sin (180^\circ - (\alpha + \beta))}$ and $c = \frac{a \sin (\alpha + \beta)}{\sin \alpha}$ ($\sin (180^\circ - (\alpha + \beta)) = \sin (\alpha + \beta)$).

515. The diagonal of the parallelogram divides it into two congruent triangles, whose angles are 30° , 45° and 105° (Fig. 177).

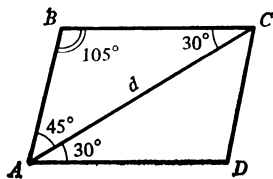


Fig. 177

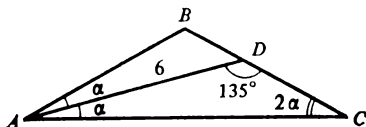


Fig. 178

Using the sine law, we find from the triangle ABC that $\frac{|AB|}{\sin 30^\circ} = \frac{d}{\sin 105^\circ}$, whence $|AB| = \frac{d \sin 30^\circ}{\sin 105^\circ}$, but $\sin 30^\circ = 1/2$, $\sin 105^\circ = \sin (90^\circ + 15^\circ) = \cos 15^\circ$, and therefore $|AB| = d/(2 \cos 15^\circ)$. From the triangle ABC we find that $|BC| = d/(\sqrt{2} \cos 15^\circ)$.

516. $6\sqrt{2}$ cm. *Hint.* If we designate the angle between the bisector AD and the base AC as α (Fig. 178), then $\hat{C} = 2\alpha$. We find from the equation $3\alpha + 135^\circ = 180^\circ$ that $\hat{C} = 30^\circ$, and then calculate $|AC|$ from the triangle ADC using the sine law.

SECTION 2.34

517. We must first divide the circle into n congruent arcs, for which purpose we construct a central angle of $360^\circ/n$ using a protractor, and then, using a pair of compasses, we lay off the resulting arc along the circle $n - 1$ times and connect successive points of division by chords or draw, through these points, tangent lines to the points at which they meet the neighbouring tangents.

518. 1. No, we cannot.

2. (a) It can. For example, this condition is satisfied by inscribed rectangle. (b) No, it cannot; if the sides of the inscribed polygon are congruent, then the arcs subtended by these sides are also congruent, and therefore each of the angles of this polygon is subtended by the same number of congruent arcs.

519. 1. The common centre of the circle inscribed in the regular polygon and the circle circumscribed about it. This point is the centre of rotations which map this regular polygon onto itself.

2. The segment of the perpendicular drawn from the centre of the regular polygon to its side. The segment $[OM]$ in Fig. 64 is the apothem which is the radius of the inscribed circle.

$$520. 1. \frac{180^\circ(5-2)}{5} = 108^\circ, \quad \frac{180^\circ(n-2)}{n}.$$

$$2. (a) \frac{360^\circ}{6} = 60^\circ, \quad (b) \frac{360^\circ}{n}.$$

3. The angle is equal to a half of the central angle,
i.e. $\frac{360^\circ}{n} \div 2 = \frac{180^\circ}{n}.$

$$521. (a) \text{ By the hypothesis, } 120^\circ = \frac{180^\circ(n-2)}{n}, \text{ where } n$$

is the number of sides of the nut. Solving this equation for n , we find that the nut has six sides, (b) eight sides.

522. (a) It can. The angles of the blocks which meet at one point must make 360° in their sum. Three blocks shaped as a regular hexagon correspond to this condition since each angle of the hexagon, equal to 120° , is contained in 360° an integral number of times. (b) No, it cannot.

$$523. 2. a_8 = 2 \cdot 1 \cdot \sin \frac{180^\circ}{8} \text{ m} = 2 \sin 22^\circ 30' \text{ m} \approx 0.77 \text{ m}, \quad a_{10} \approx 0.62 \text{ m}, \quad a_{12} \approx 0.52 \text{ m}.$$

524. From the formula $a_4 = R\sqrt{2}$, or $R = a_4/\sqrt{2}$, we find that $R = \frac{18}{\sqrt{2}} = \frac{18\sqrt{2}}{2} = 9\sqrt{2} \approx 12.7$ (cm). The diameter of the log must not be smaller than 25.4 cm.

525. From formula (3) on p. 95 we find: (a) $S_3 = \frac{1}{2} \cdot 3 \cdot 1^2 \times \sin \frac{360^\circ}{3} = 0.5 \cdot 3 \cdot \sin 120^\circ = 0.5 \cdot 3 \cdot \sin (180^\circ - 60^\circ) = 0.5 \times 3 \cdot \sin 60^\circ = 0.5 \cdot 3 \cdot 0.5 \sqrt{3} \cong 1.3 \text{ (dm}^2\text{)},$ (b) 2 dm^2 .

526. These polygons are similar since their sides are proportional and the angles of one of them are congruent to the angles of the other (each of them is equal to $\frac{180^\circ (n-2)}{n}$), and therefore their perimeters are related as $a : b$ and the areas as $a^2 : b^2$.

SECTION 2.35

527. The length of the circumference increases two times and the area of the circle, four times.

528. From the formula $a_4 = R \sqrt{2}$, or $R = \frac{a_4}{\sqrt{2}}$, we find that $R = \frac{6}{\sqrt{2}} \text{ cm} = 3 \sqrt{2} \text{ cm}; C = 2\pi R$, or $C = 2\pi \cdot 3 \sqrt{2} \text{ cm} = 6\pi \sqrt{2} \text{ cm} \approx 26.6 \text{ cm}.$

529. $a_6 = R$, $R = 5 \text{ cm}$, $C = 10\pi \text{ cm} \approx 31.4 \text{ cm}.$

530. The radius of the circumscribed circle is a median of the given triangle, and therefore, $C = 20\pi \text{ cm} \approx 62.8 \text{ cm}.$

531. We have $C = 2\pi \text{ dm}$, $P_6 = 6 \text{ dm}; C - P = (2\pi - 6) \text{ dm}$
 $0.28 \text{ dm} = 2.8 \text{ cm}.$

532. By $2\pi (R + 5) - 2\pi R = 10\pi$, i.e. by 31.4 (cm) approximately.

533. The radius of the circle is 2 cm , and therefore $S = 4\pi \text{ cm}^2 \approx 12.6 \text{ cm}^2.$

534. We assume that r is the radius of the inscribed circle, $r = 1.5 \tan 60^\circ \text{ cm} = 1.5 \sqrt{3} \text{ cm}.$ Then $S_{\text{cir}} = \pi (1.5 \sqrt{3})^2 = 6.75\pi \approx 21 \text{ cm}^2.$

535. $S_{\text{cir}} = \frac{\pi}{4} \text{ dm}^2$, $S_{\text{semicir}} = \frac{\pi}{2} \text{ dm}^2$ ($S_{\text{semicir}} > S_{\text{cir}}$).

536. $S_{\text{cir}} = \pi R^2$, $a_4 = R \sqrt{2}$, $S_{\text{sq}} = 2R^2$, $\pi R^2 - 2R^2 \approx 1.14R^2$ is the area of the trimmings, $1.14R^2$ of πR^2 is $\frac{1.14R^2 \cdot 100\%}{\pi R^2} \approx 36\%.$ Thus the trimmings constitute about 36% of the area of the circle.

537. (a) $r = d/2$, $S = \pi r^2 = \pi (d/2)^2 = \pi (d^2/4)$, (b) $C = 2\pi r$, $r = C/2\pi$, $\pi r = C/2$, $S = \pi r^2 = \pi r \cdot r = (C/2) \cdot r$, (c) $S = \frac{C}{2} \cdot r = \frac{C}{2} \cdot \frac{C}{2\pi} = \frac{C^2}{4} \cdot \frac{1}{\pi}.$

538. Solving the equation $\pi r^2 = 2\pi r$, we find that $r = 2 \text{ lin. units}.$

539. During one revolution the wheel will cover the distance equal to the length of the circumference of the wheel. Using the formula $C = \pi d$, we find that the distance is $(22/7) \cdot 0.7 \approx 2.2 \text{ m}.$ The distance covered in a minute is $2.2 \text{ m} \cdot 500 \approx 1.1 \text{ km}$, that covered in an hour is $1.1 \text{ km} \cdot 60 \approx 66 \text{ km}.$ Thus the speed of the car is $66 \text{ km/h}.$

540. The capacity of the larger pipe is equal to that of two small pipes, i.e. the sectional area of the larger pipe is equal to the sum

of the sectional areas of two smaller pipes. Solving the equation $\pi \frac{D^2}{4} = 2 \cdot \frac{\pi d^2}{4}$, we find that $D = d \sqrt{2} \approx 1.4d$.

541. Assume that a side of the square is a and the radius of the circle is r , then $S_{sq} = a^2$, $S_{c1r} = \pi r^2$. By the hypothesis, $a^2 = 1$ and $\pi r^2 = 1$, whence $a = \sqrt{1} = 1$ m, the perimeter of the square (P) is 4 m, $r = \sqrt{1/\pi}$, $C = 2\pi r = 2\pi \sqrt{\frac{1}{\pi}} = 2 \sqrt{\pi^2 \frac{1}{\pi}} = 2 \sqrt{\pi}$ m. Since $\sqrt{\pi} < 2$, it follows that $4 > 2 \sqrt{\pi}$, whence $P > C$.

SECTION 2.36

To solve problems given in this section, you must use the material presented in your school geometry book.

556. See Fig. 179.

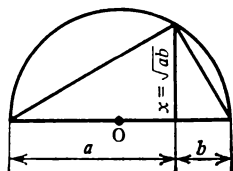


Fig. 179

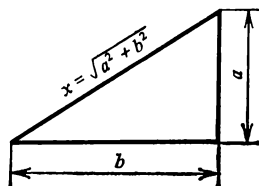


Fig. 180

557. (a) See Fig. 180, (b) see Fig. 181, (c) see Fig. 182.

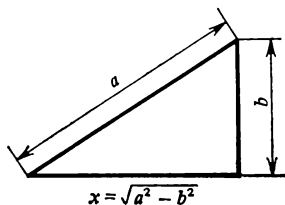


Fig. 181

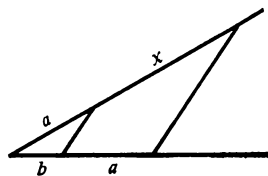


Fig. 182

Answers to Sample Examination Papers

Paper 1

$$2. 0.56(0.56 + 0.44)(0.56 - 0.44) \cdot \frac{25}{3} = 0.56 \cdot 1 \cdot 0.12 \cdot \frac{25}{3} = 0.56.$$

3. The smaller leg is 6 dm (in accordance with the property of a leg lying opposite an angle of 30°) and the larger leg is $12 \cdot \cos 30^\circ = 6\sqrt{3}$ dm, $S \approx 31$ dm²

Paper 2

2. $X \in [0.5 + \infty)$.

3. $h = 2 \cdot \sin 60^\circ = 2 \cdot \frac{\sqrt{3}}{2}$, $h = 1.7$ dm.

Paper 3

2. $(2.02 - 0.52)^2 = 1.5^2 = 2.25$.

3. Each leg is equal to $6 \cdot \sin 45^\circ = 3\sqrt{2} \approx 4.2$ (cm). The altitude is 3 cm.

Paper 4

2. $\frac{3}{2} + (2^3)^{\frac{1}{3}} + 1 = 4.5$.

3. $(\vec{AB} + \vec{BM}) + \vec{MN} = \vec{AM} + \vec{MN} = \vec{AN}$.

Paper 5

2. The inequality holds true only for those values of the variable x which satisfy at least one of the systems of simultaneous equations

$$\begin{cases} x > 0, \\ x - 1 > 0 \end{cases} \quad \text{or} \quad \begin{cases} x < 0, \\ x - 1 < 0. \end{cases}$$

The set of solutions of the given inequality is $(-\infty, 0) \cup (1, +\infty)$.

3. $(1/2)^2 + (\sqrt{2}/2)^2 + (\sqrt{3}/2)^2 = 1.5$.

Paper 6

2. $\frac{5(a-2)(a+0.6)}{3(a-2)} = \frac{5a+3}{3}$.

3. $r = 4$ cm, $C \approx 25$ cm; $S \approx 50$ cm².

Paper 7

2. If we denote the unknown number by x , then we get an equation $0.1x = 1.3 \cdot 12.8 + 13 \cdot 1.72$, or $0.1x = 1.3(12.8 + 10 \cdot 1.72)$, $0.1x = 39$, $x = 390$.

3. The hypotenuse of the triangle is unity and therefore $\sin \alpha = 0.6 \div 1 = 0.6$, $\sin \beta = 0.8$.

Paper 8

2. For $x = a + 2$ we have

$$(a+2)((a+2)-2a)+a^2=(a+2)(2-a)+a^2=4.$$

For $x = a - 2$ we have

$$(a-2)((a-2)-2a)+a^3=(a-2)(-a-2)+a^3 \\ =-(a-2)(a+2)+a^3=4.$$

3. The angle between the bisector and a side of the triangle is 30° . The obtuse angle between the bisectors is 120° .

Paper 9

2. According to the principal property of a proportion we have $10x \cdot 7 = 3.5 \cdot 0.4$, whence $x = \frac{3.5 \cdot 0.4}{10 \cdot 7} = 0.02$.

3. The altitude of the rhombus is h_p , $h_p = 5 \sin 60^\circ = \frac{5\sqrt{3}}{2}$,
 $S_p = 5 \cdot \frac{5\sqrt{3}}{2} \approx 22 \text{ (cm}^2\text{)}.$

Paper 10

2. $5 \cdot x^{0.6+0.2+1.2} = 5x^2 = 5 \cdot (1.2)^2 = 5 \cdot 1.44 = 7.2.$

3. $6 \text{ cm} \left(\text{from the formula } h = \sqrt{r^2 - \left(\frac{a}{2}\right)^2} \right).$

Paper 11

2. 0, 2.25, 5, 2.25, 0.

3. $\frac{1}{x} - 1 > 0$, or $\frac{1-x}{x} > 0$, whence (0, 1) is the set of solutions of the given inequality.

Paper 12

2. After transformations, we get an equation $4x^2 - 5x = 0$ and solve it:

$$x(4x - 5) = 0, \text{ whence } x = 0 \text{ or } x = 1.25;$$

the set of roots of the given equation is $\{0, 1.25\}$.

3. The centre of the circle is the point of intersection of the bisectors of the triangle, and the radius of the circle is the distance from the centre to any side of the triangle.

Paper 13

2. For example, $-2, -0.5, 1, 2.5 \dots$ From the formula $S_n = \frac{(2a_1 + (n-1)d)n}{2}$ we have $S_{15} = \frac{(2 \cdot (-2) + (15-1) \cdot 1.5) \cdot 15}{2} = 122.5.$

3. We have $a = d \cdot \sin 45^\circ = d \sqrt{2}/2$, and then $a = 5 \sqrt{2} \approx 7.1 \text{ cm}$; $S = (5 \sqrt{2})^2 = 50 \text{ cm}^2.$

Paper 14

2. We form the difference between the left-hand and right-hand sides of the inequality; i.e.

$$n^2 - 8(n - 2) = n^2 - 8n + 16 = (n - 4)^2.$$

For $n \neq 4$ this expression is positive and for $n = 4$ it is zero and, therefore, for any n we have $(n - 4)^2 \geq 0$; this means that $n^2 \geq 8(n - 2)$, and $n^2 = 8(n - 2)$ for $n = 4$.

3. The graph of the function is a straight line which cuts off a segment 3 units long from the x -axis and a segment 6 units long from the y -axis. The area of the triangle is 9 square units.

Paper 15

2. $2^{|x|} = 2^3$, $|x| = 3$, $x_1 = -3$, $x_2 = 3$.

3. The graph of the function is a parabola lying in the upper half-plane; (a) if $x = 2.5$, then $y \approx 3.1$, (b) if $y = 6$, then $x \approx 3.5$.

Paper 16

2. By the definition of a logarithm, $x - 3 = 10^2$, and hence $x = 103$.

3. $5 \cdot (-0.1)^3 = -0.005$ holds true, the point A belongs to the graph; $5 \cdot (-2)^3 = -50$ does not hold true, the point B does not belong to the graph.

Paper 17

2. $|2x| = 10$, $x_1 = -5$, $x_2 = 5$.

$$3. a = \left(3\frac{1}{3} - 1\frac{1}{2} \right) \div 11 = \left(3\frac{2}{6} - 1\frac{3}{6} \right) \div 11 = \left(2\frac{8}{6} - 1\frac{3}{6} \right) \div 11 = 1\frac{5}{6} \div 11 = \frac{11}{6} \div 11 = \frac{1}{6}.$$

Paper 18

2. $\{1\}$. The value $x = -2$ does not suit here since for this value of x the expression $\sqrt{x - 1}$ is meaningless.

3. See Fig. 187.

Paper 19

2. $\{-3, 7\}$.

$$3. \frac{(2^2)^{-6} \cdot (2^3)^{-4}}{2^{-22}} = \frac{2^{-12} \cdot 2^{-12}}{2^{-22}} = 2^{-24} \div 2^{-22} = 2^{(-24) - (-22)} = 2^{-2} = 1/4.$$

Paper 20

2.
$$\frac{\left(a^{\frac{1}{2}}\right)^2 - \left(b^{\frac{1}{2}}\right)^2}{\frac{1}{a^{\frac{1}{2}}} - \frac{1}{b^{\frac{1}{2}}}} - \frac{\left(a^{\frac{1}{2}}\right)^2 - \left(b^{\frac{1}{2}}\right)^2}{\frac{1}{a^{\frac{1}{2}}} + \frac{1}{b^{\frac{1}{2}}}} = \left(a^{\frac{1}{2}} + b^{\frac{1}{2}}\right) - \left(a^{\frac{1}{2}} - b^{\frac{1}{2}}\right) = 2b^{\frac{1}{2}} = 2\sqrt{b}.$$
3. $180^\circ (5-2) \div 5 = 108^\circ.$

Paper 21

2. $\log 5^2 - \log \left(3 \frac{1}{8} \cdot 8\right) = \log 25 - \log 25 = 0.$

3. The graph of the function $y = x$ is the bisector of the first and third quadrants. The graph of the function $y = x^2$ is a parabola lying in the upper half-plane.

Paper 22

1. See 475.
2. $x^2 - 100 > 0$, $(x - 10)(x + 10) > 0$, $(-\infty, -10) \cup (10, +\infty)$ is the set of solutions of the inequality.
3. This follows from the congruence of the triangles whose common side is the base of an isosceles triangle (ASA criterion).

Paper 23

1. See 475.
2. Since the value of $|x|$ is positive for all x except for $x = 0$, the domain of definition of the function is the set of all real numbers except for zero.
3. This follows from the congruence of triangles whose common side is the base of an isosceles triangle (SAS criterion).

Paper 24

2. $(a - 8)^2 - 4^2 = (a - 8 - 4)(a - 8 + 4) = (a - 12)(a - 4).$
3. The given inequality is equivalent to the inequality $(x - 3)(x + 3) < 0$, whence $x \in (-3, 3).$

Paper 25

2. $((3^2)^{0.25} - 2^1)((3^2)^{0.25} + 2^1) = (3^{0.5})^2 - 2^2 = 3 - 4 = -1.$
3. Yes, they are, since two angles of one triangle are congruent to two angles of the other.

Paper 26

3. $\log(2x + 10) = \log 2^4 + \log 3$, $\log(2x + 10) = \log(2^4 \cdot 3)$, $x + 10 = 48$, $x = 19.$

Paper 27

2. Since x^2 is positive for all x except for $x = 0$, the given inequality is equivalent to simultaneous inequalities

$$\begin{cases} x+1 > 0, \\ x \neq 0; \end{cases} \text{ hence } x \in (-1, 0) \cup (0, +\infty).$$

3. See 140.

Paper 28

2. $y = 0$ if $x = 0.5$, $y > 0$ if $x \in (0.5, +\infty)$, $y < 0$ if $x \in (-\infty, 0.5)$.

3. $b_1 = b_2 \div q = (-16) \div \frac{1}{2} = -32$, $b_3 = b_2 \cdot q = (-16) \cdot \frac{1}{2} = -8$.

Paper 29

$$2. \frac{\sqrt{x}-4}{(\sqrt{x})^2-4^2} = \frac{1}{\sqrt{x}+4}.$$

3. We add 2 to all the sides of the inequality, i.e.

$$5.4 + 2 \leq 5x \leq 7.6 + 2, \quad 7.4 \leq 5x \leq 9.6.$$

Dividing all the sides of the inequality by 5, we find that

$$1.48 \leq x \leq 1.92, \quad \text{or } x \in [1.48, 1.92].$$

Paper 30

2. We introduce the designation $x^2 = a$ and then $x^4 = a^2$. Making a substitution, we get a quadratic equation for a :

$$a^2 - 29a + 100 = 0, \quad a = 4 \text{ and } a = 25,$$

whence we have $x^2 = 4$ and $x^2 = 25$. The roots of the given equation are 2, -2, 5, -5.

Answers to Questions and Problems for Oral Examinations

1. (a) $(8 \cdot 2) \cdot 125 \cdot 250 = (8 \cdot 125) \cdot (2 \cdot 250) = 1000 \cdot 500 = 500\,000$,
(b) $(8 \cdot 8) \cdot 2.5 \cdot 12.5 = (12.5 \cdot 8) \cdot (2.5 \cdot 4 \cdot 2) = 100 \cdot 20 = 2000$.

$$2. (a) \quad 4.85 - 2\frac{4}{9} + 5\frac{3}{20} = (4.85 + 5.15) - 2\frac{4}{9} = 7\frac{5}{9},$$

(b) $(215 + 214) \cdot (215 - 214) = 429 \cdot 1 = 429$.

3. (a) $(3.7 + 6.3)^2 = 10^2 = 100$, (b) $(317 - 217)^2 = 100^2 = 10\,000$.

$$4. \quad 36.1 \cdot (2.78 + 7.22) = 26.1 \cdot (5.64 + 4.36) = 36.1 \cdot 10 = 26.1 \cdot 10 = 10 \cdot (36.1 - 26.1) = 100.$$

$$5. \quad 12.8 (12.8^2 - 7.2^2) = 448 \cdot 3.2 = 12.8 (12.8 - 7.2) (12.8 + 7.2) = 448 \cdot 3.2 = (4 \cdot 3.2) \cdot 5.6 \cdot 20 = 448 \cdot 3.2 = 3.2 (4 \cdot 5.6 \cdot 20 =$$

$$448) = 3.2 \quad (448 - 448) = 0.$$

$$6. \left(-2 \frac{1}{5}\right) \cdot \left(-1 \frac{4}{11}\right) \cdot \left(-\frac{1}{5}\right) \cdot (-5) + (1 - 7.52 \times (-8.47)) (0.4 - 0.4) = 3 + 0 = 3.$$

$$7. \frac{50}{13} \cdot \frac{26}{100} + \frac{24 \cdot 0.273}{24 \cdot 0.273} + \frac{78.3 \div 316}{78.3 \div 316} = 1 + 1 + 1 = 3.$$

8. We represent each of the given fractions as the difference of two fractions, and then $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) = 1 - \frac{1}{7} = \frac{6}{7}.$

$$9. (a) \sqrt{(3.13+3.12)(3.13-3.12)} = \sqrt{6.25 \cdot 0.01} = 2.5 \cdot 0.1 = 0.25;$$

$$(b) \sqrt{(13 \cdot 4)^2 + (13 \cdot 3)^2} = \sqrt{13^2 \cdot (4^2 + 3^2)} = 13 \cdot 5 = 65.$$

$$10. \sqrt{(12.1+8.1)(12.1-8.1) \div (0.25 \cdot (0.25+19.95))} = 4.$$

$$11. 0. \text{ Hint. First calculate the expression } 4a - 8b.$$

$$12. \frac{(a(a^2+2))^2}{(a^2+2)^2} = a^2 = \left(1 \frac{1}{3}\right)^2 = 1 \frac{7}{9}.$$

$$13. \sqrt{a^2(a-b)-b^2(a-b)} = \sqrt{(a-b)^2(a+b)} = |a-b| \sqrt{a+b} = |0.78-0.22| \cdot \sqrt{0.78+0.22} = 0.56 \cdot 1 = 0.56.$$

$$14. \sqrt{(5a+4b)^2} = |5a+4b| = |5 \cdot 0.4 + 4(-0.5)| = 0.$$

$$15. x = \frac{a-r}{b}.$$

$$16. \frac{a+b}{2}, \frac{a-b}{2}.$$

$$17. \frac{cp}{p+q}, \frac{cq}{p+q}.$$

$$18. \text{ The least value of } x + y \text{ is } -44 \text{ and the greatest value is } 5.$$

$$19. (a) \text{ For example, } (1.7 + 1.8) \div 2 = 1.75, (d) \text{ for example, } ((-0.76) + (-0.7)) \div 2 = -0.73.$$

20. The first worker can do a half of his job in an hour and the second worker a third of his job. Both workers, working together, can do $(1/2 + 1/3)$, or $5/6$ of their job in an hour. The two workers can complete the job in $1 \div \frac{5}{6} = 1.2$, i.e. in 1 h 12 min.

21. Assume that the passenger has to cover another a km. Then, while looking out of the window, he travelled twice the distance, i.e. $2a$ km. Half the journey is $2a + a = 3a$ km and the whole distance is $6a$ km. Consequently, the remainder of the journey is $a \div 6a = 1/6$.

22. $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ is divisible by 11.

$$23. S = 5.75 \cdot 0.28 \approx 1.61 \text{ dm}^2, \text{ or approximately } 1.6 \text{ dm}^2.$$

24. (a) $17 \div 2 = 8.5$, (b) $36 \div 4 = 9$, (c) $24 \div 8 = 3$,
 (d) $48 \cdot \frac{3}{4} = 36$, (e) $1/2$ of 40% is 20% .

25. (a) 27% of 39 is 39% of 27 since $\frac{39 \cdot 27}{100} = \frac{27 \cdot 39}{100}$.

26. $0.25x = 0.25$, $x = 1$.

27. On the first day 40% of the goods were sold, on the second day, 25% of 60% , i.e. 15% of the available goods. It remains to sell $100\% - (40\% + 15\%) = 45\%$ of the available goods.

28. If the original length of the side of the square is a , then the increased side is $1.5a$. The original area is a^2 and the increased area is $(1.5a)^2 = 2.25a^2$, i.e. the area increases 2.25 times or by 125% ($225\% - 100\% = 125\%$).

29. Assume that a and b are the sides of the rectangle. The original area is ab and the new one is $1.2a \cdot 0.8b = 0.96ab$. The area decreases by 4% .

30. A year later the deposited sum will be $1.03a$ roubles. Two years later it will be $1.03a + 0.3$ of $1.03a = 1.03(1 + 0.3)a = 1.03^2a$ roubles.

31. (a) $(0.2)^5 \cdot 5^5 \cdot 5 = (0.2 \cdot 5)^5 \cdot 5 = 1^5 \cdot 5 = 5$, (b) $0.25^4 \cdot 8^3 = 0.25^3 \cdot 0.25 \cdot 8^3 = (0.25 \cdot 8)^3 \cdot 0.25 = 2^3 \cdot 0.25 = 2$, (c) $2.5^5 \cdot 0.4^5 \cdot 0.4 \times 8^4 \cdot 0.125^4 = (2.5 \cdot 0.4)^5 \cdot 0.4 \cdot (8 \cdot 0.125)^4 = 1^5 \cdot 0.4 \cdot 1^4 = 0.4$,

(d) $(-1)^{(2n-1)+(1-n)+(n+2)} = (-1)^{2n+2} = 1$ ($2n+2$ is an even number), (e) $1 + \frac{1}{2} + \frac{1}{4} = 1\frac{3}{4}$, (f) $\frac{1}{2} + \frac{1}{(-2)^1} = 0$.

32. (a) $-3^2 = -9$, $(-3)^2 = 9$, $(-3)^2 > -3^2$, (b) $0.5^6 > 0.5^9$, (c) $119 \cdot 117 = (118 + 1)(118 - 1) = 118^2 - 1^2 < 118^2$, (d) $115 \times 119 = (117 - 2)(117 + 2) = 117^2 - 2^2 < 117^2$.

33. (a) $2 = \sqrt[3]{2^3} = \sqrt[3]{4}$, $5 = \sqrt[3]{5^3} = \sqrt[3]{25}$, (b) $2 = \sqrt[3]{2^3} = \sqrt[3]{8}$, $5 = \sqrt[3]{5^3} = \sqrt[3]{125}$.

34. (a) $9 < 10 < 16$, $3 < \sqrt[3]{10} < 4$, (b) $2 < \sqrt[3]{10} < 3$, (c) $2^4 < 20 < 3^4$, $2 < \sqrt[4]{20} < 3$, (d) $5^3 < 130 < 6^3$, $5 < \sqrt[3]{130} < 6$.

35. (a) $-a$, (b) $3-a$, (c) $|x-2| + |x-4|$, for $2 < x < 4$ this expression is equal to $(x-2) + (4-x) = 2$.

36. (a) $5|x|$, (b) here $x < 0$, and therefore $\sqrt{-36x^3} = \sqrt{36x^2(-x)} = -6x\sqrt{-x}$.

37. (a) $\sqrt{7b^2}$, (b) $-\sqrt{6a^2}$.

38. (a) $\sqrt[3]{3} = \sqrt[6]{3^2} = \sqrt[6]{9}$, $\sqrt[3]{2} = \sqrt[6]{2^2} = \sqrt[6]{8}$, whence $\sqrt[3]{3} > \sqrt[3]{2}$, (b) $2\sqrt[3]{4} = \sqrt[3]{8 \cdot 4} = \sqrt[3]{32}$, $3\sqrt[3]{1.5} = \sqrt[3]{27 \cdot 1.5} = \sqrt[3]{40.5}$, whence $3\sqrt[3]{1.5} > 2\sqrt[3]{4}$.

39. (a) $\sqrt{3+2\sqrt{2}} = \sqrt{(1+\sqrt{2})^2} = 1 + \sqrt{2}$, (b) $\sqrt{7-4\sqrt{3}} = \sqrt{(2-\sqrt{3})^2} = |2-\sqrt{3}| = 2-\sqrt{3}$.

$$40. (a) \sqrt[3]{3^2 - (\sqrt{5})^2} = 2, (b) \sqrt[3]{5^2 - (2\sqrt{6})^2} = 1, (c) 4\sqrt[3]{6} \approx 9.8.$$

$$41. (a) \frac{3}{\sqrt[3]{3}} = \frac{(\sqrt[3]{3})^2}{\sqrt[3]{3}} = \sqrt[3]{3}, (b) \frac{a^2 - (\sqrt{5})^2}{\sqrt{5} - a} =$$

$$\frac{(a - \sqrt{5})(a + \sqrt{5})}{-(a - \sqrt{5})} = -(a + \sqrt{5}); (c) \frac{\left(\frac{1}{x^2}\right)^2 - \left(\frac{1}{a^2}\right)^2}{\frac{1}{x^2} - \frac{1}{a^2}} =$$

$$x^{\frac{1}{2}} + a^{\frac{1}{2}}, (d) \frac{\left(a^{\frac{1}{4}}\right)^2 - \left(b^{\frac{1}{4}}\right)^2}{a^{\frac{1}{4}} - b^{\frac{1}{4}}} = a^{\frac{1}{4}} + b^{\frac{1}{4}},$$

$$(e) \frac{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}}\right)^2}{\left(\frac{1}{x^2}\right)^2 - \left(\frac{1}{y^2}\right)^2} = \frac{x^{\frac{1}{2}} + y^{\frac{1}{2}}}{\frac{1}{x^2} - \frac{1}{y^2}}.$$

$$42. (a) y = 3x^{\frac{1}{5} + 0.6 + 1.2} = 3x^2 = 3 \cdot 10^2 = 300, (b) \sqrt[3]{\sqrt[4]{2^3} \cdot \sqrt[4]{2^2}} = \sqrt[3]{2^{\frac{3}{4} + \frac{2}{4}}} = \sqrt[3]{2^{\frac{5}{4}}} = 2^{\frac{5}{12}} = 2^{\frac{3}{6} + \frac{2}{12}} = 2^{\frac{3}{6} + \frac{1}{6}} = 2^{\frac{4}{6}} = 2^{\frac{2}{3}} = 2.$$

$$43. (a) \{-3, 3\}, (b) (-3, 3), (c) (-\infty, -3) \cup (3, +\infty), (d) \{1, 5\}, (e) (-2, 4), (f) [-7, 3], (g) (-\infty, -2) \cup [6, +\infty).$$

44. (a) $(-\infty, 1) \cup (1, +\infty)$. Sometimes a student says that x is any number, forgetting that the denominator cannot be zero, (b) $(-\infty, -1.5) \cup (-1.5, +\infty)$, (c) $|x| = 3$, hence $x_1 = 3$ and $x_2 = -3$, (d) $\emptyset$, (e) $|x - 2| = 5$, $x_1 = 7$ and $x_2 = -3$, (f) $|x| = x$. If $x \geq 0$, then $x = x$, if $x < 0$, then $-x = x$; the solution of these equations is the set of all numbers of the ray $[0, +\infty)$, (g) $|y| = -y$. The solution of this equation is a ray $(-\infty, 0]$, (h) $\emptyset$. If we cancel x , we get an extraneous root $x = 0$, (i) $x = 5$. If we cancel $x - 1$, then the second root obtained, $x = 1$, is extraneous, (j) $x = 4$. Cancelling $x + 2$ does not result in an extraneous root, (k) $2 \leq x < 3$, (l) $p + 0.2$, where $p \in \mathbb{Z}$.

45. (a) $\{0\}$, (b) $\{1\}$, (c) $\{0, 2\}$, (d) $\{-1\}$. The value $x = 1$ does not suit the equation since the expression $(1 - 1)^0$ is meaningless, (e) $x^2 + 7 \neq 0$, $|x - 3| = 0$, $x = 3$, (f) $x = x^2$ or $-x = x^2$ the solution of these equations is $\{-1, 0, 1\}$, (g) we write the equations as $x + \frac{1}{x} = 2 + \frac{1}{2}$; it is easy to see that $x = 2$ or $x = 1/2$,

$$(h) \sqrt{x} + \frac{1}{\sqrt{x}} = 3 + \frac{1}{3}, \sqrt{x} = 3 \text{ or } \sqrt{x} = 1/3, \text{ hence } x = 9 \text{ or}$$

$x = 1/9$, (i) here the sum of the first and the second factor and the free term is zero ($1921 - 3902 + 1981 = 0$), therefore one of the roots $x_1 = 1$, $x_1 \cdot x_2 = \frac{1981}{1921}$ (according to the Vieta theorem), and

the other root $x_2 = \frac{1981}{1921}$, (j) $x_1 = -1$, $x_1 \cdot x_2 = -\frac{1979}{1917}$, $x_2 = -\frac{1979}{1917}$.

46. (a) $\{(0, 0)\}$. The sum of two nonnegative numbers can be zero only when each of the terms is zero, (b) $\{(3, 4)\}$, (c) since $(x-5)^2 \geq 0$, $(3x+5y)^2 \geq 0$, $|2x+y-z| \geq 0$, the given equation has a solution only for $x-5=0$, $2x+5y=0$ and $2x+y-z=0$; hence we get $x=5$, $y=-2$, $z=8$, (d) we factor the left-hand side of the equation, i.e. $x^2(x+1) + (x+1)=0$, $(x+1)(x^2+1)=0$, and get $x=-1$, (e) $x(x^2-1)=0$, $x(x-1)(x+1)=0$, whence we get $x_1=0$, $x_2=-1$ and $x_3=1$.

47. $(2, 0)$, $(-3, 0)$, $(0, -6)$. *Hint.* The ordinates of the points of intersection of the parabola and the x -axis are zero, the abscissa of the point of intersection of the parabola and the y -axis is zero.

48. $(5, 0)$, $(1, 0)$, $(-0.5, 0)$, $(-2, 0)$, $(0, 10)$.

49. (a) $a = \frac{2S}{h}$, (b) $\sin \alpha = \frac{2S}{a^2}$, (c) $a = \sqrt{S/h}$, (d) $a_1 + a_n =$

$2S \div n$, $a_n = 2S \div n - a_1$, (e) $R = \sqrt{S/\pi}$, (f) $R = \sqrt{\frac{3V}{\pi h}}$,

(g) $v_0 = S - gt$, $t = \frac{S-v_0}{g}$, (h) $m = \frac{n}{c(T-t)}$, $c = \frac{n}{m(T-t)}$

$T = n \div mc + t$, $t = T - n \div mc$.

50. (a) $(-\infty, 0) \cup (0, \infty)$, (b) $\emptyset$, (c) $(-\infty, \infty)$, (d) $|x| > 4$; *Answer:* $(-\infty, -4) \cup (4, \infty)$, (e) $(-2, 2)$, (f) the solution of the inequality reduces to a solution of the simultaneous equations

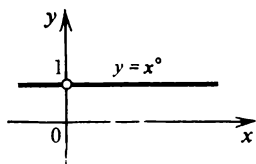


Fig. 183

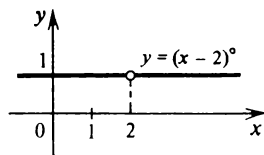


Fig. 184

$\begin{cases} (x-2)^2 \geq 0, \\ x > -1. \end{cases}$ *Answer:* $(-1, 2) \cup (2, \infty)$, (g) $(-\infty, 2)$, (h) $\log 0.1 < 0$, and therefore $2x - 1 > 0$, $x > 0.5$. *Answer:* $(0.5, \infty)$, (i) $[0, \infty)$, (j) $(-\infty, 0)$.

51. (a) $2x = y + 3$. Substituting y for x and x for y , we get $2y = x + 3$; hence the inverse of the given function is $y = 0.5x + 1.5$ (the graphs of the given function and its inverse are symmetric about the bisector of the first and the third quadrant), (b) the inverse of the given function is $y = -0.5x + 2$.

52. (a) A straight line $y = 1$ from which the point $(0, 1)$ is excluded, since the expression 0^0 is meaningless (Fig. 183), (b) see Fig. 184, (c) see Fig. 185, (d) Fig. 186, (e) Fig. 187, (f) if $x > 0$, then $y = 1$, if $x < 0$, then $y = -1$ (Fig. 188), (g) if $x \geq 0$, then $y = x^2$, if $x < 0$, then $y = -x^2$ (Fig. 189), (h) since $|y| \geq 0$,

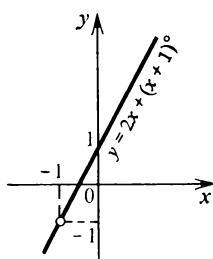


Fig. 185

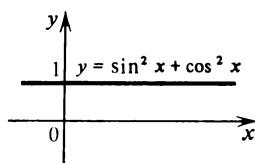


Fig. 186

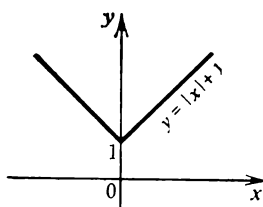


Fig. 187

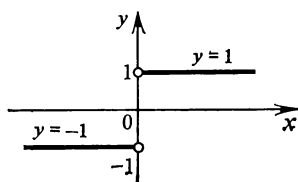


Fig. 188

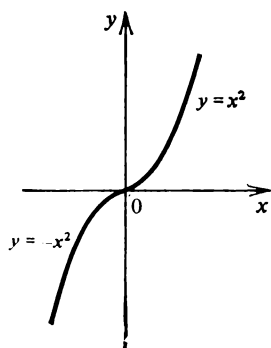


Fig. 189

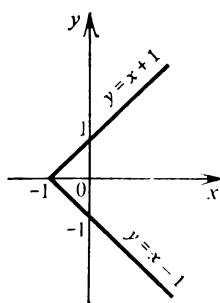


Fig. 190

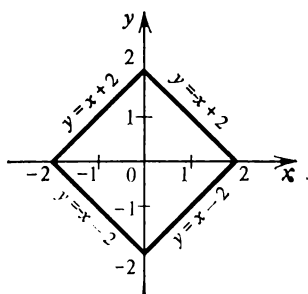


Fig. 191

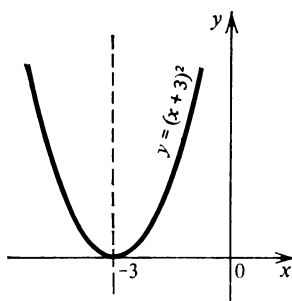


Fig. 192

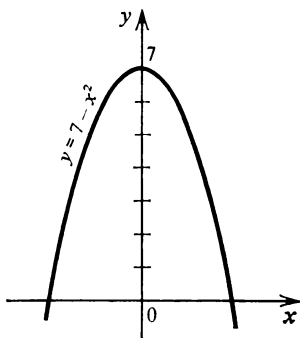


Fig. 193

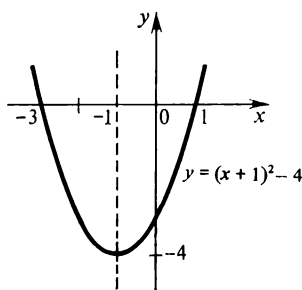


Fig. 194

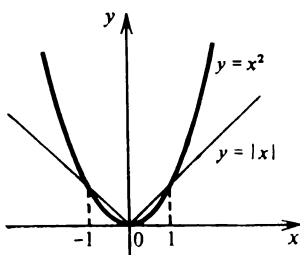


Fig. 195

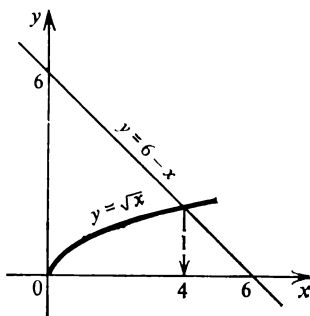


Fig. 196

we have $x + 1 \geq 0$, or $x \geq -1$. Assigning the values $x \geq -1$, we construct the graph of the function $y = x + 1$ and then the ray $y = -x - 1$, which is symmetric with respect to the graph obtained (Fig. 190).

53. This quadrilateral is a square (Fig. 191).

54. $a = -0.5$, $y = -0.5x + 4$.

55. (a) $y = x^2 - 3$, (b) $y = (x - 4)^2$, (c) $y = (x - 4)^2 - 3$.

56. (a) The least value of y is zero (Fig. 192), (b) the greatest value of y is 7 (Fig. 193), (c) $y = (x^2 + 2x + 1) - 1 - 3 = (x + 1)^2 - 4$. The least value of y is -4 (Fig. 194).

57. (a) We construct a parabola $y = x^2$ and a straight line $y = x + 2$ in the same system of coordinates; the abscissas of their intersection points $x_1 = -1$, and $x_2 = 2$ are the roots of the given equation, (b) $x_1 = -1$, $x_2 = 0$ and $x_3 = 1$ (Fig. 195), (c) $x = 4$ (Fig. 196).

58. (a) The parabola $y = x^2$ and the straight line $y = 2x$ meet at two points. The coordinates of these points $(0, 0)$ and $(2, 4)$ are the solutions of the simultaneous equations, (b) see Fig. 197, (c) $\{(0, 0), (1, 0.5)\}$.

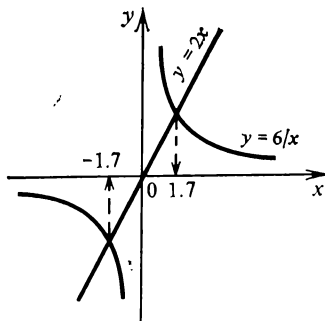


Fig. 197

59. (a) $\{(2, 1), (-2, 3)\}$. *Hint.* The solution of the given simultaneous equations reduces to the solution of two system of simultaneous equations

$$\begin{cases} x = 2, \\ x + 2y = 4 \end{cases} \quad \text{or} \quad \begin{cases} x = -2, \\ x + 2y = 4, \end{cases}$$

(b) $\{(8, 5)\}$. *Hint.* The solution of these simultaneous equations can be simplified by putting x before the brackets in the first equation and then replac-

ing $2x - 3y$ by the number 1; then we find that $x = 8$, (c) $\begin{cases} y = 0, \\ 2x + 3y = 13 \end{cases}$ or $\begin{cases} x = 2, \\ 2x + 3y = 13. \end{cases}$ Answer: $\{(6.5, 0),$

$(2, 3)\}$, (d) subtracting term by term, we get $x - y = -1$. We substitute this value into the first equation of the system, i.e. $(-1, +2)^2 + y = 8$, whence $y = 7$, $x = 6$. Answer: $\{(6, 7)\}$.

60. (a) It is the set of points of the graphs $y = x$ and $y = -x^2$, (b) the set of points of the straight line $x = 2$ and the point $(0, 0)$, (c) the set of points of the straight line $y = -x + 1$ and the point $(0, 0)$.

61. 1. The common difference of the progression $d = b - a$, the third term $a_3 = b + (b - a) = 2b - a$, and the fourth term $a_4 = 3b - 2a$.

2. The common ratio of the progression $q = b^2 : b = b$, the third term is b^3 and the fourth term is b^4 . 3. $d = a_3 - a_2 = n - m$; $a_1 = a_2 - d = m - (n - m) = 2m - n$.

62. 1. The common ratio of the progression $q = \frac{b_3}{b_2} = \frac{n}{m}$;

$$b_1 = b_2 : q = m : \frac{n}{m} = \frac{m^2}{n}.$$

2. The sum of the terms of this arithmetic progression S , in which the first term is $a_1 = 28$ and $d = -2$, is zero. Using the formula $S = \frac{(2a_1 + d(n-1))n}{2}$, we find that $\frac{(2 \cdot 28 - 2(n-1)) \cdot n}{2} = 0$; hence $(28 - (n-1))n = 0$, $n = 29$ or $n = 0$ (does not suit); $x = 28 + (-2) \cdot (29 - 1) = -28$. We can solve this

problem without resorting to the formula for the sum. There are 15 terms from 28 to 0 inclusive and 14 terms from -2 to -28 . There are 29 terms all in all.

63. (a) The difference $d = (1 + 3\sqrt{3}) - (1 + \sqrt{3}) = 2\sqrt{3}$;
 $a_{15} = 1 + \sqrt{3} + 2\sqrt{3} \cdot (15-1) = 1 + \sqrt{3} + 28\sqrt{3} = 1 + 29\sqrt{3}$.

Remark. Sometimes students erroneously replace the last expression by $30\sqrt{3}$. It should be remembered that we must first find $\sqrt{3} \approx 1.73$, then multiply this number by 29 and only then add 1, (b) $d = -3$, $a_{15} = \sqrt{2} - 1 + (-3) \cdot 14 = \sqrt{2} - 43$.

64. By the property of a geometric progression, $b_n^2 = b_{n-1}b_{n+1}$, where $n > 1$, and therefore $b_2^2 = b_1b_3$, $b_2^2 = \sqrt{2} \cdot 3\sqrt{2} = 6$; hence $b_2 = \sqrt{6}$. *Answer:* $\sqrt{2}$, $\sqrt{6}$, $3\sqrt{2}$.

65. (a) $b_{n-1} = 5 \cdot 2^{(n-1)-1} = 5 \cdot 2^{n-2}$, $b_{n+1} = 5 \cdot 2^n$, (b) $b_{n-1} = 5 \cdot \left(\frac{1}{2}\right)^{2(n-1)-1} = 5 \cdot \left(\frac{1}{2}\right)^{2n-3}$, $b_{n+1} = 5 \cdot \left(\frac{1}{2}\right)^{2(n+1)-1} = 5 \cdot \left(\frac{1}{2}\right)^{2n+1}$.

66. (a) $10^{x+1} = 10^x \cdot 10 = 3 \cdot 10 = 30$, (b) $10^{-x} = 1/10^x = 1/3$, (c) $10^{x-1} = 10^x \cdot 10^{-1} = 3 \cdot 10^{-1} = 0.3$, (d) $10^{2x} = (10^x)^2 = 3^2 = 9$.

67. (a) $-x > 0$, $x < 0$; $(-\infty, 0)$, (b) $x \leq 0$; $(-\infty, 0]$, (c) $(-\infty, 1)$, (d) $(-\infty, 1]$, (e) $(-\infty, 0) \cup (0, +\infty)$, (f) $(-\infty, \infty)$, (g) $x > 0$ and $x \neq 1$. Since $\log 1 = 0$, we have the answer: $(0, 1) \cup (1, +\infty)$, (h) $(0, +\infty)$.

68. (a) $\log 0.1 < 0$, and therefore the given expression is meaningless, (b) $\sqrt{0.1} > 0$, and therefore the given expression has sense, (c) $\log \cos 90^\circ = \log 1 = 0$.

69. (a) From the formula $10^{\log a} = a$ we have $10^{\log 8} = 8$, (b) 5, (c) $10^{\log 2 + \log 3} = 10^{\log (2 \cdot 3)} = 6$, or $10^{\log 2 + \log 3} = 10^{\log 2} \cdot 10^{\log 3} = 6$, (d) $10^{\log 2 + 2} = 10^{\log 2} \cdot 10^2 = 200$, (e) 102, (f) $10^{\log 3} \cdot 10^{-1} = 0.3$, (g) 3.1, (h) $1 \div 10^{\log 2} = 0.5$.

70. $\log 0.01 = -2$, $\log 1 = 0$, $\log 100 = 2$, $\log 10\,000 = 4$. The numbers $-2, 0, 2, 4$, form an arithmetic progression with a common difference $d = 2$. 71. (a) $\log x = 1$ or $\log x = -1$, whence $x_1 = 10$ or $x_2 = 0.1$, (b) $x^2 = 10$, $x_1 = \sqrt{10}$, $x_2 = -\sqrt{10}$, (c) $(\log x^2)^2 = 1$, whence $\log x^2 = 1$, or $\log x^2 = -1$, $x^2 = 10$ or $x^2 = 0.1$, whence $x_1 = -\sqrt{10}$, $x_2 = \sqrt{10}$, $x_3 = -\sqrt{0.1}$, $x_4 = \sqrt{0.1}$ (d) $10^{\log x^2} = x^2$, whence $2x^2 = 8$, $x^2 = 4$, $x_1 = 2$, $x_2 = -2$.

72. (a) See Fig. 198, (b) see Fig. 199.

73. (a) $\log (10a) = \log 10 + \log a = 1 + 0.1 = 1.1$, (b) $0.1 - 1 = -0.9$, (c) $1 - 0.1 = 0.9$, (d) $2 \log a = 2 \cdot 0.1 = 0.2$, (e) $0.1^2 = 0.01$.

74. By the property of the distances $|AC| \leq |AB| + |BC|$ and $|AC| \geq |AB| - |BC|$; consequently, $4 \text{ km} \leq |AC| \leq 16 \text{ km}$.

76. 0.5a.

77. Four distinct points, arbitrarily located (even if three of them and even four, lie along the same straight line), taken pairwise, define six different segments.

78. (a) All points which belong to the circle (O, r) , (b) the circle (O, r) without the points of the circumference, (c) the circle (O, r) without its centre.

79. 10 cm.

81. Yes, it can. If one of the adjacent angles is α , then the

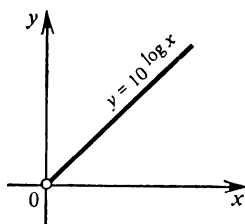


Fig. 198

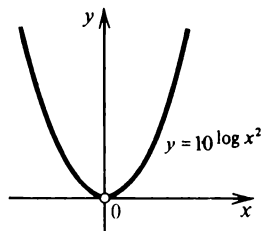


Fig. 199

other angle is $180^\circ - \alpha$. Solving the equation $(180^\circ - \alpha) - \alpha = 90^\circ$ for α , we find that one of the adjacent angles is 45° and the other is 135° .

82. Yes, it can, if the vertical angles are right angles.

83. 32° if the angle ABC is the intersection of the given angles ABD and CBD ; 112° if the angle ABC is their union.

84. Taking the point A as the centre of the circle, draw a circle which would intersect the given one. The chord which connects the points of intersection of those circles is the required one.

85. If the point C lies between A and B , then $-2 < x_C < 5$; if C does not lie between A and B , then $x_C \leq -2$ or $x_C \geq 5$; $|AB| = 7$.

86. According to the theorem on the inequality of triangles (Fig. 200) we have

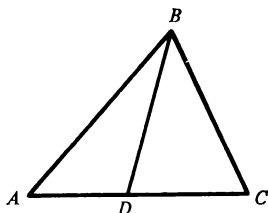


Fig. 200

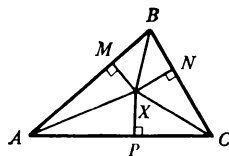


Fig. 201

$$|BD| < |AB| + |AD| \quad \text{and} \quad |BD| < |BC| + |DC|.$$

Adding these inequalities term by term, we obtain

$$|BD| + |BD| < |AB| + |AD| + |BC| + |DC|$$

or

$$2|BD| < |AB| + |BC| + |AC|; \quad |BD| < \frac{1}{2}(|AB| + |BC| + |AC|).$$

88. (a) One, (b) one, (c) three.

90. The sides of the quadrilateral obtained are pairwise parallel.

91. *Hint.* Draw a diagonal AC and then $\widehat{BAC} = \widehat{BCA}$ and, hence, $\widehat{DAC} = \widehat{DCA}$.

92. 1. If a perpendicular and an oblique line are drawn from the same point to a straight line, then the perpendicular is shorter than the oblique line.

2. No, it cannot.

93. Since $|XA| > |XM|$, $|XB| > |XN|$, $|XC| > |XP|$, it follows that $|XA| + |XB| + |XC| > |XM| + |XN| > XP$ (Fig. 201).

94. The point A_1 is the projection of the point A onto the straight line BC , the line segment A_1C is the projection of the segment AC onto that straight line, and so on (Fig. 202).

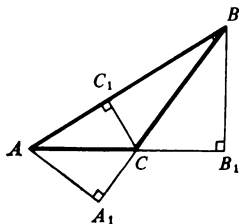


Fig. 202

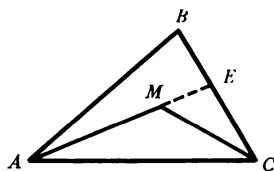


Fig. 203

95. $|MO| < |OB| + |MB|$ (inequality of triangles) or $|MA| + r < r + |MB|$, and therefore $|MA| < |MB|$.

96. We must draw a straight line through the point A at right angles to the radius OA . Since only one straight line can be drawn through the point A at right angles to the radius OA , this tangent is unique.

97. *Hint.* $\triangle ABK \cong \triangle DEL$ (a side and two adjacent angles are congruent); consequently, $[BK] \cong [EL]$.

98. No, it is not. It is true only for two congruent triangles.

99. If each of the angles of the triangle were larger than 60° , then their sum would be larger than 180° , and if they were equal to 60° each, then the triangle would not be scalene.

100. 1. An exterior angle of a triangle is equal to the sum of two angles of the triangle which are not adjacent to it, and the sum of two positive numbers is larger than each of the terms.

2. Yes, it can, if the exterior angle is adjacent to an interior angle of the triangle.

3. We extend AM (Fig. 203) until it intersects the side BC at a point E , and then $\widehat{AMC} > \widehat{AEC}$, and $\widehat{AEC} > \widehat{ABC}$ (by the property of an exterior angle of a triangle); hence $\widehat{AMC} > \widehat{ABC}$.

101. No, they cannot; in that case the sum of two angles from which the bisectors are drawn would be 180° .

102. *Hint.* Construct a triangle symmetric with respect to the given triangle about the straight line which passes through the larger leg. The union of these two triangles is an equilateral triangle.

103. 2 cm and 6 cm. *Hint.* Use twice the following property: the leg of a right triangle which lies opposite the angle of 30° is equal to a half of the hypotenuse.

104. If a right or an obtuse angle were opposite the smallest side of the triangle, then obtuse angles would be opposite the larger sides (a larger angle of a triangle lies opposite a larger side) and then the sum of the angles of the triangle would be larger than 180° .

105. The base angle of the triangle is $(180^\circ - \alpha) \div 2 = 90^\circ - \alpha/2$. This angle lies opposite the larger side of the triangle, and therefore we have an inequality $90^\circ - \frac{\alpha}{2} > \alpha$; hence $90^\circ > \frac{3}{2}\alpha$, i.e. $\alpha < 60^\circ$; in addition $\alpha > 0^\circ$; thus $\alpha \in (0^\circ, 60^\circ)$.

106. No, there is not. For instance, in a right triangle with an angle of 30° the leg lying opposite that angle is half the hypotenuse whereas the angle of 30° is equal to one third of the angle of 90° .

107. $a \neq b$. *Hint.* In the resulting rectangle the adjacent sides are equal to $a/2$ and $b/2$ (by the Thales theorem).

108. The point E_1 is the projection of the point E onto the altitude BD (Fig. 204), D is the projection of the point A onto

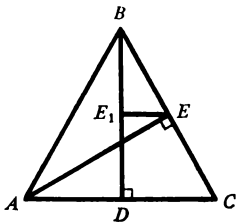


Fig. 204

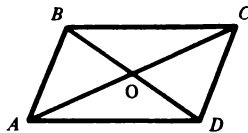


Fig. 205

this altitude. The projection of the altitude AE onto BD is E_1D . By the Thales theorem $|E_1D| = 6 \div 2 = 3$ (cm).

109. Yes, it can. The sum of the angles of a convex n -gon is equal to $2d(n - 2)$. Solving the equation $2d(n - 2) = 100d$ for n , we find that such a polygon has 52 sides.

110. If we assume that the diagonals AC and BD of the parallelogram $ABCD$ are congruent (Fig. 205), then the triangle ABD will be congruent to the triangle ADC (SSS criterion); consequently, $\angle BAD \cong \angle ADC$, and this is impossible since one of them is an acute angle and the other is an obtuse angle.

111. This angle is 30° if the altitudes are drawn from the vertex of the obtuse angle, and 150° if the altitudes are drawn from the vertex of the acute angle.

112. $6a$.

113. No, there is not.

114. No, it is not.

115. (a) Yes, there is, (b) no, there is not, (c) no, there is not.

116. No, it cannot.

117. (a) Yes, it can if the larger base of the trapezoid is equal to one of its nonparallel sides, (b) yes, it can, if the smaller base of the trapezoid is equal to one of its nonparallel sides.

118. Two opposite sides are parallel. The sum of all the interior angles is 360° . The sum of the angles adjacent to a lateral side is 180° .

119. (a) Only an isosceles trapezoid has an axis of symmetry.
(b) No, it has not.

120. No, it cannot, since otherwise the diagonals would be bisected at the point of their intersection (according to the Thales theorem), and this is valid only for a parallelogram.

121. Yes, we can. If the segments are not congruent, then they have two centres of homothety, an external centre and an internal centre; now if they are congruent, then they have only one internal centre.

122. The triangles AOB and COD , COB and AOD are homothetic (Fig. 205), the ratio of similitude $k = -1$.

123. At the point of intersection of the altitudes of the triangle; $k = 2$.

124. $\triangle ABC \sim \triangle ABD$, $\triangle ABC \sim \triangle BDC$, $\triangle ABD \sim \triangle BDC$. These three pairs of triangles are similar (two angles).

125. $\triangle ABC \sim \triangle ABD$ (two angles are equal); hence $\frac{|AB|}{|AD|} = \frac{|AC|}{|AB|} = \frac{|BC|}{|BD|}$.

126. *Hint.* $\angle E \cong \angle C$ (being inscribed angles subtended by the same arc).

127. The rhombi are similar if they have equal respective angles.

128. The hypotenuse is 10 cm. If we circumscribe a circle about the triangle, then the median will be equal to the radius of the circumscribed circle; the length of the median is 5 cm.

129. An altitude is the mean proportional between the projections of the legs onto the hypotenuse; the length of one of the projections is 8 cm and that of the other is 2 cm; the diameter is 10 cm.

130. The diameter of the circumscribed circle coincides with the hypotenuse, it is equal to 6 cm; $S = 6 \text{ cm}^2$.

131. The altitude of the triangle is 9 cm. The actual area is 108 cm^2 ($S = \frac{1}{2} \cdot 24 \cdot 9$). The area calculated erroneously is 180 cm^2 ($S_1 = \frac{1}{2} \cdot 24 \cdot 15$), and this is larger than the actual area by 72 cm^2 , which is approximately 67%.

132. *Hint.* We must divide the base of the triangle into equal parts and connect each point of division with the vertex of the triangle by a segment.

133. 0.5 dm^2 .

134. The area of the triangle ABD is $S/2$, that of the triangle AMD is $1/2$ of the area of the triangle ABD , i.e. $S/4$.

135. The cut-off triangle and the given one are similar, the ratio of similitude $k = 1/4$. The ratio of their areas is equal to the square of the ratio of similitude, i.e. $k^2 = 1/16$, the ratio of the area of the trapezoid to that of the cut-off triangle is $15/1$.

136. Yes, there is. Both the interior and the central angles of a square are equal to 90° .

138. The arcs of this circle are 60° , 120° and 180° . The first chord subtends $1/6$ of the circle and is a side of an inscribed regular hexagon; $a_6 = R$, i.e. the length of the first chord is 1 dm. The length of the second chord $a_3 = R\sqrt{3}$, it is equal to $\sqrt{3} \text{ dm} \approx 1.73 \text{ dm}$, the third chord is the diameter of the circle, it is 2 dm.

139. The hypotenuse is 5 cm. The length of the circle constructed on the hypotenuse is 5π cm, and those of the circles constructed on the legs are 3π cm and 4π cm. The area of the circle con-

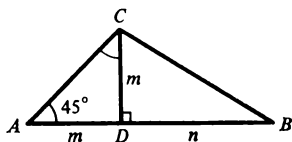


Fig. 206

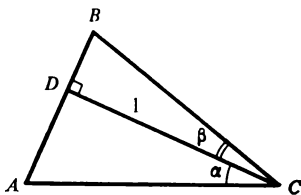


Fig. 207

structed on the hypotenuse is $\frac{25}{4}\pi$ cm², and that of the circles constructed on the legs are $\frac{9}{4}\pi$ cm² and 4π cm².

140. The hypotenuse is 13 cm. The sine of the larger acute angle is $12/13$, the cosine of that angle is $5/13$.

141. (1) In the right triangle ADC (Fig. 206) $\hat{ACD} = 45^\circ$, and therefore $|DC| = |AD| = m$. We find from the triangle BDC that $\tan \hat{B} = \frac{|DC|}{|DB|} =$

$$\frac{m}{n};$$

(2) using the Pythagorean theorem, we find from the triangle BDC that $|BC| = \sqrt{m^2 + n^2}$, $\sin \hat{B} = \frac{|DC|}{|BC|} =$

$$\frac{m}{\sqrt{m^2 + n^2}}.$$

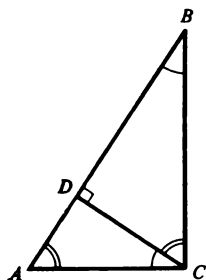


Fig. 208

142. Fig. 207. $|AD| = |CD| \tan \alpha = \tan \alpha$, $|BD| = |CD| \tan \beta = \tan \beta$, $|AC| = \frac{|CD|}{\cos \alpha} = \frac{1}{\cos \alpha}$, $|CB| = \frac{|CD|}{\cos \beta} = \frac{1}{\cos \beta}$.

143. (1) The triangles ACD and CDB (Fig. 208) are similar (two angles are equal), hence the proportionality of the corresponding sides:

$$\frac{|AC|}{|CB|} = \frac{|AD|}{|CD|} = \frac{|CD|}{|DB|} = \tan \hat{B};$$

(2) it follows from similarity of the triangles ACD and ACB that

$$\frac{|AD|}{|AC|} = \frac{|CD|}{|BC|} = \frac{|AC|}{|AB|} = \sin \hat{B}.$$

(3) This is because the area of the triangle can be expressed in two ways: either as $S = \frac{1}{2}ab$ or as $S = \frac{1}{2}ch$.

Answers to More Difficult Problems

$$1. (a) \frac{5932 \cdot 6001 - 69}{5932 + 6001 \cdot 5931} = \frac{(5931 + 1) \cdot 6001 - 69}{5932 + 6001 \cdot 5931} =$$

$$\frac{5931 \cdot 6001 + 6001 - 69}{5932 + 6001 \cdot 5931} = \frac{5931 \cdot 6001 + 5932}{5932 + 6001 \cdot 5931} = 1.$$

$$(b) \frac{254 \cdot 399 - 145}{399 \cdot 253 + 254} = \frac{(253 + 1) \cdot 399 - 145}{399 \cdot 253 + 254} = 1.$$

2. $2^n + 2^{n+1} + 2^{n+2} = 2^n (1 + 2 + 2^2) = 2^n \cdot 7$ is divisible by 7.

3. Every odd number can be written in the form $2n + 1$, where $n \in N$, and then $(2n + 1)^2 = 4n^2 + 4n + 1 = 4n(n + 1) + 1$. But of two successive natural numbers n and $n + 1$ one number is even, and therefore the first term $4n(n + 1)$ is divisible by 8 without a remainder, and the division of $4n(n + 1) + 1$ by 8 leaves unity as a remainder.

4. (a) We add m to and subtract m from the given binomial, and then we have

$$m^3 + 11m + m - m = m^3 - m + 12m$$

$$= m(m^2 - 1) + 12m = m(m - 1)(m + 1) + 12m;$$

the term $m(m - 1)(m + 1)$ is the product of three successive natural numbers, at least one of which is divisible by 2 and one by 3; consequently, $m(m - 1)(m + 1)$ is divisible by 6; the second term $12m$ is also divisible by 6 for a natural m . Hence $m^3 + 11m$ is divisible by 6. (b) We represent $8n$ as the sum of two terms $2n$ and $6n$, and then the given expression can be represented as the sum

$$(n^3 + 3n^2 + 2n) + 6n = n(n^2 + 3n + 2) + 6n.$$

The quadratic trinomial $n^2 + 3n + 2$ has roots 1 and 2; this means that $n^3 + 3n^2 + 8n = n(n - 1)(n - 2) + 6n$; the first term here is divisible by 6 (see 4 (a)), the second term $6n$ is also divisible by 6, and therefore the whole sum is also divisible by 6.

5. We designate each of the given equal ratios as k , i.e. $a/b = k$, $b/c = k$, $c/a = k$, and then $a = bk$, $b = ck$, $c = ak$. Multiplying these equations term by term, we get

$$abc = abck^3, \quad k^3 = 1, \quad k = 1,$$

and then $a = b$, $b = c$, $c = a$, and, consequently, $a = b = c$.

6. When factoring an exact cube, the exponents of the factors are divided by 3, for example, $216 = 6^3 = 2^3 \cdot 3^3$. Since $2100 = 2^3 \cdot 3 \cdot 5^2 \cdot 7$, it follows that to answer the question posed in the problem, we must multiply this number by $2 \cdot 3^2 \cdot 5 \cdot 7^2 = 4410$. We obtain the required number, i.e. $4410 = (2^2 \cdot 3 \cdot 5^2 \cdot 7)(2 \cdot 3^2 \cdot 5 \cdot 7^2) = 2^3 \cdot 3^3 \cdot 5^3 \cdot 7^3 = (2 \cdot 3 \cdot 5 \cdot 7)^3 = 210^3$.

7. (a) $x^4 + y^4 - x^3y^2 - x^2y^3 + 16xy = x^4 + y^4 - x^2y^2(x + y) + 16xy = x^4 + y^4 - x^2y^2 \cdot 2 + 16xy = (x^2 - y^2)^2 + 16xy = ((x - y)(x + y))^2 + 16xy = ((x - y) \cdot 2)^2 + 16xy = 4(x^2 - 2xy + y^2) + 16xy = 4(x^2 - 2xy + y^2 + 4xy) = 4(x + y)^2 = 4 \cdot 2^2 = 16$. (b) By the hypothesis, $x = 11$, $x + 1 = 12$; replacing 12 by $x + 1$, we find that the given expression is equal to $x^5 - (x +$

$$1) x^4 + (x+1)x^3 - (x+1)x^2 + (x+1)x - 1 = x^5 - x^6 - x^4 + x^4 + x^3 - x^3 - x^2 + x^2 + x - 1 = x - 1 = 11 - 1 = 10.$$

8. We represent each of the given fractions as the difference of two fractions, i.e.

$$\frac{1}{20} = \frac{1}{4} - \frac{1}{5}; \quad \frac{1}{30} = \frac{1}{5} - \frac{1}{6}; \quad \frac{1}{42} = \frac{1}{6} - \frac{1}{7}$$

and so on. The sum is equal to $\frac{1}{4} - \frac{1}{12} = \frac{1}{6}$.

9. *The 1st technique.* We try to represent the expression $11 + 6\sqrt{2}$ as the square of the binomial $a + b\sqrt{2}$, where a and b are natural numbers, i.e. we assume that $11 + 6\sqrt{2} = (a + b\sqrt{2})^2$ or $11 + 6\sqrt{2} = a^2 + 2b^2 + 2ab\sqrt{2}$. For this equality to hold true, it is sufficient that two conditions, $2ab = 6$ and $a^2 + 2b^2 = 11$, should be satisfied. It follows from the first equation $ab = 3$ that $a = 1, b = 3$, or $a = 3, b = 1$. We make sure by verification that the condition $a^2 + 2b^2 = 11$ is fulfilled only for $a = 3, b = 1$. This means that $11 + 6\sqrt{2} = (3 + \sqrt{2})^2$. By analogy we find that $11 - 6\sqrt{2} = (3 - \sqrt{2})^2$, and hence $\sqrt{11 + 6\sqrt{2}} + \sqrt{11 - 6\sqrt{2}} = \sqrt{(3 + \sqrt{2})^2} + \sqrt{(3 - \sqrt{2})^2} = |3 + \sqrt{2}| + |3 - \sqrt{2}| = 3 + \sqrt{2} + 3 - \sqrt{2} = 6$.

The 2nd technique. We note that $\sqrt{11 + 6\sqrt{2}} + \sqrt{11 - 6\sqrt{2}} > 0$. We seek the square of this expression:

$$(\sqrt{11 + 6\sqrt{2}} + \sqrt{11 - 6\sqrt{2}})^2 = 11 + 6\sqrt{2} + 11 - 6\sqrt{2} + 2\sqrt{11^2 - (6\sqrt{2})^2} = 36. \quad \text{Consequently,} \quad \sqrt{11 + 6\sqrt{2}} + \sqrt{11 - 6\sqrt{2}} = 6.$$

10. (a) $(\sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}})^2 = 4 + \sqrt{15} + 4 - \sqrt{15} + 2\sqrt{4^2 - (\sqrt{15})^2} = 10$, and therefore $\sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}} = \sqrt{10}$. $\sqrt{12 - 4\sqrt{5}} = (\sqrt{10} - \sqrt{2})^2 = \sqrt{10} - \sqrt{2}$. Consequently, the given expression is equal to $\sqrt{10} - (\sqrt{10} - \sqrt{2}) = \sqrt{2}$.

(b) $\sqrt{2 + \sqrt{3}} \cdot \sqrt{2 + \sqrt{2 + \sqrt{3}}} \cdot \sqrt{2^2 - (\sqrt{2 + \sqrt{2 + \sqrt{3}}})^2} = \sqrt{2 + \sqrt{3}} \cdot \sqrt{2 + \sqrt{2 + \sqrt{3}}} \cdot \sqrt{2 - \sqrt{2 + \sqrt{3}}} = \sqrt{2 + \sqrt{3}} \times \sqrt{2^2 - (\sqrt{2 + \sqrt{3}})^2} = \sqrt{2 + \sqrt{3}} \cdot \sqrt{2 - \sqrt{3}} = 1$. (c) Taking into account that $9 + 4\sqrt{5} = (2 + \sqrt{5})^2$, we have $(\sqrt{(2 + \sqrt{5})^2 + \frac{3}{2 + \sqrt{5}}}) \cdot \frac{3}{\sqrt{2 - \sqrt{5}}} = (\sqrt{\frac{3}{2 + \sqrt{5}}} + \sqrt{2 + \sqrt{5}}) \cdot \frac{3}{\sqrt{2 - \sqrt{5}}} = 2\sqrt{\frac{3}{2 + \sqrt{5}}} \cdot \sqrt{2 - \sqrt{5}} = 2\sqrt{2^2 - (\sqrt{5})^2} = -2$. (d) The given expression is equal to $(\sqrt{(2 + \sqrt{3})^2 + \sqrt{2 + \sqrt{3}}}) \cdot \sqrt{2 - \sqrt{3}} = 2\sqrt{\frac{3}{2 + \sqrt{3}}} \cdot \sqrt{2 - \sqrt{3}} = 2$.

$$\begin{aligned}
 11. \quad & \frac{4+2\sqrt{3}}{\sqrt{6}-\sqrt{3}+\sqrt{2}-1} = \frac{(\sqrt{3}+1)^2}{\sqrt{3}(\sqrt{2}-1) + (\sqrt{2}-1)} = \\
 & \frac{(\sqrt{3}+1)^2}{(\sqrt{2}-1)(\sqrt{3}+1)} = \frac{\sqrt{3}+1}{\sqrt{2}-1} = \frac{(\sqrt{3}+1)(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)} = \\
 & \frac{(\sqrt{3}+1)(\sqrt{2}+1)}{(\sqrt{2})^2-1^2} = (\sqrt{3}+1)(\sqrt{2}+1).
 \end{aligned}$$

12. (a) To simplify this expression, we must bear in mind that $\sqrt{a^2} = |a|$, $\sqrt{(a-b)^2} = |a-b|$, and know the exact definition of the absolute value. It follows from that definition that for any values of x we have $|x| \geq 0$. In particular

$$\begin{aligned}
 |x-a| &= \begin{cases} x-a, & \text{if } x-a \geq 0, \quad \text{i.e. } x \geq a, \\ a-x, & \text{if } x-a < 0, \quad \text{i.e. } x < a, \end{cases} \\
 |f(x)| &= \begin{cases} f(x), & \text{if } f(x) \geq 0, \\ -f(x), & \text{if } f(x) < 0. \end{cases}
 \end{aligned}$$

We designate the given expression as y , and then

$$y = \sqrt{(x+3)^2} + \sqrt{(x-1)^2} + \sqrt{(x+7)^2} = |x+3| + |x-1| + |x+7|.$$

Each term of the expression obtained can be written out in different forms according to the sign of $x+3$, $x-1$ and $x+7$; therefore we divide the whole set of the values of x into four intervals. The boundaries of the intervals are points at which the expressions under the sign of an absolute value vanish, i.e.

(1) $x \geq 1$, (2) $-3 \leq x < 1$, (3) $-7 \leq x < -3$, (4) $x < -7$. Let us consider these four intervals.

(1) If $x \geq 1$, then $x+3 > 0$, $x-1 \geq 0$ and $x+7 > 0$, and therefore

$$y = x+3 + x-1 + x+7 = 3x+9;$$

(2) if $-3 \leq x < 1$, then $x+3 \geq 0$, $x-1 < 0$ and $x+7 \geq 0$; and this means that

$$y = x+3 + 1-x + x+7 = 2x+11;$$

(3) if $-7 \leq x < -3$, then $x+3 < 0$, $x-1 < 0$ and $x+7 \geq 0$; consequently,

$$y = -x-3 + 1-x + x+7 = -x+5;$$

(4) if $x < -7$, then $x+3 < 0$, $x-1 < 0$ and $x+7 < 0$ consequently,

$$y = -x-3 + 1-x - x-7 = -3x-9.$$

Answer: $3x+9$ if $x \geq 1$, $2x+11$ if $-3 \leq x < 1$, $-x+5$ if $-7 \leq x < -3$, $-3x-9$ if $x < -7$.

(b) After the transformations $a+2\sqrt{a-1} = (1+\sqrt{a-1})^2$, $a-2\sqrt{a-1} = (1-\sqrt{a-1})^2$, we obtain $\sqrt{(1+\sqrt{a-1})^2} + \sqrt{(1-\sqrt{a-1})^2} = |1+\sqrt{a-1}| + |1-\sqrt{a-1}|$.

If $1 \leq a \leq 2$, then the expressions under the sign of the absolute value are nonnegative and $y = 1 + \sqrt{a-1} + 1 - \sqrt{a-1} = 2$.

13. We make the substitutions $\sqrt{a} = \sqrt{x}$, $\sqrt{b} = y$, and then $a = x^2$, $b = y^2$ and $\sqrt{ab} = xy$; the given expression assumes the form

$$\begin{aligned} & \left(\frac{x^3 + y^3}{x + y} - xy \right) \left(\frac{x + y}{x^2 - y^2} \right)^2 \\ &= \left(\frac{(x + y)(x^2 - xy + y^2)}{x + y} - xy \right) \left(\frac{x + y}{(x - y)(x + y)} \right)^2 \\ &= (x^2 - xy + y^2 - xy) \left(\frac{1}{x - y} \right)^2 = \frac{(x - y)^2}{(x - y)^2} = 1. \end{aligned}$$

14. We first eliminate the roots in the denominators of the fractions, i.e. $\frac{1}{\sqrt{1} + \sqrt{2}} = \frac{1(1 - \sqrt{2})}{(\sqrt{1} + \sqrt{2})(\sqrt{1} - \sqrt{2})} = \frac{1 - \sqrt{2}}{-1} =$

$$\sqrt{2} - 1, \quad \frac{1}{\sqrt{2} + \sqrt{3}} = \frac{1(\sqrt{2} - \sqrt{3})}{(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3})} = \sqrt{2} - \sqrt{3}, \text{ and}$$

so on. Then the given expression becomes equal to $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \dots + (\sqrt{99} - \sqrt{98}) + (\sqrt{100} - \sqrt{99}) = \sqrt{100} - 1 = 9$.

15. *The 1st technique.* It follows from the hypothesis that $a + b = -c$. We raise both sides of this equation to the third power term by term, i.e. $(a + b)^3 = (-c)^3$, $a^3 + 3ab(a + b) + b^3 = -c^3$; replacing $a + b$ by $-c$, we get $a^3 + 3ab(-c) + b^3 = -c^3$, whence we have $a^3 + b^3 + c^3 = 3abc$.

The 2nd technique. $a^3 + b^3 + c^3 = (a + b)^3 + c^3 - 3ab(a + b) = ((a + b) + c)((a + b)^2 - (a + b)c + c^2) - 3ab(a + b) = 0 \cdot ((a + b)^2 - (a + b)c + c^2) - 3ab(-c) = 3abc$.

$$\begin{aligned} 16. \quad a^3 + b^3 + c^3 - 3abc &= (a + b)^3 + c^3 - 3a^2b - 3ab^2 - 3abc \\ &= ((a + b) + c)((a + b)^2 - (a + b)c + c^2) - 3a^2b - 3ab^2 - 3abc \\ &= (a + b + c)(a^2 + 2ab + b^2 - ac - bc + c^2) - 3a^2b - 3ab^2 - 3abc \\ &= (a + b + c)(a^2 + 2ab + b^2 - ac - bc + c^2 - 3ab) \\ &= (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc). \end{aligned}$$

17. We rewrite the given equation in the form

$$a^2 + b^2 + c^2 - ac - bc - ab = 0.$$

Multiplying its both sides by 2, we obtain $2a^2 + 2b^2 + 2c^2 - 2ac - 2bc - 2ab = 0$, or $a^2 + a^2 + b^2 + b^2 + c^2 + c^2 - 2ac - 2bc - 2ab = 0$. Then we get $(a^2 - 2ab + b^2) + (a^2 - 2ac + c^2) + (b^2 - 2bc + c^2) = 0$, or $(a - b)^2 + (a - c)^2 + (b - c)^2 = 0$, but $(a - b)^2 \geq 0$, $(a - c)^2 \geq 0$ and $(b - c)^2 \geq 0$, and since the sum of nonnegative numbers is zero only if each term is zero, we have $(a - b)^2 = 0$, $(a - c)^2 = 0$ and $(b - c)^2 = 0$, whence it follows that $a = b$, $a = c$, $b = c$, i.e. $a = b = c$.

18. (a) The ordinary solution is difficult here since when the equation is reduced to a common denominator, a fourth-degree equation results. It is convenient here to use the substitution $x^2 + 2x = a$, and then the given equation is reduced to a simpler one: $\frac{1}{a} - \frac{1}{a+1} = \frac{1}{12}$. Solving it, we obtain

$$a^2 + a - 12 = 0, \quad a_1 = -4, \quad a_2 = 3.$$

Replacing a by -4 and 3 in the substitution $x^2 + 2x = a$, we get two quadratic equations $x^2 + 2x = -4$ and $x^2 + 2x = 3$. The first equation has no roots ($D < 0$); the roots of the second equation and hence those of the given one are $a_1 = -3$ and $a_2 = 1$. We can make sure by a direct verification that each of them is suitable here. (b) It is convenient first to find the product of the first and the fourth factor, the second and the third factor. Then we get two polynomials with the same coefficients in x^2 and x , i.e. $(x^2 + 3x) \times (x^2 + 3x + 2) = 5040$. This equation can be simplified by the substitution $x^2 + 3x = a$ and then we have

$$a^2 + 2a - 5040 = 0, \quad a = -72 \text{ or } a = 70;$$

$$x^2 + 3x = -72 \text{ or } x^2 + 3x = 70.$$

The first equation has no roots, the roots of the second equation and of the given equation are -10 and 7 . (c) To obtain the square of the binomial on the right-hand side of the equation, we add $4a^2x^2$ to both sides:

$$(x^2 - a^2)^2 + 4a^2x^2 = 4a^2x^2 + 4ax + 1,$$

$$(x^2 + a^2)^2 = (2ax + 1)^2,$$

$$(x^2 + a^2)^2 - (2ax + 1)^2 = 0.$$

Then we factor the left-hand side of the equation

$$(x^2 + a^2 - 2ax - 1)(x^2 + a^2 + 2ax + 1) = 0$$

and get $x^2 - 2ax + a^2 - 1 = 0$ or $x^2 + 2ax + 2a^2 + 1 = 0$, i.e. $(x - a)^2 = 1$ or $(x + a)^2 = -1$. The first equation has roots $x_1 = a + 1$ and $x_2 = a - 1$, the second equation has no solutions. Thus the given equation has roots $x_1 = a + 1$ and $x_2 = a - 1$. (d) The given equation has sense if $x - 1 \geq 0$ and $x > 0$, i.e. for $x \geq 1$; for $x > 1$ the left-hand side of the equation exceeds unity and the right-hand side is smaller than unity ($x^2 > 1$ and therefore $1/x^2 < 1$); for $x = 1$ the equation turns into a true equality, and this is the root of the equation.

19. We rewrite this equation as $(x^2 - 2x + 1) + (y^2 + 2y + 1) + (0.5z^2 - z + 1) = 0$, or $(x - 1)^2 + (y + 1)^2 + (0.5z - 1)^2 = 0$. The square of any real number is nonnegative, i.e. $(x - 1)^2 \geq 0$, $(y + 1)^2 \geq 0$ and $(0.5z - 1)^2 \geq 0$, and therefore the given equation is a true statement if each term is zero, or $(x - 1)^2 = 0$, $(y + 1)^2 = 0$, and $(0.5z - 1)^2 = 0$, whence $x = 1$, $y = -1$, $z = 2$.
Answer: $\{1, -1, 2\}$.

20. Hint. We can rewrite the equation as $(x^2 - 2\sqrt{2}x + 2) + (y^2 - 4\sqrt{2}y + 8) = 0$, or $(x - \sqrt{2})^2 + (y - 2\sqrt{2})^2 = 0$, and then $(x - \sqrt{2})^2 = 0$ and $(y - 2\sqrt{2})^2 = 0$, whence $x = \sqrt{2}$ and $y = 2\sqrt{2}$.
Answer: $\{(\sqrt{2}, 2\sqrt{2})\}$.

21. (a) *The 1st technique.* We rewrite the equation as $xy - x - y = 0$. To factor the left-hand side of the equation, we add unity to both sides, i.e. $xy - x - y + 1 = 1$, and then $(xy - x) - (y - 1) = 1$, $x(y - 1) - (y - 1) = 1$, whence $(x - 1)(y - 1) = 1$. But the product of two integers is unity if each of the numbers is equal to 1 or -1. Therefore we get two systems of equations in two variables:

$$\begin{cases} x-1=1, \\ y-1=1 \end{cases} \quad \text{or} \quad \begin{cases} x-1=-1, \\ y-1=-1. \end{cases}$$

From the first system $x_1 = 2$ and $y_1 = 2$, from the second $x_2 = 0$ and $y_2 = 0$. *Answer:* $\{(2, 2), (0, 0)\}$.

The 2nd technique. We express x in terms of y and then $x = \frac{y}{y-1}$; by the hypothesis $\frac{y}{y-1}$ must be an integer. For an odd y there are no integer solutions; when y is even, $\frac{y}{y-1}$ is

an integer only if $y-1=1$; hence $y=2$ and $x=2$. The solution $x=0$ and $y=0$ is obvious. (b) Assume that one of the numbers is x and the other y , with $x > 0$ and $y > 0$ (as distinct from the previous problem, where x and y are integers). Solving the equation $x + y = xy$ for x , we get $x = y/(y-1)$; x is positive when $y > 1$. Assigning different values to y , which are larger than unity, we get the corresponding values of x . If, for instance, $y = 1.5$, then $x = 1.5 \div 0.5 = 3$ ($3 + 1.5 = 3 \cdot 1.5$); if $y = 2$, then $x = 2$ ($2 + 2 = 2 \cdot 2$); if $y = 5$, then $x = 1.25$ ($1.25 + 5 = 1.25 \cdot 5$), and so on.

22. Note that the sum of the squares of two even numbers $(2n)^2 + (2k)^2 = 4(n^2 + k^2)$ is an even number. The sum of the squares of two odd numbers $(2n+1)^2 + (2k+1)^2 = 4n^2 + 4k^2 + 4n + 4k + 2$ is also an even number. We assume that x is an even number, and then y is an odd number, and, conversely, if y is even, then x is odd. If $x = 2n$ and $y = 2k + 1$, then $(2n)^2 + (2k+1)^2 = 1983$ or $4n^2 + 4k^2 + 4k = 1982$; cancelling 2, we get $2n^2 + 2k^2 + 2k = 991$. The resulting equation has no solutions since its left-hand side is an even number and the right-hand side is an odd number; this means that the initial equation has no solutions either.

23. (a) We divide the whole set of values of x into three intervals (see solution 12 (a)): (1) $x \geq 4$, (2) $2 \leq x < 4$, (3) $x < 2$. For $x \geq 4$ we have $x - 4 \geq 0$, $x - 2 > 0$; consequently, the equation assumes the form $x - 2 + x - 4 = 3$, whose root $x = 4.5$ satisfies the condition $x \geq 4$ and is therefore the root of the given equation. For $2 \leq x < 4$ the equation assumes the form $x - 2 + 4 - x = 3$, or $0 \cdot x = 1$; this equation has no solutions. For $x < 2$ the initial equation has the form $2 - x + 4 - x - 3$, or $-2x = -3$, whence $x = 1.5 < 2$. *Answer:* $\{1.5, 4.5\}$.

(b) If $x > 0$, then the given equation assumes the form $(x+1)(x-1) = -\frac{1}{2}$, or $x^2 - 1 = -\frac{1}{2}$; $x^2 = \frac{1}{2}$, whence $x = \sqrt{1/2} = \sqrt{2}/2 > 0$. If $x < 0$, then the given equation can be rewritten in the form $(x+1)(-x-1) = -\frac{1}{2}$ or $(x+1)^2 = \frac{1}{2}$.

Hence $x + 1 = \frac{\sqrt{2}}{2}$ or $x + 1 = -\frac{\sqrt{2}}{2}$; $x = \frac{\sqrt{2}}{2} - 1 < 0$ or $x = -\frac{\sqrt{2}}{2} - 1 < 0$. Note that $x \neq 0$ since for $x = 0$ we get a false statement $-1 = -\frac{1}{2}$. Answer: $\left\{ \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} - 1, -\left(\frac{\sqrt{2}}{2} - 1 \right) \right\}$.

24. (a) We factor the left-hand side of the equation

$$x^2(x-2)^2 + (2x^3 - 8x^2 + 8x) + (-3x^2 + 12x - 12) = 0,$$

$$x^2(x-2)^2 + 2x(x^2 - 4x + 4) - 3(x^2 - 4x + 4) = 0,$$

$$(x-2)^2(x^2 + 2x - 3) = 0.$$

Hence $x_1 = x_2 = 2$, $x_3 = 1$, $x_4 = -3$. Answer: $\{-3, 1, 2\}$.

(b) *The 1st technique.* Since $x \neq 0$, the given equation can be transformed as $x^3 + 4x - 16 = 0$, $(x^3 - 8) + (4x - 8) = 0$, $(x-2)(x^2 + 2x + 4) + 4(x-2) = 0$, $(x-2)(x^2 + 2x + 8) = 0$, whence $x = 2$, and the equation $x^2 + 2x + 8 = 0$ has no roots. Answer: $\{2\}$.

The 2nd technique. We transform the given equation as $\frac{1}{4}x^2 + 1 = \frac{4}{x}$ and construct the graphs of the functions $y = \frac{1}{4}x^2 + 1$ (a parabola) and $y = \frac{4}{x}$ (a hyperbola). The abscissa of the point of intersection of these graphs $x = 2$ is a root of the equation.

(b) By the hypothesis, $a = -\frac{1}{b^2}$; substituting this value of a into the equation, we obtain $\frac{x^3}{c} + \frac{x^2}{b} - b^2x = bc$, or $bx^3 + cx^2 - b^3cx - b^2c^2 = 0$, whence $x^2(bx + c) - b^2c(bx + c) = 0$, $(bx + c)(x^2 - b^2c) = 0$, and, finally, $x_1 = -\frac{c}{b}$, $x_2 = b\sqrt{c}$, $x_3 = -b\sqrt{c}$.

25. (a) For the given function to have sense, it is necessary that $x^3 - 5x^2 + 4 \neq 0$, or $x(x^2 - 5x + 4) \neq 0$. Solving the equation $x(x^2 - 5x + 4) = 0$, we find that the denominator of the fraction turns into zero for $x_1 = 0$, $x_2 = 1$ and $x_3 = 4$. Therefore the domain of the given function is the union of the intervals

$$(-\infty, 0) \cup (0, 1) \cup (1, 4) \cup (4, +\infty).$$

2. We express x in terms of y , i.e.

$$\frac{yx - 2y}{y - 1} = x - 3, \quad yx - x = 2y - 3, \quad x(y - 1) = 2y - 3, \quad x = \frac{2y - 3}{y - 1}.$$

This fraction has sense if $y \neq 1$, hence the domain of variation of this function is the set of all numbers except unity.

26. To calculate the absolute values, we consider four cases:

- (1) $x \geq 0$, $y \geq 1$, (2) $x < 0$, $y < 1$, (3) $x \geq 0$, $y < 1$,
- (4) $x < 0$, $y \geq 1$.

In the first case the system of equations assumes the form

$$\begin{cases} x + 3y = 7, \\ 2x + y - 1 = 3. \end{cases} \text{ From this system we get } x = 1 \geq 0, y =$$

$2 \geq 1$. In the second case the system is $\begin{cases} -x + 3y = 7 \\ 2x - y + 1 = 3. \end{cases}$

The solution of this system is $x = 2\frac{3}{5}, y = 3\frac{1}{5}$. Since $2\frac{3}{5} > 0$ and $3\frac{1}{5} > 1$, it follows that in this case the system has no

solutions. In the third case we have a system $\begin{cases} x + 3y = 7, \\ 2x - y + 1 = 3, \end{cases}$

whence $x = \frac{13}{7}, y = \frac{12}{7}$. Since $\frac{12}{7} > 1$, the system has no solutions in this case either. In the fourth case we get a system

$$\begin{cases} -x + 3y = 7, \\ 2x + y - 1 = 3, \end{cases} \text{ whence } x = \frac{5}{7}, y = \frac{18}{7}. \text{ Since } \frac{5}{7} > 0, \text{ the sys-}$$

tem has no solutions here either. *Answer:* $x = 1, y = 2$.

27. In the least favourable conditions the first key has to be tried not more than four times (if it does not fit four locks, then it fits the fifth one). It remains to try four locks. The second key is tried not more than three times (if it does not fit three locks, then it fits the fourth one); the third key is tried not more than twice, and the second key is tried only once. We obtain $4 + 3 + 2 + 1 = 10$ trials. *Answer:* not more than ten trials.

28. *The 1st technique.* Assume that initially n people were at the stadium, and the new entrance fee was p kopecks. Then the initial sum of money at the cashier's of the stadium was $150n$ kopecks. After the entrance fee was lowered, the number of spectators increased by 50 per cent, i.e. it became $1.5n$ people, the price of a ticket was p kopecks, and this means that the new raised collection of money was $1.5np$ kopecks. By the hypothesis, $1.5np$ is 125 per cent of $150n$, hence $1.5np = 1.25 \cdot 150n$, and this means that $p = 1.25 \times 150n \div 1.5n = 125$ kopecks = 1 rouble and 25 kopecks. Therefore the entrance fee became lower by 25 kopecks (150 kopecks — 125 kopecks = 25 kopecks).

The 2nd technique. The entrance fee for two people was $1.5 \cdot 2 = 3$ (roubles). After the fee was lowered, three people could come to the stadium instead of two (the number of spectators increased by 50 per cent) and those three people paid 3 roubles + 75 kopecks = 3 roubles 75 kopecks (the total sum of money increased by 25 per cent), i.e. the entrance fee is now 1 rouble and 25 kopecks ($3.75 \div 3 = 1.25$), and this means that the entrance fee was lowered by 25 kopecks.

29. Assume that the price of 25 oranges is x roubles, and then one orange costs $\frac{x}{25}$ roubles. On the other hand, we can buy x oranges for a rouble and therefore one orange costs $1/x$ rouble. Hence, solving the equation $\frac{x}{25} = \frac{1}{x}$, $x^2 = 25$, we find that $x = 5$ (oranges). Thus if we have a rouble, we can buy 5 oranges, and if we have three roubles, we can buy 15 oranges.

30. The quantity of silver in the first bar is $3.5 \cdot 0.76 = 2.66$ (kg), that in 10.5 kg of the bar is $10.5 \cdot 0.84 = 8.82$ (kg), the quantity of silver in the second bar is $8.82 - 2.66 = 6.16$ (kg), the mass of the second bar is $10.5 - 3.5 = 7$ (kg).

Thus the second bar whose mass is 7 kg contains 6.16 kg of silver. The percentage of silver in the second bar is $(6.16 \div 7) \times 100\% = 88\%$. *Answer:* 88%.

31. Eight kg of the new alloy must contain $\frac{5}{16} \cdot 8 = 2.5$ (kg) of gold. Assume that we must take x kg from the first alloy. Then x kg of the first alloy will contain $\frac{2}{5}x$ kg of gold, and $(8-x)$ kg of the second alloy will contain $\frac{3}{10}(8-x)$ kg of gold. By the hypothesis, eight kg of the new alloy contain 2.5 kg of gold, and hence the equation

$$\frac{2}{5}x + \frac{3}{10}(8-x) = 2.5.$$

From this we find that $x = 1$ (kg), $8 - x = 7$ (kg). *Answer:* One kg of the first alloy and seven kg of the second.

32. Assume that we take x kg of the first alloy and y kg of the second. Then in the first alloy there is $\frac{1}{3}x$ (kg) of zinc and $\frac{2}{3}x$ (kg) of copper. In the second alloy there is $\frac{2}{5}y$ (kg) of zinc and $\frac{3}{5}y$ (kg) of copper. The total amount of zinc in the third alloy is $(\frac{1}{3}x + \frac{2}{5}y)$ (kg) and that of copper is $(\frac{2}{3}x + \frac{3}{5}y)$. By the hypothesis, the ratio of zinc to copper in the third alloy is 5:8, and hence the equation

$$(\frac{1}{3}x + \frac{2}{5}y) : (\frac{2}{3}x + \frac{3}{5}y) = 5:8.$$

After simplification we have $(5x + 6y) : (10x + 9y) = 5:8$, whence $(5x + 6y) \cdot 8 = (10x + 9y) \cdot 5$. We find from this equation that $10x = 3y$, and the ratio of x to y is $x:y = 3:10$. *Answer:* From three parts of the first alloy and ten parts of the second.

33. Assume that the first motor-cyclist makes a trip from A to B in t hours. The second motor-cyclist spent five hours more ($7 - 2 = 5$), i.e. $(t + 5)$ hours. In t hours the first motor-cyclist will travel 60.9 t (km) and the second, 40.6 $(t + 5)$ (km). Hence the equation $60.9t = 40.6(t + 5)$, solving which we find that $t = 10$ (hours). *Answer:* The first motor-cyclist covered the distance from A to B in ten hours.

34. 1. We designate the distance AB as s (km), the time of travel from A to B as x (h), and time of travel from B to A as y (h). Then $s = 60x$ or $s = 40y$, whence $60x = 40y$, $x = 2y/3$. The average speed of travel is $\frac{2s}{x+y} = \frac{2 \cdot 40y}{2y/3 + y} = \frac{3 \cdot 80y}{5y} = 48$ (km/h). This

is the answer to the question posed in the problem. Students sometimes erroneously answer that the average speed of travel is $(60 + 40) \div 2 = 50$. This is incorrect since $\frac{2s}{x+y} \neq \left(\frac{s}{x} + \frac{s}{y}\right) \div 2$.

2. Assume that l is the length of the train, m, and v is its speed, m/s. Taking into account that the train covered l meters in 5 seconds, we get the first equation $5v = l$. The train covered 420 m in 20 s plus its own length l . Thus we have a second equation $20v = 420 + l$. Solving the system of equations $5v = l$ and $20v = 420 + l$, we find that $l = 140$ m and $v = 28$ m/s.

35. To calculate the absolute values, we consider four cases:

- (1) $x \geq 0, y \geq 1$, (2) $x < 0, y < 1$, (3) $x \geq 0, y < 1$,
(4) $x < 0, y \geq 1$.

In the first case the system assumes the form $\begin{cases} x + 3y = 7, \\ 2x + y - 1 = 3. \end{cases}$ From this system we get $x = 1 \geq 0$ and $y = 2 \geq 1$. In the second case the system is $\begin{cases} -x + 3y = 7 \\ 2x - y + 1 = 3. \end{cases}$ The solution of this sys-

tem is the numbers $x = 2\frac{3}{5}, y = 3\frac{1}{5}$. Since $2\frac{3}{5} > 0$ and $3\frac{1}{5} > 1$, the system has no solutions in this case. In the third

case we have a system $\begin{cases} x + 3y = 7, \\ 2x - y + 1 = 3, \end{cases}$ whence $x = \frac{13}{7}, y = \frac{12}{7}$. Since $\frac{12}{7} > 1$, the system has no solutions in this case either.

In the fourth case we get a system $\begin{cases} -x + 3y = 7, \\ 2x + y - 1 = 3, \end{cases}$ whence $x = \frac{5}{7}$ and $y = \frac{18}{7}$. Since $\frac{5}{7} > 0$, the system has no solutions

in this case. *Answer:* $x = 1, y = 2$.

36. (a) $3^{34} = 3^{2 \cdot 17} = (3^2)^{17} = 9^{17}$, $2^{51} = 2^{3 \cdot 17} = (2^3)^{17} = 8^{17}$. Since $9^{17} > 8^{17}$, it follows that the given inequality holds true, (b) $38^{303} = (2 \cdot 19)^{3 \cdot 101} = (8 \cdot 19^3)^{101}$, $57^{202} = ((3 \cdot 19)^2)^{101} = (9 \cdot 19^2)^{101}$; since $8 \cdot 19^3 > 9 \cdot 19^2$, it follows that $(8 \cdot 19^3)^{101} > (9 \cdot 19^2)^{101}$ as well, or $38^{303} > 57^{202}$.

37. (a) Let us take obvious inequalities $(a - b)^2 \geq 0$, $(b - c)^2 \geq 0$, $(a - c)^2 \geq 0$, or $a^2 - 2ab + b^2 \geq 0$, $b^2 - 2bc + c^2 \geq 0$, $a^2 - 2ac + c^2 \geq 0$, from which we have $a^2 + b^2 \geq 2ab$, $b^2 + c^2 \geq 2bc$, $a^2 + c^2 \geq 2ac$. Adding the resulting inequalities term by term and cancelling 2 we get $a^2 + b^2 + c^2 \geq ab + bc + ac$. We have an equality only for $a = b = c$. (b) We seek the difference between the left-hand and right-hand sides of the inequality we have to prove:

$$\frac{x^2}{1+x^4} - \frac{1}{2} = \frac{2x^2 - 1 - x^4}{2(1+x^4)} = \frac{-(x^2-1)^2}{2(1+x^4)}.$$

Since $(x^2-1)^2 \geq 0$, it follows that $-(x^2-1)^2 \leq 0$, $\frac{x^2}{1+x^4} - \frac{1}{2} \leq 0$; and therefore, $\frac{x^2}{1+x^4} \leq \frac{1}{2}$. We have an equality only for $x=1$, (c) we consider the differ-

ence $(a^2 + b^2 + c^2 + d^2 + e^2) - (ab + ac + ad + ae) = \left(\frac{a}{2} - b\right)^2 + \left(\frac{a}{2} - c\right)^2 + \left(\frac{a}{2} - d\right)^2 + \left(\frac{a}{2} - e\right)^2$. Each term of the last sum is nonnegative, and this means that the difference between the left-hand and right-hand sides of the initial inequality is also nonnegative. Equality to zero is attained for $a/2 = b = c = d = e$; consequently, $a^2 + b^2 + c^2 + d^2 + e^2 \geq ab + ac + ad + ae$,

(d) it is evident that $\frac{\sqrt{3}}{1} > 1$, $\frac{\sqrt{3}}{\sqrt{2}} > 1$, $\frac{\sqrt{3}}{\sqrt{3}} = 1$. We add these

inequalities term by term, i.e. $\frac{\sqrt{3}}{1} + \frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{3}} > 3$. We di-

vide both sides of the resulting inequality by $\sqrt{3}$ and then we have $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} > \sqrt{3}$. (e) We transform the left-hand

side of this inequality as follows: $\frac{a^2 + 5}{\sqrt{a^2 + 4}} = \frac{(a^2 + 4) + 1}{\sqrt{a^2 + 4}} =$

$$\frac{a^2 + 4}{\sqrt{a^2 + 4}} + \frac{1}{\sqrt{a^2 + 4}} = \frac{(\sqrt{a^2 + 4})^2}{\sqrt{a^2 + 4}} + \frac{1}{\sqrt{a^2 + 4}} = \sqrt{a^2 + 4} + \frac{1}{\sqrt{a^2 + 4}} > 2.$$

The last inequality holds true for any a since the sum of two mutually inverse unequal positive numbers exceeds two (see Problem 214, 1).

38. (a) By the hypothesis, $a > b$, $b > c$. Adding these inequalities term by term, we get an inequality

$$a + b > b + c. \quad (1)$$

Being given that a , b and c are terms of an arithmetic progression, we find the difference $d = b - a = c - b$, or

$$a - b = b - c. \quad (2)$$

Multiplying (1) by (2), we obtain $(a + b)(a - b) > (b + c) \times (b - c)$, or $a^2 - b^2 > b^2 - c^2$ and, from this, the inequality we had to prove: $a^2 + c^2 > 2b^2$.

(b) By the hypothesis, $a > b$, $b > c$, $a > c$, and therefore $a^2 > b^2$, $ab > bc$ and $b^2 > c^2$. We add the last three inequalities term by term, i.e.

$$a^2 + ab + b^2 > b^2 + bc + c^2,$$

multiply its both sides by the equality $a - b = b - c$ and get $a^3 - b^3 > b^3 - c^3$, whence $a^3 + c^3 > 2b^3$.

39. Assume that the smallest of the angles of this pentagon is a_1 , and the common difference of the progression is d . Since the sum of the angles of the pentagon is 540° ($180^\circ (5 - 2) = 540^\circ$), we obtain

$$a_1 + (a_1 + d) + (a_1 + 2d) + (a_1 + 3d) + (a_1 + 4d) = 540,$$

$$5a_1 + 10d = 540; \quad a_1 + 2d = 108,$$

but $a_1 + 2d$ is the third, middle, angle, and then the second angle is $108 - d$, the first angle is $108 - 2d$, the fourth angle is $108 + d$, and the fifth (the greatest) angle is $108 + 2d$. Since an interior angle of a convex polygon is smaller than 180° , we have $108 + 2d < 180$, or $2d < 72$, and hence $180 - 2d > 36^\circ$ is the smallest angle. The other angles are obviously larger than 36° .

40. The domain of the function includes only those values of x which satisfy the simultaneous inequalities

$$\begin{cases} x \geq 0, \\ 3 - x \geq 0, \\ x(x - 4) \geq 0. \end{cases}$$

The inequality $x(x - 4) \geq 0$ is satisfied for $x \geq 4$ or for $x \leq 0$, and therefore the given inequality is equivalent to two systems of simultaneous inequalities

$$\begin{cases} x \geq 0, \\ x \leq 3, \\ x \geq 4 \end{cases} \quad \text{and} \quad \begin{cases} x \geq 0, \\ x \leq 3, \\ x \leq 0. \end{cases}$$

The set of solutions of the first system is empty, the set of solutions of the second system consists of one point $x = 0$. Thus the domain of the given function consists of one element 0.

41. The quadratic trinomial $x^2 - 2x + 5$ has a negative discriminant $D = -16$, and therefore it is impossible to factor this trinomial. In addition, in this trinomial the coefficient in x^2 is larger than zero ($1 > 0$). This means that the trinomial is positive for all values of x and we can therefore neglect it. We can also neglect the factor x^2 since $x^2 > 0$ for $x \neq 0$. The initial inequality reduces to $2x - 3 > 0$, and hence $x > 1.5$.

42. (a) *The 1st technique.* Taking into account that $x^2 = |x|^2$, we rewrite the given inequality as $|x|^2 - 3|x| + 2 > 0$. We denote $|x|$ by y and then we get $y^2 - 3y + 2 > 0$. This inequality is satisfied for $y > 2$ or for $y < 1$, i.e. if $|x| > 2$ or $|x| < 1$, the solution of the first inequality is $(-\infty, -2) \cup (2, +\infty)$, and that of the second is $(-1, 1)$. The set of solutions of the given inequality is the union of these intervals, i.e. $(-\infty, -2) \cup (-1, 1) \cup (2, +\infty)$.

The 2nd technique. We can arrive at the same result, but not so easily, by uniting the solutions of the two systems of inequalities:

$$\begin{cases} x^2 - 3x + 2 > 0, \\ x > 0 \end{cases} \quad \text{and} \quad \begin{cases} x^2 + 3x + 2 > 0, \\ x < 0. \end{cases}$$

(b) *The 1st technique.* Since both sides of this inequality are nonnegative for all values of x , we have $|x + 2|^2 > |x|^2$ or $(x + 2)^2 > x^2$, whence $x > -1$, or $x \in (-1, +\infty)$.

The 2nd technique. We designate the left-hand side of the inequality as y_1 and the right-hand side as y_2 . We construct the graphs of the functions $y_1 = |x + 2|$ and $y_2 = |x|$ (Fig. 209). It can be seen from the figure that for $x < -1$ the ordinates of the graph of y_1 are larger than the respective ordinates of the graph of y_2 (this part of the graph is in a thick

line). (c) If $x \neq -1$, we have $|x+1| > 0$, and therefore we can rewrite the given inequality as $|x-2| > |x+1|$. Squaring both sides of the resulting inequality and solving the inequality $(x-2)^2 < (x+1)^2$, we find that $x < 1/2$. Taking into account that $x \neq -1$, we find the solution of this inequality $(-\infty, -1) \cup (-1, 1/2)$; its graphical solution is shown in Fig. 210.

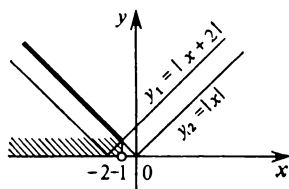


Fig. 209

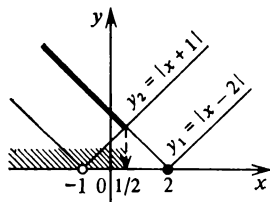


Fig. 210

43. Since the arithmetic mean and geometric mean are unequal (see 214 (3)), we have $y = \frac{1}{\sqrt{1-x}} + \frac{1}{\sqrt{1+x}} \geq \frac{2}{\sqrt{1-x}\sqrt{1+x}}$, or $y \geq \frac{2}{\sqrt{1-x^2}}$. Since $x^2 \geq 0$, it follows that $1-x^2 \leq 1$, $\sqrt{1-x^2} \leq 1$, and therefore $y \geq \frac{2}{\sqrt{1-x^2}} \geq 2$. For $x=0$ and $y=2$ this is precisely the least value of the function.

$$44. (a) x^2 - 2|x| = \begin{cases} x^2 - 2x, & \text{if } x \in [0; +\infty), \\ x^2 + 2x, & \text{if } x \in (-\infty; 0). \end{cases}$$

The graph of the function (Fig. 211) is symmetric about the y -axis;

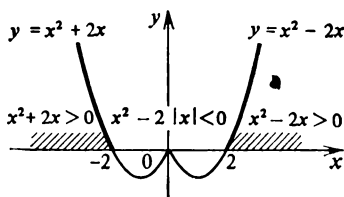


Fig. 211

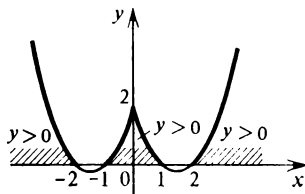


Fig. 212

as we see the solution of the inequality $x^2 - 2|x| > 0$ is the union of the sets $(-\infty, -2) \cup (2, +\infty)$, and that of the inequality $x^2 - 2|x| < 0$ is the set $(-2, 2)$. (b) The graph is shown in Fig. 212. The solution of the inequality is shown in a dash line on the x -axis.

45. (a) By the condition for the existence of the logarithm, $x > 0$. For these values of x , $y = 1$ (since $\sin^2 \alpha + \cos^2 \alpha = 1$)

(Fig. 213). (b) The domain of the given function is $x \neq 0$. After simplification we have

$$y = \frac{x \sqrt{\frac{x^4 - 2x^2 + 4x^2 + 1}{4x^2}}}{x^2 + 1} = \frac{x \sqrt{\frac{(x^2 + 1)^2}{4x^2}}}{x^2 + 1} = \frac{\frac{x(x^2 + 1)}{2|x|}}{x^2 + 1} = \frac{x}{2|x|}.$$

From this we find that $y = 1/2$ if $x > 0$, and $y = -1/2$ if $x < 0$ (Fig. 214). (c) After the transformations shown in 12 b we get $y = 2$.

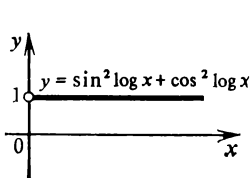


Fig. 213

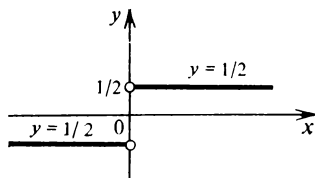


Fig. 214

Consequently, the graph of the function is the segment which is parallel to the x -axis and whose endpoints are $(1, 2)$ and $(2, 2)$.

46. (a) We construct the given set of points in each quadrant (Fig. 215):

in the first $x \geq 0, y \geq 0$, then $x + y = 2$ or $y = -x + 2$,
in the second $x < 0, y \geq 0$, then $-x + y = 2$ or $y = x + 2$,
in the third $x < 0, y < 0$, then $-x - y = 2$ or $y = -x - 2$,
in the fourth $x \geq 0, y < 0$, then $x - y = 2$ or $y = x - 2$.

The required set is the contour of the square the coordinates of whose vertices are $(2, 0)$, $(0, 2)$, $(-2, 0)$ and $(0, -2)$.

Remark. Since $|-x| = |x|$ and $|-y| = |y|$, the required set is symmetric about both axes of coordinates, and therefore it was sufficient to construct a line segment $y = -x + 2$ in the first quadrant and then construct segments which are symmetric with respect to it about both axes, (b) the given equation is equivalent to the system

$$\begin{cases} x^2 + y^2 = 0.25 \\ x > 0 \end{cases} \quad \text{or} \quad \begin{cases} x^2 + y^2 = 4, \\ x < 0. \end{cases}$$

The graph of the first system is a semicircle of radius 0.5 which lies in the right-hand half-plane, the points $(0, 0.5)$ and $(0, -0.5)$ on the y -axis exclusive. The graph of the second system is a semicircle of radius 2 which lies in the left-hand half-plane, the points $(0, 2)$

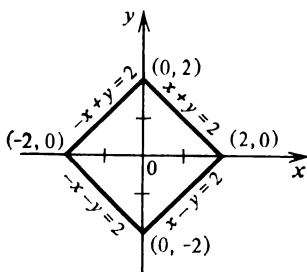


Fig. 215

and $(0, -2)$ on the y -axis exclusive (Fig. 216). (c) We transform the given equation as $x^2(y^2 - 1) - (y^2 - 1) = 0$, $(y^2 - 1) \times (x^2 - 1) = 0$, $(y - 1)(y + 1)(x - 1)(x + 1) = 0$. Hence $y = 1$, $y = -1$, $x = 1$, $x = -1$. The required set consists of two straight

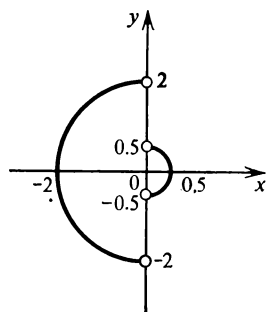


Fig. 216

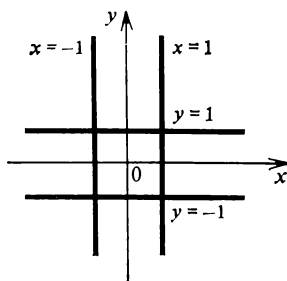


Fig. 217

lines parallel to the x -axis and two straight lines parallel to the y -axis (Fig. 217). (d) We factor the left-hand side of this equation, i.e.

$$(x^2 + y^2 - 2xy) + (2x^2y^2 - x^3y - xy^3) = 0,$$

$$(x - y)^2 - xy(x^2 + y^2 - 2xy) = 0,$$

$$(x - y)^2(1 - xy) = 0.$$

Hence $(x - y)^2 = 0$, or $1 - xy = 0$, $y = x$ or $y = 1/x$, and, consequently, the graph of the given equation is the union of the straight line passing through the origin and a hyperbola.

47. We construct the graph of the function $y = |x| - 1$ (Fig. 218). The coordinates of the points of the domain which lie below the graph satisfy the inequality $y \leq |x| - 1$. Then we construct the graph of the function $y = 3 - 3x$. The coordinates of the points of the domain which lie below that line satisfy the inequality $y < 3 - 3x$, the straight line itself not belonging to the domain. The domain which is cross hatched yields the required set of points.

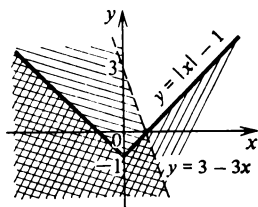


Fig. 218

48. (a) $(\log x^2)^2 = 4$, $\log x^2 = 2$ or $\log x^2 = -2$, $x^2 = 100$ or $x^2 = 0.01$, x is equal to 10 and -10 , 0.1 and -0.1 .

Answer: $\{-10, -0.1, 0.1, 10\}$, (b) $\log(-x)$ exists if $-x > 0$ or $x < 0$, but in that case $\log x^2 = 2 \log |x| = 2 \log(-x)$. We rewrite the given equation as $2 \log(-x) + \log(-x) = 3$ or $3 \log(-x) = 3$, $\log(-x) = 1$, $-x = 10$, $x = -10$. Answer: $\{-10\}$.

49. (a) $\sqrt{x} = x$ if $x = 0$ or $x = 1$, $\sqrt{x} > x$ if $0 < x < 1$ (say, $\sqrt{0.1} \approx 0.32 > 0.1$), $\sqrt{x} < x$ if $x > 1$. (b) Note that $x \geq 1$ since

if $0 < x < 1$, then $\log x < 0$ and then $\sqrt{\log x}$ has no sense. We transform the given equation as $\frac{1}{2} \log x - \sqrt{\log x} = 0$, $\frac{1}{2} (\sqrt{\log x})^2 - \sqrt{\log x} = 0$. We designate $\sqrt{\log x}$ as a , and then $\frac{1}{2} a^2 - a = 0$, whence $a = 0$ or $a = 2$, i.e. $\sqrt{\log x} = 0$ or $\sqrt{\log x} = 2$, $\log x = 0$ or $\log x = 4$, and therefore $x = 1$ or $x = 10\,000$. Solving the inequality $\frac{1}{2} a^2 - a > 0$, we find that $a > 2$ and $a < 0$, i.e. $\log x > 4$ and $\log x < 0$. The last inequality must be rejected, and then $x > 10\,000$, $\log \sqrt{x} < \sqrt{\log x}$ if $1 < x < 10\,000$.

50. We designate the length of the segment AD as x (Fig. 219), and then the length of the segment DC is $2-x$. Using the

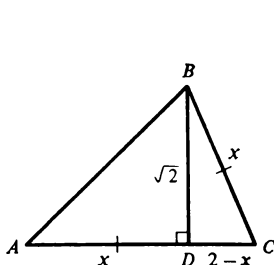


Fig. 219

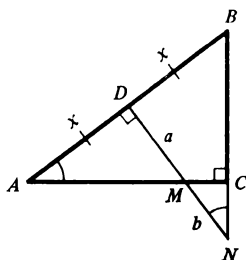


Fig. 220

Pythagoras theorem, we have $|BC|^2 = |DC|^2 + |BD|^2$ from the right triangle BDC . By the hypothesis, $|AD| = |BC| = x$. We get an equation $x^2 = (2-x)^2 + (\sqrt{2})^2$, whence $x = 1.5$, i.e. $|AD| = 1.5$ m. Using again the Pythagoras theorem, we find from the triangle ABD that $|AB|^2 = 1.5^2 + (\sqrt{2})^2 = 4.25$, $|AB| = \sqrt{4.25} =$

$$\sqrt{\frac{17}{4}} = \frac{1}{2} \sqrt{17} \text{ (m)}.$$

51. $|CD| = 2\sqrt{3}$ m, $S = 6\sqrt{3}$ m².

52. $\triangle ADM \sim \triangle DBN$ (two angles are equal) (Fig. 220),

whence $\frac{|DM|}{|DB|} = \frac{|AD|}{|DN|}$. We designate $|AD|$ as x , and then

$$\frac{a}{x} = \frac{x}{a+b}; \quad x^2 = a(a+b); \quad x = \sqrt{a(a+b)}, \quad |AB| = 2\sqrt{a(a+b)}.$$

53. $\sqrt{3}$ m. *Hint.* Draw the altitudes of the trapezoid from the vertices B and C ; they are equal to $0.5\sqrt{3}$ m each. We can find the length of the diagonal from the right triangle in which the legs are 1.5 and $0.5\sqrt{3}$ m.

54. $\angle AMB \cong \angle CBM$ (their sides extend in the same direction) (Fig. 221), and therefore $\angle BMC \cong \angle CBM$; this means that $|BC| = |CM| = a$. Using the Pythagoras theorem, we find from the triangle CMD that $|MD| = \sqrt{a^2 - b^2}$.

55. We designate the value of the inscribed angle ABC as α (Fig. 222). Then the central angle AOC is 2α (the value of an in-

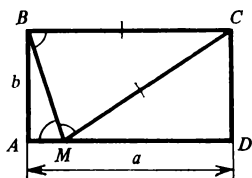


Fig. 221

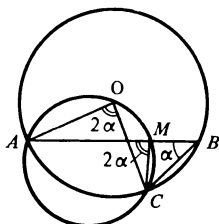


Fig. 222

scribed angle is half the angular value of the arc by which it is subtended); $\widehat{AOC} = \widehat{AMC} = 2\alpha$ (being inscribed angles subtended by the same arc AC). With respect to the triangle CMB the angle AMC is exterior, and this means that $\widehat{AMC} = \widehat{MCB} + \hat{B}$. Hence we have $\widehat{MCB} = \widehat{AMC} - \hat{B} = 2\alpha - \alpha = \alpha$; thus $\widehat{MCB} = \hat{B} = \alpha$. Hence $|MB| = |MC|$.

56. The triangle AOC is isosceles (Fig. 223) since $|OA| = |OC|$ (being the radii of the same smaller circle); consequently,

$$\angle CAO \cong \angle ACO. \quad (1)$$

In addition, $[OC] \perp [BD]$ (by the property of the radius of a circle drawn to the point of tangency) and $[AD] \perp [BD]$ (the inscribed angle ADB rests on a diameter), and therefore $[OC] \parallel [AD]$ and

$$\angle CAD \cong \angle ACD. \quad (2)$$

It follows from (1) and (2) that $\angle DAC \cong \angle CAO$. Hence we infer that AC is the bisector of the angle BAD .

57. We draw, through the point A (Fig. 224), a common tangent to both circles to the point D of its intersection with (BC) , and then $|DB| = |DA| = |DC|$ (the lengths of the tangents drawn

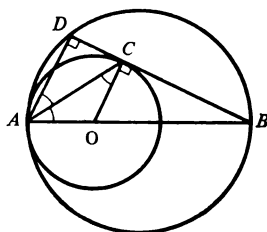


Fig. 223

to a circle from the same point are equal). We designate $\widehat{DAB} = \alpha$, $\widehat{DAC} = \beta$, and then $\widehat{BAC} = \alpha + \beta$. In the triangle ABC we have $\widehat{BAC} + \hat{B} + \hat{C} = 180^\circ$. Hence $(\alpha + \beta) + \alpha + \beta = 180^\circ$, or $2(\alpha + \beta) = 180^\circ$; $\alpha + \beta = \widehat{BAC} = 90^\circ$.

58. Assume that E is the midpoint of the side AB (Fig. 225). We draw a median EF of the trapezoid $ABCD$. Since $[EF] \parallel [AD]$, it follows that $\widehat{ADE} = \widehat{DEF}$, $\widehat{ADF} = \widehat{EDF}$ (by the hypothesis). Hence we infer that $\widehat{DEF} = \widehat{EDF}$, and this means that $|EF| = |FD| = \frac{1}{2} |CD| = 5 \text{ cm}$; $|EF| = \frac{|AD| + |BC|}{2}$, whence we have $|AD| = 2|EF| - |BC|$ or $|AD| = 2 \cdot 5 - 2 =$

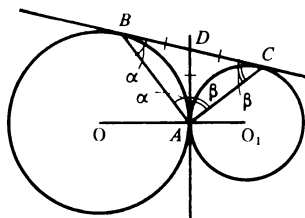


Fig. 224

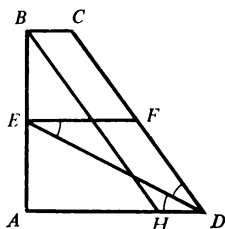


Fig. 225

8 (cm). We draw $[BH] \parallel [CD]$, and then the quadrilateral $BCDH$ is a parallelogram, $|BH| = 10 \text{ cm}$, $|AH| = 8 - 2 = 6 \text{ (cm)}$. Thus in the triangle ABH the sides are 6, 8 and 10 cm. By the inverse of the Pythagoras theorem this triangle is right-angled, i.e. AB is an altitude of the given trapezoid and the area of the trapezoid is 40 cm^2 ($5 \cdot 8 = 40$).

59. We extend the nonparallel sides AB and DC of the trapezoid to the point M they meet at (Fig. 226). By the hypothesis,

$\widehat{AMD} = 90^\circ$. According to the Pythagoras theorem we find from the triangle AMC that $|AC|^2 = |AM|^2 + |MC|^2$. From the triangle BMC we have $|BD|^2 = |BM|^2 + |DM|^2$. We add the resulting equations term by term, i.e. $|AC|^2 + |BD|^2 = (|AM|^2 + |DM|^2) + (|MC|^2 + |MB|^2)$; but $|AM|^2 + |DM|^2 = |AD|^2$ (from the triangle AMD) and $|MC|^2 + |MB|^2 = |BC|^2$ (from the triangle BMC), and therefore $|AC|^2 + |BD|^2 = |AD|^2 + |BC|^2$. Remark. The last equation is valid for any quadrilateral $ABCD$ whose sides AB and CD are perpendicular (we have not used the parallelism of the sides BC and AD).

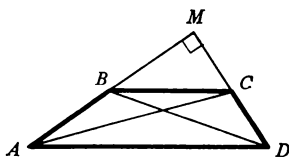


Fig. 226

60. We draw a segment EA (Fig. 227), and then $\widehat{AEC} = \widehat{ABC}$ (being inscribed angles subtended by the same arc AC), $\widehat{ACE} = \widehat{BCE}$ (by the hypothesis), and therefore $\triangle DBC \sim \triangle AEC$ (two

angles are equal). Consequently, we have a proportion $\frac{|AC|}{|CD|} = \frac{|CE|}{|BC|}$, and hence the equality $|AC| \cdot |BC| = |CD| \cdot |CE|$ we had to prove.

61. We take an arbitrary point M on the side AB of the equilateral triangle ABC (Fig. 228) and draw perpendiculars MD and

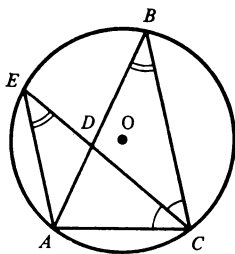


Fig. 227

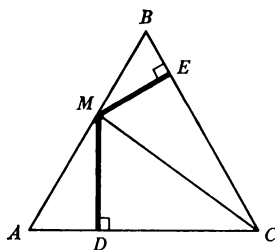


Fig. 228

ME to the sides AC and BC . We have to prove that $|MD| + |ME|$ is a constant quantity. We connect M and C by a line segment and designate the length of the side of the triangle ABC as a , and then the area S of the triangle is equal to the sum of the areas of the triangles AMC and MBC , or $S = \frac{1}{2} a |MD| + \frac{1}{2} a |ME| = \frac{1}{2} a (|MD| + |ME|)$. From this we get $|MD| + |ME| = \frac{2S}{a}$ which is a constant quantity for the given triangle and is equal to its altitude.

62. *Hint.* $[OC]$ and $[OD]$ are the bisectors of the angles BCD and ADC (Fig. 229), the triangle COD is right-angled, $|CD| =$

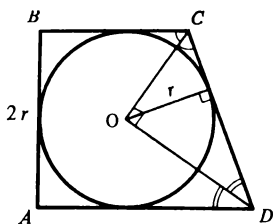


Fig. 229

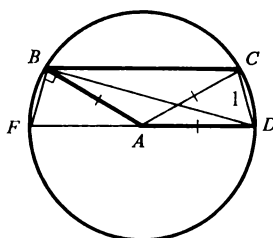


Fig. 230

10 cm. From the formula $h_c = \frac{ab}{c}$ we find the altitude of the triangle COD which is also a radius of the inscribed circle, $h_c = r = 4.8$ cm. The length of the altitude of the trapezoid $|AB| = 2r = 9.6$ cm. By the property of the sides of the circumscribed

quadrilateral $|AB| + |CD| = |AD| + |BC|$, and therefore $|AD| + |BC| = 10 \text{ cm} + 9.6 \text{ cm} = 19.6 \text{ cm}$. The perimeter of the trapezoid is 39.2 cm.

63. The vertex A of the trapezoid $ABCD$ (Fig. 230) is at the same distance from the vertices B , C and D . We draw a circle with centre at the point A and radius equal to 2 cm. Then we connect the vertex B with the endpoint of the diameter F and get an isos-

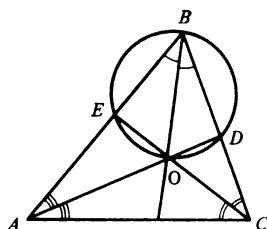


Fig. 231

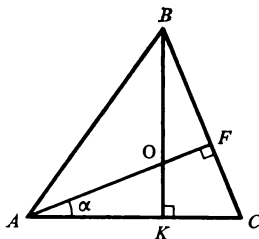


Fig. 232

celes trapezoid $FBCD$ (the arcs included between parallel chords are congruent). We find $|BD|$ from the right triangle FBD : $|BD| = \sqrt{|FD|^2 - |FB|^2} = \sqrt{4^2 - 1^2} = \sqrt{15} \approx 3.9 \text{ cm}$; $|BD| \approx 3.9 \text{ cm}$.

64. 1. $\widehat{BAC} + \widehat{BCA} = 180^\circ - 60^\circ = 120^\circ$ (Fig. 231), and therefore $\widehat{OAC} + \widehat{OCA} = \frac{1}{2}(\widehat{BAC} + \widehat{BCA}) = 60^\circ$, and hence

$\widehat{AOC} = \widehat{EOD} = 120^\circ$. Since the sum of the opposite sides of the quadrilateral $BEOD$ is 180° , we can circumscribe a circle about this quadrilateral. The segment BO is the bisector of the angle B (three bisectors of a triangle meet at one point). The point O bisects the arc ED (congruent arcs correspond to congruent inscribed angles), i.e. $\widehat{EO} = \widehat{OD}$. Hence we infer that $|OE| = |OD|$ (if two arcs are congruent, then the lengths of the chords which subtend those arcs are equal).

2. We designate $\widehat{ABC}$ as x (Fig. 231), and then $\widehat{BAC} + \widehat{BCA} = 180^\circ - x$. In the triangle OAC we have $\widehat{OAC} + \widehat{OCA} = (180^\circ - x) \div 2 = 90^\circ - x/2$; $\widehat{AOC} = \widehat{EOD} = 180^\circ - (90^\circ - x/2) = 90^\circ + x/2$. Since a circle is circumscribed about the quadrilateral $BDOE$, it follows that $\widehat{EOD} + \widehat{DBE} = 180^\circ$, whence $90^\circ + x/2 + x = 180^\circ$, $x = 60^\circ$, i.e. the required angle is 60° .

65. 1 m². *Hint.* The triangles ABC and BCD have equal areas since they have a common base BC and equal altitudes drawn to that base.

66. *Hint.* We introduce the designation $\widehat{FAC} = \alpha$ (Fig. 232),

and then $\hat{C} = 90^\circ - \alpha$, $\widehat{FOK} = 90^\circ + \alpha$ (by the property of an exterior angle of the triangle AOK); $\widehat{FOK} + \hat{C} = 180^\circ$ and therefore a circle can be circumscribed about the quadrilateral $FOKC$.

67. (a) We take an obvious inequality $(a - b)^2 \geq 0$ or $a^2 + b^2 - 2ab \geq 0$, whence we have

$$ab \leq \frac{a^2 + b^2}{2}. \quad (1)$$

In the triangle ABC $0 < \hat{C} < 180^\circ$, and therefore

$$0 < \sin \hat{C} < 1. \quad (2)$$

Multiplying term by term inequalities (1) and (2) whose left-hand and right-hand sides are positive, we obtain

$$ab \sin \hat{C} \leq \frac{a^2 + b^2}{2} \quad \text{or} \quad \frac{1}{2} ab \sin \hat{C} \leq \frac{a^2 + b^2}{4},$$

but the area of the triangle ABC is $S = \frac{1}{2} ab \sin \hat{C}$, and therefore $S \leq \frac{a^2 + b^2}{4}$. (b) *Hint.* Multiply two inequalities $ab \leq \frac{(a+b)^2}{4}$ and $\sin \hat{C} \leq 1$ term by term.

68. We designate the side of the triangle as a and that of the square as b , and then $S_\Delta = \frac{a^2 \sqrt{3}}{4}$ and the area of the square equal to it is b^2 , $S = b^2$; hence

$$\frac{a^2 \sqrt{3}}{4} = b^2, \quad \text{or} \quad a^2 \sqrt{3} = 4b^2.$$

Extracting an arithmetic square root of both sides of the last equation, we get $a \sqrt{\sqrt{3}} = 2b$, i.e. $a \sqrt[4]{3} = 2b$. Let us prove that $3a > 4b$. We have

$$[a = \frac{2b}{\sqrt[4]{3}}, \quad 3a = \frac{2b \cdot 3}{\sqrt[4]{3}} = \frac{2b \sqrt[4]{3^4}}{\sqrt[4]{3}} = 2b \sqrt[4]{27}.$$

We consider the difference $3a - 4b = 2b \sqrt[4]{27} - 4b = 2b (\sqrt[4]{27} - 2) = 2b (\sqrt[4]{27} - \sqrt[4]{16}) > 0$, whence $3a > 4b$. **Remark.** You can use a requisite table or your pocket calculator to establish that $\sqrt[4]{27} = \sqrt{\sqrt{27}} \approx \sqrt{5.196} \approx 2.279 > 2$.

69. Since the sum of two positive numbers is larger than each of the terms, we have $c^3 > a^3$ and $c^3 > b^3$, whence $c > a$ and $c > b$. By the cosine law $c^2 = a^2 + b^2 - 2ab \cos C$, or $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$. We multiply the numerator and denominator of the fraction by c and get $\cos C = \frac{a^2 c + b^2 c - c^3}{2abc}$; we replace c^3 by the expression

$a^3 + b^3$ which is equal to it, i.e.

$$\cos C = \frac{a^2c + b^2c - (a^3 + b^3)}{2abc} = \frac{a^3(c-a) + b^3(c-b)}{2abc}.$$

Here $a^2 > 0$, $c - a > 0$, $b^2 > 0$, $c - b > 0$, $2ab > 0$; consequently, $\cos C > 0$ and the angle C is acute.

70. We can find the area of the triangle from the formula $S = \frac{1}{2} \times ab \sin C$, and therefore, bearing in mind the hypothesis, we get an equation $\frac{1}{4}(a^2 + b^2) = \frac{1}{2}ab \sin C$. We multiply both sides of the equation by 4 and obtain

$$a^2 + b^2 = 2ab \sin C, \quad a^2 + b^2 - 2ab \sin C = 0,$$

$$a^2 + b^2 - 2ab + 2ab - 2ab \sin C = 0,$$

$$(a - b)^2 + 2ab(1 - \sin C) = 0,$$

since $(a - b)^2 \geq 0$, $0 < \sin C \leq 1$, it follows that $1 - \sin C \geq 0$. But the sum of two nonnegative numbers is zero if each term is zero, and therefore the last equality is possible if

$$\begin{cases} (a - b)^2 = 0, \\ 1 - \sin C = 0, \end{cases}$$

i.e. if $a = b$. This means that $\hat{A} = \hat{B}$ and $\sin C = 1$, whence $\hat{C} = 90^\circ$ and then $\hat{A} = \hat{B} = 45^\circ$.

71. We draw a perpendicular to the given parallel lines from the vertex A of the triangle ABC (Fig. 233). Assume that their points

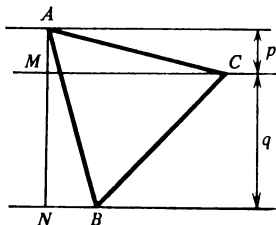


Fig. 233

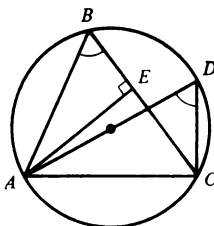


Fig. 234

of intersection are M and N . By the hypothesis $|AM| = p$ and $|MN| = q$. In the right triangle ABN we have $|AB| > |AN|$, i.e. $a > p + q$, but $p + q > 2\sqrt{pq}$ (the arithmetic mean of two unequal positive numbers is larger than their geometric mean), and therefore $a > 2\sqrt{pq}$. If $p = q$ and the side AB is perpendicular to the given straight lines, then $a = p + q = 2\sqrt{pq}$.

72. We draw a diameter AD (Fig. 234), connect the points D and C and draw $[AE] \perp [BC]$, and then the triangle ABE is similar to the triangle ADC ($\angle B \cong \angle D$ as inscribed angles subtended by

the same arc AC , $\widehat{BEA} = \widehat{ACD} = 90^\circ$), whence we have

$$\frac{|AD|}{|AB|} = \frac{|AC|}{|AE|}, \quad \frac{2R}{c} = \frac{b}{h_a}, \quad 2R = bc \div h_a,$$

but $\frac{1}{2} ah_a = S$, $h_a = \frac{2S}{a}$, and therefore $2R = bc \div \frac{2S}{a} = \frac{abc}{2S}$;

$$R = \frac{abc}{4S}.$$

73. Assume that the angle between the given sides is C , and then $S = \frac{1}{2} ab \sin C$. Since $\sin C$ assumes the greatest value when the angle C is a right one ($\sin 90^\circ = 1$), the area is the greatest when the given sides are mutually perpendicular.

74. Assume that two sides of the triangle are a and b , with $a > b$, and h_a and h_b are the corresponding altitudes. We must prove that $h_a < h_b$. We have $S = \frac{1}{2} ah_a = \frac{1}{2} bh_b$, $ah_a = bh_b$, whence $h_a/h_b = b/a$; since $a > b$, it follows that $b/a < 1$, and therefore $h_a/h_b < 1$, whence $h_a < h_b$.

75. Through the midpoint K of the side AB (Fig. 235) we draw

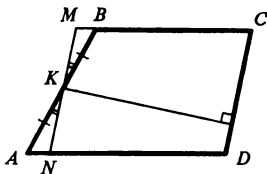


Fig. 235

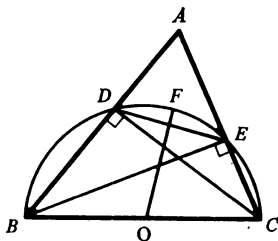


Fig. 236

a straight line parallel to the lateral side CD and assume that N and M are the points of its intersection with AD and with the extension of the side BC . The triangles AKN and KBM are congruent (a side and two adjacent angles are congruent), and therefore the areas of these triangles are equal. The area of the trapezoid $ABCD$ is equal to that of the parallelogram $NMCD$ (they consist of the same pentagon $NKBCD$ and the triangles AKN and KMB of the same area). The base CD of this parallelogram is equal to a , the altitude KL to b , and its area S is equal to ab , hence the area of the trapezoid is ab .

76. Assume that in the triangle ABC (Fig. 236) BE and CD are altitudes and F is the midpoint of the segment DE ; by the hypothesis, $[FO] \perp [DE]$ ($O \in [BC]$). We must prove that $|BO| = |OC|$. The side BC is the diameter of the circle drawn through the points D and E which are the feet of the altitudes CD and BE (the vertices of the right triangles with the hypotenuse BC lie on the circumference of a circle with diameter BC). The segment DE is a chord of this circle, and therefore FO , which is a perpendicular

drawn from the midpoint of DE , passes through the centre of the circle, i.e. through the point O , which is the midpoint of the side BC .

77. 1. We draw a median CO from the vertex of the right angle (Fig. 237), and then $|CO| = |AO| = \frac{1}{2}c$ and $\widehat{ACO} = \widehat{A} = 15^\circ$.

By the property of an exterior angle of a triangle, $\widehat{COB} = 30^\circ$.

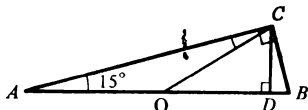


Fig. 237

Assume that CD is an altitude drawn from the vertex of the right angle, and then in the triangles COD we have $|CD| = \frac{1}{2}|CO| =$

$$\frac{1}{4}c; S = \frac{1}{2}|AB| \cdot |CD| = \frac{1}{2}c \cdot \frac{1}{4}c = \frac{1}{8}c^2.$$

2. This follows from the product of the legs of a triangle being twice the area of this triangle. On the other hand, we have proved that the doubled area of the triangle is $2 \cdot \frac{1}{8}c^2 = \left(\frac{1}{2}c\right)^2$.

78. By the hypothesis, $\angle AMB \cong \angle AMD$ (Fig. 238), but $\angle AMB \cong \angle MAD$ (being interior alternate angles at parallel lines), and therefore, $\angle MAD \cong \angle AMD$ and $|MD| = |AD| = \frac{1}{2}|CD|$. Thus in the right triangle MCD we have $|CD| = \frac{1}{2}|MD|$, and therefore $\widehat{CMD} = 30^\circ$, $\widehat{AMB} = \widehat{AMD} = (180^\circ - 30^\circ) \div 2 = 75^\circ$.

79. The areas of the triangles ABD and ACD are equal since they have a base AD in common and equal altitudes (Fig. 239),

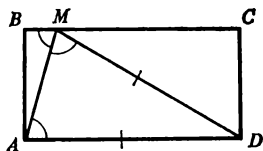


Fig. 238

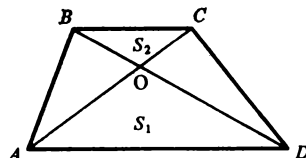


Fig. 239

i.e. $S_{ABD} = S_{ACD}$, and so we have $S_{ABD} - S_{AOD} = S_{ACD} - S_{AOD}$ or $S_{AOB} = S_{COD}$; $\frac{S_{BOC}}{S_{COD}} = \frac{|BO|}{|OD|}$, $\frac{S_{AOB}}{S_{AOD}} = \frac{|BO|}{|OD|}$ (these triangles have an altitude in common). Hence $\frac{S_{BOC}}{S_{COD}} = \frac{S_{AOB}}{S_{AOD}}$;

as we have proved, the means of this proportion are equal, and therefore

$$S_{AOB}^2 = S_{BOC} S_{AOD} \quad \text{or} \quad S_{AOB}^2 = S_1 S_2, \quad S_{AOB} = \sqrt{S_1 S_2}.$$

$$\begin{aligned} S_{ABCD} &= S_{AOD} + S_{BOC} + 2S_{AOB} = S_1 + S_2 + 2\sqrt{S_1 S_2} \\ &= (\sqrt{S_1})^2 + (\sqrt{S_2})^2 + 2\sqrt{S_1 S_2} (\sqrt{S_1} + \sqrt{S_2})^2. \end{aligned}$$

80. The right triangles AEC and BDC (Fig. 240) are similar since they have an acute angle C in common, and therefore the corresponding sides of these triangles are proportional, i.e.

$$\frac{|BC|}{|AC|} = \frac{|DC|}{|CE|}.$$

In the triangles ABC and DCE the sides which include the angle C are proportional, and therefore $\triangle ABC \sim \triangle DCE$ (SAS criterion). In similar triangles congruent angles

lie opposite the corresponding sides, and therefore $\widehat{EDC} = \widehat{BAC} =$

α and $\widehat{DEC} = \widehat{ABC} = \beta$.

81. Assume that the length of the diagonal is d , and then $S_{sq} = \frac{1}{2} d^2$. The diagonals divide the rectangle into four triangles of the same area. If we designate the angle between the diagonals as α , then the area of one of these triangles is $\frac{1}{2} \cdot \frac{d}{2} \cdot \frac{d}{2} \cdot \sin \alpha = \frac{1}{8} d^2 \sin \alpha$. We express the area of the whole rectangle as

$$S_{rec} = \left(\frac{1}{8} d^2 \sin \alpha \right) \cdot 4 = \frac{1}{2} d^2 \sin \alpha.$$

If this rectangle is not a square, then its area is smaller than that of the square since then $0 < \sin \alpha < 1$ and $S_{sq} > S_{rec}$.

82. Assume that in the isosceles triangle ABC (Fig. 241), where

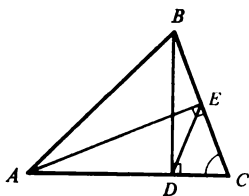


Fig. 240

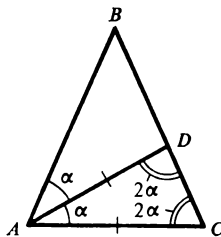


Fig. 241

$|AB| = |BC|$, the bisector AD cuts off a triangle ACD which is similar to the triangle ABC . This triangle is also isosceles since $|AD| = |AC|$ in it (otherwise the corresponding sides would

not be proportional). We designate $\widehat{DAC}$ as α , and then $\widehat{BAC} = \widehat{C} = \widehat{ADC} = 2\alpha$. Taking into account that the sum of the angles

of the triangle ADC is 180° , we get an equation $\alpha + 2\alpha + 2\alpha = 180^\circ$, whence $\alpha = 36^\circ$; $\hat{C} = \widehat{BAC} = 36^\circ \cdot 2 = 72^\circ$, $\hat{B} = 36^\circ$.

83. If $\hat{C} = 90^\circ$, then $m_c = \frac{1}{2}c$; if $\hat{C} < 90^\circ$, then $m_c > \frac{1}{2}c$; if $\hat{C} > 90^\circ$, then $m_c < \frac{1}{2}c$. *Hint.* We can verify this if we construct

a circle on the side AB as a diameter. In the first case the vertex belongs to the circle (Fig. 242), in the second case it lies outside of the circle, and in the third case it lies inside the circle.

84. 1. $a = 2\sqrt{r^2 - d^2}$, $b = 2\sqrt{r^2 - 4d^2}$ (Fig. 243).

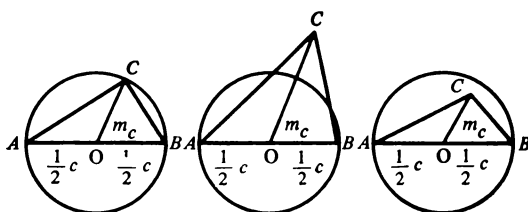


Fig. 242

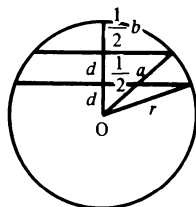


Fig. 243

2. For the inverse proportionality it is necessary that the ratio of the lengths of the chords be equal to the inverse ratio of their distances from the centre, i.e. $a/b = 2$, or

$$\frac{2\sqrt{r^2 - d^2}}{2\sqrt{r^2 - 4d^2}} = 2, \quad \frac{r^2 - d^2}{r^2 - 4d^2} = 4; \quad \begin{cases} r^2 - d^2 = 4r^2 - 16d^2, \\ r \neq 2d, \end{cases}$$

$5d^2 = r^2$, $d^2 = r^2/5$, $d = r/\sqrt{5}$. Thus, if the distance to the larger chord is $r/\sqrt{5}$ and to the smaller one $2r/\sqrt{5}$, then the lengths of the chords are in inverse proportion to their distances from the centre.

85. Assume that a is the smallest side of the triangle and d is the common difference of the progression, and then the sides of the triangle are equal to a , $a + d$, $a + 2d$, using the Pythagoras theorem, we get $a^2 + (a + d)^2 = (a + 2d)^2$, whence $a = 3d$ i.e. the legs are $3d$, $4d$ and the hypotenuse is $5d$.

The radius of the circle inscribed in this triangle can be found from the formula $r = S/p$ (see p. 266), where S is the area of the triangle and p is its half-perimeter: $S = \frac{1}{2} \cdot 3d \cdot 4d = 6d^2$, $p = (3d +$

$4d + 5d) \div 2 = 6d$. Answer: $r = \frac{6d^2}{6d} = d$.

86. The area of the triangle $S = \frac{ah_a}{2}$, where a is a side of the triangle and h_a is the altitude drawn to that side. It follows from relation that $\frac{1}{2}h_a = S/a$. On the other hand, the radius of the circle

inscribed in this triangle $r = S/p$ (see p. 266). We compare the fractions S/a and S/p . Since half the perimeter $p > a$, it follows that $S/p < S/a$, or $r < \frac{1}{2} h_a$. We can prove by analogy that $r < \frac{1}{2} h_b$ and $r < \frac{1}{2} h_c$.

87. According to the Pythagoras theorem

$$a^2 + b^2 = c^2, \quad (1)$$

and, in addition, in a right triangle

$$2ab = 2ch_c, \quad (2)$$

since each side of this equation is four times the area of the triangle. It follows from (1) and (2) that $a^2 + b^2 + 2ab = c^2 + 2ch_c$, or $(a + b)^2 = c^2 + 2ch_c + h_c^2 - h_c^2$; $(a + b)^2 = (c + h_c)^2 - h_c^2$; hence $(a + b)^2 < (c + h_c)^2$ and $a + b < c + h_c$.

88. According to the cosine law, in this triangle $c^2 = a^2 + b^2 - 2ab \cos \hat{C}$. We multiply both sides of this equation by c , and then $c^3 = a^2c + b^2c - 2abc \cos \hat{C}$. Using the hypothesis, we replace c^3 by the expression $a^3 + b^3$ equal to it and obtain

$$a^3 + b^3 = a^2c + b^2c - 2abc \cos \hat{C}.$$

We express $\cos \hat{C}$ in terms of a , b and c , i.e.

$$2ab \cos \hat{C} = a^2c + b^2c - (a^3 + b^3),$$

$$\cos \hat{C} = \frac{a^2(c-a) + b^2(c-b)}{2abc}.$$

Since $c^3 = a^3 + b^3$, we have $c > a$, $c > b$ and $c - a > 0$, $c - b > 0$, whence $\cos \hat{C} > 0$ and therefore the angle C is acute.

89. Assume that $|MB| = 2$ cm and $|MC| = 11$ cm (Fig. 244).

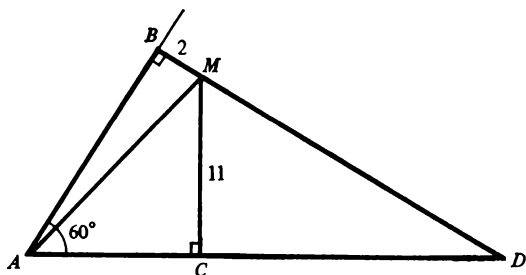


Fig. 244

We extend the segment BM to the point D of its intersection with the side AD of the given angle. In the right triangle MCD we have

$\widehat{MDC} = 30^\circ$, and consequently $|MD| = 22$ cm and $|BD| = 24$ cm. From the triangle ABD we have $|AB| = |BD| \tan 30^\circ =$

$24 \frac{\sqrt[3]{3}}{3} = 8\sqrt[3]{3}$ (cm). Using the Pythagoras theorem, we have from the triangle ABM that $|AM| = \sqrt{|BM|^2 + |AB|^2} = \sqrt{2^2 + (8\sqrt[3]{3})^2} = 14$ (cm).

90. Assume that BM and BN are altitudes of the rhombus $ABCD$ (Fig. 245). The triangles ABM and BNC are congruent, and there-

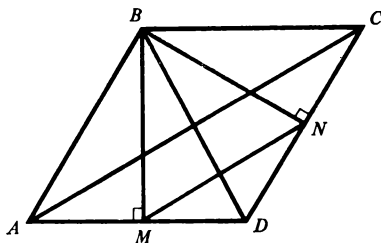


Fig. 245

fore $|AM| = |CN|$ and $|MD| = |DN|$. In addition, in the isosceles triangle ACD we have $|MN| = \frac{1}{2}|AC|$ (by the hypothesis), and therefore MN is the median of this triangle and N is the midpoint of CD . Consequently, BN is a median and an altitude of the triangle BCD and therefore $|BD| = |BC| = |CD|$. This means that the triangle BCD is equilateral, its angles are 60° each and therefore the angles of the rhombus are 60° and 120° .

91. Assume that $ABCD$ is the given trapezoid (Fig. 246). We

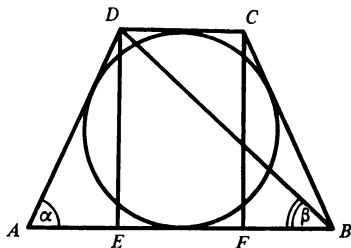


Fig. 246

construct E and F , which are the projections of the vertices D and C onto the larger base AB . By the property of the sides of a circumscribed quadrilateral $|AD| + |BC| = |AB| + |DC|$. Since $|AD| + |BC|$, it follows that

$$2|AD| = |AB| + |DC|. \quad (1)$$

On the other hand, $|BC| = |DC| + |FB|$, $2|BC| = 2|DC| + 2|FB|$, or $2|BC| = |DC| + |DC| + 2|FB|$, whence

$$2|BC| = |DC| + |AB|. \quad (2)$$

From (1) and (2) we have $|AD| = |BE|$. From the triangle ADE we have $\sin \alpha = \frac{|DE|}{|AD|}$. From the triangle DEB we have $\tan \beta = \frac{|DE|}{|BE|}$, but $|AD| = |BE|$, and consequently $\sin \alpha = \tan \beta$.

Reference Material

BASIC IDENTITIES AND FORMULAS

I. Laws of Arithmetic

- $a + b = b + a$ (commutative law of addition);
- $(a + b) + c = a + (b + c)$ (associative law of addition);
- $ab = ba$ (commutative law of multiplication);
- $(ab)c = a(bc)$ (associative law of multiplication);
- $(a + b)c = ac + bc$ (distributive law of multiplication).

II. Special Cases

- $a + 0 = 0 + a = a$ (addition of zero);
 - $a - 0 = a$ (subtraction of zero);
 - $0 - a = -a$ (subtraction from zero);
 - $a \cdot 1 = 1 \cdot a = a$ (multiplication by unity);
 - $a \div 1 = a$ (division by unity);
 - $a \div a = 1$ (division of equal numbers);
 - $1 \div a = 1/a$ (division of unity);
 - $a \cdot 0 = 0 \cdot a = 0$ (multiplication by zero);
 - $0 \div a = 0$ ($a \neq 0$) (division of zero).
- We cannot divide by zero.*

III. Relationship between Given Numbers and the Results of Operations on Them

1. If $a + b = c$, then $a = c - b$; $b = c - a$ (finding the unknown term).
2. If $a - b = c$, then $a = b + c$; $b = a - c$ (finding the unknown minuend and subtrahend).
3. If $ab = c$, then $a = c \div b$, $b = c \div a$ (finding the unknown factor).
4. If $a \div b = c$, then $a = bc$, $b = a \div c$ (finding the unknown minuend and subtrahend).

IV. Per cent

1. $p\%$ of the number a constitute $\frac{ap}{100}$ (finding the per cent of a given number).

2. If $p\%$ of the unknown number x is a , then $x = \frac{a \cdot 100}{p}$ (finding a number from its given percentage).

3. The per cent ratio of a and b is $\frac{a}{b} \cdot 100\%$ (finding the per cent ratio of two numbers).

V. Proportion

1. If $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$ (the principal property of a proportion).

2. If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$, $\frac{d}{b} = \frac{c}{a}$, $\frac{d}{c} = \frac{b}{a}$ (transposition of the terms of a proportion).

VI. Absolute Value of a Number

$$|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0, \end{cases} \quad |x - a| = \begin{cases} x - a, & \text{if } x \geq a, \\ a - x, & \text{if } x < a \end{cases}$$

(finding the absolute value of a number).

VII. Fractions

$$\frac{a}{b} = \frac{a \cdot m}{b \cdot m}, \quad m \neq 0 \quad (\text{the principal property of a fraction});$$

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}, \quad \frac{a}{b} - \frac{c}{b} = \frac{a-c}{b} \quad (\text{addition and subtraction of fractions having the same denominator});$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \quad \frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd} \quad (\text{addition and subtraction of fractions having different denominators});$$

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \quad (\text{multiplication of fractions});$$

$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc} \quad (\text{division of fractions});$$

$$\frac{a}{b} = 0 \quad \text{if } a = 0 \text{ and } b \neq 0 \quad (\text{condition of equality of a fraction to zero}).$$

VIII. Identities of Abridged Multiplication

$$(a-b)(a+b) = a^2 - b^2 \quad (\text{the product of the difference of two expressions by their sum});$$

$$(a+b)^2 = a^2 + 2ab + b^2, \quad (a-b)^2 = a^2 - 2ab + b^2 \quad (\text{the square of a binomial});$$

$$\begin{aligned}
(a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3, (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \\
&\quad \text{(the cube of a binomial);} \\
a^2 - b^2 &= (a-b)(a+b) \text{ (the difference of squares of numbers);} \\
a^3 - b^3 &= (a-b)(a^2 + ab + b^2); \text{ (the difference and the sum} \\
a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \text{ of the cubes of numbers).}
\end{aligned}$$

IX. Powers with Rational Exponents

$$\begin{aligned}
a^n &= a \cdot a \dots a, \quad n \in \mathbb{N}, \quad n \neq 1 \text{ (finding a power with a natural} \\
&\quad \text{exponent);} \\
a^1 &= a \text{ (finding a power with exponent 1);} \\
a^0 &= 1, \quad a \neq 0 \text{ (finding a power with exponent zero);} \\
a^{-n} &= \frac{1}{a^n}, \quad a \neq 0, \quad n > 0 \text{ (finding a power with a negative} \\
&\quad \text{exponent);} \\
a^{m/n} &= \sqrt[n]{a^m}, \quad a > 0, \quad m \in \mathbb{Z}, \quad n \in \mathbb{N} \text{ (finding a power with a frac-} \\
&\quad \text{tional exponent);} \\
a^m \cdot a^n &= a^{m+n} \text{ (multiplication of powers with the same base);} \\
a^m \div a^n &= a^{m-n}, \quad a \neq 0 \text{ (division of powers with the same base);} \\
(ab)^n &= a^n \cdot b^n \text{ (the power of a product);} \\
\left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n} \text{ if } b \neq 0 \text{ (the power of a fraction);} \\
(a^m)^n &= a^{mn} \text{ (the power of a power).}
\end{aligned}$$

X. Properties of Arithmetic Square Roots

$$\begin{aligned}
\text{If } \sqrt{a} &= b, \text{ then } a = b^2, \text{ if } a \geq 0 \text{ and } b \geq 0 \text{ (finding an arithme-} \\
&\quad \text{tic square root);} \\
\sqrt{a^2} &= |a| \text{ (a square root of a complete square);} \\
(\sqrt{a})^2 &= a \text{ if } a \geq 0 \text{ (the basic property of a square root);} \\
\sqrt{ab} &= \sqrt{a} \sqrt{b} \text{ if } a \geq 0 \text{ and } b \geq 0 \text{ (extracting a root of a prod-} \\
&\quad \text{uct);} \\
\sqrt{\frac{a}{b}} &= \frac{\sqrt{a}}{\sqrt{b}} \text{ if } a \geq 0 \text{ and } b > 0 \text{ (extracting a root of a frac-} \\
&\quad \text{tion).}
\end{aligned}$$

XI. The Arithmetic Mean and the Geometric Mean

$$\begin{aligned}
\frac{a+b}{2} &\text{ (finding the arithmetic mean of the numbers } a \text{ and } b\text{);} \\
\sqrt{ab}, \quad a > 0, \quad b > 0 &\text{ (finding the geometric mean of the numbers} \\
&\quad a \text{ and } b\text{);} \\
\frac{a+b}{2} &\geq \sqrt{ab} \text{ if } a > 0 \text{ and } b > 0 \text{ (comparing an arithmetic mean} \\
&\quad \text{and a geometric mean).}
\end{aligned}$$

XII. Quadratic Equations

$ax^2 + bx + c = 0$, $a \neq 0$ (quadratic equation of a general form);

$D = b^2 - 4ac$ (the discriminant of a quadratic equation);

$x = \frac{-b \pm \sqrt{D}}{2a}$ (formula for the roots of a quadratic equation).

If $D > 0$, then $X = \left\{ \frac{-b - \sqrt{D}}{2a}, \frac{-b + \sqrt{D}}{2a} \right\}$ (two roots).

If $D = 0$, then $X = \left\{ -\frac{b}{2a} \right\}$ (one root).

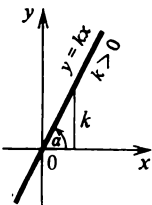
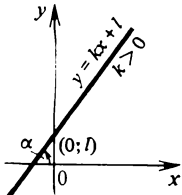
If $D < 0$, then $X = \emptyset$ (the equation has no roots).

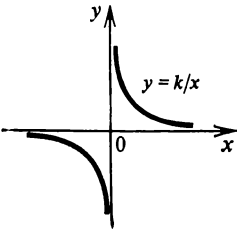
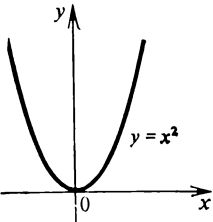
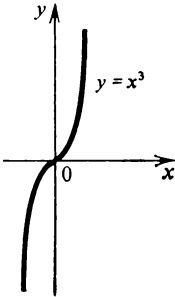
If $D > 0$, then $x_1 + x_2 = -\frac{b}{a}$, $x_1 \cdot x_2 = \frac{c}{a}$ (the Vieta theorem).

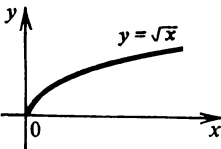
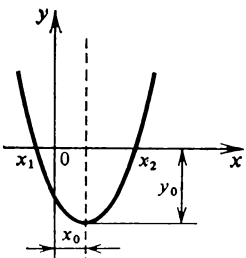
If $D \geq 0$, then $ax^2 + bx + c = a(x - x_1)(x - x_2)$ (expansion of a quadratic trinomial).

XIII. Elementary Functions and Their Graphs

(X is the domain of a function, Y is the range of a function).

Function defined by a formula	Graph of a function	Definition
1. $y = kx$, $k \neq 0$ (direct proportionality) $X = (-\infty, +\infty)$ $Y = (-\infty, +\infty)$		A straight line passing through the origin. The proportionality factor $k = \tan \alpha$
2. $y = kx + l$ (a linear function) $X = (-\infty, +\infty)$ $Y = (-\infty, +\infty)$		A straight line passing through the point $(0, l)$ not parallel to the y-axis

Function defined by a formula	Graph of a function	Definition
3. $y = \frac{k}{x}$, $k \neq 0$ (inverse proportionality) $X = (-\infty, 0) \cup (0, +\infty)$ $Y = (-\infty, 0) \cup (0, +\infty)$		Hyperbola
4. $y = x^n$ (a power function) (1) $y = x^2$ for $n=2$ $X = (-\infty, +\infty)$ $Y = [0, +\infty]$		Parabola with vertex at the origin
(2) $y = x^3$ for $n=3$ $X = (-\infty, +\infty)$ $Y = (-\infty, +\infty)$		Cubic parabola

Function defined by a formula	Graph of a function	Definition
(3) $y = \sqrt[n]{x}$ for $n = \frac{1}{2}$ $X = [0, +\infty)$ $Y = [0, +\infty)$		Half a parabola
$y = ax^2 + bx + c$, $a \neq 0$ a quadratic function) $X = (-\infty, +\infty)$ $Y = \left[-\frac{D}{4a}, +\infty\right)$ if $a > 0$		A parabola with vertex at the point (x_0, y_0), where $x_0 = \frac{x_1 + x_2}{2} = -\frac{b}{2a},$ $y_0 = -\frac{D}{4a},$ x_1 and x_2 are the roots of the function

XIV. Arithmetic Progression

$a_n = a_1 + d(n-1)$ (formula for the n th term);

$S_n = \frac{a_1 + a_n}{2} \cdot n$, $S_n = \frac{2a_1 + d(n-1)}{2} \cdot n$ (formulas for the sum of the first n terms).

XV. Geometric Progression

$b_n = b_1 q^{n-1}$, $b_1 \neq 0$, $q \neq 0$ (formula for the n th term);

$S_n = \frac{b_n q - b_1}{q - 1}$, $S_n = \frac{b_1 (q^n - 1)}{q - 1}$, $q \neq 1$ (formulas for the sum of the first n terms).

XVI. Logarithms

The equation $\log a = x$ means that $10^x = a$ (definition of a logarithm).

$10^{\log a} = a$ (the basic logarithmic identity).

$\log_{10} b = \log b$ (decimal logarithm of the number b).

$\log(ab) = \log a + \log b$, $a > 0$, $b > 0$ (logarithms of a product).

$\log \frac{a}{b} = \log a - \log b$, $a > 0$, $b > 0$ (logarithm of a quotient).

$\log a^n = n \log a$, $a > 0$ (logarithm of a power).

Powers of the Numbers 2, 3 and 5

$2^0 = 1$	$3^0 = 1$	$5^0 = 1$
$2^1 = 2$	$3^1 = 3$	$5^1 = 5$
$2^2 = 4$	$3^2 = 9$	$5^2 = 25$
$2^3 = 8$	$3^3 = 27$	$5^3 = 125$
$2^4 = 16$	$3^4 = 81$	$5^4 = 625$
$2^5 = 32$	$3^5 = 243$	$5^5 = 3025$
$2^6 = 64$		
$2^7 = 128$		
$2^8 = 256$		
$2^9 = 512$		
$2^{10} = 1024$		

BASIC DEFINITIONS, THEOREMS AND FORMULAS OF GEOMETRY

Definitions and Theorems

1. A *line segment* is a set consisting of two different points and all the points between them.

2. A *broken (polygonal) line* is a union of segments such that the end of each segment (except for the last one) is the beginning of the next segment and the adjacent segments do not lie on the same straight line.

3. A broken line is longer than the distance between its ends.

4. A broken line is said to be *closed* if the end of its last segment is the beginning of the first.

5. A *polygon* is the union of a closed broken line and its interior.

6. An *angle* is a figure formed by two different rays with a common origin and a part of the plane bounded by them.

7. A *bisector of an angle* is a ray which begins at the vertex of the angle and divides the angle in two.

8. *Adjacent angles* are two angles whose union is a straight angle and whose intersection is a ray. *The sum of adjacent angles is 180° .*

9. *Vertical angles* are two angles which are symmetric about their common vertex, or two angles (smaller than a straight angle) the sides of one of which are rays lying opposite the sides of the other angle. *Vertical angles are congruent.*

10. Two straight lines which belong to the same plane are said to be *parallel* if they have no points in common or coincide.

11. Not more than one straight line parallel to the given straight line can be drawn through a given point of a plane (axiom of parallelism).

12. Not more than one straight line can be drawn through a given point of the plane at right angles to a given straight line.

13. If two straight lines are perpendicular to the same straight line, then they are parallel.

14. Centro-symmetric straight lines are parallel.

15. If some two corresponding angles (or two alternate angles) formed by the intersection of two straight lines by a third straight line are congruent, then these straight lines are parallel.

16. Segments of two parallel straight lines included between two parallel straight lines are congruent.

17. The set of points of a plane which are equidistant from the endpoints of a segment is the *perpendicular bisector* of that segment.

18. The set of points of an angle which are equidistant from its sides is the *bisector* of that angle.

19. A *triangle* is a polygon which has three sides.

20. A *bisector of a triangle* is a segment of the bisector of an angle of the triangle drawn from its vertex to the opposite side.

21. A *median of a triangle* is a segment which connects a vertex of the triangle with the midpoint of the opposite side.

22. An *altitude of a triangle* is a segment of the perpendicular drawn from a vertex to the opposite side.

23. In an isosceles triangle (1) the base angles are congruent, (2) an altitude drawn from a vertex is a bisector and a median at the same time.

24. If three sides of one triangle are congruent to three sides of the other triangle, then the triangles are congruent.

25. If two sides and the angle between them of one triangle are congruent to two sides and the angle between them of the other triangle, then the triangles are congruent.

26. If a side and two adjacent angles of one triangle are congruent to a side and two adjacent angles of the other triangle, then the triangles are congruent.

27. The sum of the angles of a triangle is 180° .

28. Each angle of an equilateral triangle is 60° .

29. The leg of a right triangle opposite an angle of 30° is equal to a half of the hypotenuse.

30. An angle adjacent to an interior angle of a triangle is known as its exterior angle.

31. An exterior angle of a triangle is equal to the sum of two angles of the triangle which are not adjacent to it.

32. An exterior angle of a triangle is larger than each of the angles of the triangle which are not adjacent to it.

33. The sum of the interior angles of a convex n -gon is $180^\circ (n - 2)$.

34. The sum of the exterior angles of a convex polygon, taken one at each vertex, is 360° .

35. A larger angle lies opposite the larger side of a triangle.

36. A larger side lies opposite the larger angle of a triangle.

37. Equal angles lie opposite equal sides of a triangle, and conversely, equal sides lie opposite equal angles of a triangle.

38. If we lay off several congruent segments along one side of an angle and draw through them parallel straight lines which cut the other side of the angle, then they will intercept congruent segments on it.

39. A *median of a triangle* is a segment which connects the mid-points of two sides of the triangle.

40. A median of a triangle is parallel to the third side and its length is equal to a half of the length of that side.

41. Parallel straight lines which cut the sides of an angle intercept proportional segments on them.

42. If three sides of one triangle are proportional to three sides of the other, then the triangles are similar.

43. If two sides of one triangle are proportional to two sides of the other triangle and the angles between these sides are congruent, then the triangles are similar.

44. If two angles of one triangle are congruent to two sides of the other, then the triangles are similar.

45. The square of the hypotenuse of a right triangle is equal to the sum of the squares of its legs.

46. The square of a side of a triangle is equal to the sum of the squares of the other two sides minus the doubled product of those sides by the cosine of the angle between them.

47. The sides of a triangle are proportional to the sines of the opposite angles.

48. A *circumference* is a set of all points of a plane which are at a given distance from a given point lying in that plane.

49. A *circle* is a set of all points of a plane the distance from each of which to a fixed point of the plane does not exceed a given distance.

50. If two chords of a circle are congruent, then the arcs which they subtend are also congruent.

51. Chords of equal lengths are equidistant from the centre of a circle, and conversely, chords which are equidistant from the centre are equal in length.

52. Of two noncongruent chords of a circle the longer one lies closer to the centre.

53. The set of all the vertices of right triangles with a specified hypotenuse is a circle whose diameter is that hypotenuse.

54. An inscribed angle is equal to a half of the angular measure of the arc which subtends it.

55. The straight line which is perpendicular to the diameter of a circle and passes through its endpoint is a tangent to that circle.

56. A tangent to a circle is perpendicular to the diameter which passes through the point of tangency.

57. A circle can be circumscribed about any triangle and that circle is unique. The centre of the circle is the point of intersection of the perpendicular bisectors of the sides of the triangle.

58. A circle can be inscribed in any triangle and that circle is unique. The centre of the circle is the point of intersection of the bisectors of the triangle.

59. The area of a triangle is equal to a half of the product of the base by the altitude.

60. The area of a triangle is equal to a half of the product of its two sides by the sine of the angle between them.

61. A *parallelogram* is a quadrilateral whose opposite sides are pairwise parallel.
62. The midpoint of a diagonal of a parallelogram is its centre of symmetry.
63. The opposite sides of a parallelogram are pairwise congruent.
64. The opposite angles of a parallelogram are pairwise congruent.
65. The sum of the angles which are adjacent to one side of a parallelogram is 180° .
66. If the opposite sides of a quadrilateral are pairwise congruent, then the quadrilateral is a parallelogram.
67. If two opposite sides of a quadrilateral are congruent and parallel, then the quadrilateral is a parallelogram.
68. The area of a parallelogram is equal to the product of its base by the altitude.
69. A *rectangle* is a parallelogram whose all angles are right angles.
70. The perpendicular bisector of a side of a rectangle is its *axis of symmetry*.
71. The diagonals of a rectangle are congruent.
72. A *rhombus* is a parallelogram whose all sides are congruent.
73. The straight line which contains a diagonal of a rhombus is its axis of symmetry.
74. The diagonals of a rhombus are mutually perpendicular and divide the angles of the rhombus in two.
75. A *square* is a rectangle whose all sides are congruent.
76. A *trapezoid* is a quadrilateral whose two sides are parallel and the other two are nonparallel.
77. The *median of a trapezoid* is a segment which connects the midpoints of the nonparallel sides of the trapezoid.
78. The median of a trapezoid is parallel to the bases and its length is equal to a half of the sum of the lengths of the bases.
79. The area of a trapezoid is equal to the product of a half of the sum of its bases by the altitude.
80. The ratio of the perimeters of similar polygons is equal to their ratio of similitude.
81. The ratio of the areas of similar polygons is equal to the square of the ratio of similitude.
82. The sum of the opposite angles of an inscribed quadrilateral is 180° .
83. If the sum of two opposite angles of a quadrilateral is 180° , then a circle can be circumscribed about that quadrilateral.
84. The sums of the lengths of the opposite sides of a circumscribed quadrilateral are equal.
85. If the sums of the lengths of the opposite sides of a quadrilateral are equal, then a circle can be inscribed in that quadrilateral.
86. A *regular polygon* is a polygon whose all sides are congruent and all the angles are congruent.
87. A circle can be inscribed in any regular polygon.
88. The area of a regular polygon is equal to a half of the product of its perimeter by the radius of the inscribed circle.

Geometry Formulas

(S is the area everywhere)

Name of a formula	Formula	Elucidation of formulas
Triangle inequality	$a < b + c$	a, b, c are lengths of sides of the triangle
The sum of the angles of a triangle	$\hat{A} + \hat{B} + \hat{C} = 180^\circ$	$\hat{A}, \hat{B}, \hat{C}$ are the values of the angles of the triangle
The area of a triangle	$S = \frac{1}{2} ah$ $S = \frac{1}{2} ab \sin \gamma$	a is the length of the base, h is the altitude drawn to that base a, b are the lengths of sides, γ is the angle between the sides a and b
The area of a right triangle	$S = \frac{1}{2} ab$ $S = \frac{1}{2} ch_c$	a, b are the lengths of the legs, c is the hypotenuse h_c is the altitude drawn to the hypotenuse
The Pythagoras theorem	$c^2 = a^2 + b^2$	c is the length of the hypotenuse, a, b are the lengths of legs
The radius of the circle circumscribed about a right triangle	$R = \frac{1}{2} c$	R is the radius of the circumscribed circle, c is the length of the hypotenuse
The area of an equilateral triangle	$S = \frac{1}{4} a^2 \sqrt{3}$	a is the length of a side
The altitude of an equilateral triangle	$h = \frac{1}{2} a \sqrt{3}$	h is the altitude, a is the length of a side

Name of a formula	Formula	Elucidation of formulas
The altitude of a right triangle	$h_c = \frac{ab}{c}$	a, b are the lengths of the legs, c is the length of the hypotenuse, h_c is the altitude of the triangle
The radius of the circle circumscribed about an equilateral triangle	$R = 2r = \frac{a}{3} \sqrt{3}$	R is the radius of the circumscribed circle, r is the radius of the inscribed circle, a is the length of a side
The radius of the circle circumscribed about a triangle	$R = \frac{a}{2 \sin \alpha}$	R is the radius of the circumscribed circle, a is the length of a side, α is the angle opposite a
The radius of the circle inscribed into a triangle	$r = \frac{S}{p}$	S is the area, p is half the perimeter
The cosine law	$c^2 = a^2 + b^2 - 2ab \cos \gamma$	a, b, c are the lengths of sides of the triangle, γ is the angle opposite the side c
The sine law	$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$	a, b, c are the lengths of sides of the triangle, α, β, γ are the values of the opposite angles
The area of a square	$S = a^2$	a is the length of a side
The area of a rectangle	$S = ab$	a, b are the lengths of sides

Name of a formula	Formula	Elucidation of formulas
The area of a parallelogram	$S = ah$ $S = ab \sin \alpha$	a is the length of a side, h is the altitude to that side a, b are the lengths of sides, α is the value of the angle between the sides
The area of a rhombus	$S = ah$ $S = \frac{1}{2} d_1 d_2$	a is the length of a side, h is the altitude d_1 and d_2 are the lengths of diagonals
The area of a trapezoid	$S = \frac{1}{2} (a + b) h$ $S = ch$	a, b are the lengths of sides, h is the altitude c is the length of the median
The sum of the angles of a polygon	$S_n = 180^\circ (n - 2)$	S_n is the sum of the angles, n is the number of sides of the polygon
The area of a regular polygon	$S = \frac{1}{2} Pr$	P is the perimeter, r is the radius of the circle inscribed in the polygon (apothem)
The sides of regular polygons	$a_n = 2R \sin \frac{180^\circ}{n}$ $a_3 = R \sqrt{3},$ $a_4 = R \sqrt{2},$ $a_6 = R$	a_n is the length of a side of a regular inscribed n-gon, R is the radius of the circumscribed circle a_3, a_4, a_6 are the lengths of sides of the regular inscribed triangle, quadrilateral, and hexagon, respectively
The equation of a circle with centre at the origin	$x^2 + y^2 = R^2$	R is the radius of the circle

Name of a formula	Formula	Elucidation of formulas
The length of circle	$C = 2\pi R$ $C = \pi D$	C is the length of the circle, R is the radius D is the diameter, $\pi \approx 3.14$
The length of the arc of a circle	$l = \frac{\pi R \alpha}{180}$	l is the length of the arc, R is the radius, α is the angle in degrees
The area of a circle	$S = \pi R^2$ $S = \frac{\pi D^2}{4}$	R is the radius, D is the diameter
The area of a sector	$S = \frac{\pi R^2 \alpha}{360}$	R is the radius, α is the arc in degrees

TRIGONOMETRIC FORMULAS

Relationship Between the Sides and the Angles in a Right Triangle

(a and b are the lengths of the legs, c is the length of the hypotenuse, $\hat{A}$ is the angle lying opposite a , $\hat{B}$ is the angle lying opposite b)

$$\sin \hat{A} = \frac{a}{c}, \quad a = c \cdot \sin \hat{A}; \quad c = \frac{a}{\sin \hat{A}},$$

$$\cos \hat{A} = \frac{b}{c}, \quad b = c \cdot \cos \hat{A}; \quad c = \frac{b}{\cos \hat{A}},$$

$$\tan \hat{A} = \frac{a}{b}, \quad a = b \cdot \tan \hat{A}; \quad b = \frac{a}{\tan \hat{A}}.$$

Trigonometric Identities

$$\sin^2 \alpha + \cos^2 \alpha = 1 \quad \cos (90^\circ - \alpha) = \sin \alpha$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} \quad \sin (90^\circ + \alpha) = \cos \alpha$$

$$\sin (180^\circ - \alpha) = \sin \alpha \quad \cos (90^\circ + \alpha) = -\sin \alpha$$

$$\cos (180^\circ - \alpha) = -\cos \alpha \quad \sin (-\alpha) = -\sin \alpha$$

$$\sin (90^\circ - \alpha) = \cos \alpha \quad \cos (-\alpha) = \cos \alpha$$

Values of Trigonometric Functions of Some Angles

	0°	30°	45°	60°	90°	180°
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	—	0

Table of Square Roots

(the radicand is a complete square)

$\sqrt{0}=0$	$\sqrt{289}=17$	$\sqrt{1156}=34$
$\sqrt{1}=1$	$\sqrt{324}=18$	$\sqrt{1225}=35$
$\sqrt{4}=2$	$\sqrt{361}=19$	$\sqrt{1296}=36$
$\sqrt{9}=3$	$\sqrt{400}=20$	$\sqrt{1369}=37$
$\sqrt{16}=4$	$\sqrt{441}=21$	$\sqrt{1444}=38$
$\sqrt{25}=5$	$\sqrt{484}=22$	$\sqrt{1521}=39$
$\sqrt{36}=6$	$\sqrt{529}=23$	$\sqrt{1600}=40$
$\sqrt{49}=7$	$\sqrt{576}=24$	$\sqrt{1681}=41$
$\sqrt{64}=8$	$\sqrt{625}=25$	$\sqrt{1764}=42$
$\sqrt{81}=9$	$\sqrt{676}=26$	$\sqrt{1849}=43$
$\sqrt{100}=10$	$\sqrt{729}=27$	$\sqrt{1936}=44$
$\sqrt{121}=11$	$\sqrt{784}=28$	$\sqrt{2025}=45$
$\sqrt{144}=12$	$\sqrt{841}=29$	$\sqrt{2116}=46$
$\sqrt{169}=13$	$\sqrt{900}=30$	$\sqrt{2209}=47$
$\sqrt{196}=14$	$\sqrt{961}=31$	$\sqrt{2304}=48$
$\sqrt{225}=15$	$\sqrt{1024}=32$	$\sqrt{2401}=49$
$\sqrt{256}=16$	$\sqrt{1089}=33$	$\sqrt{2500}=50$

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